Power Markets: Technical Onboarding Notes

v0.41.0

> Technical onboarding notes for learning power markets through physical constraints, market institutions, contracts, numerical intuition, and applied decision cases.

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These notes are for a fast, practical ramp into power markets, especially around renewable portfolios. They are written for analysts, business developers, quants, modelers, and other professionals entering power-market work without much prior domain experience. The volumes move from physical and institutional basics, to commercial examples and analytical workflows, to current strategic questions. The aim is to make key terms, units, constraints, and recurring trade-offs easier to recognize. The references provide the path for deeper study and follow-up.

Download Course PDF **Download Casebook PDF** **Open Commercial Workbench**

Methodology

These notes were built with a learning method designed for entering technical domains quickly: gather strong primary sources, identify the core guiding questions, work back to first principles, and build intuition through orders of magnitude, scaling relationships, and worked decision frames.

The workflow was accelerated by a custom AI research and knowledge-management system I built for semantic retrieval, source organization, drafting, and the editorial loops.

These notes are evolving. As I learn more and receive feedback, I keep improving the material.

Feedback is very welcome. Select any passage and click 'Send feedback' to send me your input on the spot.

How To Use These Notes

- Start each part with its Q&A Overview PDF to see the main questions and the shape of the argument.
- Use Volume I for the basic physical and institutional map: energy, capacity, timing, networks, roles, balancing, and prices.
- Use Volume II to practice commercial and analytical examples: PPAs, storage, dispatch, tariffs, forecasting, valuation, contracts, and portfolio control.
- Use Volume III to get oriented in strategic debates and claims that depend on market conditions: congestion, cannibalization, storage duration, AI and data-centre load, adequacy, and market design.
- Use the Casebook when a concept feels familiar enough to apply, then compare your answer with the worked decision frame.
- Return to the Q&A Overview PDF after finishing a part to check what stuck and what still feels unclear.

Course Structure

The sequence moves from foundations to commercial examples to strategic questions, then into applied cases and reference material.

Volume I

Power System Fundamentals And Market Institutions

Electricity must balance moment by moment through a constrained network. The first volume gives the basic map: balance, frequency, wires, actors, market sequence, and the first economics of solar and storage.

- Understand why TSOs, BRPs, BSPs, bidding zones, balancing, and ancillary services exist.
- Connect MW, MWh, voltage, congestion, curtailment, and settlement to economic pressure.
- Read market explanations with the grid constraints visible.

Open Volume I **Q&A Overview PDF**

Volume II

Commercial Analysis, Contracts, And Asset Decisions

The second volume works through commercial and analytical examples: site assessments, tariffs, PPAs, BESS sizing, dispatch, forecasting, valuation, risk, and portfolio controls.

- Decompose a PPA or battery case into price, shape, imbalance, grid, credit, optionality, and control rights.
- Ask what a model is allowed to know, what it is optimizing, and which constraints are binding.
- Practice the questions that sit behind investment and operating decisions.

Open Volume II Q&A Overview PDF

Volume III

Market Design, Portfolio Strategy, And Operating Models

The third volume surveys strategic questions under uncertainty: congestion, cannibalization, duration choice, data-centre flexibility, AI, adequacy, market power, and changing rules.

- Separate dated market color from durable mechanisms.
- Use evidence packs, option values, constraints, and assumption flips as reading tools.
- Separate durable mechanisms from dated market claims.

Open Volume III

Q&A Overview PDF

Casebook

Applied Decision Memos And Models

The Casebook gives bounded exercises around quote stacks, duration tests, grid-aware site assessments, partner scorecards, customer-product tests, and operating-model roadmaps.

- Use the lectures to produce a recommendation.
- Name the binding constraint, retained risk, and assumption that would change the decision.
- Use the concepts in a constrained scenario before moving to project-specific diligence.

Open Casebook

Q&A Overview PDF

Companion

Commercial Workbench

Interactive companion for the shaped solar-storage cases.

- Use it to test delivery coverage, residual gaps, indicative premium, BESS value, and partner evidence while working through Cases 1, 4, and 5.

Open Workbench

Glossary

Glossary

An alphabetical reference for the terms, acronyms, market roles, and technical objects used throughout the notes.

- Use it for quick recall.
- Return to the relevant lecture when the term needs mechanism, economics, or context.

Open Glossary

References

References

Books, reports, official pages, and selected practitioner analyses used across the notes.

- Market figures and rule-specific details include dates where timing matters.

Open References

Volume I: Power System Fundamentals And Market Institutions

Chapter 1. Energy, Power, And Delivery Timing

References: [1] [2] [3]

A customer buys electricity, but delivery happens at a rate, in a place, and at a time. A factory needs power in this hour. A city has an evening peak. A wire can carry only so much at a moment. Power-market reasoning therefore starts by separating the amount of energy from the rate at which it arrives. Once those are separate, familiar words like generation, load, capacity, average, and peak become less slippery. A megawatt-hour (MWh) is a real amount of energy, but it is useful only when its timing, place, and reliability match the need it is supposed to serve.

Opening Puzzle

A generator reports: “We generated 1 MWh.”

Before using that sentence in a power-market decision, ask for the missing coordinates: over what interval, at which connection point, and under what obligation?

One megawatt-hour could mean:

- 1 megawatt (MW) delivered steadily for 1 hour.
- 0.25 MW delivered for 4 hours.
- 4 MW delivered for 15 minutes.
- Energy delivered in the wrong place, at the wrong hour, when nobody can use it.

The unit `MWh` gives the energy total. A market product also needs the delivery interval, location, and performance obligation.

First-Principles Model

Start with the accounting identity that every power-market model eventually uses.

Energy is accumulated over an interval. Power is the rate of transfer during that interval.

In the units used here:

```
energy = power x time
```

If power is measured in MW and time in hours, energy is measured in MWh:

```
1 MW x 1 hour = 1 MWh
```

A project with high annual energy may still fail a commercial obligation if its hourly power arrives at the wrong times. Factories, cities, and grids need power in specific intervals, not only annual totals.

Definitions

Power: the rate at which energy is produced, consumed, transferred, charged, or discharged. Typical units are watt (W), kilowatt (kW), megawatt (MW), and gigawatt (GW).

Energy: the accumulated quantity over time. Typical units are watt-hour (Wh), kilowatt-hour (kWh), megawatt-hour (MWh), gigawatt-hour (GWh), and terawatt-hour (TWh).

Megawatt-hour (MWh): one megawatt delivered or consumed for one hour. It is an energy unit.

Generation: electrical energy produced by a generator over a period, or power output at a moment. The unit tells you which meaning is intended.

Load: electrical demand. Load can be the demand of a device, building, portfolio, bidding zone, or whole system.

Peak load: the maximum load over a defined period and geography.

Capacity: the maximum output or transfer capability under stated conditions. Capacity is usually a power number, such as MW.

Capacity factor: actual energy over a period divided by the energy that would have been produced if the asset ran at full capacity for the whole period.

Numerical Intuition

These are unit anchors:

```
1 MWh = 1 MW x 1 hour
1 gigawatt-year (GW-year) = 1 GW x 8,760 hours = 8.76 terawatt-hours (T
```

These conversions are useful because electricity conversations constantly move between annual energy and hourly power.

Worked Estimate

A customer consumes `1 TWh/year`.

Average power is:

```
1,000,000 MWh/year / 8,760 hours/year ~= 114 MW
```

A flat 114 MW load would consume the same annual energy. The customer's actual load can still move far above and below that average.

If the customer's peak is 200 MW, a 114 MW average describes only one dimension of the load. If a solar project produces much of its energy at noon, annual matching does not guarantee hourly matching.

The numerical lesson: annual MWh is a total; power-market risk lives in the shape of delivery.

Mechanism

Power systems must serve demand continuously. Energy can be counted after the fact, but system operation depends on the rate of injection and withdrawal during each delivery interval.

Storage can shift electricity across time, but it is limited by power capacity, energy capacity, efficiency, state of charge, location, and cost. The system therefore cares about both annual energy and the rate at which energy is injected or withdrawn.

A market participant may therefore ask:

- How many MWh?
- In which hour?
- At which node, zone, meter, or connection point?
- With what confidence?
- With what obligation if actual delivery differs?

A complete market question cannot stop after the first item.

Example

A 10 MW generator running at full output for 3 hours produces:

```
10 MW x 3 h = 30 MWh
```

A 30 MW generator running for 1 hour also produces:

```
30 MW x 1 h = 30 MWh
```

The energy total is the same, but the power profile is different.

If the buyer needs 10 MW for three hours, the first generator naturally matches the buyer better. The second generator creates a surplus in one hour and a shortage in two other hours unless something else absorbs and redelivers the energy.

In Practice

For an independent power producer (IPP), this unit distinction changes how a product is priced and promised.

When someone proposes a solar power purchase agreement (PPA), ask both “How many MWh per year?” and “What hourly shape are we selling, and who carries the mismatch between production and customer demand?”

In Sweden, production and consumption statistics are based on measured hourly values reported to eSett. The operational record is hourly because running and settling the system requires more than annual totals.

Misconceptions

“A MWh is a MWh.”

Physically, the energy unit is the same. Commercially and operationally, a MWh at 03:00 in one place may not solve the same problem as a MWh at 18:00 somewhere else.

“Capacity is energy.”

Capacity is a rate or capability, usually MW. Energy is the accumulated result, usually MWh.

“Average power is the same as deliverable power.”

Average power is a summary. Deliverability depends on timing, location, availability, and grid constraints.

Recap

Power-market reasoning begins with rate. Energy tells you how much arrived over time; power tells you how fast it had to arrive in each hour or moment. Two assets can produce the same MWh and still solve different problems if one matches the buyer’s shape and the other creates surplus and shortage. Annual MWh is a useful total. Product definition also needs delivery rate, location, timing, reliability, and the obligation if reality differs.

Chapter 2. Real-Time Power System Balancing

References: [4] [5]

A solar portfolio is scheduled to deliver power during the next hour. Cloud cover arrives earlier than forecast, output drops, and the system has to replace the missing MW in real time. The schedule remains the commercial reference; physical balance is handled continuously. This chapter separates those two layers: the grid's immediate balancing need and the later settlement of the portfolio's deviation from its position.

Opening Puzzle

A `500 MW` solar portfolio is scheduled for the delivery hour.

Cloud cover arrives sooner than forecast and the portfolio produces `1%` less than scheduled.

What has to move in real time, and who carries the deviation after the meter data arrives?

First-Principles Model

A power grid is a shared physical system made of rotating and electronic devices. At each moment, electrical supply and electrical demand must be kept close enough to balance for the system to remain secure.

A simple accounting version is:

```
net imbalance = actual generation + imports − actual consumption − expo
```

For one portfolio, a settlement version is:

```
portfolio imbalance = actual delivery or consumption − scheduled positi
```

They answer different questions. The first is physical system balance. The second is financial responsibility for a market participant’s deviation from its position.

The market exists partly because the physical system needs someone to be responsible when plans and reality differ.

Definitions

System balance: the physical condition in which supply and demand, including imports/exports and losses depending on boundary, are close enough for secure operation.

Frequency: in an alternating-current (AC) grid, the system electrical frequency. In the Nordic and broader European context the nominal frequency is 50 hertz (Hz), but it does not use that as a control-band claim.

Imbalance: the difference between allocated physical volumes and final traded or scheduled volumes for a balance responsible party over an imbalance settlement period, using the eSett terminology developed further in Lecture 5.

Balance responsible party (BRP): party financially responsible for imbalances for a delivery point or portfolio.

TSO: transmission system operator. Transmission system operators (TSOs) organize balancing activities and maintain system balance within their mandates under the EU balancing guideline.

Balancing capacity: power capability kept available for balancing service under product rules, usually measured in MW. The asset may be paid for readiness even before energy is activated.

Balancing service provider (BSP): party qualified to provide balancing services to the transmission system operator.

Balancing energy: energy delivered or absorbed after activation for balancing, usually measured in MWh and provided by balancing service providers in the eSett terminology.

Numerical Intuition

A simple forecast-error anchor:

```
Portfolio size = 500 MW
Forecast error = 1%
Power error = 500 MW x 0.01 = 5 MW
If it lasts 1 hour: 5 MW x 1 h = 5 MWh
```

A 1% forecast miss on a 500 MW portfolio is a 5 MW real-time problem.

If the error lasts only 15 minutes:

```
5 MW x 0.25 h = 1.25 MWh
```

The power error is still 5 MW. The energy volume is smaller because the error lasts for less time.

Worked Estimate

Suppose an independent power producer (IPP) has 500 MW of weather-dependent generation in a bidding zone. The forecast misses by 1% during a cloudy hour.

The missing or excess power is about 5 MW. If the imbalance price exposure were hypothetically `EUR 100/MWh` for that hour, the one-hour financial exposure would be:

```
5 MWh x EUR 100/MWh = EUR 500
```

That number is illustrative, not a market claim. The scaling is:

- double the portfolio, double the MW error;
- double the time length of the error, double the MWh exposure;
- errors that are correlated across many assets matter more than errors that cancel inside the portfolio.

Mechanism

The grid is always moving from plan to reality.

Before delivery, market participants forecast demand, renewable output, plant availability, interconnector flows, and prices. They trade or schedule positions. As real time approaches, better information arrives. Intraday markets and operational corrections narrow the gap.

At delivery, any remaining physical gap still has to be handled. TSOs use balancing resources and operational tools to keep the system secure. Later, imbalance settlement assigns financial responsibility for deviations.

This sequence turns forecast error into:

- a physical balancing need;
- a market adjustment problem;
- a settlement exposure;
- a governance question if a BRP, BSP, trader, or optimizer is acting on behalf of an asset owner.

Example

Take a solar portfolio scheduled to deliver 100 MW from 12:00 to 13:00.

At 12:20, cloud cover arrives faster than forecast. Actual output falls to 85 MW.

The system cannot say, “We will true-up at the end of the month.” Someone else must increase injection, reduce demand, release stored energy, or change flows. Later, settlement determines who pays or receives for the deviation, depending on the market rules and prices.

The physical problem comes first. The financial problem is the institutional wrapper around it.

In Practice

For a renewable IPP with outsourced trading, the core oversight question is:

Did the partner manage the forecast-to-delivery chain well, given the information available at the time?

A bad benchmark would compare the partner against perfect hindsight. A useful benchmark asks:

- What forecast was available before gate closure?
- What could have been traded intraday?
- What constraints did the asset have?
- What imbalance remained?
- Was the residual error normal weather uncertainty, poor forecasting, slow trading, or misaligned incentives?

This becomes the starting point for later BRP/BSP monitoring.

Misconceptions

"Forecast error is just noise."

In a power system, forecast error becomes power that must be balanced and money that must be settled.

"Balancing is only a market."

Balancing begins as a physical need. The market design is the way responsibility, incentives, and costs are organized.

"Storage solves balancing completely."

Storage can respond quickly and provide important services, but it has finite power, finite energy, efficiency losses, state-of-charge constraints, and opportunity costs.

Recap

System balance and portfolio imbalance answer different questions. System balance is the physical need to keep supply and demand close enough for secure operation; portfolio imbalance is the participant's difference between actual delivery or consumption and its scheduled position. Forecasts and schedules help narrow the gap before delivery, but real-time MW decide what the grid must handle. Settlement then assigns the financial result. A small percent forecast error on a large renewable portfolio is therefore power to balance, energy to settle, and a partner-governance question when someone else trades or optimizes for the asset owner.

Chapter 3. Load Shape And Time Variation

References: [1] [2] [6]

One hour of metering gives a number; a year of metering gives a curve. That curve is the object the power system has to serve: how much energy was used, and when each MW was needed. Two customers can have the same annual MWh and require different wires, generators, batteries, trading actions, and contracts. A useful first step is to treat demand as a time series rather than a single total.

Opening Puzzle

Two portfolios each consume `87,600 MWh/year`.

Portfolio A consumes exactly 10 MW every hour.

Portfolio B consumes 20 MW in some hours and nearly nothing in others.

The two portfolios have the same annual energy and different grid requirements.

System operation is built around the load curve.

First-Principles Model

Load is demand over time:

```
load(t) = electrical demand at time t
```

Generation is also a time series:

```
generation(t) = electrical output at time t
```

The mismatch is where power-market structure begins.

For a simplified system with variable renewable generation:

```
residual load(t) = demand(t) − variable renewable generation(t)
```

This definition depends on the convention: you must state which resources are subtracted. If you subtract solar only, residual load means “load net of solar.” If you subtract wind and solar, it means “load net of wind and solar.”

Definitions

Load: electricity demand at a point, portfolio, zone, or system over time.

Load shape: the pattern of load across hours, days, seasons, or years.

Peak load: the maximum load over a defined period and geography.

Average load: total energy divided by total time.

Residual load: demand remaining after subtracting a specified category of generation, usually variable renewables.

Capacity factor: actual energy produced during a period divided by the energy that would have been produced at full capacity for the entire period:

```
capacity factor = actual MWh / (MW capacity x hours)
```

Shape risk: the risk that production or consumption timing differs from a contracted, hedged, or desired delivery shape.

Numerical Intuition

If annual consumption is `87,600 MWh/year`, average load is:

```
87,600 MWh/year / 8,760 h/year = 10 MW
```

But peak could be 12 MW, 20 MW, or 50 MW. The average does not tell you.

For capacity factor, a 100 MW generator producing 175,200 MWh in a non-leap year has:

```
capacity factor = 175,200 MWh / (100 MW x 8,760 h)
capacity factor = 175,200 / 876,000 = 20%
```

This calculation is a unit example rather than a technology benchmark.

Worked Estimate

Consider a buyer who wants a flat 10 MW every hour for one day:

```
10 MW x 24 h = 240 MWh/day
```

Now imagine a solar asset that also produces 240 MWh that day, but only between 08:00 and 18:00.

If production is spread evenly over those 10 daylight hours:

```
240 MWh / 10 h = 24 MW during daylight
0 MW during the other 14 hours
```

Annual or daily MWh match. Hourly shape does not.

To turn that solar shape into a flat product, someone must handle surplus daylight energy and missing evening/night energy. That “someone” might be the buyer, seller, trader, BRP , battery, market, or a contract settlement formula.

Mechanism

Load shape matters because the grid has to meet the curve, not the average.

Daily shape: mornings, evenings, working hours, and industrial schedules all create different ramps and peaks.

Seasonal shape: heating, cooling, daylight, hydrology, and industrial activity shift demand and generation across the year.

Renewable shape: solar is correlated with sunlight; wind is weather-driven; hydro depends on water availability and reservoir strategy.

Residual load is the curve left for other resources to serve. If solar output rises strongly at midday, residual load may fall at midday and rise again in the evening. That creates flexibility needs and can change price shape.

This connects physics to PPAs: a contract promises both annual MWh quantity and ownership of the shape.

Example

A buyer consumes:

```
Hour 1: 10 MWh
Hour 2: 10 MWh
Hour 3: 10 MWh
Hour 4: 10 MWh
```

A generator produces:

```
Hour 1: 0 MWh
Hour 2: 20 MWh
Hour 3: 20 MWh
Hour 4: 0 MWh
```

Both total `40 MWh`. But hourly mismatch is:

```
Hour 1: −10 MWh
Hour 2: +10 MWh
Hour 3: +10 MWh
Hour 4: −10 MWh
```

The total mismatch over four hours nets to zero. The operational and commercial mismatch does not disappear unless the market, storage, or contract absorbs it.

In Practice

For a solar-plus-battery energy storage system (BESS) power purchase agreement (PPA), the product question is:

Are we selling as-produced solar, a shaped profile, a flat block, or a customer-matched product?

Each answer changes the model:

- as-produced shifts more shape exposure to the buyer;
- shaped delivery shifts more responsibility to the seller or optimizer;
- a battery can reduce some mismatch, but only within its power, energy, efficiency, degradation, and state-of-charge limits;
- the BRP/trading partner may manage residual imbalance, but the asset owner still needs to know what risk remains.

The same annual MWh can be cheap or expensive depending on the clock.

Misconceptions

"If annual MWh match, the product is matched."

Annual matching can hide hourly surplus and shortage.

"Peak load is just a bigger average."

Peak is a constraint event. It can drive capacity needs, tariffs, grid reinforcement, and reliability planning.

"Capacity factor tells you when energy arrives."

Capacity factor compresses time into one ratio. It does not show hourly shape.

Current Debates Or Caveats

Solar-heavy hours can lower residual load and realized solar prices when demand, exports, storage, or flexible load do not absorb the output. The mechanism is portable; price levels, capture-price figures, and country-year counts need their own date, market area, and data definition.

Recap

Annual MWh gives the energy total; load shape gives the timing requirement. Once demand and generation are drawn as time series, peak load becomes a constraint, residual load becomes the curve other resources must serve, and capacity factor becomes a compressed view of a richer pattern. The same total energy can leave hourly surplus, shortage, or a contract exposure for someone to absorb. Power-market reasoning improves when the clock stays in the model.

Chapter 4. Networks, Bidding Zones, Tariffs, And Congestion

References: [7] [8] [9] [10] [11]

A solar map can make a site look obvious: strong production, clean layout, healthy MWh. Then the model reaches the connection point. The local wire may not have room, the export path may bind, or the available right may come with tariff and curtailment terms that change the economics. In power markets, location is the path by which each MWh becomes deliverable, priced, and paid for.

Opening Puzzle

A project model says a solar plant is profitable.

The next question determines whether the model is usable:

Can it actually inject power where it wants to connect?

If the answer is no, the project remains a land option with a production forecast.

First-Principles Model

Electricity needs wires. Wires have limits.

At any location, a simplified surplus equation is:

```
local surplus = local generation − local demand − export capability
```

If local generation rises above local demand and export capability, the system has a bottleneck. Something must change:

- prices separate by area;
- generation is curtailed;
- consumption increases locally;
- storage charges;
- grid capacity is reinforced;
- connection rights are limited or conditional.

Location is part of the asset.

Definitions

Transmission: transport of electricity on extra-high-voltage and high-voltage interconnected systems, excluding supply, using the ACER/EU definition.

Distribution: transport of electricity on high-, medium-, and low-voltage distribution systems for delivery to customers, excluding supply, using the ACER/EU definition.

Bidding zone / bidding area: a market area in which the market clears a price, with area prices able to differ when transmission constraints bind.

Congestion: a condition where desired power flows exceed available transmission or network capability.

Connection agreement: an agreement governing the right, technical conditions, capacity, costs, and process for connecting a facility to a grid.

Grid tariff / network tariff: recurring charges paid for use of the network. ACER distinguishes these from one-off connection charges.

Connection charge: typically a one-off charge covering part or all of the cost of connecting or upgrading a user's connection.

Flexible or limited connection agreement: an agreement that includes conditions to limit and control injection into, or withdrawal from, the grid.

Loss charge: a charge or credit linked to electrical losses caused or reduced by network use.

Reactive power / reactive energy charge: a charge related to withdrawing or injecting reactive power outside allowed limits.

Numerical Intuition

The unit habit matters more than any single tariff value.

Capacity charges often scale with subscribed power:

```
annual capacity cost = subscribed MW x EUR/MW-year
```

Energy or loss charges scale with actual energy:

```
annual energy charge = delivered MWh x EUR/MWh
```

To compare them in a project model, convert everything to an energy basis:

```
grid cost per MWh = annual grid cost / annual MWh delivered
```

Example:

```
annual grid cost = EUR 1,000,000/year
annual delivery = 200,000 MWh/year
grid cost = EUR 5/MWh
```

This illustrative number is a scenario value; country tariff claims need current official sources.

Worked Estimate

Suppose a project subscribes for 100 MW of grid capacity and expects 200,000 MWh/year of net delivery.

If recurring grid costs are `EUR 10,000/MW-year`, annual cost is:

```
100 MW x EUR 10,000/MW-year = EUR 1,000,000/year
```

Converted to energy:

```
EUR 1,000,000/year / 200,000 MWh/year = EUR 5/MWh
```

If the same project delivers only 100,000 MWh/year because of curtailment or lower production, the same annual capacity cost becomes:

```
EUR 1,000,000/year / 100,000 MWh/year = EUR 10/MWh
```

The scaling lesson: MW-based costs punish low utilization. Curtailment removes revenue and can raise fixed grid cost per delivered MWh.

Mechanism

A grid constraint has several commercial faces.

First, market price. Nord Pool explains that when available transmission capacity between bidding areas is constrained, different area prices can be established. Power flows from lower-price areas toward higher-price areas when constraints bind.

Second, connection feasibility. A generator needs a connection point, technical design, and allocated capacity. In Svenska kraftnät's 2025 connection guidance for the Swedish national grid, connection should be made at the lowest suitable voltage level, there must be a tangible need, and Svenska kraftnät decides the connection point and technical design.

Third, tariff exposure. In the Swedish national-grid example used here, subscribers pay capacity fees, usage fees, and possible excess charges. The usage fee covers the cost of buying electricity losses in the network. Temporary subscriptions can be granted subject to available network capacity and may be suspended if Svenska kraftnät assesses that they may lead to congestion.

Fourth, curtailment or limited access. ACER defines flexible connection agreements as agreed conditions that can limit and control injection or withdrawal. Svenska kraftnät's network development material discusses conditional connection agreements as a temporary measure to connect actors before network measures are in place, where the connecting party must accept conditions.

Example

Compare two identical 100 MW solar projects.

Project North:

- high generation potential;
- weak local demand;
- constrained export path;
- high risk of area-price separation or curtailment;
- capacity-based grid costs.

Project Near Load:

- slightly lower generation potential;
- better local demand or grid headroom;
- lower expected curtailment;
- different tariff and connection terms.

The higher-MWh project may not be the higher-value project. The wires decide how much of the MWh can become revenue.

In Practice

For a project model or investment committee memo, grid work should not sit in a forgotten cost row.

Ask:

- What is the connection point?
- Is capacity firm, reserved, conditional, temporary, or still speculative?
- Which voltage level and network owner are involved?

- Are charges MW-based, MWh-based, loss-based, reactive-power-related, or one-off connection charges?
- What is the tariff year and geography?
- What happens if the asset overruns subscribed capacity?
- What happens if the grid constrains export during high-generation hours?

For a solar-plus-BESS project, the battery may reduce some grid stress if it charges during local surplus and discharges when export headroom or prices are better. But that value depends on the actual connection agreement, metering, tariff design, and dispatch rights.

Misconceptions

"Grid cost is just a line item."

Grid terms can change sizing, layout, dispatch, contracting, financing, and go/no-go feasibility.

"A price zone is just an administrative area."

Bidding areas are market representations of network constraints and regional conditions. They are institutional, but they point back to physical limits.

"A connection request equals a connection right."

Connection processes usually require technical studies, maturity evidence, agreements, capacity allocation, and cost responsibility. The exact rules are jurisdiction-specific.

Current Debates Or Caveats

Tariff and connection claims depend on the rulebook, network owner, voltage level, user class, and tariff year. The Swedish national-grid details above are one dated example, not a portable tariff model. Project-specific distribution-grid tariff treatment still needs the relevant distribution system operator (DSO) source.

This chapter does not compare Nordic tariff regimes country by country. ACER's tariff taxonomy is enough to show why casual comparison is risky: network charges combine different capacity, energy, loss, reactive-power, connection, voltage-level, user-type, and national design choices.

Recap

A project has a resource side and a wire side. The resource side estimates how many MWh the asset can produce; the wire side decides whether those MWh can be injected, where they are priced, what network charges attach, and when output may be limited. Once the wire side is visible, a high-production site can become a weaker investment than a lower-production site with better grid headroom and cleaner connection terms.

Chapter 5. Market Participants And Responsibilities

References: [3] [12]

Consider a 100 MW asset that promised 100 MW for the hour and delivered 96 MW. The missing 4 MWh is small enough to fit in one line of arithmetic, but several obligations attach to it. Someone must keep the physical system balanced. Someone must report the meter data. Someone must compare the schedule with reality. Someone must settle the money. Once the deviation is traced through the market process, the acronyms become a map of responsibilities. Each actor exists because one obligation has to land somewhere.

Opening Puzzle

An asset sells `100 MW` for the hour and delivers `96 MW`.

The physical deviation is easy to compute. The responsibility chain is harder. The transmission system operator (TSO) sees a balancing problem, the metering party reports the data, the balance responsible party (BRP) carries the imbalance exposure, a balancing service provider (BSP) may have delivered reserve, and the commercial owner sees the cash effect.

Which actor owns each obligation created by the missing `4 MWh`?

First-Principles Model

Follow one MWh from asset to settlement.

1. A generator injects electricity into a grid.
2. A grid operator measures or receives metered data.
3. A market position says what was sold or scheduled.
4. The system operator keeps the physical system balanced.
5. Settlement compares actual physical volumes with traded or scheduled positions.
6. Someone pays or receives money for the difference.

The market roles are the institutions needed to make that chain work.

Definitions

TSO: transmission system operator. In the eSett Nordic handbook, the TSO is responsible for security of supply, real-time coordination of supply and demand, operation of the high-voltage grid, and ultimate responsibility for imbalance settlement under national laws.

DSO: distribution system operator. In the eSett terminology, a DSO owns or operates a distribution grid, distributes electricity from producers to customers, meters production/consumption/exchange, and reports metered data to relevant stakeholders.

BRP: balance responsible party. In the eSett terminology, a BRP has valid agreements with eSett and a TSO and manages a balance obligation for itself or others.

BSP: balancing service provider. In the eSett terminology, a BSP has agreements with eSett and a TSO or is bound by TSO terms, and has reserve-providing units or groups able to provide balancing services to TSOs.

Balance obligation: the BRP's obligation to continuously plan for and achieve balance between electricity supplied and withdrawn, and to perform financial settlement of imbalances.

NEMO: nominated electricity market operator. In the eSett terminology, a power exchange for delivery trades that receives bids and determines market energy prices after technical constraints from the system operator.

Retailer / supplier: the actor selling electricity to end users and buying electricity directly, via other retailers, or via a NEMO. Terminology varies by country and source.

Aggregator: an actor combining multiple customer loads or generated electricity for sale, purchase, or auction in electricity markets.

Offtaker: the buyer under a commercial contract such as a PPA. This is a commercial role, not necessarily a grid or settlement role.

Regulator: the authority setting or approving rules, methodologies, licenses, tariffs, or terms depending on jurisdiction.

Numerical Intuition

Map one 100 MW asset for one hour.

If it is scheduled for 100 MW but delivers 96 MW:

```
power deviation = 96 MW − 100 MW = −4 MW
energy deviation over 1 hour = −4 MWh
```

The first question is which artifact fixes each obligation. In a `−4 MWh` miss, the schedule, meter, balancing instruction, settlement statement, and commercial contract can point to different actors.

Question created by the `−4 MWh` miss	Evidence to inspect	Natural role in the simplified map	Why it matters
What was promised?	final schedule, trades, nominations	BRP / trader interface	sets the reference volume
What physically happened?	metered generation	DSO or metering party	sets actual delivery
Who corrected the physical system?	balancing activation and delivery data	TSO / BSP	explains real-time correction
Who carries the cash effect?	imbalance settlement and commercial contracts	BRP, owner, offtaker , or supplier depending on agreement	separates formal role from economic exposure

The same `−4 MWh` therefore appears as physics, data, settlement, and governance. The useful skill is tracing it through the evidence chain before assigning blame or value.

Mechanism

The TSO owns the real-time system problem.

The DSO owns much of the distribution-grid and metering interface problem.

The BRP owns the balance-responsibility problem: plan, submit, verify, and settle positions so that deviations are financially accounted for.

The BSP owns the balancing-service delivery problem: qualify resources, offer services, deliver activated reserves, and verify delivered volumes according to the relevant terms.

The NEMO or power exchange owns the organized market-clearing and trade-reporting interface.

The supplier or retailer owns the customer supply relationship.

The aggregator owns the portfolio assembly problem, turning many small flexible loads or generators into something that can participate in markets.

The offtaker owns the commercial demand or procurement problem.

The regulator sets the permitted rules of the game.

Role-Obligation Table

Actor	Physical obligation	Financial obligation	Data / legal obligation
TSO	Secure high-voltage system operation and real-time balance	Procure/pay for balancing according to rules	Submit data to settlement systems; follow national/EU rules
DSO	Operate distribution grid; connect and meter users	Country-specific network/tariff obligations	Meter and report production, consumption, and exchange data
BRP	Plan balanced schedules for its	Financially liable for imbalances	Agreements with eSett/TSO; submit

Actor	Physical obligation	Financial obligation	Data / legal obligation
	responsibility area		and verify trade/plan data
BSP	Provide balancing services from qualified resources	Settlement for balancing services and regulation imbalances	Agreements with eSett/TSO; verify balancing-service data
NEMO / exchange	No direct grid operation in this simplified frame	Market clearing / trade interface	Report day-ahead/intraday trade data
Retailer / supplier	Supply customer portfolio commercially	Buy/sell energy for customers	Maintain BRP relationship and customer obligations
Aggregator	Coordinate flexible loads/generation	Depends on market role and contracts	Aggregate resource data and comply with participation rules
Offtaker	Consume or financially contract for energy	Pay under contract; carry allocated risks	Contractual performance and reporting obligations
Regulator	No operational dispatch role	Approves or oversees market/tariff frameworks	Licenses, decisions, methodologies, compliance

This table is a teaching simplification. Exact duties vary by country, voltage level, market role, and agreement.

Example

A solar asset sells energy under a PPA.

The offtaker wants renewable electricity or price certainty. The supplier or trader may handle market access. The BRP carries imbalance responsibility for the relevant metering points or portfolio. The DSO or grid operator supplies metered data. The TSO maintains system balance. If the asset or battery provides reserves, the BSP role applies.

If production misses forecast, the asset owner needs the responsibility map:

- Who had balance responsibility?
- Who controlled bidding?
- Who received metered data?
- Who could update the position before delivery?
- Who earned balancing-service revenue, if any?
- Who bore imbalance cost?

These questions form a first partner-oversight check.

In Practice

For renewable-IPP outsourced trading and balancing oversight, the BRP/BSP distinction is central.

A BRP's core role is balance responsibility and financial settlement of deviations. A BSP's core role is providing balancing services from qualified resources. One company may perform multiple roles, but the obligations are conceptually different.

In Sweden, BSP and BRP role separation started contractually on 2024-05-01, while full independent BSP implementation remains a staged process expected by 2028 at the latest in the cited Svenska kraftnät material. That timeline anchors the Swedish governance example.

Misconceptions

"A BRP and BSP are the same thing."

They can be performed by the same business entity in some settings, but the obligations differ: balance responsibility versus balancing-service provision.

“The exchange runs the grid.”

The exchange clears trades. The TSO runs the transmission system and manages real-time balance within its mandate.

“The asset owner can outsource the whole problem.”

Execution can be outsourced. Economic exposure, governance, and strategic understanding cannot be outsourced completely.

Current Debates Or Caveats

Actor roles are jurisdiction-specific and changing. Swedish BSP/BRP implementation is explicitly date-specific; Nordic/eSett definitions are the Nordic context, not a universal legal map for all markets.

Recap

Market participants become easier to understand when each acronym is tied to an obligation. Start with one MWh, then ask what has to happen physically, financially, legally, and in the data trail. The TSO keeps the real-time system secure, the DSO handles distribution and metering interfaces, the BRP owns balance responsibility, the BSP provides balancing services, and the NEMO clears organized trades. Suppliers, aggregators, offtakers, and regulators sit around that chain with customer, portfolio, contract, or rule-setting duties. When a schedule and actual delivery differ, ask which obligation each actor carried.

Chapter 6. Electricity Market Timeframes

References: [13] [14] [3] [4]

A solar plant has several time-stamped descriptions during a market day. Yesterday's schedule says one thing, the latest forecast says another, the control room sees the physical system, and the meter later records what actually happened. The market has to support a plan before delivery, updates as information improves, real-time balancing, and post-delivery settlement. That sequence is the market clock.

Opening Puzzle

A solar plant promises tomorrow's output today. At noon today, tomorrow's weather forecast says 50 MW at 13:00. At 09:00 tomorrow, a cloud front moves. At 12:55, the plant is producing 38 MW. At 13:15, meters record what happened.

Which number should the market believe: the forecast, the latest trade, the physical flow, or the meter?

Each number belongs to a different point in the market sequence.

First-Principles Model

Three properties shape the market clock:

1. It must balance physically in real time.
2. It is scheduled financially before real time.
3. Forecasts improve as real time approaches.

The market therefore uses a sequence of corrections:

- Day-ahead: make the main plan.
- Intraday: repair the plan as information improves.
- Balancing: let the TSO correct the residual physical error.
- Imbalance settlement: allocate the financial consequence after meters and schedules are compared.

Each stage exists because the previous stage is useful but imperfect.

Definitions

- **Day-ahead market:** auction before delivery day where volumes clear for the next day's market time units.
- **Intraday market:** trading after day-ahead clearing and before delivery, used to adjust positions closer to real time.
- **Balancing:** TSO-controlled action to maintain system balance when the market's self-correction is no longer enough.
- **Imbalance settlement:** post-delivery financial settlement comparing final position with allocated or metered reality.
- **Market time unit:** the time granularity of traded or settled energy, such as an hour or 15 minutes depending on market and date.
- **Bidding zone:** an area where a market price is formed subject to network constraints.

Numerical Intuition

The market clock turns a power error into an energy exposure by multiplying by the settlement interval:

```
energy error = power error x settlement-interval hours
```

A 3 MW residual error over one hour is 3 MWh. The same 3 MW residual error over a 15-minute market time unit is 0.75 MWh, because 15 minutes is 0.25 hours.

Worked Estimate

Suppose a 100 MW solar plant sells 60 MW for one hour day-ahead.

One hour before delivery, the forecast falls to 45 MW. If the plant buys back 15 MW intraday, its final schedule becomes:

```
60 MW − 15 MW = 45 MW
```

If actual metered output is 42 MW, the remaining deviation is:

```
42 MWh actual − 45 MWh final schedule = −3 MWh imbalance
```

The example separates the gross forecast miss from the residual imbalance after intraday repair. A 15 MW forecast miss can become a 3 MWh imbalance if the intraday market lets the asset repair most of the error before delivery.

Stage	Best available description	Action still possible	Error left for next stage
Day-ahead	forecast and planned position	create the baseline schedule	forecast error
Intraday	updated forecast and available trades	repair the position	residual forecast and liquidity error

Stage	Best available description	Action still possible	Error left for next stage
Balancing	physical system close to delivery	TSO correction	settlement exposure
Imbalance settlement	meter and final position	allocate money	no physical correction left

Equations

Final scheduled position:

```
final position = day-ahead position + intraday buys/sells + accepted bilate
```

Units: MW for a time interval, or MWh after multiplying by interval length.

Imbalance volume:

```
imbalance = actual allocated or metered energy − final scheduled energy
```

Units: MWh per settlement interval. Sign conventions vary by market, so the equation is conceptual unless a specific settlement handbook is being used.

Mechanism

Day-ahead is the big coordination step. Nord Pool's day-ahead source describes a closed auction for next-day delivery, matched through SDAC and Euphemia while accounting for network constraints. It creates a shared baseline for tomorrow's schedules and positions.

Intraday trading updates positions as forecasts, outages, demand, and cross-border capacity change. EU Regulation 2017/2195 explicitly values the ability for market participants to balance themselves as close as possible to real time.

Balancing is the control room layer. When delivery is close enough that normal trading cannot solve the residual mismatch, TSOs activate balancing energy and reserves.

Imbalance settlement is the accounting layer. eSett's Nordic handbook frames the Nordic model as a common settlement interface where balance responsible parties' (BRPs') imbalances are calculated from physical volumes and traded energy volumes, with eSett acting as the central settlement operator for Denmark, Finland, Norway, and Sweden.

Examples

A wind farm sells day-ahead. The forecast improves after the auction. It trades intraday to reduce exposure. If it still over- or under-produces relative to its final position, the BRP sees an imbalance settlement consequence.

A battery may do the opposite. It can choose whether to remain flexible for intraday opportunities or commit capacity to reserves. The market clock is therefore also an option clock: every stage closer to real time has better information but less freedom.

In Practice

For an asset owner with outsourced trading, the practical question is:

- What was the day-ahead position?
- What intraday trades were available and feasible?
- What final position was carried into delivery?
- What actual metered production occurred?
- What imbalance remained, and who economically bears it under the contract?

This forms the basis of a shadow-trading benchmark. It compares actual decisions with feasible decisions using information and control rights available at the time, rather than perfect hindsight.

Misconceptions

- "The spot price is the market." The spot auction is one stage in a time sequence.

- “Real-time balancing is just another exchange trade.” Not quite. Balancing is TSO-controlled system operation with market mechanisms around it.
- “A better day-ahead forecast removes imbalance risk.” It reduces risk, but the last forecast error is usually born close to delivery.

Current Debates And Caveats

Shorter market time units, such as 15 minutes, can reduce structural mismatch between flat schedules and changing physical output. Nord Pool’s MTU source describes a 2024-2025 rollout and stated European SDAC 15-minute MTU go-live on trading day `2025-09-30`, for delivery day `2025-10-01`.

Continuous intraday design is also a live market-design topic. Practitioner sources emphasize liquidity and incentives; official market rules define the settled rule facts.

Recap

The market clock is a sequence of corrections. Day-ahead gives the system a shared baseline, intraday lets participants repair that baseline as forecasts change, balancing handles the residual physical error when delivery is too close for ordinary trading, and imbalance settlement compares the final position with metered or allocated reality. Forecast, trade, physical flow, and meter each matter because each belongs to a different moment on the clock.

Chapter 7. Merit Order And Price Formation

References: [13] [15] [16]

Consider one sunny hour in one bidding zone. Solar plants are producing, demand is finite, and the auction still has to choose enough offers to cover the hour. The solar plant's own fuel cost alone cannot decide what it earns. It may be paid a price set by a more expensive plant at the edge of the stack, or it may earn a low price when similar low-cost output pushes the marginal unit down. The relevant question is which unit is marginal in that hour.

Opening Puzzle

If a solar plant has almost zero fuel cost, why does it sometimes earn a high price? And if solar is cheap, why can adding more solar make solar owners poorer in sunny hours?

First-Principles Model

Start with one hour in one bidding zone.

Demand wants 1,000 MWh. Generators offer energy:

- wind and solar at very low variable cost,
- nuclear and hydro depending on opportunity cost and constraints,
- coal, gas, or imports at higher variable cost,
- demand response or scarce reserves at high opportunity cost.

The market accepts offers until 1,000 MWh is covered. In the simplest merit-order model, the last accepted unit sets the clearing price for accepted MWh.

This simplified model gives the first lever of intuition for European market design.

Definitions

- **Merit order:** ranking of available supply offers from low to high offer price.
- **Marginal unit:** the accepted resource at the edge of meeting demand.
- **Clearing price:** the price produced by the auction for a market time unit and bidding zone.
- **Scarcity:** condition where supply or flexibility is tight relative to demand and reliability needs.
- **Negative price:** market price below zero, usually reflecting inflexibility, constraints, policy incentives, or willingness to pay to keep producing or consuming.
- **Capture price:** generation-weighted average price received by an asset or technology.

Numerical Intuition

Use a simplified stack:

Resource	Available volume	Offer
Wind/solar	300 MWh	0 EUR/MWh
Nuclear/hydro	300 MWh	20 EUR/MWh
Gas	500 MWh	80 EUR/MWh

If demand is 700 MWh, gas is marginal and price is about 80 EUR/MWh in the simplified model. If solar output rises by 300 MWh and demand stays fixed, the gas unit may no longer be needed and the clearing price can fall toward the next unit's offer.

These are illustrative numbers only, not market data.

Worked Estimate

Take a 100 MW solar portfolio for one sunny hour.

Case A: price is 70 EUR/MWh.

```
100 MWh x 70 EUR/MWh = 7,000 EUR
```

Case B: solar is abundant and the price in that sunny hour is 5 EUR/MWh.

```
100 MWh x 5 EUR/MWh = 500 EUR
```

The same plant, sun, and 100 MWh produce 14 times less revenue in that hour. Low variable cost and high realized value are different quantities.

Equations

Simplified market-clearing condition:

```
accepted supply at or below clearing price >= demand
```

Plain meaning: the auction accepts enough supply to serve demand. The final accepted offer is the marginal offer in the simplified model.

Generation-weighted capture price:

```
capture price = sum(generation_t x price_t) / sum(generation_t)
```

Units: EUR/MWh if prices are EUR/MWh and generation is MWh. The equation matters because solar produces disproportionately in sunny hours, not average hours.

Mechanism

The market stack teaches three things.

First, price is about the marginal MWh. A plant can be cheap on average but receive a price set by a more expensive marginal resource.

Second, renewables shift the residual problem. When wind or solar output is high, less conventional generation is needed to meet the same demand. That can push the marginal unit down the stack.

Third, revenue is weighted by production timing. A solar plant earns the prices in the hours when it generates.

This mechanism leads to cannibalization. When many solar plants produce at the same time, their output can depress the prices in the hours that drive their revenue.

Examples

Low renewable output and high demand: the marginal unit may be expensive, so market prices can be high.

High solar output and moderate demand: the marginal unit may be cheap, or the system may hit export, storage, demand, or must-run constraints.

Negative-price hour: negative prices can occur when supply is inflexible relative to demand, export, storage, and curtailment options, and some participants have contractual, technical, tax, or subsidy reasons to keep producing. The specific cause is market-specific.

In Practice

For a solar PPA, ask both “what is the average market price?” and “what price does the project capture when it generates?”

A shaped or hybrid solar-plus-BESS offer is partly an answer to this problem. The battery or contract shape tries to move value from low-price solar hours into higher-value customer or market hours. But that creates opportunity cost, degradation, imbalance exposure, and dispatch decisions.

For portfolio modeling, the merit-order picture also warns against adding assets that all earn money in the same hours and all suffer in the same low-price hours.

Misconceptions

- “Cheap renewables always mean cheap revenue for renewable owners.” Cheap variable cost can lower market prices in production hours.
- “The marginal price is the full truth.” It is a teaching model. Real clearing includes network constraints, coupled zones, block products, and detailed bidding rules.
- “Negative prices prove the market is broken.” They may indicate inflexibility or policy interactions, but the interpretation depends on the system and rule context.

Current Debates And Caveats

Practitioner sources discuss rising negative-price hours and solar capture erosion in several European markets. The mechanism is the transferable part; counts and capture rates need dated market data.

The durable mechanism is narrower and stronger: correlated low-variable-cost output can reduce the realized value of that same output.

Recap

The stack explains why an asset’s realized price can diverge from the average. In the simplified market, the price comes from the last accepted MWh needed to meet demand, so a low-variable-cost plant can earn a high price when a costly unit is marginal. The same plant can earn little in the hours it produces most if similar output pushes the marginal unit down the stack, or if the system runs into export, storage, demand, or must-run constraints. Capture price asks what the asset earns in its own production hours, not what the market averaged across all hours.

Chapter 8. Forecast Error And Imbalance Settlement

References: [3] [4] [12]

A control room can be calm while the meter quietly disagrees with the plan. A solar forecast said one thing, the market position carried that expectation into delivery, and the plant produced something else. The grid still has to balance in real time, and the settlement system still has to decide who was long or short after the fact. A forecast error left in the final position becomes an imbalance, and imbalance settlement gives it an owner and a price.

Opening Puzzle

A solar plant forecasts 50 MWh for a settlement interval, sells 50 MWh, and produces 45 MWh. The weather was weaker than expected, so the delivered volume is below the market position.

The system still received `5 MWh` less than the market position said it would receive.

Who is short, who balances the system, and what price is attached to that difference?

First-Principles Model

The physical grid only sees injection and withdrawal. The market sees positions. Settlement connects the two.

There are three layers:

- Forecast: what you expected.
- Schedule or position: what you committed through markets and nominations.
- Actual: what meters or allocation data say happened.

The imbalance is the difference between the last financial position and physical reality, under a market's sign convention.

Definitions

- **Forecast error:** difference between expected and actual generation or consumption.
- **Final position:** the market position carried into delivery after day-ahead, intraday, bilateral, and other accepted adjustments.
- **Imbalance:** difference between final scheduled/traded position and actual allocated or metered energy.
- **Imbalance price:** price used to settle imbalance volumes; EU balancing rules tie it to the value/cost of balancing energy or avoided activation, with market-specific implementation.
- **BRP:** balance responsible party, financially responsible for imbalances in the settlement framework.
- **BSP:** balancing service provider, a party that can offer balancing or ancillary services to the TSO when qualified.

Numerical Intuition

- A 5 MW error lasting 1 hour is 5 MWh.
- A 5 MW error lasting 15 minutes is 1.25 MWh.
- A 20 EUR/MWh unfavorable imbalance spread on 5 MWh is 100 EUR.

These are scale anchors, not market-price claims.

Worked Estimate

Suppose a 100 MW solar plant is 5 MW short for one hour after intraday trading closes.

```
imbalance volume = -5 MW x 1 h = -5 MWh
```

If the effective cost relative to the alternative position is 40 EUR/MWh:

```
5 MWh x 40 EUR/MWh = 200 EUR
```

A single hour is small. Repeated across 200 difficult hours, the same 5 MWh error becomes:

```
5 MWh/hour x 200 hours x 40 EUR/MWh = 40,000 EUR
```

The structural lesson: small forecast errors can matter when they are frequent, correlated with stressed prices, or allocated poorly in contracts.

Equations

Conceptual imbalance:

```
imbalance volume = actual allocated or metered energy - final scheduled ene
```

Conceptual cost:

```
imbalance cost = imbalance volume x imbalance settlement price
```

Better commercial version:

```
economic impact = imbalance volume x (imbalance price - value of the positi
```

The last form is often more useful for partner monitoring because it separates settlement mechanics from avoidable execution loss.

Mechanism

Forecast error begins as uncertainty. It becomes a schedule problem when the asset takes a market position. It becomes a system problem if the error remains at delivery. It becomes an economic problem when settlement assigns a price to the deviation.

eSett's Nordic material describes imbalance settlement as using trades, consumption, production, MGA imbalance, and imbalance adjustments. The exact algebra is handbook-specific, but the mechanism is robust: the settlement operator reconstructs who was long or short after the fact.

EU Regulation 2017/2195 frames imbalance pricing around balancing energy and avoided activation. Imbalance settlement is part of the incentive system that nudges BRPs to help keep the system balanced.

Svenska kraftnät's BRP/BSP material shows why roles matter. A BRP carries balance responsibility. A BSP provides balancing services. The same company may do both, but the roles are conceptually different.

Examples

Solar shortfall: a plant schedules 50 MWh and produces 45 MWh. Someone must supply the missing 5 MWh physically or financially.

Wind surplus: a wind farm schedules 20 MWh and produces 28 MWh. The surplus may be valuable or costly depending on the system's need and the imbalance pricing rule.

Battery misdelivery: a battery committed to provide balancing service may be judged against delivered activation. That can create separate BSP settlement or misdelivery consequences, depending on market rules.

In Practice

For outsourced BRP execution, the asset owner should ask:

- Did the BRP have the forecast data in time?
- Were intraday corrections feasible?
- Was the final position reasonable given information at the time?
- Were imbalance costs caused by market conditions, forecast quality, data failures, risk appetite, or poor execution?
- Does the contract pass imbalance cost through, share it, cap it, or include it in a fee?

Outsourcing balancing does not necessarily outsource economic risk. The legal role, operational task, and final commercial exposure can be split by contract.

Misconceptions

- "Forecast error is only a forecasting problem." Forecast error can become a market access, data, timing, and contract problem.
- "The BRP always eats the imbalance cost." Not necessarily; contractual pass-through can put exposure back on the asset owner.
- "A shadow benchmark should use perfect hindsight." It should use the feasible information set available before each decision.

Current Debates And Caveats

Swedish BRP/BSP role separation is date-specific. The Svenska kraftnät source used here states that terms were decided in May 2023, the role split started on 2024-05-01, and full independent BSP implementation is expected by 2028 at latest.

Imbalance settlement period and market time unit claims are also date-specific, especially during 15-minute rollout.

Recap

Forecast, final position, and actual delivery are three different objects. The grid only receives the physical difference, but settlement compares that difference to the position carried into delivery and prices the remaining imbalance. From there, the BRP role explains who is responsible inside the settlement framework, while the contract explains whether the asset owner, partner, or another party ultimately bears the cost. The 5 MWh miss is therefore a chain of timing, execution, settlement, and risk allocation.

Chapter 9. Frequency, Reserves, And Ancillary Services

References: [4] [17] [18] [19] [20] [21] [5]

At 14:00, a battery may sit still while the grid keeps moving around it. Its value may be in the MWh it delivers later, or in the MW it promises to hold ready now. That distinction matters because frequency changes when generation and consumption drift apart, and the TSO needs different layers of response to contain, restore, and replace reserves across time scales. The obligation starts with the promise the asset made: how fast it can respond, how long it can sustain delivery, and what state of charge, availability, and alternative value that promise consumes.

Opening Puzzle

A battery is idle at 14:00 and still earns revenue.

The asset sold availability in that interval: the right for the system operator to call on fast MW if frequency moves.

What did the battery promise, and what energy headroom does that promise consume?

First-Principles Model

In a synchronous AC system, frequency is the common electrical speed of the system. It moves when generation and consumption do not match. If load is greater than supply, frequency tends to fall. If supply is greater than load, frequency tends to rise.

The TSO therefore needs layers of response:

- very fast containment to stop the deviation getting worse,
- automatic restoration to move the system back toward target,
- manual or centrally activated restoration to replace fast reserves and manage larger or longer deviations.

These products serve different time scales and system needs.

Definitions

- **Ancillary service:** service procured to support secure operation of the power system, beyond ordinary energy delivery.
- **Reserve capacity or balancing capacity:** MW held available for possible activation under product rules.
- **Balancing energy:** MWh delivered or absorbed when a reserve is activated.
- **Activation:** instruction or automatic response that turns reserved capability into physical action.
- **FCR:** frequency containment reserve, fast response to contain frequency deviations.
- **aFRR:** automatic frequency restoration reserve, automatically activated to restore frequency or control-area balance.
- **mFRR:** manual frequency restoration reserve, manually or centrally activated restoration reserve.
- **State of charge:** battery energy stored, in MWh or percent of usable capacity.

Numerical Intuition

Reserve products differ first by time, then by obligation. A battery that can move `10 MW` is not automatically a `10 MW` reserve product in every market, because the product also defines response speed, endurance, symmetry, telemetry, prequalification, and state-of-charge management.

Use a four-field test:

Field	Commercial meaning
response time	how quickly MW must appear
sustain duration	how much MWh must be held available
activation logic	whether the asset moves automatically, by TSO signal, or through market activation
opportunity cost	which arbitrage, PPA-shaping, or other reserve use is blocked

The Nordic and Danish sources make this concrete. The Nordic TSOs' 2025 battery paper says batteries can participate in all five Nordic reserve products it reviews: fast frequency reserve (FFR), FCR-D, FCR-N, aFRR, and mFRR. Product eligibility still depends on prequalification, bid size, endurance, control system, state of charge, grid connection, and BRP/BSP setup. Energinet tender conditions effective 2025-09-23 give dated Danish examples: FCR in DK1 reaches full response within `30 seconds`, FFR activation can be as fast as `0.7 seconds` in low-inertia situations, and mFRR full delivery can be `12.5 minutes` for daily procurement or up to `90 minutes` for monthly procurement. These are product/date examples, not universal timing rules.

For a current product decision, use the controlling product terms, not the summary table in this chapter.

Worked Estimate

A 10 MW / 20 MWh battery has 2 hours of duration at full power.

If it reserves 5 MW upward for one hour, it must be able to deliver if called. If its state of charge is too low, the promise is not credible.

Simple opportunity-cost frame:

```
reserve value = reserve capacity payment + expected activation value −
```

If the battery could earn 600 EUR by arbitrage in that hour but earns 400 EUR from reserve capacity and expected activation, the reserve choice costs 200 EUR before considering strategic or risk benefits.

The exact numbers are illustrative. The decision compares reserve revenue with the best feasible use of the same MW and MWh.

Equations

Reserve opportunity cost:

```
opportunity cost = best feasible alternative value − reserve commitment val
```

Battery duration:

```
duration (h) = usable energy capacity (MWh) / power rating (MW)
```

Reserve feasibility check:

```
required energy buffer >= committed reserve MW x required delivery duration
```

This is deliberately simple. Real rules include activation profiles, baselines, symmetric/asymmetric bids, state-of-charge management, prequalification, and penalties.

Mechanism

FCR is the first containment layer for frequency deviations. It reacts quickly to limit the movement before slower restoration processes take over.

aFRR is a controlled automatic correction. It helps restore the system after the first response.

mFRR is slower and more deliberate. It relieves faster reserves, handles remaining imbalances, and can be activated through balancing energy markets.

Capacity and activation are different. A provider can be paid for standing ready, then separately settled for energy when activated, depending on the product design.

For batteries, readiness consumes optionality. A battery cannot spend the same MW and MWh on every product at once. A committed reserve position constrains state of charge, availability, degradation, intraday trading, and arbitrage.

The Nordic TSO battery paper also separates prequalification from actual supply. A battery can be prequalified in several markets and in both directions, so prequalified MW is not the same as installed capacity or active daily offer. For a project model, treat prequalification as access to a product; the dispatch model still decides whether bidding the product beats the feasible alternatives in that interval.

Examples

Battery providing FCR: it must keep enough headroom and footroom to respond up or down if the product requires it.

Battery doing energy arbitrage: it buys or charges in low-price hours and sells or discharges in high-price hours.

Battery doing both: possible only if market rules and physical constraints allow it. “Revenue stacking” means constrained portfolio optimization inside one asset.

In Practice

For a solar-plus-BESS project, reserve-market revenue may look attractive in a model. The review question is:

- Was capacity reserved?
- Was activation likely?
- What state-of-charge buffer was required?
- What arbitrage or PPA-shaping value was forgone?
- Were degradation and availability obligations priced?
- Who controls dispatch: owner, optimizer, BRP, BSP, or another party?

For partner oversight, a scorecard should separate market conditions from execution. Low reserve revenue may be due to saturated prices, infeasible state of charge, poor bidding, excessive conservatism, or contractual limits.

Misconceptions

- "Ancillary services are free upside." They use scarce flexibility.
- "A MW is a MW." A MW in 0.7 seconds, 30 seconds, 5 minutes, and 15 minutes are commercially different promises.
- "Capacity payment means no energy risk." Activation, baselines, delivery failure, and state-of-charge constraints can still matter.

Current Debates And Caveats

Ancillary-service saturation is plausible when many fast flexible assets chase a small product, but timing and magnitude are market-specific. For a project model, test saturation product by product: market size, eligible capacity, qualification rules, activation patterns, and the opportunity cost of using the same battery elsewhere.

European balancing platforms such as PICASSO and MARI are important, but platform status and key performance indicators (KPIs) are dated claims. The PICASSO KPI report used here covers 2025 operational results in its source; any platform statement should state the report date and scope.

Recap

Reserve value begins with readiness as well as delivery. Energy markets settle MWh after physical delivery; reserve and ancillary-service markets pay for MW held under rules, then may turn that readiness into balancing energy through activation. The idle battery at 14:00

may therefore be earning money, but it is not unconstrained. A reserve commitment ties up state of charge, availability, degradation headroom, and the chance to use the same asset for arbitrage or PPA shaping.

Chapter 10. Voltage, Reactive Power, Inertia, And Inverter-Based Resources

References: [10] [22] [23] [5]

A solar project can look complete in a market model and still fail the grid conversation. The annual MWh are attractive, the day-ahead price is known, and the connection capacity has a MW label. Then the grid study asks a different set of questions: can the local voltage stay inside limits, can the network survive one important outage, can protection systems see faults clearly, and can the inverter provide reactive support without giving up too much active-power export? The missing layer is the electrical envelope underneath the commercial model: whether a site is limited by energy value, by active-power capacity, or by the AC grid’s need for voltage, fault strength, and fast control.

Opening Puzzle

Suppose total generation equals total load in every market interval.

Can the grid still be in trouble?

First-Principles Model

The market model usually starts with active power:

```
MW exported over time -> MWh sold
```

The grid study sees a larger object:

```
safe operation =
  active-power balance
  + voltage control
  + frequency control
  + fault detection and protection
  + thermal and stability limits
  + secure operation after credible outages
```

Active power `P` transfers energy. Reactive power `Q` helps create and maintain the voltage fields that let AC equipment operate. Apparent power `S` is the combined electrical loading of equipment:

```
S^2 = P^2 + Q^2
```

This small triangle is useful because it shows a real tradeoff. An inverter or transformer with a fixed MVA capability cannot always provide maximum MW export and maximum MVAr support at the same time.

Definitions

- **Voltage:** electrical potential at a node. Equipment and stability require voltage to stay within acceptable limits.
- **Active power:** real power that transfers energy, measured in W or MW.
- **Reactive power:** AC power component associated with electric and magnetic fields and voltage support, measured in var or MVAr.
- **Apparent power:** combined active and reactive loading, measured in VA or MVA.
- **Inertia:** resistance of rotating mass to sudden frequency change after a disturbance.
- **Rate of change of frequency:** how fast frequency moves after the system loses generation or load.
- **Fault level / short-circuit strength:** the network's ability to supply fault current and maintain a clear voltage reference during faults.
- **Short-circuit ratio:** a diagnostic concept comparing grid strength with converter capacity; definitions and thresholds depend on the study method.
- **Grid-following inverter:** inverter that synchronizes to an existing grid waveform.
- **Grid-forming inverter:** inverter controlled to establish or support voltage and frequency behavior more actively.
- **N-1:** security concept under which the system should remain within defined limits after the loss of one relevant element.

Numerical Intuition

Start with a `100 MVA` inverter.

If it exports `100 MW` at unity power factor, the apparent-power budget is fully used:

```
Q = sqrt(100^2 - 100^2) = 0 MVAr
```

If the same inverter exports `80 MW`, it has reactive headroom:

```
Q = sqrt(100^2 - 80^2) = 60 MVAr
```

The exact capability curve is equipment-specific, but the tradeoff is the lesson. Voltage support can consume electrical headroom that the energy model may have counted as export capability.

Now put money on the tradeoff. If a project curtails `20 MW` for `3` sunny hours so the inverter can provide required voltage support, and the foregone energy would have sold for `60 EUR/MWh`, the lost energy value is:

```
20 MW x 3 h x 60 EUR/MWh = EUR 3,600
```

That number shows the scale of a modeling error if the grid obligation is treated as free.

What The Grid Study Is Really Testing

An energy model asks:

```
How many MWh can the asset sell, and at what price?
```

A connection or stability study asks a different set of questions.

Test	Question	Commercial consequence
Thermal limit	Can lines, transformers, and cables carry the current?	Export cap, curtailment, reinforcement cost, or delayed connection.
Voltage limit	Can the node stay within voltage range in normal and stressed conditions?	Reactive obligations, inverter settings, static synchronous compensator (STATCOM), transformer tap strategy, or curtailment.
Fault strength	Is the network electrically stiff enough for protection and converter controls?	Additional equipment, lower export, grid-forming requirements, or connection risk.
Frequency response	Can the system arrest and correct frequency deviations?	Reserve obligations, system-service opportunity, or operating constraints.
N-1 security	Does the system survive a defined outage?	Reinforcement, remedial actions, limited access, or changed dispatch rights.

Kirschen and Strbac's power-system-operation material treats operational reliability, ancillary services, N-1 security, transient stability, and reactive support as system-operation problems, not as afterthoughts. Fingrid's 2026-2035 development plan similarly treats inverter-connected generation and storage as part of a changing stability problem, with grid-forming storage and static synchronous compensators (STATCOMs) among possible solutions.

Mechanism

Voltage is local. Frequency is shared across a synchronous area, but voltage can be weak at one bus, feeder, or load pocket while the wider system frequency looks normal.

Reactive power is one control lever. Synchronous machines, capacitor banks, reactors, STATCOMs, synchronous condensers, transformers, and inverters can all support voltage in different ways. Long lines, cables, large motors, changing load, and distributed generation can all create reactive-power needs.

Inertia is about the first seconds after a disturbance. When a large unit trips, more synchronous inertia slows the initial frequency movement. Lower-inertia systems need faster response, better controls, or different operating margins.

Fault strength is about electrical stiffness. In a strong grid, an inverter has a clear voltage waveform to follow and protection systems see large fault currents. In a weak grid, converter controls become more important and can interact. Practitioner material on short-circuit-ratio diagnostics is useful for intuition, but detailed threshold claims belong in engineering studies, not in a general lecture.

Inverter-based resources are programmable, which creates both opportunity and risk. A battery or solar inverter can be very fast, but the value depends on control mode, qualification, grid-code obligations, measurement, and whether providing the service conflicts with energy, PPA, reserve, or warranty use.

Example

Consider a 100 MW solar plant connecting to a weak local grid.

The energy-only view says:

```
noon output = 100 MW
price = 55 EUR/MWh
revenue for one hour = EUR 5,500
```

The connection study adds:

- voltage rises near the connection point during high export
- reactive absorption is required in some sunny hours
- one line outage makes the local voltage limit bind
- inverter settings must be coordinated with the TSO or DSO
- some support may be mandatory rather than separately paid

The commercial model now needs two extra columns:

```
active MW allowed after grid constraint
reactive MVAr obligation or headroom
```

Without those columns, the model may overstate energy revenue and understate connection risk.

In Practice

For a renewable or BESS project, ask for a grid-physics diagnostic before treating the connection as a simple MW pipe.

Diagnostic question	Evidence to request
What is the connection capacity in MW and under what operating assumptions?	Connection offer, grid study, curtailment or limited-access conditions.
What reactive-power or power-factor requirements apply?	Grid-code clause, capability curve, transformer and inverter data.
Does the node look electrically weak?	Fault-level or short-circuit diagnostics, local TSO/DSO study language.
What happens under N-1?	Contingency study, remedial action scheme, reinforcement requirement.
Can the asset provide services while fulfilling the PPA or dispatch plan?	Control-rights matrix, qualification rules, state-of-charge and availability constraints.
Is the service paid, mandatory, or only useful for connection?	Tariff, tender, grid-code, or connection-agreement evidence.

The practical distinction is simple: a project can be energy-rich and grid-weak. In that case the right commercial lever may be a different connection design, inverter capability, grid service, curtailment assumption, or site choice, not a better day-ahead price forecast.

Misconceptions

“If energy balances, the grid is fine.” Energy balance is necessary, but voltage, stability, protection, and N-1 constraints still matter.

“Reactive power is wasted energy.” Reactive power supports voltage and can determine whether MWh can move, even though it is not delivered MWh.

“A MW connection right is enough.” The usable value of the connection depends on operating limits, voltage obligations, curtailment rules, tariffs, and control rights.

“All inverters behave the same.” Their behavior depends on control mode, firmware, settings, grid strength, and grid-code requirements.

"Grid-forming solves the problem." It can help in some systems, but implementation, verification, interaction with other controls, and procurement rules still matter.

Current Debates And Caveats

Procurement of voltage support, inertia-like services, fault-current capability, and grid-forming behavior is still evolving. Named products and eligibility need current TSO or grid-code evidence.

Technical diagnostics such as short-circuit ratio are useful for framing the grid study. They are not a substitute for the project's interconnection study, equipment capability curves, protection study, or TSO/DSO requirements.

Recap

MW and MWh describe only part of a grid-connected asset. The AC grid also needs voltage support, reactive headroom, fault strength, frequency behavior, thermal security, and credible-outage resilience. The practical model is the MVA triangle plus the grid-study checklist: every promise to export active power is made inside an electrical envelope. If that envelope binds, the commercial question changes from "what is the energy price?" to "which constraint is limiting the asset, what does compliance cost, and does any grid-service value offset it?"

Chapter 11. Generation Technologies And Operating Characteristics

References: [1] [24]

Consider six assets, each labeled 100 MW. One burns fuel, one holds water, one waits for wind, one follows daylight, one stores electricity, and one is a factory that can briefly turn demand down. The numbers look identical, but the promises are not. Some assets can choose when to move. Some move only when nature allows. Some run out of fuel, water, stored energy, or customer patience. Before a model asks what the asset earns, it has to ask what kind of physical behavior the asset can safely promise.

Opening Puzzle

Six assets are advertised as “100 MW”: a gas turbine, a hydro plant, a wind farm, a solar farm, a battery, and a factory that can turn down load. Are they equivalent?

Commercially, no. The label says how fast energy can flow at one instant. It does not say when the asset can produce, how long it can keep going, how quickly it can move, what it costs to move, or whether output depends on weather or stored fuel.

For asset modeling, a MW is a size label, not a promise.

First-Principles Model

An electricity asset is defined by the operating promise it can support.

The physical question is:

```
When can this asset inject, withdraw, or avoid power?
```

The commercial question is:

```
What promise can I safely sell without being punished when reality arri
```

A solar plant can promise cheap energy when there is sunlight, but it cannot promise a flat midnight delivery by itself. A battery can promise fast movement across time, but only within its stored energy and efficiency losses. A hydro reservoir can often promise controllability, but only inside water, environmental, and reservoir constraints. Demand response can reduce demand, but only if the customer process allows it.

Definitions

- **Power:** rate of energy flow, measured in MW.
- **Energy:** accumulated power over time, measured in MWh.
- **Nameplate capacity:** rated maximum power under defined conditions.
- **Capacity factor:** actual energy over a period divided by energy that would have been produced at full nameplate power for every hour in that period.
- **Dispatchable:** controllable by an operator within technical constraints.
- **Weather-dependent:** output depends materially on weather conditions.
- **Energy-limited:** power can be delivered only until an energy store, fuel budget, water budget, or demand-flexibility window is exhausted.
- **Round-trip efficiency:** energy out divided by energy in for a storage cycle, using the same energy units.

Numerical Intuition

- `1 MW x 1 hour = 1 MWh` by definition.
- `1 GW average for 1 year = 8.76 TWh/year` by calculation.
- Sweden PV: IEA PVPS Task 1 reports total Swedish PV capacity of 4,830.7 MW at end-2024 and estimated PV generation of 4,087 GWh in 2024.
- Sweden power-system anchor: IEA PVPS reports Swedish electricity consumption of 136.6 TWh and generation of about 169.6 TWh in 2024, with PV equal to 2.4% of consumption.

Worked Estimate

Start with the full-output envelope behind the Swedish PV anchor:

```
4,830.7 MW x 8,760 h/year = about 42.3 TWh/year if every MW ran at full
```

IEA PVPS estimated Swedish PV generation at 4.087 TWh in 2024. That is roughly one-tenth of the full-output envelope, but because the capacity figure is end-2024 installed capacity, treat it as a scale check rather than an official capacity factor.

Now take a `100 MW` solar asset. Use the Swedish fleet scale check as a rough magnitude anchor rather than a site forecast:

```
Energy = 100 MW x 8,760 h/year x 0.10
       = about 88 GWh/year
```

The scale matters more than the extra decimal: a `100 MW` solar plant is not a `100 MW` average power source. The gap between nameplate power and average output is the shape of sunlight, not a performance defect.

Equations

```
Expected energy = nameplate capacity x capacity factor x hours
```

Meaning: if a plant with `P` MW nameplate produces as if it were at full output for fraction `CF` of the period, over `H` hours it produces `P x CF x H` MWh.

```
Battery duration = energy capacity / power rating
```

Meaning: a `200 MWh / 100 MW` battery can discharge at full power for about 2 hours before considering usable-energy and operating limits.

Mechanism

Technology	Energy source	Controllability	Main correlation	Limiting constraint	Commercial implication
Thermal	Fuel	Usually high if available	Fuel and outage risk	fuel, ramping, emissions, heat-rate economics	Can sell firmer shapes, but fuel and carbon exposure matter.
Reservoir hydro	Stored water	Often high within limits	Hydrology and reservoir policy	inflows, reservoir levels, environmental rules	Valuable flexibility; not infinite energy.
Run-of-river hydro	Water flow	Lower than reservoir hydro	hydrology	current flow	More like weather-dependent generation.

Technology	Energy source	Controllability	Main correlation	Limiting	Commercial implication
Wind	Wind	Low over resource, some curtailment control	regional weather	wind speed, grid	Energy is cheap when windy; shape risk is large.
Solar PV	Sunlight	Low over resource, some curtailment control	daylight and weather	irradiance, grid, inverter capacity	Generation is concentrated; realized value depends on sunny-hour prices.
Battery	Stored electricity	High within state-of-charge limits	market operation, not fuel weather	MWh, MW, efficiency, degradation, warranty	Sells time-shifting and fast flexibility, not net energy creation.
Demand response	Avoided consumption	Depends on process	customer activity	comfort, production, process constraints	Can be very valuable flexibility, but availability is contractual and behavioral.

Use a control-and-scarcity distinction: who controls the energy flow, and what runs out first?

Promise Test

Before comparing revenue, ask what each asset can safely promise.

Promise test	Solar PV	Battery	Reservoir hydro	Demand response
Can it choose the hour?	only partly	yes, within state of charge	often, within water constraints	only if the customer process allows
What runs out first?	sunlight	stored MWh / warranty headroom	water / environmental limits	customer tolerance or process flexibility
What is risky to sell alone?	flat delivery outside sunny hours	long-duration energy	unlimited energy	unlimited interruption rights

This test is deliberately simple. It keeps a model from treating all `100 MW` labels as equivalent before the delivery promise has been specified.

Examples

A 100 MW gas turbine and a 100 MW solar plant can both inject 100 MW at an instant. But the gas turbine's next question is fuel, start time, ramp rate, and emissions cost. The solar plant's next question is irradiance and hour of day. If a customer asks for 100 MW every hour overnight, the solar plant alone cannot make that promise.

A 100 MW / 200 MWh battery and a 100 MW / 400 MWh battery have the same power rating. They can both move at 100 MW. One has roughly a 2-hour energy tank; the other has roughly a 4-hour tank. Calling both "100 MW batteries" hides the whole economic difference.

In Practice

For a renewable IPP working in Nordic markets, technology characteristics enter the model before finance begins.

In Sweden, IEA PVPS reports that PV produced an estimated 4,087 GWh in 2024 against 4,830.7 MW installed by year-end. As a 2024 Swedish scale anchor, that number is useful without becoming a universal performance rule. It also shows why PV matters commercially before it dominates total energy: even a few percent of consumption can be concentrated into the same hours and locations.

For project modeling, start with:

```
What physical promise can this asset make, and what market or contract sett
```

For portfolio modeling, the second question is:

```
Does this asset add the same risk I already have, or does it add a differen
```

Misconceptions

- “All MW are equal.” MW is power rating. Commercial value depends on timing, controllability, duration, and constraints.
- “Low marginal cost means low risk.” Weather-dependent output can create shape, capture, curtailment, and imbalance risk.
- “A battery creates energy.” A battery stores energy and returns less than was charged, but it may return it at a much more valuable time.
- “Capacity factor is value.” Capacity factor is energy quantity relative to nameplate. Value also depends on price during production.

Current Debates And Caveats

Technology cost, battery deployment, and battery performance benchmarks are date-specific. IEA reports that battery costs fell sharply through 2023, but it avoids turning those global cost facts into Nordic project assumptions.

Claims that storage becomes essential above a particular renewable-penetration threshold are system-specific rather than universal tripwires. Use them as prompts to ask about residual load, grid constraints, flexibility, interconnection, and demand response, not as a standalone planning rule.

Recap

A technology is a set of operating limits around a commercial promise. The same 100 MW label can mean firm controllable production, weather-shaped output, stored flexibility, or avoided demand. The next questions are physical before they are financial: what controls the energy flow, what is it correlated with, what runs out first, and how quickly can it move? Only then does the commercial question become clear: which promise can be sold without being punished when delivery, weather, constraints, or customer behavior arrive?

Chapter 12. Solar Production, Capture Price, And Cannibalization

References: [24] [25] [16]

A solar plant sells when the sun is up, when clouds allow output, and when the market clears the price for those hours. A yearly average price can look comfortable while the plant's own production sits mostly in cheaper hours. Capture price measures that timing problem by weighting the price curve by the plant's generation. The same mechanism explains why adding more solar can change the value of solar itself. When many plants produce together, they can push down the prices in the same hours they need to sell.

Opening Puzzle

A solar plant sells all its power into a market whose average price is `60 EUR/MWh`. Does the plant earn `60 EUR/MWh`?

First-Principles Model

The simple average price is sufficient only if generation is spread like the market average. Solar revenue is computed hour by hour:

```
revenue = sum over hours of generation in that hour x price in that hou
```

The capture-price form divides that revenue by generated MWh:

```
capture price = sum_t(generation_t x price_t) / sum_t(generation_t)
```

If solar produces mostly at noon, then noon prices matter more than midnight prices. A high evening price does not help much if the plant is not producing then.

Definitions

- **Capture price:** generation-weighted average market price for an asset or technology over a period.
- **Capture rate:** capture price divided by the simple average spot price for the same geography and period.
- **Cannibalization:** value erosion that can occur when many similar generators produce at the same time and depress prices during their own production hours.
- **Residual load:** load remaining after subtracting specified variable renewable generation, usually wind and solar. The exact subtraction convention must be stated.
- **Curtailment:** reduction of output below available generation, for economic, grid, or system reasons.
- **Negative price:** market price below zero, meaning injection pays the market rather than receiving money, before considering subsidies, hedges, or contract settlements.
- **Shape risk:** risk that production or consumption timing differs from the contracted or hedged shape.

Numerical Intuition

- Sweden PV scale: IEA PVPS reports 4,830.7 MW installed PV capacity at end-2024.
- Sweden PV output: IEA PVPS estimates 4,087 GWh PV generation in 2024, equal to 2.4% of Swedish electricity consumption.
- Illustrative numbers below are invented for teaching and are not market data.

Worked Estimate

Illustrative two-hour market:

Hour	Spot price	Solar generation
Noon	20 EUR/MWh	1 MWh
Evening	100 EUR/MWh	0 MWh

Time-weighted average price:

```
(20 + 100) / 2 = 60 EUR/MWh
```

Solar capture price:

```
(1 x 20 + 0 x 100) / (1 + 0) = 20 EUR/MWh
```

Capture rate:

```
20 / 60 = 33%
```

The solar plant earns the prices in its own production hours, not the simple average price.

Equations

```
Capture price = sum_t(generation_t x price_t) / sum_t(generation_t)
```

Units: if generation is MWh and price is EUR/MWh, the numerator is EUR and the denominator is MWh, so the result is EUR/MWh.

```
Capture rate = capture price / average spot price
```

This is dimensionless. A capture rate below 1 means the plant earned less than the simple average market price over the same period.

Mechanism

Solar has three features that matter commercially:

1. It has very low short-run marginal cost once built.
2. It produces in a correlated pattern: daylight, weather, season.
3. New solar plants in the same region often add output in the same hours as existing solar plants.

When solar output is small relative to load and flexibility, solar may simply displace more expensive generation. But as solar output becomes large in some hours, residual load falls. If demand, exports, storage, and flexible load cannot absorb the energy, the clearing price in those hours can fall.

This mechanism is cannibalization: the asset partly helps create the prices it is exposed to.

Curtailment is related but not identical:

- Economic curtailment: the plant is out of the money at the clearing price or self-curtails to avoid bad prices.
- Grid curtailment: output is reduced because local or system network constraints bind.
- Flexibility-driven curtailment: output is reduced to provide balancing or reserve value.

Those three have different economics. Treating all curtailed MWh as the same hides whether the problem is market price, network physics, or flexibility value.

Examples

A merchant solar plant cares about capture price directly. A baseload PPA seller cares because it may need to buy power in non-solar hours and sell solar output in solar hours. A customer buying an as-produced solar PPA cares because the contract may deliver cheap clean energy, but not necessarily when the customer consumes.

Add a battery and the question changes:

```
Can the battery move enough solar MWh from low-price solar hours to hig
```

The dispatch model gives the mechanism.

In Practice

For a Swedish solar developer, 2024 PV was still small in annual-energy terms: IEA PVPS estimates PV at 2.4% of Swedish electricity consumption. But annual share can understate shape pressure. Solar output is concentrated in sunny hours, and Swedish grid constraints are geographically uneven.

For a PPA product discussion, the important questions are:

- Is the buyer taking as-produced solar shape, or is the seller reshaping it?
- If reshaped, who pays for the missing hours?
- Is the shape cost priced with realistic capture, imbalance, curtailment, and battery assumptions?
- Are capture-rate numbers dated, zonal, and defined consistently?

Misconceptions

- “Average spot price is the solar price.” Solar earns a generation-weighted price.
- “Cannibalization means solar is bad.” Cannibalization means timing matters. Cheap noon energy can still be valuable if a buyer, battery, export path, or flexible load can use it.

- “Negative prices and curtailment are one thing.” Negative prices are a market price. Curtailment is reduced output. They can interact, but they are different mechanisms.
- “A battery automatically fixes capture risk.” A battery helps only within MW, MWh, efficiency, degradation, tariff, and market-spread limits.

Current Debates And Caveats

Practitioner sources cite market-specific capture-rate episodes. Use those episodes to frame dated market analysis; broader claims need the market area, period, source dataset, and capture-rate definition.

Negative-price hour counts are also date-specific. Avoid saying “Sweden has X negative-price hours” unless the source, market area, period, and methodology are locked.

Recap

Solar economics starts by replacing the market average with the plant’s own hours. Capture price asks what prices the plant actually touched, weighted by its MWh in each hour. Cannibalization is the next step: when many similar plants touch the same hours, those hours can clear lower unless demand, storage, exports, or contracts absorb the shape. Any revenue estimate, PPA shape, or battery case that began with average price has to be rebuilt around hourly overlap.

Chapter 13. Battery Storage And Time Shifting

References: [5]

A control room may see prices fall at noon and rise again in the evening. A battery can take energy from one hour and return it in another. The first number people quote, MW, only states the power limit. It does not say how long useful discharge can last, how much energy survives the round trip, or what must be held back for another obligation. The first test is what kind of time-shifting asset the battery is, and what constraints travel with every cycle.

Opening Puzzle

A project description says, “100 MW battery.”

Which specification is missing?

A 100 MW / 100 MWh battery, a 100 MW / 200 MWh battery, and a 100 MW / 400 MWh battery can all discharge at the same maximum power. They cannot sustain that discharge for the same length of time.

First-Principles Model

A battery is a constrained time-shifting asset.

It takes electrical energy from one interval, stores it chemically, and gives electrical energy back later. The useful function is changed timing, bounded by losses, degradation, control rights, and market rules.

Three numbers define the basic state:

```
MW: how fast the battery can charge or discharge
MWh: how much energy it can store
round-trip efficiency (RTE): how much energy survives the round trip
```

Then come the operating constraints: state of charge, usable depth of discharge, degradation, warranty, availability, grid limits, and market rules.

Definitions

- **Battery energy storage system (BESS):** a battery system, power conversion equipment, controls, and supporting infrastructure used to store and return electrical energy.
- **Power rating:** maximum charge or discharge rate, in MW.
- **Energy capacity:** amount of stored energy, in MWh. Nameplate and usable capacity may differ.
- **Battery duration:** energy capacity divided by power rating, in hours.
- **State of charge (SoC):** stored energy at a moment, expressed in MWh or percent of usable capacity.
- **Round-trip efficiency (RTE):** energy discharged divided by energy charged over a cycle.
- **Depth of discharge:** fraction of stored energy that is used in a discharge.
- **Cycle life:** number of charge/discharge cycles before evident deterioration under stated conditions.
- **Degradation:** loss of capacity or performance over time from calendar aging, cycling, temperature, and operating regime.
- **Augmentation:** adding or replacing battery equipment to restore or maintain usable capacity.

Numerical Intuition

- `100 MW / 200 MWh = 2 hours` by definition.
- `100 MW / 400 MWh = 4 hours` by definition.
- For illustration, assume `90%` round-trip efficiency. Project work should use the original equipment manufacturer (OEM) warranty, test data, or a dated technical benchmark.
- Michaelides' storage chapter defines round-trip efficiency as the product of charge and discharge efficiency and gives an 80% storage example. This chapter uses 90% as a round scenario assumption, not as a universal battery benchmark.

Worked Estimate

Suppose a battery charges with `100 MWh` when power is cheap. If RTE is 90%, it can later discharge:

```
Energy out = 100 MWh x 0.90 = 90 MWh
```

If it bought at `20 EUR/MWh` and sold at `80 EUR/MWh`, ignoring degradation, tariffs, availability, fees, and bidding risk:

```
Charge cost = 100 MWh x 20 EUR/MWh = 2,000 EUR
Discharge revenue = 90 MWh x 80 EUR/MWh = 7,200 EUR
Gross spread value = 5,200 EUR
```

The break-even sell price in this illustrative case is:

```
2,000 EUR / 90 MWh = 22.2 EUR/MWh
```

With 90% efficiency, buying at 20 does not require selling above 20; it requires selling above about 22.2 before other costs. Add degradation and tariffs, and the required spread rises.

Equations

```
Duration (h) = energy capacity (MWh) / power rating (MW)
```

Meaning: duration is how long the battery can sustain full power, before usable-energy limits.

```
Energy out = energy in x round-trip efficiency
```

Meaning: if the battery charges 100 MWh and RTE is 90%, the useful discharged energy is 90 MWh.

```
State of charge_next = state of charge_now + charged energy x charge effici
```

Meaning: charging and discharging move the stored-energy state, but losses make the state change asymmetrical.

Mechanism

A battery has four commercial powers:

1. It shifts energy from low-value hours to high-value hours.
2. It responds quickly, which can make it useful for reserves and balancing.
3. It reshapes renewable output, which can support PPA products.
4. It creates optionality: the owner can decide later which market or obligation is worth serving.

Each value source consumes a physical or contractual constraint:

- MWh limits how long the battery can act.
- MW limits how fast it can act.
- RTE means every cycle loses energy.
- Degradation means cycling has a cost even if the market price looks attractive.
- State of charge creates opportunity cost: a full battery cannot absorb cheap energy; an empty battery cannot discharge into a spike.
- Warranty and grid rules may forbid the theoretical optimum.

Examples

A 2-hour battery can be excellent for short evening peaks, reserve products, or smoothing sharp solar ramps. A 4-hour battery may be better when the valuable spread is wider in time or the contract requires longer firming. Neither is intrinsically superior.

If a solar farm produces excess at noon and a PPA promises evening delivery, the battery can shift part of the production into the delivery window. MW defines the maximum shift rate, MWh defines how much energy can be shifted, RTE defines the loss, and degradation defines the operating cost.

In Practice

For an IPP moving from solar into solar-plus-BESS, a common shortcut is to price the battery as a black-box revenue stack.

Start with the physical ledger:

- What is the grid connection import/export limit?
- Is the battery allowed to charge from grid, solar, or both?
- What is usable MWh after reserve margins and warranty limits?
- What SoC must be held for reserve obligations or PPA firming?
- Who controls dispatch: owner, BRP, BSP, optimizer, or customer?

Only then ask which revenues can be stacked. Two revenues that need the same SoC at the same time are not additive.

Misconceptions

- “A 100 MW battery is enough information.” You need MWh, duration, SoC limits, efficiency, degradation, and control rights.
- “Efficiency is a small detail.” It changes the spread required to make time-shifting profitable.
- “Revenue streams are additive.” They compete for the same physical battery, state of charge, grid rights, warranty budget, and control window.

- “Longer duration is always better.” Longer duration costs more and only pays if the market or contract values the extra hours.

Current Debates And Caveats

Battery costs, deployment, and performance change quickly. IEA reports very large global lithium-ion cost declines through 2023 and rapid 2023 power-sector deployment, but those facts do not determine a Swedish project’s economics.

Cycle-life and degradation are central to investment decisions, but no universal LFP cycle-life number is used here. Any later analysis using cycle-life as a hard input should extract original equipment manufacturer (OEM), NREL/ATB, or technical reference support.

Recap

A battery should be read as a physical ledger before it is read as a revenue stack. MW sets the charge or discharge speed, MWh sets the stored energy, duration links the two, and SoC, RTE, degradation, warranty, grid limits, and control rights decide what the battery can actually do in a given hour. The opening question, “For how long?”, is the first test because it exposes whether a battery can serve the obligation being priced or only looks large on a single power rating.

Chapter 14. Nordic Market Structure, Settlement, And Grid Evidence

References: [7] [3] [26] [10] [22] [11] [24]

The Nordic market is useful to study because it combines regional integration with strong local differences. Nord Pool is a major Nordic nominated electricity market operator (NEMO) and price-reference source; Nordic day-ahead trading is coupled through the European market-coupling framework and local bidding areas. eSett gives a common settlement layer for Denmark, Finland, Norway, and Sweden. Each country still has its own TSO, grid planning, tariff design, connection process, balancing-product details, and local network constraints. For a solar or battery project, the word "Nordic" is therefore only a starting geography. The investable object is a specific site in a specific bidding area with specific grid rights, settlement roles, and contract exposure.

Opening Puzzle

A developer brings a `100 MW` solar-plus-storage opportunity in Sweden. The resource estimate is ready, and the first revenue case uses a regional price curve.

Before valuation, the model needs five location facts:

```
country and bidding area
area-price exposure and hedge basis
grid owner, voltage level, and connection status
import/export rights and tariff treatment
BRP/BSP setup, data rights, and settlement perimeter
```

The same physical project can face different economics in SE2, SE4, DK2, southern Finland, or Norway because the binding constraint may be a bidding-area separation, a local grid bottleneck, an import limit for the battery, a tariff classification, or a balance-responsibility arrangement.

First-Principles Model

Start with one delivery hour in one bidding area:

```
local balance =
  local generation
  + imports
  − local demand
  − exports
  − losses
```

Nord Pool's system price is calculated from sale and purchase orders while disregarding available transmission capacity between Nordic bidding areas. Area prices are calculated when transmission constraints and local supply-demand conditions matter. The project therefore has two related price questions:

```
energy value = delivered MWh x relevant area price
basis exposure = area price − financial reference price
```

That basis exposure is why an asset paid an area price can still carry risk under a system-price hedge or a contract priced from a different reference.

Settlement adds a second layer. eSett's Nordic model calculates and settles imbalances for balance responsible parties (BRPs) from trades and realized consumption or production, handles balancing-service settlement for balancing service providers (BSPs), and runs collateral, invoicing, reporting, and market-behaviour monitoring functions. The commercial model has to know where the asset sits in that settlement structure.

Definitions

- **Nordic system price:** Nord Pool reference price calculated without considering available transmission capacity between Nordic bidding areas.
- **Area price:** bidding-area price reflecting local supply, demand, and available transmission capacity.
- **Bidding area:** market area for which a local price is calculated; Nordic bidding-zone definitions sit inside national TSO operation and applicable EU/regulatory review processes.
- **Basis risk:** difference between the price used for physical revenue and the price used for a hedge, PPA, or financial contract.
- **BRP:** balance responsible party; financially responsible for imbalances under its responsibility in the relevant settlement model.
- **BSP:** balancing service provider; qualified to provide balancing capacity or balancing energy to the TSO under the relevant product rules.
- **Connection right:** technical and contractual permission to inject, withdraw, or reserve capacity at a grid connection point.
- **Grid tariff:** network charge whose structure depends on country, network owner, voltage level, tariff year, user type, and charge basis.

Numerical Intuition

The first Nordic numbers to keep in view are structural anchors.

- Nord Pool's bidding-area source captured on 2026-05-16 states that Sweden has four bidding areas, Denmark is split into western and eastern Denmark, Finland is one bidding area, and Norway has five bidding areas at that time.
- IEA PVPS reports Swedish electricity consumption of `136.6 TWh` in 2024 and generation of about `169.6 TWh`.
- IEA PVPS reports Swedish PV generation of `4.087 TWh` in 2024, equal to `2.4%` of Swedish electricity consumption.
- IEA PVPS reports Swedish installed PV capacity of `4.8307 GW` at end-2024.

Convert the Swedish annual anchors into average power:

```
136.6 TWh/year / 8,760 h/year = 15.6 GW average load
4.087 TWh/year / 8,760 h/year = 0.47 GW average PV output
```

The annual PV average is small relative to national average load. The project issue is timing and location: PV arrives in daylight hours, unevenly by season and weather, and in a grid where bidding areas and local connection conditions can separate the commercial value of similar MWh.

Two-Site Comparison

Take two illustrative `100 MW` solar projects with the same annual generation and capex.

Driver	Site A	Site B
Annual generation	100 GWh	100 GWh
Area capture value	45 EUR/MWh	47 EUR/MWh
Expected curtailment/grid loss	2 EUR/MWh	7 EUR/MWh
Grid/tariff/balancing cost	4 EUR/MWh	3 EUR/MWh

Driver	Site A	Site B
Net before opex/capex	39 EUR/MWh	37 EUR/MWh

Site B has the higher area-price value and the lower first-pass net value:

```
Site A: 100,000 MWh x 39 EUR/MWh = EUR 3.9 million/year
Site B: 100,000 MWh x 37 EUR/MWh = EUR 3.7 million/year
```

The numbers are illustrative. The Nordic point is specific: a price-area uplift can be offset by curtailment, grid terms, balancing exposure, or contract basis.

Nordic Structure

The Nordic map has four layers that matter commercially.

Market Layer

Nord Pool is one major route into Nordic day-ahead trading through bidding areas. The system price is a reference for many financial contracts, while area prices can differ when transmission constraints bind. The classic Nordic risk-management problem includes both price level and system-to-area basis.

Physical Layer

The region is integrated, but its physical systems are not interchangeable. Hydropower has historically been central in Norway and northern Sweden. Denmark has a different wind-and-interconnector profile. Finland’s single bidding area does not make a Finnish project commercially equivalent to a Swedish or Danish one. A storage asset near a bottleneck, a load pocket, or an interconnector constraint is exposed to a different problem than the same battery in a hydro-dominated surplus area.

Settlement Layer

eSett provides a harmonized Nordic settlement interface for the participating TSOs. Its model is built around BRPs, BSPs, metered data, trades, imbalance settlement, balancing-service settlement, invoicing, collateral, and market-behaviour monitoring. The asset owner's governance problem must map commercial decisions to settlement evidence.

Grid And Tariff Layer

Grid rights remain national and local. Svenska kraftnät, Fingrid, Energinet, Statnett, DSOs, regulators, and connection agreements decide the usable MW, connection timing, tariff class, loss treatment, curtailment conditions, and metering requirements. ACER is useful for tariff-design categories across Europe, but it does not turn Nordic grid economics into a common tariff table.

Evidence Checklist

A first Nordic project assessment should separate what is common regional structure from what is local evidence.

Layer	Minimum evidence	Why it matters
Market	country, bidding area, system-price or area-price reference, hedge basis	revenue, PPA settlement, and hedge fit
Physical	hydro/wind/solar/load context, interconnectors, suspected bottleneck	capture, curtailment, and storage use case
Grid	network owner, voltage level, connection status, import/export right	usable MW and timing to revenue
Tariff	tariff year, capacity/energy/loss/reactive-power basis, storage treatment	net value of charging, export, and co-location
Settlement	BRP, BSP if relevant, metering grid area, data flow, collateral or invoicing exposure	imbalance cost, reserve settlement, and partner oversight

Layer	Minimum evidence	Why it matters
Contract	PPA zone, certificate claim, financial hedge, dispatch rights, firming source	whether the contract matches the asset's location and control rights

If a memo says "Nordic" but cannot fill this table, it is still at the geography stage rather than the valuation stage.

Example

Compare two project labels:

```
Project 1: Swedish solar-plus-BESS, SE4, firm export, limited import, a
Project 2: Danish solar-plus-BESS, DK2, separate import treatment, cust
```

The labels are already enough to ask different questions.

For Project 1, the first concern may be system-to-area basis, southern Swedish grid constraints, and whether the battery can charge from the grid or only from co-located solar. For Project 2, the first concern may be the DK2 price and interconnector context, the tariff treatment of storage import/export, and whether the product is really solar shaping, customer firming, reserve participation, or merchant arbitrage.

Neither project can be valued from a Nordic average. Each needs an area-price view, a connection-right view, a tariff view, and a settlement/contract view.

In Practice

For a first investment conversation, prepare a one-page Nordic evidence card.

Field	Minimum acceptable answer
Location	country, bidding area, grid owner, voltage level

Field	Minimum acceptable answer
Price exposure	area price, system-price basis if relevant, expected capture issue
Physical constraint	export, import, curtailment, voltage, queue, interconnector, or local load
BESS relevance	shape, arbitrage, reserve, grid constraint, PPA firming, or portfolio hedge
Tariff risk	charge basis and tariff year to verify before quote or investment committee
Settlement and partner	BRP/BSP setup, data rights, and imbalance or activation exposure
Contract implication	what the location changes in the PPA, hedge, certificate claim, or dispatch right

The compact card keeps regional labels from entering a project model as economic assumptions.

Misconceptions

“Nordic means one market.” The region has common market and settlement architecture, while bidding areas, grid rights, tariffs, and product rules remain local enough to change project economics.

“The system price is the asset price.” A physical asset earns or settles against the relevant area, contract, or settlement reference. The system price can be a hedge reference without eliminating basis risk.

“Hydro makes the Nordic flexibility problem simple.” Hydro is central in parts of the region, but location, reservoir conditions, interconnectors, price areas, grid constraints, and product rules still decide value.

“A country surplus means every project is in a surplus location.” Sweden can be a net exporter while a specific bidding area, grid node, or connection point is constrained.

“BRP/BSP is operations detail.” It defines financial responsibility, balancing-service eligibility, data rights, settlement evidence, and partner governance.

Current Claims To Refresh

Refresh these before a live memo:

- bidding-area definitions and any changes by TSO
- tariff year, voltage level, network owner, and storage classification
- connection queue status and conditional-access terms
- imbalance settlement period, gate closures, and settlement handbook version
- reserve-product qualification, procurement, and activation rules
- current area-price, system-price, and hedge-basis data for the delivery period

The structural point is durable: Nordic integration creates common references, while local constraints decide project value.

Recap

A Nordic project becomes model-ready when the regional label is decomposed into bidding area, area/system price exposure, grid rights, tariff treatment, settlement role, and contract basis. Nord Pool explains the system-price and area-price structure. eSett explains the Nordic settlement layer. TSOs, DSOs, tariff documents, and connection agreements explain the usable MW. A serious model keeps those layers separate before it prices the asset.

Chapter 15. Power Market System Map

References: [13] [3] [4] [27]

A solar farm sells tomorrow's output before tomorrow's weather exists. The forecast becomes a day-ahead position, the position becomes a schedule, the hour arrives, meters record what happened, and settlement assigns the difference to someone. The physical MWh and the financial obligation travel through different clocks. Following one forecast error through those clocks shows the market, grid, roles, data, balancing, settlement, and governance in one chain.

Opening Puzzle

A solar farm forecasts `100 MWh` for an hour and sells `100 MWh` day-ahead.

Clouds arrive. The plant produces `70 MWh`.

Where did the missing `30 MWh` go?

First-Principles Model

Electricity has two clocks.

```
physical clock: supply and demand must balance continuously
market clock: forecast -> trade -> schedule -> adjust -> operate -> met
```

Markets organize responsibility around the physical clock under uncertainty.

The basic identity is:

```
measured reality - scheduled or contracted position = deviation to expl
```

Sign conventions and settlement details vary by market rule. The mental model does not: a position is made before the hour, reality is measured after the hour, and the difference becomes money and evidence.

Definitions

- **Forecast:** estimate of future production, consumption, prices, or constraints.
- **Position:** commercial exposure created by trades, contracts, hedges, or schedules.
- **Schedule / plan:** committed or reported position for a delivery period.
- **Day-ahead market:** auction before delivery day that sets prices and traded volumes for future delivery periods.
- **Intraday market:** later trading window used to adjust positions closer to delivery.
- **Balancing:** TSO action and market process used to keep or restore system balance.
- **Imbalance:** difference between scheduled/traded position and actual metered production or consumption under the market rules.
- **BRP:** party financially responsible for imbalances under its responsibility.
- **BSP:** party qualified to provide balancing capacity or balancing energy to the TSO.
- **Metering:** measurement of actual production, consumption, and exchanges used for settlement.
- **Settlement:** financial reconciliation after delivery.

Numerical Intuition

Illustrative one-hour miss:

```
Scheduled sale = 100 MWh
Actual generation = 70 MWh
Shortfall = 30 MWh
```

If the day-ahead price is `50 EUR/MWh`, the scheduled sale is:

```
100 MWh x 50 EUR/MWh = EUR 5,000
```

If the shortfall is corrected or settled at an illustrative `90 EUR/MWh`, the gross cost of the miss is:

```
30 MWh x 90 EUR/MWh = EUR 2,700
```

The net outcome depends on the actual market, contract, and settlement rules. The scale point is enough for Volume I: a forecast miss moves through trading, balancing, settlement, and partner performance.

Delivery-Hour Evidence Trail

A well-governed organization should be able to reconstruct one delivery hour as a dossier.

Layer	Question	Evidence
Forecast	What did we expect before trading?	Weather forecast, production model, load forecast, uncertainty band.
Day-ahead position	What did we sell or buy?	Orders, cleared volumes, area price, hedge/PPA position.

Layer	Question	Evidence
Schedule and nomination	What position entered the market/settlement process?	Nominations, BRP data, market-operator and settlement records.
Intraday repair	What changed closer to delivery?	Updated forecasts, intraday trades, available liquidity, decision logs.
Real-time operation	What did the system and asset do?	Metered output, curtailment instructions, reserve activation, outages.
Balancing	What did the TSO need?	Activated balancing energy, reserve data, BSP settlement if relevant.
Metering	What was measured?	Metered generation/consumption, metering grid area, validation status.
Imbalance settlement	Who paid or received for the deviation?	BRP imbalance calculation, price, fees, collateral effect.
Governance	Was performance explainable?	Benchmark, attribution, partner scorecard, data-quality flags.

The eSett Nordic Imbalance Settlement Handbook supports this chain at institutional level: imbalance settlement establishes financial balance after the operation hour, BRPs are financially liable for imbalances under their responsibility, TSOs procure balancing services from BSPs, and settlement relies on metering, reporting, invoicing, collateral, and market-behaviour monitoring.

Mechanism

Trace the missing `30 MWh`.

1. The forecast creates a trading expectation.
2. The day-ahead trade creates a commercial position.
3. The schedule or nomination makes the position visible to the settlement structure.
4. Intraday trading can reduce the error if the forecast changes early enough and liquidity exists.
5. Real-time balancing handles the residual physical mismatch.
6. Metering turns actual production into settlement data.
7. Imbalance settlement assigns the financial consequence to the BRP under the rules.
8. Governance asks whether the miss was forecast error, market condition, asset constraint , data problem, partner execution, or contract design.

The same MWh can appear in several conversations. The meteorologist sees forecast error. The trader sees a position. The TSO sees system balance. The BRP sees imbalance responsibility. The asset owner sees retained economics. The customer sees whether the PPA promise was kept. The risk function sees evidence.

Example

Add a battery.

The solar farm still produces `70 MWh` instead of `100 MWh`, but a co-located battery has `20 MWh` available and the right to discharge for contract delivery.

The remaining shortfall becomes:

```
100 MWh scheduled − 70 MWh solar − 20 MWh battery = 10 MWh residual
```

That looks better, but it raises new questions:

- Was the battery already reserved for a balancing service?
- Did discharging violate state-of-charge requirements for the next hour?
- Who had dispatch control?
- Does the PPA allow battery-backed delivery?
- What degradation or opportunity cost was incurred?
- Did the BRP see the battery state in time?

Flexibility changes the chain, but it does not remove the need to trace the chain.

In Practice

For an asset owner using outsourced trading or balancing, the governance question is:

```
Given forecasts, prices, asset constraints, contract rights,
and data available at the time, what should a feasible operator have done?
```

That defines the basis for shadow trading. The benchmark must use information available at the time, not perfect hindsight. It also needs the same constraints as the partner: market access, gate closures, SoC, grid limits, PPA obligations, reserve commitments, and risk limits.

The same chain explains why data rights matter. Without forecasts, schedules, bids, trades, meter data, constraint logs, settlement records, and partner decisions, the asset owner cannot tell whether money was lost to weather, market conditions, a physical constraint, a bad contract, poor execution, or missing data.

Misconceptions

“Markets and operations are separate.” Market positions become physical obligations and settlement exposures.

“The BRP fixes everything.” The BRP carries defined financial responsibility, but the asset owner may retain commercial exposure through contracts and partner agreements.

“A BSP revenue is free upside.” Reserve commitments consume availability, state of charge, control rights, and sometimes energy headroom.

“Settlement is only back-office accounting.” Settlement is where physical deviations become money and governance evidence.

“A forecast error is just a model-quality issue.” It becomes a commercial issue when it changes trades, imbalance costs, customer delivery, or partner performance.

Current Debates And Caveats

Exact market time units, gate closures, platform participation, reserve products, imbalance pricing components, and settlement data models are dated claims. Operational work should cite the current official market-operator, TSO, and settlement sources for the relevant market and delivery period.

The illustrative imbalance calculation uses simplified signs and prices. Nordic settlement has specific data models and sign conventions, so operational work must use the current official definitions.

Recap

One missing MWh can be followed through the system: forecast, trade, schedule, repair, operate, meter, settle, and review. The missing MWh is a difference between a position and measured reality. Markets make that difference governable by assigning responsibility, pricing corrections, and creating evidence. Once the reader can draw that chain, the separate pieces of Volume I become one operating sequence.

Volume II: Commercial Analysis, Contracts, And Asset Decisions

Chapter 1. Asset Revenue And Cash Flow

References: [11] [24] [15]

A customer wants a smoother power shape than a solar plant naturally produces. The site can be technically sound, the panels can deliver, and the headline price can look reasonable, yet the deal can still disappoint once weather, timing, grid costs, balancing exposure, debt service, and the merchant tail pass through the model. For an independent power producer (IPP) developing solar plus battery storage, the model has to show how physical performance becomes cash that can pay costs, support debt, and leave value for equity. It connects MWh, realized prices, contracts, grid constraints, and financing promises.

Opening Puzzle

Two solar sites each expect `100 GWh/year` of output.

Site A has higher yield. Site B has lower yield, but better realized prices, lower grid charges, stronger contract cover, and a lender case that supports more debt.

The investment committee prefers Site B.

Which variables made the lower-energy site the stronger financial asset?

First-Principles Model

Start with one hour.

```
cash_in_hour_h = generation_h x realized_price_h
```

Units:

- `generation_h`: MWh
- `realized_price_h`: EUR/MWh
- `cash_in_hour_h`: EUR

Then add the costs and exposures that change collectible cash:

```
project_revenue = contracted_revenue + merchant_revenue + certificate_r

operating_cash_before_debt =
 project_revenue
 - grid_and_balancing_costs
 - operating_costs
 - land_and_insurance
 - taxes_or_fees
```

The physical asset produces MWh; finance cares which euros come from a PPA, merchant sale, guarantee of origin (GO) or certificate settlement, or battery service, because the uncertainty and control rights differ.

Definitions

Capex is the upfront investment required to build or acquire the asset. For solar this includes modules, inverters, mounting, electrical works, development, grid-connection-related costs where applicable, and contingency. For a battery energy storage system (BESS), it includes the battery system, power conversion, controls, safety systems, civil works, grid interface, and often augmentation planning.

Opex is recurring cost: operations and maintenance, asset management, insurance, land lease, monitoring, and other ongoing costs.

EBITDA is earnings before interest, taxes, depreciation, and amortization. In project-model language, it is often a rough operating-profit layer before financing and tax details.

Net present value (NPV) is the discounted value of future cash flows minus the initial investment:

```
NPV = sum_t(FCF_t / (1 + r)^t) − initial_capex
```

Units:

- `FCF_t`: EUR in period `t`
- `r`: discount rate per period
- `NPV`: EUR

Internal rate of return (IRR) is the discount rate that makes NPV equal zero. It is the rate implied by the modeled cash-flow path.

Debt service coverage ratio (DSCR) is:

```
DSCR_t = cash_available_for_debt_service_t / scheduled_debt_service_t
```

Units:

- **numerator:** EUR per period
- **denominator:** EUR per period

- **DSCR:** ratio

Merchant tail is the part of the asset life after the contracted phase, or the uncontracted slice during the contracted phase, where revenues depend more directly on market prices, capture, curtailment, balancing, and tariff exposure.

Cost of capital is the return required by the people supplying money to the project. Jenny Chase’s solar-finance discussion makes the important renewable point: when most costs are upfront, financing cost can be one of the main determinants of energy cost.

Debt is money lent to the project. It is usually cheaper than equity because debt is paid before equity, but it also creates fixed obligations.

Equity is ownership capital. Equity receives what remains after operating costs, debt service, reserves, and other senior claims. It is riskier, so it usually requires more upside.

Weighted average cost of capital (WACC) combines the cost of equity and debt after weighting by their share in the capital structure. The number changes with market rates, contract quality, project risk, tax, country, and lender/investor appetite.

Levelized cost of energy (LCOE) is a discounted cost-per-MWh metric. It is useful for cost comparison, but it does not by itself price shape, capture, credit, imbalance, certificates, grid constraints, or merchant-tail risk.

Numerical Intuition

Use these as project-model sanity checks:

- `1 MW x 1 hour = 1 MWh`.
- `1 GW average x 8,760 hours/year = 8.76 TWh/year`.
- IEA PVPS reports Sweden's 2024 average PV yield at `850 kWh/kWp`; a round `1 GWh/year per megawatts peak (MWp)` is a first-pass simplification near the same order of magnitude. Site-specific irradiation, losses, degradation, and availability replace this round anchor in an investment model.

The commercial anchor is sensitivity:

```
annual cash-flow sensitivity = exposed MWh/year x assumption change
```

The numbers in this section are unit and sensitivity anchors. Deal benchmarks for DSCR, leverage, PPA tenor, WACC, capex, and opex have to come from the specific project, lender, country, contract, and date.

Worked Estimate

Illustrative asset:

- Solar size: `100 MWp`
- Annual yield assumption: `1 GWh/year per MWp`
- Annual generation: `100 MWp x 1 GWh/MWp-year = 100 GWh/year`
- Realized price assumption: `50 EUR/MWh`

```
gross_revenue = 100 GWh/year x 1,000 MWh/GWh x 50 EUR/MWh
 = EUR 5,000,000/year
```

Now test sensitivity. A `5 EUR/MWh` difference in realized price is:

```
100,000 MWh/year x 5 EUR/MWh = EUR 500,000/year
```

Capture price and shape deserve explicit sensitivities. A small price-per-MWh error becomes a large annual cash-flow error when multiplied by project volume.

Mechanism

The cash-flow model has seven conversion layers.

1. Physical production: irradiance, wind, availability, degradation, clipping, losses, and curtailment decide MWh.
2. Market realization: the timing and location of MWh decide capture price alongside average price.
3. Contract conversion: PPA, hedge, contract-for-difference (CfD)-like settlement, floor, cap, or merchant exposure decide which cash flows are stabilized and which remain open.
4. Grid conversion: connection cost, tariffs, subscribed capacity, loss charges, excess charges, and curtailment rights convert location into cost and operational constraints.
5. Operating conversion: opex, insurance, land, asset management, and balancing costs remove cash before debt.
6. Financing conversion: debt service takes priority over equity distributions.
7. Residual conversion: taxes, reserve accounts, distributions, refinancing, repowering, and merchant tail decide equity value.

The model reveals which assumptions matter enough to change the decision.

Example: The Lower-Yield Site Wins On Realized Cash Flow

Suppose Site A produces `100 GWh/year` and realizes `45 EUR/MWh`. Site B has better timing, grid position, and contract cover, so it realizes `55 EUR/MWh`.

The break-even generation for Site B is:

```
break_even_generation_B =
  100,000 MWh/year x 45 EUR/MWh / 55 EUR/MWh
  = 81,818 MWh/year
  = 81.8 GWh/year
```

Site B can be a stronger financial asset even with lower yield:

Site	Generation	Realized Price	Gross Revenue
A	`100 GWh/year`	`45 EUR/MWh`	`4.50 million EUR/year`
B	`90 GWh/year`	`55 EUR/MWh`	`4.95 million EUR/year`

The lower-yield site wins when realized-price gain, grid-cost saving, contract quality, and debt headroom outweigh the lost MWh. The investment memo should therefore show both expected MWh and break-even MWh.

In Practice

For a solar-plus-BESS IPP, the project model has to answer a practical question:

What changes if the commercial team promises a customer a smoother shape than the solar plant naturally produces?

The model should separate:

- project value: what the asset earns under feasible operation

- customer value: what the buyer gets from price predictability, certificate treatment, and load matching
- portfolio value: what the owner gains from netting, optionality, hedge fit, and shared grid capacity
- lender value: what part of cash flow is contracted enough to support debt

An investment committee should receive the link from physical assumptions to risk-bearing cash flows alongside the IRR.

For a shaped customer offer, the same logic applies to the promise rather than the site. Show the MWh that the plant naturally produces, the MWh the customer wants in each period, the cost of filling the gap, and the portion of cash flow stable enough to support financing.

Misconceptions

IRR is the model's answer to the assumptions, not a property of the asset.

LCOE can help compare cost of production, while a PPA price must also reflect shape, credit, imbalance, tenor, settlement, certificates, and risk allocation.

Contracted revenue is not risk-free revenue. A contract can fix price while leaving volume, shape, balancing, curtailment, credit, or operational risk exposed.

Grid cost can change the optimal size, connection design, BESS duration, curtailment strategy, and PPA product.

Caveats / Current Debates

Finance thresholds are project-, lender-, jurisdiction-, contract-, and period-specific. Generic claims such as “bankable DSCR is X” or “normal leverage is Y” would be false precision without deal-specific evidence.

Merchant-tail value is especially assumption-sensitive. It depends on price formation, capture erosion, curtailment, technology degradation, tariff design, repowering options, and residual contractability.

Recap

A project model translates physical performance into risk-bearing cash flow. Start with MWh, multiply by realized EUR/MWh, then pass the result through contracts, grid costs, opex, balancing exposure, debt service, taxes, reserves, and merchant-tail assumptions. NPV, IRR, and DSCR are outputs of that chain, not independent facts about the asset. The smoother customer shape from the opening problem becomes a cash-flow conversion test: which physical, market, contract, grid, and financing assumptions decide whether the offer creates value or only moves risk into a place the model has not shown?

Chapter 2. Grid Connection And Tariff Economics

References: [8] [9] [28] [29] [11] [30]

A project can clear the revenue model and still fail the grid-access test. The connection offer fixes more than a cable route: it decides how many MW the site can inject or withdraw, which charges sit on capacity rather than energy, and when access can be limited. For a solar, battery, or hybrid project, those terms change the value of the same physical asset. The commercial question is whether the site has bought the right kind of grid access for the product it wants to sell.

Opening Puzzle

Two identical `50 MW / 100 MWh` batteries sit in the same bidding zone.

Battery A can import and export `50 MW` on a firm connection. Battery B can export `50 MW`, import only `10 MW` from the grid, and faces a capacity charge on subscribed import.

Before either asset trades a single MWh, their revenue stacks and operating constraints differ.

Which line in the connection offer changed the value?

First-Principles Model

Ignore tariff labels first. Ask what the project is buying.

A generator buys the right to put power into a network at a point. A battery may buy both the right to inject and the right to withdraw. A load buys the right to withdraw. A hybrid site may share infrastructure but still face rules that separate generation, storage, and consumption.

The annual grid-cost equation can be sketched as:

```
annual_grid_cost =
 annualized_connection_cost
 + fixed_subscription_charge
 + subscribed_capacity_MW x capacity_charge_per_MW_year
 + exported_MWh x export_charge_per_MWh
 + imported_MWh x import_charge_per_MWh
 + loss_charge
 + reactive_power_charge
 + excess_or_overrun_charge
 + curtailment_or_interruption_cost
```

Units:

- connection cost: EUR/year after annualization
- capacity charge: EUR/MW-year
- energy charge: EUR/MWh
- reactive power: often EUR/MVAr or related tariff unit
- curtailment cost: lost margin, EUR/year

The exact terms vary by country, voltage level, tariff year, and whether the asset is connected to transmission or distribution.

Definitions

Connection agreement: the agreement that defines how an asset connects to the grid, including technical requirements, capacity, cost responsibility, timing, and operational conditions.

Allocated capacity: capacity assigned to a project or connection under a grid process. In the Swedish national-grid source, allocated capacity is conditional and remains valid only if the corresponding production or consumption power is subscribed to in the usage agreement before the deadline in the connection agreement.

Subscribed capacity: the MW level contracted for grid use or tariff purposes.

Construction contribution: a one-off contribution toward grid works needed for connection or reinforcement.

Loss charge: a charge intended to recover network energy losses. Svenska kraftnät's tariff source states that from 1 January 2020, its energy-loss price is based on actual hourly market price per bidding zone plus a supplement charge.

Limited grid access: an arrangement where a user accepts disconnection, interruption, or constrained access under specified conditions, often in return for lower access cost.

Reactive power: AC power component associated with voltage support, measured in var or MVAr. It can matter commercially where tariffs or technical requirements penalize or compensate reactive-power behavior.

Numerical Intuition

Use these as dated examples, not universal tariff levels:

- Sweden, Svenska kraftnät connection process source: feasibility study fee described as `SEK 100,000` fixed plus `SEK 1,200/MW` variable for 2025.
- Sweden, Svenska kraftnät tariff source: from 1 January 2020, energy losses are priced from actual hourly bidding-zone market prices plus a supplement charge.
- Denmark, Energinet tariff catalogue 2026 source: a 2026 consumption-tariff cap is described as `11.5 Danish øre/kWh` after use of saved bottleneck revenues; large consumers above `100 GWh/year` have a reduced energy-based system tariff in that source. This is a transmission-tariff example, not a total delivered-energy network charge.
- Finland, Fingrid main grid service pricing 2026 source: all energy-based fee components are calculated hourly by summing 15-minute interval meter data; facilities of at least `1 MW` are central to the generation, storage, and hybrid tariff categories in the excerpt. Project work still needs the current tariff sheet, connection category, and metering setup.

These examples cannot be compared directly without a country sheet: currency, voltage level, user class, tariff year, and inclusion/exclusion of DSO charges differ.

Worked Estimate

Illustrative connection:

- One-off connection/reinforcement cost: `EUR 3 million`
- Economic life for annualization: `20 years`
- Simple straight-line annualization for intuition: `EUR 3 million / 20 = EUR 150,000/year`
- Project output: `100 GWh/year`

```
connection_cost_per_MWh = 150,000 EUR/year / 100,000 MWh/year
 = 1.5 EUR/MWh
```

Now change only output. If curtailment or a lower-capacity-factor design reduces output to `60 GWh/year`:

```
150,000 EUR/year / 60,000 MWh/year = 2.5 EUR/MWh
```

Fixed grid costs become heavier when fewer MWh carry them. Grid cost changes the economic value of production shape, curtailment, and battery operation.

Mechanism

Grid economics has four layers.

The first layer is access. Can the project connect, and by when? A cheap site with no timely capacity can be more expensive than a worse-looking site with a viable connection.

The second layer is size. MW is not a cosmetic input. A 100 MW export right, a 100 MW import right, and a 100 MW bidirectional right may create different network costs and different operating freedom.

The third layer is usage. MWh-based tariffs, losses, and energy charges turn operating patterns into cost. A battery that imports during some hours and exports during others can face both sides of the tariff design unless the regime nets or classifies flows differently.

The fourth layer is conditionality. Limited or conditional access may be cheaper but changes dispatch. Compare the full net value of each access design:

```
net_value = market_value − tariff_cost − cost_of_constraints
```

Connection timing belongs inside the same assessment. A higher-margin site that connects late can lose to a lower-margin site that can be built sooner.

Illustrative timing test:

Site	Annual gross margin	Firm connection	Timing implication
A	4.0 million EUR/year	2030	higher margin but delayed
B	3.8 million EUR/year	2028	lower margin but executable earlier

The undiscounted gross margin delayed by choosing A over B is:

```
2 years x 4.0 million EUR/year = 8.0 million EUR timing exposure
```

This first-pass alarm ignores capex timing, discounting, taxes, development cost, financing, and option value. Its purpose is narrower: a `0.2 million EUR/year` margin advantage cannot be discussed without the two-year delay, grid-right certainty, and opportunity cost of development resources.

Example: Co-Located Solar Plus BESS

Suppose a solar plant has `100 MWp` of panels but only `80 MW` of export capacity. Without a battery, some high-output hours may be clipped or curtailed at the grid limit. With a battery, some clipped energy can be shifted.

The battery is valuable if:

```
value_of_shifted_energy + flexibility_revenue
>
battery_capex_and_opex
+ efficiency_loss
+ degradation_cost
+ additional_grid_tariffs
+ opportunity_cost
```

The grid term can flip the answer. If charging from solar behind the same connection is treated differently from importing from the grid, or if storage capacity is charged separately, the optimal BESS size can change.

In Practice

For a solar or hybrid project, and especially when comparing Nordic countries, turn “what is the tariff?” into specific evidence questions:

- Is this transmission or distribution connected?
- Is the project a generator, load, storage asset, or hybrid under the relevant tariff?
- Are import and export measured separately or netted over a defined interval?
- Is the connection firm, conditional, limited, interruptible, or subject to curtailment?
- Is the cost mostly one-off, annual MW, variable MWh, loss-linked, reactive-power-linked, or overrun-linked?
- Does co-location reduce network use or merely move cost into another tariff category?

Turn the answers into a site assessment before ranking projects.

Site-Assessment Field	What To Record
net MWh	production after expected curtailment or constraint losses
net EUR/MWh	market value after grid, tariff, loss, and balancing costs
annual margin	first-pass gross margin before capex and opex
firm connection date	when the MW right can actually be used
export right	firm, conditional, staged, or constrained injection capacity
import right	whether BESS charging from the grid is allowed, limited, or prohibited
tariff/capacity exposure	fixed MW charge, MWh charge, loss charge, overrun, or special storage treatment
portfolio fit	whether the site adds or reduces zone, shape, and partner concentration
decision meaning	why this site is priority, hold, redesign, or reject

The commercial model should show the site as a grid-rights object, not as a latitude-longitude point alone.

The IFC utility-scale solar guide adds the development diligence layer that often decides whether the grid assessment is real. A useful site memo should attach the grid connection feasibility evidence, land and access constraints, geotechnical and environmental studies, water and soiling assumptions, permitting path, and construction access. Those items do not replace the tariff model; they decide whether the tariff and connection assumptions can be executed on the ground.

Misconceptions

“The tariff is small” can be false when the tariff is capacity-based and the project has low utilization.

“Co-location avoids grid cost” is too broad. Co-location may reduce some costs, create other costs, or be neutral depending on metering, legal classification, voltage level, and tariff design.

“A connection right is permanent value” is too broad. Some capacity is conditional, milestone-based, or tied to subscription and usage agreements.

“Country A is cheaper than country B” is unsafe without matching year, voltage level, customer class, currency, MW/MWh basis, and whether DSO costs are included.

Caveats / Current Debates

ACER’s 2025 tariff report frames European tariff reform as an active design problem: power-based charges, time signals, locational signals, flexible connection agreements, and cost-reflectivity are all in motion. That means a project model using current tariffs should show a change-case alongside the base-case.

The examples above are transmission-level or national-grid examples. They do not settle distribution-level connection costs, so a project model needs the relevant DSO tariff, connection, and metering rules.

Recap

Grid access is a priced operating right. The connection determines how much power can move, when that right can be constrained, and which costs attach to MW, MWh, losses, reactive power, or overruns. Two identical batteries can diverge before their first trade. A useful project model annualizes one-off costs, separates capacity from usage charges, and prices curtailment or interruption as part of the dispatch problem.

Chapter 3. Solar Asset Cash Flow

References: [24] [15] [30] [31] [16]

An investment committee receives three commercial designs for the same solar project. The first sells raw solar output. The second sells a shaped product and buys missing hours from the market. The third adds a battery so some output can be shifted into the buyer's higher-value hours. The annual generation line barely changes across the three cases. The cash-flow risk changes everywhere: capture price, curtailment, grid charges, balancing, contract penalties, battery degradation, financeability, and the merchant tail.

The investment question is which design turns this site, grid connection, and contract opportunity into the best risk-adjusted cash flow.

Opening Puzzle

A `100 megawatts peak (MWp)` solar project can be offered three ways:

- sell as-produced solar
- sell a shaped daytime/evening profile
- add a `40 MW / 80 MWh` BESS and sell a higher-value shaped profile

The solar-only case has the lowest capex. The shaped case earns a higher contract price, but it must buy replacement energy when the plant is short. The hybrid case adds capex and degradation, but it can reduce curtailment and source less replacement energy in expensive hours.

The investment question is whether the next obligation or asset added to the project improves cash flow after pricing the risk it creates.

First-Principles Model

Start with the project cash-flow stack:

```
project_cash_flow =
  contracted_revenue
  + merchant_revenue
  + certificate_or_attribute_revenue
  - grid_and_tariff_cost
  - balancing_and_imbalance_cost
  - curtailment_value_loss
  - shape_and_firming_cost
  - operating_cost
  - BESS_degradation_and_augmentation_cost
  - debt_service_or_capital_charge
```

The hourly revenue term is still the foundation:

```
hourly_market_value = generation_h x spot_price_h
capture_price = sum_h(generation_h x spot_price_h) / sum_h(generation_h
capture_rate = capture_price / average_spot_price
```

For a shaped PPA, add the obligation:

```
shape_shortfall_h = max(contracted_shape_h - delivered_generation_h - B
shape_surplus_h   = max(delivered_generation_h + BESS_discharge_h - con

shape_cost =
  sum_h(shape_shortfall_h x replacement_price_h)
  - sum_h(shape_surplus_h x resale_price_h)
  + imbalance_and_transaction_costs
```

For a battery, the dispatch must respect physical limits:

```
0 <= charge_h <= import_limit_MW
0 <= discharge_h <= export_limit_MW
```

```
0 <= state_of_charge_h <= usable_energy_MWh
state_of_charge_{h+1} = state_of_charge_h + charge_h x efficiency - dis
```

The BESS creates value only where its feasible dispatch beats its cost and opportunity cost:

```
incremental_BESS_value =
  avoided_shape_cost
  + recovered_curtailment_value
  + merchant_or_reserve_value
  - degradation_and_augmentation
  - extra_grid_and_market_costs
  - capital_charge
```

Commercial Test

Compare designs on cash-flow quality as well as annual MWh.

Design	Main value source	New risk created	First test
Solar-only as-produced	low-cost generation sold as produced	capture discount, curtailment, merchant-tail exposure	does capture price after grid and balancing clear the project hurdle?
Solar shaped without BESS	higher contract usefulness	replacement energy, basis, imbalance, delivery penalties	does the buyer premium exceed expected shape cost plus risk margin?
Solar plus BESS	shape value, curtailment recovery, optionality	capex, degradation, dispatch conflict, control rights	does the battery clear its own marginal hurdle under realistic dispatch rights?

The model should show the reason each case wins or loses. A hybrid design that only wins because all battery revenues are stacked without conflict should be rejected.

The IFC utility-scale solar guide is useful here because it ties the project model to development evidence. Before an investment case accepts annual yield, capex, opex, or schedule, it should show the solar resource assessment, loss assumptions, grid-connection feasibility, engineering, procurement, and construction (EPC) allocation, operations and maintenance (O&M) allocation, permitting status, construction risk, and financing due diligence. A clean cash-flow table is weak if the underlying development packet cannot support the inputs.

Numerical Intuition

Scenario assumptions for one investment test. They are illustrative; current market inputs must be refreshed before use in a real case. The Swedish PV deployment and generation scale anchors come from IEA PVPS Task 1 Sweden 2024; the design sensitivities below are scenario assumptions for teaching.

Item	Assumption
Solar size	`100 MWp`
Annual yield	`1 GWh/year per MWp`
Annual solar generation	`100,000 MWh/year`
Average spot price	`60 EUR/MWh`
Solar capture rate, base	`0.80`
Grid and balancing cost, as-produced	`4 EUR/MWh`
Solar-only annual operating cost	`1.0 million EUR/year`
Solar-only annual capital charge	`3.2 million EUR/year`

Solar-only base case:

```
capture_price = 0.80 x 60 EUR/MWh = 48 EUR/MWh
gross_market_revenue = 100,000 MWh x 48 EUR/MWh = 4.80 million EUR/year
grid_and_balancing = 100,000 MWh x 4 EUR/MWh = 0.40 million EUR/year

cash_flow_before_tax =
  4.80 - 0.40 - 1.00 - 3.20
  = 0.20 million EUR/year
```

The project is alive, but thin. A capture assumption that moves by a few points changes the investment answer. The comparison below adds an explicit curtailment-value loss to all three designs, so the solar-only result moves from the first test's `0.20m EUR/year` to `0.00m EUR/year`.

Three-Design Comparison

Add two contract alternatives.

Item	Solar-only	Shaped PPA	Solar + BESS shaped
Contracted / monetized volume	`100,000 MWh`	`100,000 MWh`	`102,000 MWh`
Realized contract or market price	`48 EUR/MWh`	`62 EUR/MWh`	`64 EUR/MWh`
Gross revenue	`4.80m EUR`	`6.20m EUR`	`6.53m EUR`
Curtailment value loss	`0.20m EUR`	`0.20m EUR`	`0.08m EUR`
Grid, balancing, transaction costs	`0.40m EUR`	`0.55m EUR`	`0.70m EUR`
Residual firming / replacement cost	`0.00m EUR`	`1.05m EUR`	`0.45m EUR`
Operating cost excluding BESS wear	`1.00m EUR`	`1.05m EUR`	`1.25m EUR`
BESS degradation / augmentation allowance	`0.00m EUR`	`0.00m EUR`	`0.35m EUR`

Item	Solar-only	Shaped	Solar + shaped
Annual capital charge	3.20m EUR	3.20m EUR	4.25m EUR
Cash flow before tax	**0.00m EUR**	**0.15m EUR**	**-0.55m EUR**

Calculations:

```
solar_only = 4.80 - 0.20 - 0.40 - 1.00 - 3.20 = 0.00
shaped = 6.20 - 0.20 - 0.55 - 1.05 - 1.05 - 3.20 = 0.15
solar_BESS = 6.53 - 0.08 - 0.70 - 0.45 - 1.25 - 0.35 - 4.25 = -0.55
```

The shaped product wins in this base case because the buyer premium is large enough to cover residual firming. The battery reduces replacement and curtailment cost, but the annual capital charge and degradation allowance are too high for this contract.

Sensitivities That Decide The Case

The first sensitivity is solar capture.

Solar capture rate	Solar-only cash flow	Shaped PPA cash flow
0.90	0.60m EUR/year	0.30m EUR/year
0.80	0.00m EUR/year	0.15m EUR/year
0.70	-0.60m EUR/year	-0.05m EUR/year

The shaped product is less exposed to merchant capture, but it is more exposed to replacement cost. It is safer only when the contract premium pays for that exposure.

The second sensitivity is residual firming cost.

```
shaped_cash_flow = 1.20m EUR/year before residual_firming_cost
```

If residual firming rises from `1.05m` to `1.35m EUR/year`, shaped cash flow falls from `0.15m` to `-0.15m EUR/year`. A seller should not quote shape without a replacement-energy and imbalance sensitivity.

The third sensitivity is BESS marginal value.

```
required_BESS_extra_value =
  annual_capital_charge_delta
  + degradation_allowance
  + extra_operating_and_grid_cost
```

In the scenario:

```
required_BESS_extra_value =
  (4.25 - 3.20) + 0.35 + (1.25 - 1.05) + (0.70 - 0.55)
  = 1.75 million EUR/year
```

The battery supplies `1.05m EUR/year` of visible improvement before extra BESS costs in this simple case:

```
price_and_volume_uplift = 6.53 - 6.20 = 0.33
avoided_curtailment = 0.20 - 0.08 = 0.12
avoided_firming = 1.05 - 0.45 = 0.60
extra visible value = 1.05 million EUR/year
```

After extra costs, the battery still misses its hurdle. The recommendation changes if a credible reserve, grid-service, or customer-premium value can be added without double-counting the same MW, MWh, state of charge, or cycle budget.

The fourth sensitivity is merchant tail. A 15-year PPA on a 30-year solar asset leaves half the life exposed to future prices, capture, curtailment, repowering, and degradation. The base case should show contracted-period cash flow and residual value separately:

```
project_value = contracted_period_value + merchant_tail_value
```

The merchant tail is usually a scenario range, not a single optimistic number.

Decision Frame

The investment recommendation should answer five questions.

1. Does the site clear its hurdle under conservative capture, grid, and curtailment assumptions?
2. Does the shaped PPA premium exceed residual firming, basis, imbalance, and transaction costs?
3. Does the battery clear its marginal hurdle after degradation, augmentation, and opportunity cost?
4. Which cash flow can a lender or investment committee treat as contracted, observable, or merchant upside?
5. Which assumption flips the recommendation?

For the scenario above, recommend the shaped PPA without BESS, with three conditions:

- cap residual firming exposure through pass-through, collar, buyer tolerance band, or portfolio hedge
- keep the battery option alive only if dispatch rights and an additional value pool are secured
- show the investment committee (IC) a merchant-tail range that does not use today's capture rate as a permanent fact

Common Failure Modes

"Use average spot price" overstates revenue when solar captures below the time-weighted average.

"Sell shape because the PPA price is higher" hides replacement energy, basis, balancing, and penalty exposure.

“Add BESS because flexibility has value” skips the marginal hurdle. Flexibility has value only when the project owns the right to use it and the value exceeds capital, wear, fees, and opportunity cost.

“The merchant tail is upside” gives the project credit for risk without pricing future capture erosion, curtailment, repowering needs, and market-design uncertainty.

Caveats / Current Debates

Capture rates, curtailment, negative-price exposure, tariffs, and balancing costs are market-, zone-, and period-specific. The mechanisms are durable; the inputs must be dated and sourced in a real investment case.

The IEA PVPS Sweden 2024 figures are useful Swedish scale anchors for PV deployment and generation. They do not establish forward capture rates, curtailment levels, BESS costs, or PPA premiums. Those require current project-specific sources.

Recap

Solar cash-flow decisions depend on hourly production, price, grid, and contract exposure even when the model reports annual totals. The right design is the one whose contract premium, captured value, grid position, firming cost, financing treatment, and merchant-tail range survive sensitivity testing. In the scenario, a shaped PPA beats solar-only because the premium covers residual firming, while the BESS waits for stronger marginal evidence. The conclusion should always name the assumption that can overturn it.

Chapter 4. PPA Risk Allocation

References: [6] [32] [33] [25]

A customer asks for a ten-year solar PPA. The first real decision is what the seller promises in each hour, what the buyer receives as evidence, and whose cash flow moves when generation, load, market prices, certificates, curtailment, payment, or delivery do not line up. The same plant can support very different contracts: all metered output, a fixed schedule, or a financial settlement against a reference price. Each version assigns quantity, shape, balancing, credit, certificates, data, and fallback exposure to different parties. A usable PPA assessment therefore starts with the mismatches before it argues about the label.

Opening Puzzle

A customer, lender, trader, and asset manager all approve the phrase “ten-year solar PPA.”

The first negative-price hour, curtailment event, certificate delay, or payment dispute reveals that each party assumed a different obligation.

When generation, load, market price, certificates, curtailment, payment, or delivery diverge from the plan, which cash flow moves and who owns it?

First-Principles Model

Start with four objects.

```
G_t = asset generation in interval t
Q_t = contract quantity in interval t
P_t = market or reference price in interval t
A_t = attributes or certificates assigned to interval or period t
```

The PPA defines how these objects settle. It may say the buyer takes all metered output. It may say the seller delivers a fixed schedule. It may settle financially against a reference price. It may include certificates in the power price or price them separately. It may require collateral, fallback prices, termination amounts, conditions precedent, commercial operation milestones, or data handover.

A PPA is therefore a bundle of smaller promises:

```
energy promise
price promise
shape promise
attribute promise
credit promise
operational promise
evidence promise
```

Each promise needs an owner.

Definitions

Contract quantity is the MWh specified by the contract. It can be all metered output, part of metered output, or a fixed schedule.

Physical settlement means the contract is tied to physical delivery arrangements. It still requires market roles, metering, nominations, and balancing rules.

Financial settlement means the parties settle a price difference against a reference price. The buyer may still buy physical power elsewhere.

Reference price is the market index or formula used for settlement, such as a day-ahead price over a calculation period.

Certificate treatment defines whether Guarantees of Origin or similar certificates are included, separately priced, transferred later, or excluded.

Credit support means collateral, guarantee, parent support, letter of credit, or another mechanism that reduces counterparty default risk.

Termination amount is the value due if the contract ends early under defined conditions.

Numerical Intuition

Credit risk can look abstract until you put size on it.

Suppose a PPA covers:

```
100 GWh/year
50 EUR/MWh
10 years
```

The contract value is:

```
100,000 MWh/year x 50 EUR/MWh x 10 years
= 50,000,000 EUR
```

RE-Source's July 2025 guide gives about 10 percent of total contract value as a credit-support scale example. Live credit support depends on replacement cost, mark-to-market exposure, potential future exposure, tenor, thresholds, collateral form, parent support, and counterparty quality. In this illustrative case:

```
credit support scale = 50,000,000 EUR x 10%
                     = 5,000,000 EUR
```

A credit-support number at this scale changes the sales conversation. The customer may like the PPA price but dislike posting support. The seller may like the offtaker but dislike unsecured exposure. A "green energy" product has become a balance-sheet negotiation.

The Risk Ledger

For any PPA, build this ledger before arguing about labels.

Risk bucket	Question	Typical consequence if ignored
Quantity	Is the buyer taking all output, part of output, or a fixed schedule?	Shape and volume exposure.
Price	Fixed, floating, indexed, hybrid, floor, cap, or collar?	Wrong hedge value or upside sharing.
Shape	Does `Q_t` follow generation, load, or a schedule?	Market purchases in expensive hours, surplus in cheap hours.
Balancing	Who nominates, who is BRP, who pays deviations?	A fixed price hides imbalance cost.
Curtailment	Is curtailed energy deemed delivered, excused, replaced, or lost?	Revenue and certificate disputes.
Certificates	Are guarantees of origin (GOs) bundled, unbundled, separately priced, or excluded?	Double-claiming or customer-evidence failure.
Credit	What collateral or guarantee supports payment and delivery?	A good price with bad credit may not finance.
Tenor	How long is the promise, and what merchant tail remains?	Revenue certainty may end before asset risk does.
Termination	How is early exit valued?	Mark-to-market exposure.
Data/control	Who has schedules, meter data, forecasts, dispatch rights, and audit trail?	No one can prove performance afterward.

This ledger defines the PPA economics. The legal drafting matters, but the business question is simpler: when production, prices, certificates, or settlement differ from the base case, whose cash flow moves?

Mechanism

The EFET/RE-Source template is useful because it makes the allocation choices visible.

First, settlement must be selected. Physical and financial settlement create different interfaces. Physical settlement creates operational interfaces. Financial settlement creates reference-price and basis questions.

Second, contract quantity must be selected. All metered output, an agreed part of output, and a fixed delivery schedule create different products. The quantity clause is where a large part of shape risk is born.

Third, pricing must be selected. A single fixed price, a period-specific schedule, or an alternative formula will behave differently under volatility, cannibalization, and inflation.

Fourth, certificates must be selected. A certificate can be separately priced or embedded in the energy price. If the buyer needs evidence for a claim, the evidence chain is part of the product.

Fifth, fallback and termination mechanics must be selected. Long contracts need rules for market disruption, early termination, and valuation. These terms allocate payment when the expected settlement process fails.

Example

Take the same 100 MW solar plant.

Contract A says the buyer takes all metered output at a fixed price, with bundled certificates. The buyer receives solar shape. The seller has a clearer revenue line but still cares about availability, curtailment, balancing terms, credit, and certificate issuance.

Contract B says the seller delivers 20 MW flat every hour. The seller now owns the shape problem. In non-solar hours it must buy, hedge, dispatch storage, use portfolio supply, or financially settle. The physical plant is unchanged; the seller has accepted a different delivery obligation.

Contract C is a virtual PPA against a reference price. Now physical supply is separate. The buyer may get hedge value and certificates, but basis risk and accounting treatment matter. The PPA may hedge one price while the buyer pays another.

The same plant can therefore produce three different businesses.

Residual Firming Test

For Contract B, make the delivery obligation visible before pricing:

```
20 MW flat every hour =
  20 MW x 8,760 h/year
  = 175,200 MWh/year
  = 175.2 GWh/year
```

A `20 MW` flat delivery obligation is larger than a `100 GWh/year` solar output assumption before hourly shape is considered. The term sheet must say who supplies the residual `75.2 GWh/year` equivalent, who pays for expensive hours, and whether the exposure is capped, indexed, passed through, or a no-bid condition.

Residual term	Commercial decision
Uncapped seller firming	Seller prices a short position in difficult hours.
Cap	Seller accepts defined exposure; buyer owns the tail beyond the cap.
Index or pass-through	Buyer carries an observable residual cost driver; fixed PPA price should fall.
Reopener or no-bid	Product is not financeable or controllable under current terms.

For a shaped or fixed-schedule product, the term sheet also needs a residual-firming rule. The seller can carry expected residuals inside the fixed price, but the tail cannot be left vague. Common commercial mechanisms are:

Mechanism	What it does	When it is useful
Cap	Seller covers firming up to a defined EUR or MWh limit.	Buyer wants a fixed product but seller will not accept uncapped tail exposure.
Collar	Buyer and seller share outcomes outside a defined band.	Both sides want budget stability without pretending hard-hour costs are fixed.
Index or pass-through	Residual firming follows a market or formula.	The cost driver is observable and neither side wants to warehouse it.
Reopener	Terms are renegotiated if a trigger is breached.	Tariff, certificate, rule, or liquidity changes can alter the product.
Fallback price	A defined price applies when the primary settlement	Market disruption, missing data, or delivery failure

Mechanism	What it does	When it is useful
	path fails.	needs a calculable rule.
Termination amount	Early exit is valued by a defined method.	Credit or performance failure creates a mark-to-market exposure.

These terms decide whether a shaped offer is a bounded product or an uncapped short position in difficult hours.

In Practice

Before a PPA offer goes to pricing, answer these questions:

- What exactly is `Q_t`: metered output, fixed schedule, profile, or buyer load?
- Which market price or reference price settles the contract?
- Who owns positive and negative mismatch?
- Who is BRP, and who pays imbalance?
- Are certificates bundled, separate, or excluded?
- What credit support is required, and can the buyer post it?
- What happens under curtailment, outage, negative prices, and market disruption?
- What data proves performance after delivery?
- Which right does the seller need: dispatch, nomination, curtailment, storage use, or partner audit?

A priced PPA needs these assignments before the offer leaves the commercial team.

Misconceptions

“A PPA fixes revenue.” It fixes some cash-flow rules. Other risks may remain open.

“Physical is safer than financial.” Physical and financial settlement create different exposures. Neither is automatically safer.

“Certificates are a side issue.” If the buyer is paying for a claim, certificate treatment is part of the product.

“Baseload is just a flat price.” Baseload is a delivery-shape obligation. Someone must make the flat shape exist.

“The template standardizes the deal.” A template standardizes places where choices are made. The choices still determine the economics.

Recap

Read a PPA as a ledger of promises against reality. `G_t`, `Q_t`, `P_t`, and `A_t` will rarely line up perfectly, so the contract must say who owns the quantity gap, how the price settles, which certificates prove the claim, what credit support stands behind the promise, and what fallback applies when delivery, payment, curtailment, or market disruption breaks the expected path. Once those assignments are visible, the commercial discussion can focus on the mismatches that are priced, controlled, capped, transferred, or still open.

Chapter 5. PPA Structure And Contract Design

References: [6] [32] [33] [34] [35] [36] [37] [25]

A buyer buys a hedge, a renewable claim, a budget shape, a sustainability evidence package, or some mix of those. The seller accepts volume risk, shape risk, basis risk, certificate obligations, credit exposure, residual firming cost, and operational control requirements. A good PPA structure is the one that matches the buyer’s actual problem while leaving the seller with risks it can price, observe, and govern.

Opening Puzzle

A manufacturer asks for renewable electricity, price certainty, and a product that tracks its load better than raw solar. Its load is flatter than solar and has an evening shoulder. The buyer also wants credible certificate evidence for its public claims.

The seller can offer:

- as-produced solar
- a shaped/profile PPA
- a virtual PPA against a market reference
- a portfolio PPA
- a solar-plus-BESS shaped product

The most attractive name may be the wrong structure if it leaves the buyer with basis risk it does not understand or leaves the seller with firming exposure it cannot control.

First-Principles Model

Every PPA structure answers six questions.

```
1. What volume is contracted?
2. Which price reference settles value?
3. How does the contract relate to the buyer's physical load?
4. Which certificates or evidence move with the energy or hedge?
5. Who owns residual firming, balancing, basis, and volume mismatch?
6. What credit support protects the promised cash flows?
```

The mismatch map:

```
load_mismatch_h       = buyer_load_h − contracted_delivery_h
generation_mismatch_h = contracted_delivery_h − asset_generation_h
basis_mismatch_h      = buyer_exposure_price_h − contract_reference_pri
evidence_mismatch_h   = buyer_claim_requirement_h − certificates_or_evi
credit_exposure       = replacement_cost_if_counterparty_defaults
control_mismatch      = promised_shape − dispatch_and_data_rights_held
```

Structure selection is therefore a risk-allocation decision:

```
recommended_structure =
  highest buyer value
  that the seller can price, hedge, evidence, finance, and govern
```

Buyer Load Before Product Name

Use the buyer’s load and claim requirement to reduce the menu.

Buyer feature	Why it matters	Structure implication
Flat load	raw solar leaves many non-solar hours open	shaped, portfolio, or financial hedge may be more useful

Buyer feature	Why it matters	Structure implication
Daytime-heavy load	solar shape may naturally match part of exposure	as-produced or profile PPA may be enough
Evening shoulder	seller may need BESS, portfolio sourcing, or market purchases	price residual firming explicitly
Exposure in a different bidding zone	hedge may miss delivered-cost risk	basis sensitivity is mandatory
Hourly or strict evidence claim	annual certificates may be insufficient for the buyer's goal	define certificate timing, geography, and audit standard
Weak credit or long tenor	default replacement cost can dominate pricing	credit support, parent guarantee, collateral, or shorter tenor matter

The first pass should make the buyer's residual position visible. If the contract covers `70%` of annual MWh but misses the buyer's expensive hours, the buyer may still be exposed to the risk it wanted to reduce.

Structure Matrix

Structure	Buyer value	Seller exposure	Use when...
As-produced / pay-as-produced	asset link, simple renewable procurement, often lower price	lower shape obligation; still production, availability, certificate, balancing, and credit risk	buyer can absorb renewable shape through supplier, portfolio, or flexible load

Structure	Buyer value	Seller exposure	Use when...
Fixed schedule / baseload	simple volume and budget shape	seller owns generation shortfall and sourcing risk	seller has portfolio, hedge, or storage support and prices the obligation
Profile / shaped PPA	better match to expected load or value hours	forecast, update-rule, residual firming, imbalance, and tolerance-band risk	shape is valuable, measurable, and bounded
Virtual / financial PPA	hedge against reference price plus certificate transfer if agreed	basis risk, accounting treatment, collateral, mark-to-market exposure	buyer wants financial hedge and understands the reference price
Cross-border PPA	access to resource outside buyer location	basis, certificate acceptance, legal and tax complexity	buyer values the resource and accepts cross-border evidence rules
Portfolio PPA	smoother profile from multiple assets	allocation, correlation, transparency, and data-right risk	diversification genuinely reduces residual exposure
Solar-plus-BESS PPA	stronger shape, possible evening coverage	capex, degradation, dispatch priority, opportunity cost, data rights	customer premium beats battery opportunity cost
Threshold hourly-matching product	stronger time-based clean-energy claim	evidence standard, residual firming, fallback procurement, audit burden	buyer will pay for measured hourly performance

Contract terms can blend rows, but the risk questions do not disappear.

Nordic And EU Structure Checks

The 2025 Nordic Energy Research / AFRY PPA report makes the structure menu more precise. It separates several routes that can look similar from the seller's side but allocate balancing, supplier interface, certificate handling, and basis risk differently.

Route	Practical form	Main check before pricing
Utility-buyer PPA	producer sells to a utility or trader	confirm whether the buyer manages balancing, supplier interface, and resale
Utility-seller PPA	utility or trader sells power onward to a corporate buyer	identify which risks remain with the producer after the intermediary shapes or supplies the product
Corporate PPA with sleeving	producer, corporate, and supplier or sleeving entity share settlement roles	map title transfer, imbalance, supplier top-up, certificate treatment, and surplus handling
Financial PPA	parties settle a difference against a reference price while physical supply is handled separately	test basis risk, collateral, accounting, reference-price fit, and certificate transfer

Commission Recommendation (EU) 2026/917 uses the same distinction for policy purposes. Physical PPAs bring delivery, scheduling, balancing, and volume exposure into the contract perimeter. Financial PPAs settle against a reference price and leave physical supply and imbalance management to other arrangements. Pay-as-produced structures move more volume and profile risk to the offtaker; baseload or shaped structures move more of that work to the producer, intermediary, portfolio, or battery.

For a seller, the structure choice should name both the buyer and the route-to-market actor. A shaped corporate PPA with a sleeving supplier can be easier to deliver than a direct physical promise if the sleeving contract clearly assigns residual energy, imbalance, and supplier duties. A financial PPA can be simpler operationally but more exposed to reference-price and collateral questions.

Numerical Intuition

Scenario buyer:

Item	Assumption
Annual load	`120,000 MWh/year`
Solar PPA volume offered	`80,000 MWh/year`
Buyer current expected market cost	`70 EUR/MWh`
As-produced PPA price	`48 EUR/MWh`
Shaped PPA price	`60 EUR/MWh`
Virtual PPA fixed strike	`55 EUR/MWh`
Expected basis cost for virtual PPA	`4 EUR/MWh`
Certificate/evidence budget	included only if specified in term sheet

As-produced solar covers `80,000 MWh/year` on paper, but only `45,000 MWh/year` overlaps the buyer's load hours in this scenario. The remaining `35,000 MWh/year` is paid-for surplus that needs a resale, settlement, basis, and certificate treatment. The buyer still buys `75,000 MWh/year` from its supplier or the market:

```
residual_load = 120,000 - 45,000 = 75,000 MWh/year
```

Before valuing surplus resale or settlement, the buyer's gross cash cost is:

```
ppa_cost = 80,000 x 48 = 3.84m EUR
residual_cost = 75,000 x 70 = 5.25m EUR
gross_cost_before_surplus_value = 9.09m EUR
```

The final average cost then depends on the surplus value:

```
average_cost =
  (9.09m EUR - 35,000 MWh x surplus_settlement_value)
  / 120,000 MWh
```

As-produced buyer economics therefore need both the PPA payment and the surplus settlement line.

A shaped PPA delivers `75,000 MWh/year` into the buyer's target hours at `60 EUR/MWh`, leaving `45,000 MWh/year` residual:

```
shaped_cost = 75,000 x 60 = 4.50m EUR
residual_cost = 45,000 x 70 = 3.15m EUR
total_cost = 7.65m EUR
average_cost = 63.75 EUR/MWh
```

The shaped product is more expensive in expected average cost, but it may reduce volatility and peak-hour exposure. That buyer value should be measured rather than assumed.

For the seller, the shaped premium is:

```
premium = 75,000 MWh x (60 - 48) EUR/MWh = 0.90m EUR/year
```

If residual firming, imbalance, and transaction costs are `0.62m EUR/year`, the shaped structure leaves `0.28m EUR/year` before extra risk margin:

```
net_shape_margin = 0.90 - 0.62 = 0.28m EUR/year
```

If those costs rise above `0.90m EUR/year`, the seller should redesign the product, add a pass-through or collar, reduce the shape, or decline.

For a virtual PPA, basis can erase expected value:

```
expected_hedge_saving = 80,000 x (70 - 55) = 1.20m EUR/year
basis_cost = 80,000 x 4 = 0.32m EUR/year
```

```
net_before_collateral_and_accounting = 0.88m EUR/year
```

The buyer may still prefer it, but the recommendation must state that the hedge settles against a reference price rather than the buyer's delivered cost.

Recommendation Test

Use a weighted decision table when the buyer has multiple objectives.

Criterion	As-produced solar	Shaped PPA	Virtual PPA	Solar+BESS shaped
Load fit	low	medium-high	financial only	high if dispatch rights are secured
Price certainty	medium	high within tolerance bands	high against reference price	high within battery limits
Basis risk	low if same zone physical exposure	low to medium	medium to high	low to medium
Certificate/evidence fit	term-sheet dependent	term-sheet dependent	term-sheet dependent	term-sheet dependent
Residual firming burden	mostly buyer	shared or seller	buyer/supplier	seller/BESS/portfolio
Seller governability	high	medium	medium	medium to low
Credit/collateral complexity	medium	medium	higher mark-to-market risk	medium to high

Scenario recommendation: offer a shaped PPA with tolerance bands and explicit residual-firming rules; keep solar-plus-BESS as an upgrade only if the buyer pays for evening coverage and the seller retains dispatch rights needed to manage degradation and opportunity cost.

The term sheet should specify:

- hourly or monthly shape definition and update rules
- same-zone or cross-zone reference price and basis treatment
- certificate type, geography, vintage, transfer timing, and audit evidence
- residual firming responsibility and fallback procurement rules
- imbalance and balancing cost allocation
- credit support, collateral triggers, and replacement-cost language
- data rights for metering, schedules, certificate tracking, and settlement

Decision Frame

Recommend the structure only after these gates clear.

Gate	Pass condition	Redesign signal
Buyer load fit	product reduces the exposure the buyer actually cares about	annual MWh coverage hides expensive residual hours
Reference-price fit	hedge settles close to buyer's economic exposure	basis dominates expected savings
Evidence fit	certificates and data support the buyer's stated claim	marketing claim outruns evidence
Residual firming	seller can price, hedge, pass through, or cap the obligation	shape premium is smaller than firming cost range
Credit	default and collateral exposure are sized and allocated	long tenor with weak support
Control rights	seller can dispatch, source, or audit what it promises	product relies on partner or battery rights the seller lacks

The final memo should say: “recommend shaped PPA under these terms,” “recommend virtual PPA despite basis caveat,” or “decline until the buyer accepts a narrower promise.” Avoid a menu with no recommendation.

Misconceptions

“The PPA label tells you the economics.” The economics live in volume, reference price, settlement, evidence, firming, credit, and control terms.

“As-produced means the seller has no risk.” Production, availability, certificates, balancing, buyer credit, and market rules still matter.

“A virtual PPA hedges the buyer’s bill.” It hedges a reference price. The buyer’s delivered bill may move differently.

“Hybrid means firm.” Storage improves the toolset. Firmness depends on duration, state of charge, dispatch priority, grid rights, degradation budget, and penalties.

“Certificates are an appendix.” For many buyers, certificate timing, geography, and auditability are part of the product.

Caveats / Current Debates

PPA accounting, certificate acceptance, support-scheme interaction, collateral treatment, and cross-border claims are jurisdiction- and date-specific. RE-Source Platform’s Corporate PPA Guide, July 2025, is useful for corporate procurement framing, while ECTI’s 2026 PPA analysis is useful for structure language such as baseload delivery. A live term sheet still needs current legal, accounting, certificate, and market advice.

Recap

PPA structure selection starts with the buyer’s load, reference-price exposure, evidence requirement, and credit position. The seller then tests whether the promised shape can be produced, sourced, hedged, certified, financed, and audited. In the scenario, a shaped PPA with tolerance bands is the base recommendation because it solves more of the buyer’s problem than raw solar while keeping residual firming governable. Solar+BESS becomes attractive only when the customer premium pays for the battery capacity and dispatch rights needed to deliver it.

Chapter 6. PPA Pricing, Shape Risk, And Customer Value

References: [38] [6] [32] [33] [34] [25]

A customer asks for solar at a fixed price with less hourly variability. Sales needs a EUR/MWh price, the project model has a revenue requirement, and the battery model can smooth some hours but not for free. Pricing starts by specifying what the offer promises: raw generation, a shaped profile, a hedge, certificates, simplicity, or some mixture of all of them. Once the promise is clear, the price has to show who owns each transformation and what it costs. Otherwise the quote can look neat while the risk sits unnamed in the margin.

Opening Puzzle

A customer asks for “solar at a fixed price, but smoother.”

Sales wants a number. The project team has an LCOE. The battery team has an optimizer. The buyer has a budget. The pricing memo needs to answer:

```
What exactly are we promising, and which EUR/MWh pays for each part of the
```

PPA pricing is a quote stack built from more than LCOE plus margin.

First-Principles Model

Use three hourly curves.

```
G_t = asset generation, MWh
Q_t = contracted delivery or settlement quantity, MWh
P_t = market/reference price, EUR/MWh
```

Raw generation has value:

```
raw generation value = sum_t(G_t x P_t)
```

The contract promise has value:

```
contract profile value = sum_t(Q_t x P_t)
```

The gap is the economic work required to transform one curve into another.

```
shape/firming cost
  = market purchases
  + storage losses
  + degradation
  + imbalance and balancing
  + grid and tariff effects
  + hedge cost
  + foregone optionality
  + risk margin
```

Definitions

Quote stack is the decomposition of an offered price into components that can be debated, sourced, and stress-tested.

Seller all-in cost is the cost and retained risk of delivering the promise.

Customer willingness-to-pay is the buyer’s value from the hedge, budget certainty, decarbonization claim, evidence quality, simplicity, and alternatives.

Firming cost is the cost of converting variable output into a firmer or more useful shape.

Reference price is the market index or formula against which a financial or indexed structure settles.

Levelized cost of energy (LCOE) is a cost-of-production metric. It is useful for asset comparison, but a PPA price also has to cover shape, settlement, evidence, credit, balancing, optionality, and retained risk.

Guarantee of origin (GO) is a certificate used in Europe to document renewable attributes. If certificates are included in a product, the term sheet should state vintage, geography, allocation, cancellation, and evidence rules.

Indexation links price to inflation, market indices, capture-rate models, imbalance costs, GO prices, or other defined variables.

Optionality cost is the value of flexibility the seller gives up by committing an asset, battery, hedge, or portfolio to a customer promise.

Numerical Intuition

Build an illustrative quote stack in EUR/MWh.

Component	Scenario value	Why it exists
Raw energy required price	45.0 EUR/MWh	Asset revenue requirement or alternative route-to-market.
Shape/firming	9.0 EUR/MWh	Market purchases, storage use, or portfolio netting.
Imbalance/balancing	2.0 EUR/MWh	Forecast error, nominations, BRP cost, settlement risk.
Grid/tariff/losses	1.5 EUR/MWh	Import/export, losses, reserved capacity, or local charges.
Certificates/evidence	1.0 EUR/MWh	GO or evidence treatment, if included.
Credit/tenor	1.5 EUR/MWh	Collateral, default risk, tenor, termination exposure.
Optionality cost	3.0 EUR/MWh	Flexibility reserved for the product rather than other uses.
Risk margin	2.0 EUR/MWh	Residual uncertainty after modeled costs.

```
offer price = 45 + 9 + 2 + 1.5 + 1 + 1.5 + 3 + 2
            = 65 EUR/MWh
```

If the buyer's willingness-to-pay is 68 EUR/MWh, the buyer surplus is:

```
68 - 65 = 3 EUR/MWh
```

That margin is thin. The offer may be worth making, but it is sensitive to shape cost, grid charges, and credit support.

The shape line should not be guessed. A first derivation comes from residual buckets.

Illustrative shaped product:

- contracted volume: `100,000 MWh/year`
- normal residual energy to firm: `7,500 MWh/year`
- normal residual premium: `45 EUR/MWh`
- stress residual energy to firm: `5,000 MWh/year`
- stress residual premium: `120 EUR/MWh`
- storage losses, degradation, and dispatch fees assigned to shaping: `100,000 EUR/year`

```
base_expected_shape_cost =
  7,500 x 45 + 100,000
  = 437,500 EUR/year
```

The stress bucket is a downside case, not an extra expected line unless the pricing method assigns a probability or risk margin to it:

```
stress_residual_cost = 5,000 x 120 = 600,000 EUR/year
```

The table shows which hours create the price. Price the base expected shape cost separately from the tail: through probability weighting, a risk margin, a cap, a collar, a pass-through, or a reopener. If the stress bucket is wrong, the quote changes; if the battery cannot legally or economically serve the stress bucket, the offer has to change.

Sensitivity

Now change only the shape/firming component.

Shape/firming cost	Offer price	Decision signal
6 EUR/MWh	62 EUR/MWh	Product has room to negotiate.
9 EUR/MWh	65 EUR/MWh	Viable but tight.
12 EUR/MWh	68 EUR/MWh	Break-even against buyer value.
15 EUR/MWh	71 EUR/MWh	No-bid, redesign, or change structure.

This sensitivity table shows which assumption controls the deal.

Pricing Mechanism

Seller cost and buyer value are different.

Seller cost asks:

```
What must we spend, reserve, hedge, collateralize, or risk to deliver this
```

Buyer value asks:

```
What does this product save or solve compared with the buyer's alternatives
```

A product can be cheap to provide but valuable to the buyer. A product can be expensive to provide and still not valuable enough to buy. The commercial task is to find products where the seller’s controllable cost is below the buyer’s willingness-to-pay.

Use this test:

```
seller margin = offer price − seller all-in cost
buyer surplus = buyer willingness-to-pay − offer price
```

A good offer has both. If the seller margin is positive while buyer surplus is negative, the product is unlikely to sell. If buyer surplus is positive while seller margin is negative, the product may sell and then hurt.

For hourly or 24/7-style products, add a premium stack before showing the price:

```
hourly_product_premium =
  expected_firming_cost
  + evidence_and_reporting_cost
  + imbalance_and_basis_cost
  + partner_or_supplier_fee
  + credit_and_collateral_cost
  + risk_capital_charge
  + seller_margin
```

If the buyer pays a premium for hourly matching but the premium covers only expected firming, the seller may still be underpricing the evidence, credit, partner, and tail-risk work. A stronger public claim requires a more expensive control and proof system.

Pricing Structures

Fixed price gives clarity. It also concentrates risk if costs move.

Floating or indexed price moves with a reference. The reference can be a day-ahead price, GO price, inflation index, capture-rate model, imbalance-cost formula, or multi-factor index. The key question is whether the reference tracks the risk being transferred.

Hybrid pricing shares risk. A floor, cap, collar, or upside-sharing formula can make a deal possible by moving tail outcomes between buyer and seller. ECTI's 2026 study gives an example of a capped-upside structure where a buyer accepts a lower strike price but receives only a small share, such as 5 percent, of market upside. Use that as a risk-sharing example, not as a standard term.

Indexation protects long contracts from pretending that money, costs, and risk stay frozen.

Certificate pricing must be explicit. If the certificate value is embedded in the power price, the seller and buyer should still know what evidence is being transferred.

Reference Price And Nordic Liquidity

The Nordic Energy Research / AFRY report gives a useful pricing discipline for Nordic PPAs. For a baseload PPA, the reference price is the baseload electricity price: a time-weighted price for the period. For a pay-as-produced PPA, the reference price is the relevant capture price: a generation-weighted price for that technology and profile.

That distinction changes the quote stack.

Contract shape	Pricing reference	Main residual question
Baseload	time-weighted market price over the delivery period	who sources or sells deviations from the flat block
Pay-as-produced	generation-weighted capture price	whether the buyer accepts the asset's profile and cannibalization exposure
Shaped or profile PPA	value of the promised profile hour by hour	whether shape cost is priced, capped, indexed, or passed through
Financial PPA	reference-market settlement price	how basis, collateral, and accounting treatment affect the buyer's real hedge

The same report notes that forward markets and fundamental models are the two main pricing approaches. Nordic forward-market liquidity is weaker at longer tenors than the term of many PPAs, so a ten- or fifteen-year price usually cannot be read from one clean market curve. A pricing memo should say which inputs come from observable forwards, which come from a fundamental view, and which remain negotiated risk premia.

Example

The buyer wants smoother solar.

Option A is as-produced solar at 51 EUR/MWh. The buyer takes shape risk and uses its supplier or portfolio to cover non-solar hours.

Option B is shaped solar at 65 EUR/MWh. The seller keeps more shape risk and uses BESS, portfolio netting, or market purchases.

Option C is a hybrid formula: 58 EUR/MWh fixed for the as-produced component plus an indexed firming charge based on actual non-solar-hour sourcing cost with a cap.

Which is best?

The answer depends on the buyer’s problem. If the buyer wants a simple budget hedge and can manage shape, Option A may be better. If the buyer wants a cleaner evening profile and values simplicity, Option B may work. If both parties distrust fixed firming assumptions, Option C may be the most honest.

The model should show what each structure makes explicit or hides.

In Practice

A PPA pricing memo should include:

- product definition: as-produced, shaped, fixed, indexed, hybrid, or portfolio-backed
- settlement reference and calculation period
- expected generation and contract profile
- quote stack in EUR/MWh
- owner of shape, imbalance, curtailment, certificates, and credit risk
- sensitivity table for shape cost, price spread, imbalance, grid/tariff, and credit support
- buyer-value argument
- seller margin and retained-risk argument
- assumption that flips the recommendation

If the memo cannot explain the price, the price is not yet ready for a customer.

Misconceptions

“The PPA price is LCOE plus margin.” LCOE is only one input.

“Shape is a sales feature.” Shape is an obligation with a cost function.

“A battery makes firming cheap.” A battery changes the cost function and spends optionality.

“Customer value equals seller cost.” They are two sides of the negotiation and need separate estimates.

“A risk margin fixes uncertainty.” A risk margin without named risks is an unexplained plug.

Recap

The quote stack is the offer’s pricing structure: start with the asset’s raw energy value, add each cost of turning that curve into the contracted profile, then test whether the resulting offer leaves both seller margin and buyer surplus. The same customer request can become as-produced solar, a shaped fixed-price product, or a hybrid formula, but each version moves shape, imbalance, credit, optionality, and evidence risk differently. A defensible price names those movements in EUR/MWh and exposes the assumption that could make the deal fail.

Chapter 7. Investment Committee Proposals

References: [11] [6] [32] [33] [39]

Sales has a customer conversation before finance has a model. The customer wants solar, evening coverage, a price collar, certificates, and budget stability. The commercial team needs to turn that sentence into an offer that can survive modeling, legal drafting, lender questions, operations, and partner execution. The translation layer runs from customer promise, to term-sheet variables, to hourly economics, to an investment committee (IC) packet with a recommendation and a clear assumption that can flip it.

Opening Puzzle

Sales asks:

```
Can we quote a solar PPA with evening coverage and a price collar?
```

The proposal team should first write the promise in operational form:

```
which MWh, in which hours, at which reference price,
with which certificate treatment, backed by which asset or portfolio,
under which dispatch rights, and with which risks retained?
```

Once the promise is written that way, the IC question becomes concrete: approve, redesign, no-bid, or ask for more evidence.

First-Principles Model

Every IC proposal connects four objects:

```
customer promise
asset and portfolio capability
market and grid interface
risk allocation and governance
```

The product is investable only when those objects become hourly cash flows and explicit obligations:

```
scenario_gross_margin =
  contracted_revenue
  + merchant_revenue
  + certificate_revenue
  − market_purchases
  − balancing_and_imbalance_cost
  − grid_and_tariff_cost
  − BESS_degradation_and_opportunity_cost
  − opex
  − credit_and_collateral_cost
```

Each line needs units: MWh, MW, EUR/MWh, EUR/MW-period, fixed EUR, or a contract formula. If a line has no owner or unit, it is not ready for IC.

Definitions

- **Proposal test:** quick check of whether a customer promise is physically, commercially, and contractually plausible.
- **Term-sheet variable:** contract choice that changes cash flow or risk, such as quantity, settlement index, collar, curtailment rule, certificates, credit support, or termination right.
- **Risk memo:** map of what can go wrong, who owns it, how large it could be, and what control or price buffer exists.
- **IC case:** decision package combining commercial terms, model outputs, sensitivities, risks, mitigants, and go/no-go questions.
- **Partner-execution plan:** practical operating model for BRP, BSP, optimizer, supplier, metering, forecasting, nominations, dispatch, and reporting.
- **Scenario gross margin:** revenue minus direct operating, market, balancing, tariff, degradation, credit, and execution costs under a named scenario.

Numerical Intuition

Use the quote-miss equation:

```
annual miss = contract volume x omitted EUR/MWh component
```

For a `250 GWh/year` customer product:

```
250 GWh/year = 250,000 MWh/year
```

An omitted `4 EUR/MWh` shape, balancing, grid, or credit component is:

```
250,000 MWh/year x 4 EUR/MWh = EUR 1,000,000/year
```

The number is simple because the implication is large. A small unit-price omission can become an IC-level error.

Mini IC Case

Illustrative customer request:

- `250,000 MWh/year` shaped solar product.
- Fixed strike with a collar on market settlement.
- Evening coverage from a co-located battery or portfolio.
- Certificates included.
- Seller retains imbalance exposure.

First quote stack:

Component	EUR/MWh	Comment
Project revenue requirement	50	Base asset economics.

Component	EUR/MWh	Comment
Expected shape and sourcing	6	Evening coverage, residual market purchase, storage losses.
Balancing and imbalance	2	Forecast and settlement exposure retained by seller.
Battery opportunity and degradation	3	Flexibility reserved for the customer instead of merchant use.
Grid/tariff and losses	1.5	Import/export, losses, and connection use.
Certificates and evidence	1	Included in product promise.
Credit/collateral and tenor	1.5	Counterparty and duration cost.
Residual risk margin	2	Unmodeled tail and execution risk.

```
offer price = 50 + 6 + 2 + 3 + 1.5 + 1 + 1.5 + 2
            = 67 EUR/MWh
```

Annual contracted revenue at this price:

```
250,000 MWh x 67 EUR/MWh = EUR 16.75 million/year
```

If sales quotes `60 EUR/MWh` because the base project looked competitive, the missing margin is:

```
250,000 MWh x (67 − 60) EUR/MWh = EUR 1.75 million/year
```

The IC discussion should now focus on which component can be reduced honestly: relax evening shape, pass through imbalance, change collar width, reserve less battery capacity, shorten tenor, or change certificate/evidence treatment.

The Translation Stack

1. Write The Promise

Translate customer language into contract variables.

Customer phrase	Proposal variable
“Solar”	Asset source, certificate treatment, additionality/evidence claim.
“Evening coverage”	Hourly quantity, residual sourcing, BESS dispatch priority, market-purchase fallback.
“Fixed price”	Strike price, settlement index, inflation/indexation, imbalance pass-through.
“Collar”	Floor, cap, sharing formula, reference price, settlement period.
“Budget stability”	Volume tolerance, credit support, termination rights, force-majeure and curtailment treatment.

2. Map The Supply

Identify how each promised hour is served:

- project generation
- co-located BESS discharge

- portfolio netting
- market purchase
- hedge or financial settlement
- curtailment or force-majeure rule

The same battery MWh cannot simultaneously be reserved for customer shape, merchant arbitrage, and reserve capacity unless the priority rules and opportunity costs are explicit.

3. Assign Risk Ownership

Use a risk-allocation table before pricing.

Risk	Seller retains	Buyer retains	Shared or indexed
Volume	Contract shortfall and firming	As-produced variability	Tolerance band or true-up
Shape	Evening profile and residual purchase	Raw solar shape	Indexed firming charge
Price	Fixed strike and collar exposure	Floating market exposure	Floor/cap/upside share
Balancing	Forecast error and imbalance	Pass-through imbalance	Shared fee or cap
Curtailment	Deemed delivery or replacement	Lost volume	Rule by cause
Certificates	Delivery and evidence	Separate procurement	Split certificate product
Credit	Counterparty default exposure	Collateral posting	Mutual support thresholds
Dispatch	Seller optimizer	Buyer control	Priority and override rules

RE-Source and EFET/RE-Source template material support the basic contract discipline: corporate PPAs are negotiated instruments where quantity, price mechanism, certificate treatment, settlement, fallback, credit, and termination terms must be explicit. The proposal model must therefore follow the term sheet, not the other way around.

The IC Packet

A clean IC packet should be short but complete.

Page	Content	Approval question
1. Recommendation	Approve, redesign, no-bid, or hold for evidence.	What decision is being asked?
2. Product terms	Quantity, shape, price, collar, certificates, tenor, settlement index.	What exactly is sold?
3. Supply plan	Asset, BESS, portfolio, hedge, market purchase, fallback.	Can we deliver each hour?
4. Economics	Quote stack, base/downside/upside margin, NPV/IRR if project-linked.	Is return adequate for retained risk?
5. Sensitivities	Shape cost, residual purchase, imbalance, tariff, availability, credit.	Which assumption flips the decision?
6. Risk allocation	Risk owner, cap/pass-through, collateral, termination, data rights.	Are unpriced risks acceptable?
7. Execution plan	BRP/BSP/optimizer roles, dispatch priority, reporting, scorecard.	Can operations perform the contract?
8. Open evidence	Tariff source, grid connection, customer load, certificate	What must be verified before binding offer?

Page	Content	Approval question
	standard, legal clauses.	

The IC packet fails if the recommendation could be made without reading the sensitivity page. Its work is to expose the decision, the retained risk, and the assumption that would change the answer.

Decision Frame

Use four gates.

Gate	Pass condition	Failure response
Product clarity	Every customer promise maps to a quantity, price, evidence, or control term.	Rewrite the term sheet.
Deliverability	Each promised hour has a feasible supply or settlement route.	Redesign shape or fallback.
Risk-adjusted value	Seller margin survives the first plausible downside.	Raise price, pass risk through, or no-bid.
Governance	BRP/BSP/optimizer/data rights make performance observable.	Add reporting rights or do not approve.

The final IC answer should name the first assumption that changes the recommendation. Examples: residual firming cost above `10 EUR/MWh`, tariff treatment worse than assumed, customer load shape materially different, BESS dispatch rights unavailable, or credit support unacceptable.

In Practice

For a solar-plus-BESS IPP, the proposal must connect sales, modeling, legal, operations, and partner governance.

If the customer values evening coverage, the model must show when the BESS is reserved for the product and when it is free for merchant or reserve use. If it does both, the term sheet must define dispatch priority and opportunity cost.

If the site is constrained, the proposal must show whether the contract assumes export rights, import rights for charging, curtailment compensation, or a grid/tariff pass-through.

If execution is outsourced, IC should require a benchmark and reporting plan before approval. A good price can still become a bad deal if nominations, dispatch rights, data, or incentives make delivery ungovernable.

Misconceptions

"Sales owns the product and finance owns the model." Contract wording decides cash flows, so product and model have to be built together.

"A risk memo is a list of bad things." A useful risk memo assigns owner, magnitude, mitigant, price, and evidence.

"IC wants one answer." IC needs the tradeoff: what changes if the shape is relaxed, the collar is widened, tenor changes, or dispatch rights move.

"Partner execution can be fixed later." Dispatch rights, nominations, balancing, metering, and reporting can change product economics.

Caveats / Current Debates

PPA practice, hybrid structures, guarantee schemes, certificates, and accounting treatment are date- and contract-specific. Practitioner material is useful for market color and product patterns, but binding terms require the actual contract, legal review, current certificate rules, and current market/settlement sources.

The numerical case is an illustrative investment-committee (IC) test. It is designed to teach structure, not to imply a market quote.

Recap

A sales idea becomes an IC proposal when every part of the customer promise is translated into hourly delivery, settlement, risk ownership, operating control, and evidence. The proposal is ready when IC can see the offer price, the retained risks, the partner-execution plan, and the first assumption that would change the recommendation.

Chapter 8. BESS Revenue Stacks

References: [17] [20] [38] [40] [23] [41] [21] [39]

The project model looks better every time another BESS revenue line is added. Arbitrage fills one row, reserve capacity another, balancing activation another, and PPA firming another. In a real interval, those rows compete for one export limit, one import limit, one state of charge, one cycle budget, one grid agreement, and one control setup. A bankable stack is the set of obligations that can fit together at this site, under this contract, after pricing the best use each commitment gives up.

Opening Puzzle

A project model shows five possible BESS revenues: day-ahead arbitrage, intraday trading, reserve capacity, balancing activation, and PPA firming.

Which of these can one battery actually underwrite together at this site, under this contract, with this grid connection and degradation budget?

A revenue stack is a feasible portfolio of obligations on the same MW, MWh, state of charge, cycles, grid rights, and control rights.

First-Principles Model

A battery earns by moving electricity across time or by being ready to move it. Both actions consume scarce resources.

Scarce resource	What it constrains	Typical revenue affected
Import MW	How fast the battery can charge	arbitrage, curtailment recovery
Export MW	How fast the battery can discharge	peak discharge, upward reserves, PPA shaping
Usable MWh	How long it can sustain a position	arbitrage duration, firming, reserve sustain
State of charge (SoC)	Whether the asset is ready before uncertainty arrives	frequency response, balancing, evening delivery
Cycle budget	How much lifetime is consumed by dispatch	intraday trading, activation, arbitrage
Grid rights	What import/export is physically and contractually allowed	co-location, tariffs, connection value
Control rights	Who is allowed to dispatch or reserve the asset	optimizer, BRP/BSP, offtaker, owner

Arbitrage means charging when electricity is lower value and discharging when it is higher value. Reserve capacity means being paid to keep MW available in case the system needs a response. Activation energy is the energy actually moved if that response is called.

Those definitions are simple. The constraint is that the same battery cannot promise the same scarce resource twice.

If `40 MW` is reserved for a frequency product, that `40 MW` is no longer freely available for an evening discharge in the same interval. If `80 MWh` must be held as upward reserve headroom, that state of charge is not available for a different PPA-shaping promise. If high-frequency trading consumes cycles, those cycles are no longer free in the degradation model.

The physical stack is therefore:

```
feasible revenue stack =
  obligations that fit inside MW, MWh, SoC, cycle, grid, and control co
```

not:

```
sum of every revenue source the asset could theoretically access
```

Nordic reserve markets show why the resource accounting has to be product-specific. The Nordic TSOs’ 2025 battery reserve paper says batteries can participate in fast frequency reserve (FFR), FCR-D, FCR-N, aFRR, and mFRR, but each product asks for a different mix of response speed, endurance, prequalification, and state-of-charge management.

Product	What the battery must reserve	Commercial implication
FFR	very fast upward response for low-inertia periods, with short endurance options	mainly a speed and availability product; hours procured vary with inertia conditions
FCR-D	automatic disturbance response outside the normal frequency band; at least `20 min` endurance at full activation	modest energy need, but power reservation and state of charge still reduce other uses

Product	What the battery must reserve	Commercial implication
FCR-N	symmetric normal-operation response with `1 h` endurance in each direction	more energy-intensive for BESS because up and down capability must both be credible
aFRR	automatic restoration service, procured separately up and down, with Nordic endurance requirements in the `1–4 h` range	competes directly with energy inventory and sustained delivery promises
mFRR	manual restoration with `12.5 min` full activation in the standard Nordic description and shorter energy activation market (EAM) endurance windows	ramp speed alone is insufficient; bid size, activation path, and energy window matter

The same TSO paper warns that prequalified battery volumes do not equal active market supply, because one asset can be prequalified for several products and for both directions. It also states the practical stacking rule plainly: the same capacity cannot be bid into multiple markets for the same delivery period. Product eligibility is therefore the starting point; interval-level opportunity cost is the decision.

Numerical Intuition

Take a `100 MW / 200 MWh` battery with `90%` round-trip efficiency.

One evening has a clean arbitrage opportunity:

- charge `200 MWh` at `25 EUR/MWh`
- discharge `180 MWh` after losses at `95 EUR/MWh`
- degradation cost: `12 EUR/MWh discharged`
- charge-side grid and market fees: `3 EUR/MWh charged`

The arbitrage value is:

```
discharge revenue = 180 MWh x 95 EUR/MWh = 17,100 EUR
charge cost = 200 MWh x 25 EUR/MWh = 5,000 EUR
gross spread value = 12,100 EUR
degradation = 180 MWh x 12 EUR/MWh = 2,160 EUR
fees = 200 MWh x 3 EUR/MWh = 600 EUR

net arbitrage value = 12,100 - 2,160 - 600 = 9,340 EUR
```

Now add a reserve product in the same evening:

- reserved capacity: `40 MW`
- reservation window: `4 h`
- reserve capacity price: `25 EUR/MW/h`
- upward service must be sustainable for `2 h`

The capacity payment is:

```
40 MW x 4 h x 25 EUR/MW/h = 4,000 EUR
```

But the reserve commitment requires headroom:

```
reserve_energy_headroom = 40 MW x 2 h = 80 MWh
```

Those `80 MWh` are not free. If the best alternative use of that energy headroom is worth `45 EUR/MWh`, the opportunity cost is:

```
80 MWh x 45 EUR/MWh = 3,600 EUR
```

The reserve looks plausible before activation, extra wear, and imbalance effects:

```
capacity payment - opportunity cost = 4,000 - 3,600 = 400 EUR
```

If the same evening has a stronger arbitrage or firming opportunity worth `70 EUR/MWh`, the opportunity cost becomes:

```
80 MWh x 70 EUR/MWh = 5,600 EUR
```

Now the same reserve contract is weak:

```
capacity payment - opportunity cost = 4,000 - 5,600 = -1,600 EUR
```

The general test is:

```
incremental_stack_value =
  capacity_payment
  + expected_activation_margin
  - opportunity_cost
  - extra_degradation
  - grid_and_imbalance_costs
```

A product clears the stack only when its value exceeds the best feasible use the battery gives up.

Revenue Lines Become Obligations

Each revenue line asks for a different claim on the asset.

Revenue source	Main obligation	What can break the stack
Day-ahead or intraday arbitrage	Move MWh across price intervals	losses, fees, degradation, spread compression
Reserve capacity	Hold MW and state of charge available	opportunity cost of withheld MW/MWh
Balancing activation	Move energy if called	activation uncertainty, imbalance exposure, wear
PPA shaping or firming	Deliver or absorb MWh in specified hours	conflict with merchant dispatch or reserve readiness
Curtailment recovery	Store otherwise-spilled renewable energy	grid limit, timing mismatch, full battery
Local flexibility or grid service	Be available at a constrained location	contract availability, measurement, qualification
Capacity-like value	Be dependable during stress hours	duration credit, SoC management, availability rules

The table is useful only if it is read as a conflict map. A high reserve price can be attractive in a quiet spread environment and unattractive during a volatile week. A PPA-firming obligation can be valuable if it reduces shape cost, but expensive if it blocks dispatch during the same high-value hours. Curtailment recovery can be real value at one congested solar site and almost irrelevant at another.

The site, contract, and optimizer decide which rows can coexist.

Bankable Revenue And Expected Value

A project owner can care about expected value. A lender also needs enforceable downside protection.

Two stacks can have the same expected annual value and different investment value:

Revenue type	Financing treatment intuition
Long-term contracted availability with strong counterparty	Most bankable
Regulated or auctioned capacity payment with clear rules	Often partly bankable
Repeated observable market revenue with liquid history	Evidence value with haircut
Optimizer forecast of merchant spreads	Upside unless strongly evidenced
New product, thin liquidity, changing rules	Strategy option, not financing foundation

Suppose one model shows `5 million EUR/year` of contracted floor and another shows `7 million EUR/year` of forecast merchant upside. The second may have higher expected value and still support less debt if the revenue disappears under saturation, rule change, or competitor entry.

A BESS investment case should separate:

```
underwritten_floor
base_case_optimization
merchant_upside
stress_case_value
```

Putting all four into one “revenue stack” number hides the risk allocation.

The contract route decides what part of the stack can be underwritten.

Route	What is sold	Control-right effect	Investment reading
Full merchant optimization	The optimizer pursues energy, reserve, and balancing value.	Owner keeps upside and downside if it has visibility and controls.	Highest optionality, weakest downside certainty.

Route	What is sold	Control-right effect	Investment reading
Floor or revenue-share	A counterparty provides a minimum payment, often with upside sharing.	Some dispatch discretion, data, or revenue priority moves to the counterparty.	Lower expected upside may support more financeable value.
Toll or availability-style contract	The counterparty pays for access or availability.	The counterparty may control dispatch within agreed limits.	Good downside readability if performance obligations are clear.
Partial hybrid PPA	A defined slice of BESS flexibility supports a customer shape.	The reserved slice cannot be freely used for merchant or reserve value.	Attractive only if the customer premium beats opportunity cost.
Combined hybrid PPA	Renewable output and BESS are sold as one customer product.	More flexibility is locked into the customer obligation.	Useful when buyer value, evidence, and control rights justify the lock-up.

Illustrative financeability test:

Route	Expected value	Downside value	Lender-counted share in scenario	Financeable value
Full merchant	1.70 million EUR/year	0.70 million EUR/year	20%	0.34 million EUR/year
Floor contract	1.20 million EUR/year	0.90 million EUR/year	100%	0.90 million EUR/year
Partial hybrid PPA	1.55 million EUR/year	contract-specific	scenario-specific	depends on customer obligation and residual rights

The percentages are scenario assumptions rather than lender rules. They teach the comparison: expected value, downside value, and financeable value are different columns.

Saturation Changes The Opportunity Cost

Reserve and flexibility markets can be small relative to the batteries that want to enter them.

If a product needs `X MW` and many similar assets can provide `X MW`, the marginal value of the next asset falls. The market-structure risk is:

```
small product need + fast entry + similar assets = revenue compression risk
```

Energy arbitrage can compress too. If many batteries charge in the same low-price hours and discharge in the same high-price hours, they flatten the spread they came to harvest. The first batteries may earn by exploiting scarcity across time. The later batteries may compete that scarcity away.

For a project model, saturation changes two things:

- expected price: future reserve or spread revenue may be lower than recent history.
- opportunity cost: if one market compresses faster than another, the optimal stack can shift.

That makes static stacking unreliable. The stack should be tested under price compression, activation changes, degradation-cost sensitivity, and grid/tariff changes.

Decision Frame

For a proposed BESS stack, ask these questions in order.

1. What is physically committed in each interval?

```
reserved_export_MW + scheduled_discharge_MW <= export_limit_MW
reserved_import_MW + scheduled_charge_MW <= import_limit_MW
```

If this fails, the stack is not feasible.

2. What state of charge is required before uncertainty arrives?

```
required_SoC_for_reserve
+ required_SoC_for_PPA_or_firming
+ operating_buffer
<= usable_energy_MWh
```

If this fails, the stack is double-counting energy readiness.

3. What is the opportunity cost of each commitment?

```
commitment_value >= best_feasible_alternative_value
                    + incremental_wear
                    + grid_and_imbalance_costs
                    + risk_margin
```

If this fails, the revenue line is real but not attractive.

4. Who controls dispatch?

A market can exist and still be unavailable to the project if the offtaker, optimizer, BRP, BSP, warranty, or grid agreement assigns control elsewhere.

5. Which part can be underwritten?

Split the stack into:

```
contracted_or_rule-based_floor
observable_repeatable_market_value
optimizer_forecast
unpriced_option_value
```

The first category can carry more investment weight than the last.

6. What breaks first in stress?

Run at least four stresses:

- lower spreads or reserve prices
- higher degradation or augmentation cost
- lower availability or more activation
- tighter grid/tariff/connection limits

The best stack is the one whose value survives the constraints most likely to bind at that site.

Recap

A BESS revenue stack is a scarce-resource allocation problem before it is a revenue forecast. Each line claims MW, MWh, state of charge, cycles, grid rights, or control rights, so the commercial test is whether the obligations can coexist and whether each one beats the best feasible alternative use of the asset. Once that opportunity cost is priced, the model should separate underwritten floor, repeatable market value, optimizer forecast, and option value. Extra revenue rows create no value unless the asset, contract, and investment case can carry them together.

Chapter 9. Battery Duration Selection

References: [11] [38] [40] [41] [42] [21] [39]

A developer is comparing two designs for the same battery: `100 MW / 200 MWh` and `100 MW / 400 MWh`. The second design can hold more solar, cover longer delivery blocks, and support longer scarcity events. It also carries more capital, more degradation exposure, more tariff questions, and more risk that stored energy cannot be used when the portfolio needs it. The extra MWh has to earn its cost under this product, this site, and this portfolio.

Opening Puzzle

“Should we build a two-hour or four-hour battery?”

Answering the question requires product, grid, contract, and capital-allocation evidence, not duration labels alone.

Two hours for what?

- catching one price spike
- holding reserve headroom
- shaping solar into evening blocks
- reducing curtailment
- supporting a fixed PPA delivery schedule
- contributing to capacity during stress hours
- preserving optionality in a market whose rules may change

Duration is the price paid for extending the battery’s useful delivery window.

First-Principles Model

A battery has power and energy:

```
power = MW
energy = MWh
duration_h = usable_energy_MWh / discharge_power_MW
```

Examples:

```
100 MW / 100 MWh = 1 h
100 MW / 200 MWh = 2 h
100 MW / 400 MWh = 4 h
```

That identity is only the unit check. The decision is about the next MWh.

The commercial question is:

```
Does the extra MWh earn more than it costs?
```

More explicitly:

```
annual_incremental_value
  = extra_energy_MWh
    x useful_cycles_per_year
    x net_value_per_MWh
```

The word `useful` matters. It is capped by the tightest real constraint:

```
useful_cycles_per_year =
  min(physical_opportunity_cycles,
      warranty_compatible_cycles,
      thermal_and_data_compatible_cycles,
```

```
        insurance_permitted_cycles,
        grid_and_product_eligible_cycles)
```

The extra duration clears the hurdle only if:

```
annual_incremental_value
>
annualized_extra_capex
+ extra_degradation
+ extra_tariffs_and_fees
+ extra_risk_cost
+ opportunity_cost
```

Definitions

Usable energy: energy available for operation after depth-of-discharge, reserve, warranty, thermal, and operating limits.

Incremental duration: additional MWh added while holding MW roughly fixed.

Project policy constraint: a possible insurance, lender, warranty, or operating-policy limit on cycling, throughput, depth of discharge, duty cycle, or approved operation. Read it from project documents as a diligence question, not as a universal BESS market term.

Net value per MWh: value of discharging an additional MWh after charging cost, efficiency loss, degradation, fees, and opportunity cost.

Dead capex: energy capacity that is physically present but rarely earns enough incremental value.

Duration hedge: extra energy capacity bought partly to handle uncertainty: longer price events, longer PPA shortfalls, reserve calls, curtailment windows, or grid constraints.

The Cost Scale

The International Energy Agency’s 2024 battery report gives a useful global scale anchor. In the STEPS scenario, average total upfront costs for four-hour utility-scale battery storage decline from `USD 290/kWh` in 2022 to `USD 175/kWh` in 2030. The same report warns that individual projects can vary widely, from over `USD 500/kWh` to below `USD 200/kWh`, depending on site, technology, and regulation.

Use `USD 175/kWh` as a scale anchor rather than a project quote. It is an all-in four-hour system-cost anchor, not the marginal cost of adding duration to a specific project. Marginal duration pricing needs vendor or engineering, procurement, and construction (EPC) quotes separating cells, containers, power-conversion-system sharing, balance of plant, augmentation, warranty, grid, tax, and financing effects.

Numerical Intuition

The Extra Two Hours

Compare:

```
100 MW / 200 MWh = 2 h
100 MW / 400 MWh = 4 h
```

The four-hour version has `200 MWh` more energy capacity.

Using the all-in `USD 175/kWh` figure only as an order-of-magnitude test:

```
200 MWh = 200,000 kWh
extra energy block scale = 200,000 x 175 = USD 35,000,000
```

Use a rough `10%` annual carrying charge for the capital hurdle:

```
annual hurdle = 35,000,000 x 10% = USD 3,500,000/year
```

Now ask how often the extra `200 MWh` must be valuable.

At `USD 50/MWh` net value:

```
cycles_needed = 3,500,000 / (200 x 50) = 350 cycles/year
```

At `USD 25/MWh` net value:

```
cycles_needed = 3,500,000 / (200 x 25) = 700 cycles/year
```

At `USD 75/MWh` net value:

```
cycles_needed = 3,500,000 / (200 x 75) = about 233 cycles/year
```

This test tells the team what evidence must support the duration decision:

```
show me the cycles, the margin, the contract, and the constraint.
```

What Makes Duration Valuable?

Extra duration is valuable when the marginal MWh has something to do.

Driver	Why duration helps
Long price spreads	The high-price block lasts longer than the short battery can serve
Solar shifting	Midday energy needs to reach evening demand or contract hours
Curtailment	The project would otherwise spill useful MWh during constrained hours
Shaped PPA	The seller must deliver a fixed or shaped block beyond natural production
Reserve sustain requirement	The product needs energy headroom as well as fast MW
Capacity or adequacy value	Stress events last long enough that 1h or 2h storage has limited credit
Portfolio interaction	Extra MWh reduces exposure across several assets or contracts

The IEA scenario logic fits this intuition. As solar penetration rises, the share of four-to-eight-hour battery systems rises relative to shorter systems, and the global average storage fleet duration rises from around two hours in the IEA report baseline to `2.7 h` in 2030 and `3.4 h` by 2050 in STEPS.

Use that as system-level direction. The site-level answer still depends on the local product, connection, tariff, degradation, warranty, and portfolio need.

What Makes Duration Wasteful?

Extra duration is wasteful when the marginal MWh sits idle or destroys value.

Driver	Why duration can fail
Narrow reserve product	Product pays for MW, not much extra MWh
Short price events	The spread is exhausted after one or two hours
High capex	The extra energy block raises the annual hurdle too much
High degradation	Useful cycles become expensive
Warranty envelope	SoC, charge/discharge rate (C-rate), temperature, throughput, data, or repair terms reduce feasible operation
Grid import limits	The battery cannot refill when it needs to
Export constraints	Extra stored energy cannot reach market or PPA delivery
Tariffs and fees	Charging or peak usage charges consume the spread
Contract priority	PPA obligations crowd out merchant optimization

Longer duration also adds forecasting and control complexity. A one-hour decision is often a choice about now. A four-hour decision is a choice about a sequence: charge now, hold inventory, preserve optionality, deliver later, and still be ready for the next product.

A Three-Lens Sizing Method

Use three lenses before declaring the duration.

1. Product Lens

What is the battery paid to do?

Product need	Duration clue
Fast frequency response	MW and response speed may dominate MWh
Reserve capacity with sustain requirement	Duration must cover the energy obligation
Intraday/DA arbitrage	Duration follows the shape and persistence of spreads
Solar-plus-storage PPA	Duration follows the contracted delivery shape
Curtailment recovery	Duration follows timing and length of curtailment
Capacity-style value	Duration follows stress-event method and capacity-credit rules

The Nordic TSO battery-reserve paper gives concrete examples of the spread between "fast" and "long enough." FFR is very fast and upward-only, with short endurance options. FCR-D needs at least `20 min` of full activation. FCR-N needs `1 h` of endurance in each direction. aFRR can require `1–4 h` depending on country and market path. mFRR uses a `12.5 min` full-activation standard in the Nordic description, while the energy activation market (EAM) endurance window is tied to one or two market quarters. A one-hour battery can be excellent for one product and weak for another; duration is a product requirement before it is an asset label.

2. Site Lens

What does the grid connection allow?

A co-located battery may be attractive because it can use solar energy that would otherwise be curtailed, reduce export peaks, or make better use of a limited connection. The same battery may struggle if it cannot import enough, if tariffs punish charging, or if export rights are tight when discharge would be valuable.

Do not write “grid rights checked” in the memo. Ask for the evidence:

Evidence	Question It Answers
export right	How many MW can the site inject, and when is that right firm or conditional?
import right	Can the BESS charge from the grid, only from co-located solar, or not at all?
metering boundary	Are generation, storage, and load measured separately or netted?
tariff trigger	Which action creates capacity, MWh, loss, or overrun charges?
curtailment rule	Who decides curtailment, and is curtailed energy compensated, excused, or lost?
flexible or limited access terms	Does cheaper or earlier access come with interruptibility that changes dispatch value?

The site can make two hours better than four. It can also make four hours better than two. The evidence is in the connection terms, tariff rules, curtailment pattern, and product rights.

3. Portfolio Lens

What does the rest of the portfolio already do?

If wind, solar, PPAs, hedges, and other batteries already cover a shape risk, the marginal value of another hour may be low. If the portfolio has a persistent evening short, the same hour may be valuable.

Duration is both an asset property and a portfolio property.

Worked Decision Test

Suppose a developer is comparing two designs for a solar-plus-storage project.

Item	2h design	4h design
Power	100 MW	100 MW
Energy	200 MWh	400 MWh
Extra energy	-	200 MWh
Annual hurdle for extra energy	-	USD 3.5 million/year

Now build three scenarios for the extra two hours.

Scenario	Useful cycles/year	Net value/MWh	Annual value	Decision signal
Weak spread	150	USD 25	USD 0.75 million	Extra duration fails
Strong shaped-PPA need	260	USD 75	USD 3.90 million	Extra duration may clear
Merchant-only base	250	USD 50	USD 2.50 million	Extra duration fails before option value

This sanity gate tells you where to spend modeling time before building a valuation model.

If the only rationale for the 4h design is “batteries are moving to longer duration,” reject it.

If the rationale is “this contract has a persistent evening delivery obligation, this site has solar curtailment at noon, this grid connection allows charging, and the extra MWh survives degradation, insurance, and tariff costs,” then model it properly.

The same method works for `1h`, `2h`, and `4h` when each block is valued at the margin. The next table is an illustrative scenario test for sensitivity.

Incremental block	Extra energy	Useful cycles/year	Net value/MWh	Annual value	Annual hurdle	Signal
First hour	30 MWh	310	55 EUR/MWh	0.51 million EUR/year	0.50 million EUR/year	just clears before extra value
Second hour	30 MWh	240	45 EUR/MWh	0.32 million EUR/year	0.35 million EUR/year	needs curtailment, PPA, or option value
Third and fourth hours	60 MWh	120	25 EUR/MWh	0.18 million EUR/year	0.70 million EUR/year	fails without a different product

This table is deliberately marginal. It blocks the common averaging error: using the valuable first-hour spread to justify the weak later hours.

In Practice

A duration memo should include:

- asset: MW, MWh, usable energy, round-trip efficiency, availability assumptions
- site: export/import rights, co-location rules, tariff treatment, curtailment pattern
- product: PPA shape, reserve sustain requirement, arbitrage horizon, capacity-credit method
- economics: incremental capex, fixed O&M, degradation, warranty envelope, insurance-policy constraints, augmentation, fees, taxes if relevant
- uncertainty: spread compression, rule changes, forecast error, counterparty/control rights
- sensitivity: the assumption-flip test that changes the recommendation

The recommendation should say:

```
2h is better under assumptions A, B, C.
4h becomes better if X happens.
Both fail if Y is true.
```

Site-specific evidence is more useful than a global fleet-average duration because the project decision depends on the site, contract, grid rights, and portfolio need.

Recap

Battery duration is a marginal investment choice. The extra hour only works when the added MWh repeatedly earns more through useful cycles and net value than it costs through capital, degradation, insurance restrictions, tariffs, risk, and opportunity cost. That turns the opening `2h` versus `4h` question into an investment test: show the contract shape, spread persistence, curtailment window, grid rights, warranty and insurance limits, and portfolio need that will keep the marginal MWh busy. Once duration clears that test, the next decision is how to dispatch the battery.

Chapter 10. Dispatch Optimization

References: [17] [11] [23] [43] [40] [38] [44] [21]

An analyst receives a battery schedule: charge at noon, discharge in the evening, hold some capacity for reserve, finish the day with a chosen state of charge. Evaluating that schedule starts with the optimization problem that produced it. The relevant inputs are the information set, prices or forecasts, MW and MWh limits, reserve commitments, contract rights, grid import/export limits, degradation cost, and terminal state of charge. Dispatch optimization is useful because it makes those assumptions explicit, then shows the asset behavior implied by them.

Opening Puzzle

A battery schedule charges at noon, discharges in the evening, holds reserve headroom, and ends the day with a target state of charge. The schedule is not enough to judge the model. Evaluation starts by reconstructing the problem statement:

- information available at decision time;
- price or forecast source;
- reserve products and sustain requirements;
- degradation, warranty, and terminal state-of-charge costs;
- import/export limits and grid charging rights;
- contract delivery obligations and penalties;
- dispatch and control rights.

The evaluation question is:

```
What problem did we ask it to solve?
```

First-Principles Model

Dispatch is constrained choice over time.

At each interval `t`, choose:

```
c_t = charge power, MW
d_t = discharge power, MW
r_up_t = upward reserve capacity, MW
r_down_t = downward reserve capacity, MW
soc_t = state of charge, MWh
```

Given:

```
price_t = energy price, EUR/MWh
reserve_price_t = capacity price, EUR/MW/h
delta_t = interval length, h
P_ch_max = maximum charge power, MW
P_dis_max = maximum discharge power, MW
E_min, E_max = usable energy bounds, MWh
```

The dispatch model chooses actions to maximize value subject to physics, market rules, contracts, and risk limits.

Definitions

Objective function: what the model is trying to maximize or minimize.

Decision variable: something the model is allowed to choose, such as charge, discharge, or reserve quantity.

Constraint: a rule the model must obey.

State variable: a value that carries memory across time, such as state of charge.

State of charge (SoC): the stored energy in the battery at a point in time, measured in MWh or as a percentage of usable energy.

Terminal value: value assigned to the battery's ending state of charge so the model does not empty the battery merely because the model horizon ends.

Shadow price: the marginal value of relaxing a binding constraint by one unit. If one extra MWh of energy capacity would increase optimal profit by `60 EUR`, the energy constraint has a shadow price of `60 EUR/MWh` in that interval.

Perfect-foresight dispatch: optimization using actual future prices. Useful as an upper bound; operational benchmarks should use information that was tradable at the decision time.

Forecast-based dispatch: optimization using information available at the time. Necessary for fair partner and model benchmarking.

The Core Objective

A useful battery objective starts simple:

```
maximize sum_t [
  energy_revenue_t
  + reserve_capacity_revenue_t
  + activation_revenue_t
  + contract_value_t
```

```
  - charge_energy_cost_t
  - degradation_cost_t
  - grid_and_market_fees_t
  - penalties_t
]
+ terminal_soc_value
```

For energy arbitrage only:

```
energy_revenue_t = price_t x d_t x delta_t
charge_energy_cost_t = price_t x c_t x delta_t
degradation_cost_t = deg_cost_per_MWh x d_t x delta_t
```

Units:

- `price_t`: EUR/MWh
- `c_t`, `d_t`: MW
- `delta_t`: hours
- `c_t x delta_t`: MWh
- objective: EUR

The objective is where commercial choices are encoded. If the model excludes degradation, it will over-cycle. If it excludes penalties, it will ignore delivery obligations. If it excludes terminal value, it may end the day empty.

Core Constraints

Power limits:

```
0 <= c_t <= P_ch_max
0 <= d_t <= P_dis_max
```

State-of-charge balance:

```
soc_t+1 =
  soc_t
  + eta_ch x c_t x delta_t
```

```
  − (d_t x delta_t) / eta_dis
  + activation_effect_t
```

Energy bounds:

```
E_min <= soc_t <= E_max
```

Reserve headroom:

```
d_t + r_up_t <= P_dis_max
c_t + r_down_t <= P_ch_max
```

Reserve sustain requirement:

```
r_up_t x sustain_hours <= soc_t − E_min
r_down_t x sustain_hours <= E_max − soc_t
```

Nordic reserve products give `sustain_hours` real content. In the Nordic TSO battery paper, FCR-D requires at least `20 min` of full activation, FCR-N requires `1 h` in each direction, aFRR endurance can sit in the `1–4 h` range depending on the country and market path, and mFRR energy-activation endurance is described around one or two market quarters. The optimizer should therefore parameterize reserve commitments by product and geography instead of treating “reserve” as one generic MW line.

Grid import/export:

```
c_t + site_load_t <= import_limit_t
d_t + solar_export_t <= export_limit_t
```

Contract delivery:

```
delivered_MWh_t + market_purchase_MWh_t >= contracted_schedule_MWh_t
```

For a solar-plus-BESS shaped PPA, the contract constraint often has three sources:

```
solar_to_contract_t
+ bess_discharge_to_contract_t
+ market_or_portfolio_purchase_t
>= contracted_schedule_t
```

and the battery state must still respect losses:

```
soc_t+1 =
  soc_t
  + eta_ch x charge_MWh_t
  − discharge_MWh_t / eta_dis
```

If the contract needs `20 MWh` in an evening hour and the BESS can discharge only `12 MWh` without violating state of charge or reserve headroom, the remaining `8 MWh` is a firming purchase, a buyer pass-through, a portfolio allocation, or a delivery failure. The optimizer identifies where the shortfall sits under the chosen rules.

The exact model changes by product. The discipline does not: every symbol needs a unit and every constraint needs an economic reason.

Numerical Intuition

Information, Terminal SoC, And Shadow Price

Power-system economics has a simple lesson that matters a lot for batteries: energy and reserve are linked. Kirschen and Strbac show that the sale of reserve and energy should be optimized jointly because providing reserve can reduce energy sales and vice versa. The opportunity cost is real even when no energy moves.

For a battery, the same discipline also appears at the horizon boundary. A schedule that empties the battery can look profitable if the model ignores the value of stored energy after the modeled day.

Illustrative example:

- battery state of charge at 18:00: `90 MWh`
- minimum terminal state of charge at 06:00 for next-morning delivery: `50 MWh`
- evening discharge candidate: `50 MWh`
- forecast evening margin: `65 EUR/MWh`
- terminal value of stored energy: `80 EUR/MWh`

The visible evening dispatch appears to earn:

```
50 MWh x 65 EUR/MWh = 3,250 EUR
```

But only `40 MWh` is free above the terminal requirement:

```
free discharge above terminal SoC = 90 MWh − 50 MWh = 40 MWh
terminal shortfall from 50 MWh discharge = 50 MWh − 40 MWh = 10 MWh
```

The last `10 MWh` would be sold below the terminal value:

```
10 MWh x (65 − 80) EUR/MWh = −150 EUR
```

In this illustrative setup, the terminal state of charge has an economic shadow value of `80 EUR/MWh`. The commercial test compares the visible evening margin with the stored-energy value under the information available at dispatch time.

If the actual 22:00 price later clears at `140 EUR/MWh`, that can improve future forecasts. For judging the 18:00 schedule, use only information that was tradable at 18:00. A dispatch model therefore evaluates one battery across obligations and time instead of ranking markets or hours one by one.

Shadow Prices And Binding Constraints

A dispatch schedule says what to do.

A shadow price says what the bottleneck is worth.

Examples:

Binding constraint	Shadow-price meaning
Energy capacity	Value of one more MWh of usable battery energy
Discharge power	Value of one more MW of inverter/export power
Import limit	Value of one more MW of charging access
Reserve sustain	Value of relaxing the energy headroom requirement
PPA delivery	Cost of one more MWh of firm obligation
Terminal SoC	Value of ending with more stored energy

This can be more useful than the dispatch itself. If the shadow price of export capacity is high for many hours, the problem may be grid rights. If the shadow price of energy capacity is high only a few days per year, the answer may be a contract or market-purchase hedge rather than a bigger battery.

Perfect Foresight As An Upper Bound

A perfect-foresight optimization can be useful. It gives an upper bound:

```
what could have been earned if we knew the future and had perfect execution
```

But a partner, trader, or optimizer cannot be judged against that number without adjustment.

A fair benchmark must use:

- forecasts available before gate closure
- prices available at the decision time
- asset availability known at the time
- actual grid and contract constraints
- actual control rights
- realistic bid sizes and liquidity
- ex-ante risk limits

The same boundary appears in practitioner storage tools. DNV describes a hybrid-storage optimizer that calculates dispatch on a `24-hour` lookahead at `15-minute` intervals, uses mixed-integer linear programming (MILP), embeds technical loss trees, and includes interconnection limits, battery cycling, throughput requirements, warranty, and capacity-maintenance strategies. It then adjusts ideal perfect-foresight revenue toward operational reality. Use it here as model-scope evidence. Making a universal industry-standard claim would need broader sources.

Shinde's work on short-term electricity trading is useful here. Multistage stochastic models can beat deterministic equivalents in some settings, but the result depends on transaction costs, liquidity, transmission capacity, imbalance prices, scenario design, and computation time. Complexity has value only when it changes a feasible decision enough to pay for itself.

A Minimal Dispatch Stack

A practical first dispatch model for a BESS project can be built in layers.

Layer	Add
Physical	MW, MWh, SoC, efficiency

Layer	Add
Technical losses	renewable profile, storage loss tree, availability, auxiliary load
Economic	prices, fees, degradation, augmentation
Contractual	PPA delivery, reserve obligations, penalties
Grid	import/export limits, interconnection limits, curtailment, tariff trigger
Uncertainty	forecasts, scenarios, risk limits
Governance	control rights, audit trail, override rules

If the model is used for investment, it must be conservative about future revenues.

If the model is used for operations, it must respect information timing.

If the model is used for partner oversight, it must benchmark only what the partner could actually control.

For a multi-product BESS, add a priority order before running the optimizer.

Priority Item	What It Protects
safety and grid compliance	the asset cannot violate technical or connection limits
contracted PPA delivery or penalty rule	customer promise and fallback procurement
reserve obligation and sustain rule	market qualification and activation readiness
degradation and warranty envelope	future asset value and availability
terminal state of charge	next-day obligations and option value
merchant arbitrage	residual optionality after obligations are met

The order is a business decision with mathematical consequences. If the optimizer is allowed to treat all revenues as freely interchangeable, it may sell the same flexibility twice.

In Practice

When reviewing a dispatch result, ask:

1. Is the objective aligned with the business decision?

For investment, expected revenue sits beside downside and bankability. For operations, real-time feasibility matters. For oversight, fairness of the benchmark matters.

2. Are all binding constraints visible?

If the model only outputs profit and schedule, you cannot tell whether the bottleneck is energy, power, grid, contract, or market access.

3. Are losses and degradation inside the dispatch, not patched afterward?

If degradation is added after optimization, the dispatch may be too aggressive.

4. Is the model using the information it would have had at the time?

Hindsight is useful for learning. A performance score needs the information available at decision time.

5. Does the ending SoC make economic sense?

A model that empties the battery at the horizon end may be omitting terminal value.

Recap

Battery dispatch is constrained choice over time, so the schedule is only the visible output of a specified objective, constraint set, and information set. The review starts by identifying the problem the optimizer solved. Once charge, discharge, reserve, contracts, SoC, grid limits, degradation, terminal value, and control rights are inside the model, shadow prices expose the real bottleneck: more MW, more MWh, more grid rights, more control rights, better forecasts, or a better contract.

Chapter 11. Forecasting, Error, And Commercial Value

References: [3] [4] [15] [43] [45] [46]

A forecasting vendor promises lower solar error. A partner says its own forecast is good enough. The trading team wants a price signal for intraday decisions. The battery optimizer wants a view of evening spreads. Finance wants to know whether any of this is worth paying for. Accuracy alone cannot answer. A forecast earns budget only when its errors are weighted by the cash-flow decisions they change.

Opening Puzzle

Two models forecast the same solar plant.

Model	Physical mean absolute error (MAE)	Value-weighted error
Model A	`4.0 MWh/hour`	`260 EUR/hour`
Model B	`4.8 MWh/hour`	`180 EUR/hour`

Which forecast is more useful for day-ahead nomination, intraday trading, BESS dispatch, and partner oversight?

First-Principles Model

Start with physical error:

```
forecast_error_t = actual_t − forecast_t
```

The commercial error attaches a value to the MWh error:

```
value_weighted_error_t =
  abs(forecast_error_t)
  x decision_value_weight_t
```

The weight depends on the decision:

```
decision_value_weight_t =
  expected_imbalance_cost_t
  or intraday_spread_t
  or BESS_opportunity_cost_t
  or PPA_delivery_penalty_t
  or partner_benchmark_impact_t
```

Forecast value is incremental:

```
forecast_tool_value =
  value(decisions with tool)
  − value(decisions with current process)
  − license_or_build_cost
  − data_and_operations_cost
  − model_risk_and_governance_cost
```

The tool matters only where the organization can act before the relevant gate closes.

Forecasts By Decision

Decision	Forecast object	Deadline	Value test
Day-ahead nomination	production/load volume by delivery interval	before day-ahead gate closure	reduced expected imbalance and better contract position
Intraday rebalancing	updated production/load and price/liquidity	before intraday liquidity disappears or gate closes	trade cost beats expected imbalance cost
BESS dispatch	price path, activation risk, state-of-charge need	before dispatch or reserve commitment	dispatch value beats degradation and opportunity cost
PPA delivery	contract shape versus expected asset/portfolio delivery	before sourcing window closes	avoided firming cost or penalty exceeds transaction cost
Partner oversight	feasible benchmark inputs available at the time	before scorecard attribution	separates market conditions from partner execution

A single model may serve several rows, but each row has its own clock and cash-flow weight.

Numerical Intuition

Scenario: `50 MWp` solar portfolio, 30 days, hourly settlement. Current process and vendor tool are compared on the same historical month using only information available at the time.

Decision bucket	Current value-weighted error	Tool value-weighted error	Gross improvement
Day-ahead nomination	`54,000 EUR/month`	`42,000 EUR/month`	`12,000 EUR/month`
Intraday rebalancing	`46,000 EUR/month`	`27,000 EUR/month`	`19,000 EUR/month`
BESS dispatch	`38,000 EUR/month`	`31,000 EUR/month`	`7,000 EUR/month`
Partner oversight attribution	`18,000 EUR/month`	`12,000 EUR/month`	`6,000 EUR/month`
Total	**`156,000 EUR/month`**	**`112,000 EUR/month`**	**`44,000 EUR/month`**

Tool costs:

Cost item	Assumption
Vendor license and data feeds	`22,000 EUR/month`
Internal operations and validation	`8,000 EUR/month`
Total recurring cost	`30,000 EUR/month`

Base-case net value:

```
net_value = 44,000 − 30,000 = 14,000 EUR/month
```

```
annualized_net_value = 14,000 x 12 = 168,000 EUR/year
```

The tool clears a narrow hurdle in this scenario. The decision still depends on whether the improvement is stable, actionable, and contractually usable.

Value-Weighted Error In Practice

Physical errors should not be discarded. They diagnose the model. They should be paired with financial weights.

Metric	What it catches	What it misses
Mean absolute error (MAE) / root mean square error (RMSE) in MWh	average physical forecast quality	price, liquidity, penalties, and ability to act
Bias	systematic over- or under-forecasting	timing of expensive errors
Calibration	whether probability ranges are honest	whether the decision uses the range well
Value-weighted error	financial damage of errors	physical model cause unless decomposed
Decision uplift backtest	actual cash-flow change under feasible actions	can overfit if governance is weak

Use the physical metrics to improve the model. Use value-weighted metrics to decide whether to fund it.

Build, Buy, Or Ignore

The forecast-tool decision should be made by capability, not by vendor enthusiasm.

Option	Use when...	Risk
Ignore / keep current process	value-weighted error is small, no action window exists, or partner already bears and evidences the risk	exposure can grow with volumes or contracts
Buy forecast service	improvement is measurable, data rights exist, and internal team can validate outputs	vendor black box, weak fit to PPA/BESS decisions
Build internal model	forecasting is strategic, data advantage exists, and enough volume justifies staff and governance	fixed cost, model risk, maintenance burden
Hybrid: buy base forecast and build decision layer	external weather/model feed is good enough, value comes from portfolio, contract, or dispatch translation	unclear accountability unless interfaces are defined

For the scenario above, recommend buying a forecast service for production and price inputs while building the internal value-weighting, backtest, and partner-oversight layer. The proprietary edge sits in translating forecast updates into PPA, BESS, and partner decisions.

Kill or pause the tool if any of these conditions appear for two consecutive review cycles:

- net value falls below zero after data and operating cost
- forecast updates arrive after the relevant action window
- partner contracts block use of the forecast in dispatch or attribution
- improvement is concentrated in a small sample of non-repeatable events
- the team cannot reconcile forecast, schedule, trade, meter, and settlement data

Day-Ahead And Intraday

A day-ahead forecast is useful because it shapes the initial position. An intraday forecast is useful because it can repair that position before settlement.

Illustrative intraday decision:

```
expected_shortfall = 3 MWh
intraday_buy_price = 72 EUR/MWh
expected_imbalance_cost = 105 EUR/MWh
transaction_cost = 2 EUR/MWh

rebalance_value =
  3 x (105 - 72 - 2)
  = 93 EUR
```

The trade is small, but repeated errors in high-cost periods accumulate. If liquidity is unavailable, or if the update arrives too late, the same forecast has explanatory value rather than trading value.

BESS Dispatch

A battery forecast must be judged against opportunity cost.

```
dispatch_gain =
  expected_discharge_later_value
  - discharge_now_value
  - efficiency_loss
  - degradation
  - reserve_or_PPA_opportunity_cost
```

A price forecast that correctly identifies the highest-price hour can still fail if the battery must hold state of charge for reserve availability or PPA delivery. The backtest should include actual state of charge, grid limits, warranty constraints, and control rights.

Partner Oversight

For BRP/BSP or optimizer governance, the forecast is part of the evidence chain. A fair benchmark uses:

- forecast versions available at the time
- market prices and liquidity visible at the time

- schedules, bids, trades, and activations with timestamps
- asset constraints and outages known at the time
- contract rights and risk limits actually assigned
- settlement data and corrections

The benchmark should attribute error before action. A partner cannot be blamed for a weather miss inside an agreed tolerance band, a grid constraint, or a missing data feed the owner failed to require. A partner can be challenged when data was available, action was feasible, limits allowed it, and the dispatch still missed the feasible benchmark.

Decision Frame

Approve a forecast tool only if the memo can fill this table.

Question	Required answer
Which decision changes?	day-ahead nomination, intraday trade, BESS dispatch, PPA sourcing, or partner scorecard
What is the action deadline?	market gate, liquidity window, dispatch time, scorecard cycle
What is the value-weighted baseline error?	current process measured in EUR, with physical error alongside it
What improvement is evidenced?	backtest using time-available data, not hindsight
What rights are needed?	data, dispatch, partner reporting, settlement, and audit rights
What is the kill rule?	net value, timeliness, data reconciliation, or sample-stability threshold

For the scenario: buy the tool for one year, require monthly value-weighted backtests, build the internal decision layer, and kill it if the net value is negative for two consecutive quarters after excluding non-repeatable events.

Caveats / Current Debates

The eSett Nordic imbalance settlement handbook is the relevant anchor for how imbalances become settlement quantities in the Nordic context. It does not make any specific forecast vendor valuable.

Priyanka Shinde's short-term electricity-market thesis is useful for illustrating how imbalance-price estimate errors can change simulated intraday trading behavior. It is a model context rather than a live benchmark.

Practitioner ranges for renewable forecast error, such as the `5–20%` range cited in Danthine's article, should be treated as market color unless backed by asset-specific validation. Forecast performance is technology-, horizon-, geography-, and data-quality-specific.

Market time units, gate closures, intraday liquidity, imbalance-pricing rules, and reserve products change by market and date. Verify operational rule claims against current official market-operator and TSO sources before using them in a tool business case.

Recap

Forecasting value is decision-weighted error reduction. Physical accuracy matters, but budget should follow the errors that move cash flow before a deadline: day-ahead nomination, intraday rebalancing, BESS dispatch, PPA firming, and partner attribution. In the scenario, the sensible path is to buy external forecast inputs and build the internal layer that weights errors by value, tests feasible actions, and governs partners. The tool survives only while measured net value stays positive.

Chapter 12. Battery Degradation, Warranties, And Operating Constraints

References: [38] [40] [41] [47] [42] [48]

A dispatch case shows a clean trade: buy low, sell higher, cycle the battery, book the spread. But each MWh moved through the battery also moves heat through the system, consumes some useful life, and may use contractual headroom that the project needs later. A bid that looks profitable before losses and wear can shrink to a thin margin or become uneconomic once the battery's operating limits are priced. The commercial test is whether this specific cycle pays enough for energy losses, degradation, fees, opportunity cost, and the warranty, insurance, or augmentation consequences it creates.

Opening Puzzle

A battery optimizer selects one attractive cycle.

It charges when power costs `40 EUR/MWh` and discharges when power sells for `70 EUR/MWh`. The gross spread is positive.

The same cycle also creates losses, consumes cycle life, moves the asset closer to augmentation, and may use warranty, insurance, state-of-charge, or availability headroom needed for another obligation.

Should the cycle run?

First-Principles Model

A battery is a stock of usable energy with power limits and a finite useful life.

The energy state evolves:

```
soc_t+1 =
  soc_t
  + eta_ch x charged_MWh_t
  − discharged_MWh_t / eta_dis
```

The asset state also evolves:

```
health_t+1 =
  health_t
  − wear_from_time_t
  − wear_from_operation_t
```

We rarely know the exact health equation from public sources. But we know the commercial consequence:

```
net_dispatch_value =
  market_value
  − energy_losses
  − degradation_cost
  − grid_and_market_fees
  − opportunity_cost
  − warranty_insurance_or_availability_penalties
```

Gross spread is one input.

Definitions

Degradation: loss of battery capability over time and use, such as usable capacity, efficiency, power capability, or reliability.

Cycle aging: degradation associated with charge-discharge use: throughput, depth of discharge, C-rate, current, and operating pattern.

Calendar aging: degradation associated with time, temperature, and state-of-charge conditions even when the asset is not cycling heavily.

Throughput: cumulative energy charged or discharged, usually measured in MWh.

Depth of discharge: how much of the battery's usable energy window is discharged in a cycle.

C-rate: charge or discharge power relative to energy capacity. A `1C` discharge empties the battery's usable energy in about one hour; `0.5C` in about two hours; `0.25C` in about four hours.

Capacity warranty: promise about retained usable capacity over time, usually conditional on how the battery is operated and measured.

Warranty envelope: the operating conditions under which promised performance, remedies, or availability commitments remain valid.

Insurance operating covenant: condition in an insurance policy or endorsement that limits how the asset may be operated while keeping coverage, pricing, or claim eligibility intact.

Augmentation: adding, replacing, or refurbishing capacity or components to maintain contracted or economic performance.

Degradation-adjusted spread: spread after charging cost, efficiency losses, degradation, fees, tariffs, and opportunity cost.

Why Wear Is Physical

The storage textbooks are useful here because they stop the analyst from treating degradation as a flat cost line.

Chemical batteries change internally as they charge and discharge. Storage references emphasize that cycle life depends on depth of discharge, discharge/charge rate, and environmental conditions. Michaelides describes structural change, electrode deformation, dendrite and filament formation, self-discharge, and thermal limits as physical mechanisms with accounting consequences.

The practical lesson:

```
the same MWh can have different wear cost depending on how, when, and where
```

A shallow cycle inside a comfortable operating window has a different wear profile from a deep, high-power cycle in poor thermal conditions.

Numerical Intuition

Dispatch Threshold For Cycling

Suppose a battery analyst uses a simple degradation allowance.

Illustrative assumptions:

- augmentation or life-consumption pool: `16 million EUR`
- lifetime discharged throughput allocated to that pool: `1,000,000 MWh`

Then:

```
degradation_cost = 16,000,000 / 1,000,000
degradation_cost = 16 EUR/MWh discharged
```

Now take a one-cycle dispatch:

- charge price: `40 EUR/MWh`
- round-trip efficiency: `90%`
- degradation cost: `16 EUR/MWh discharged`
- grid and market fees: `3 EUR/MWh discharged`

To discharge `1 MWh`, the battery must buy:

```
1 / 0.90 = 1.111 MWh
```

Charging energy cost per discharged MWh:

```
40 / 0.90 = 44.44 EUR/MWh
```

Break-even discharge price:

```
44.44 + 16 + 3 = 63.44 EUR/MWh
```

So selling at `70 EUR/MWh` leaves only:

```
70 − 63.44 = 6.56 EUR/MWh
```

The headline spread was:

```
70 − 40 = 30 EUR/MWh
```

The degradation-adjusted margin is:

```
6.56 EUR/MWh
```

Those economics support a different decision.

If degradation cost is `30 EUR/MWh` instead of `16`, the break-even price becomes:

```
44.44 + 30 + 3 = 77.44 EUR/MWh
```

Now the same positive spread is uneconomic.

The Warranty Is A Control Surface

A warranty shapes the dispatch problem.

DNV's storage-warranty review is useful because it turns warranty language into model variables. It stresses that storage warranties vary across projects and that degradation depends on use, rest time, state of charge, temperature, cycling, and other operating metrics. A warranty can therefore change feasible dispatch before any legal remedy is discussed.

Common dimensions to read into the model include:

Warranty or performance dimension	Dispatch implication
Usage or cycling degradation	Annual cycles or discharged MWh may consume the capacity curve faster
Calendar degradation	Underuse still consumes time-based capacity headroom
Warranty term	Project life beyond the term needs augmentation, replacement, or residual-value treatment
Temperature	Thermal-management limits can bind operation and auxiliary-power needs
Average SoC	A high average SoC can make a revenue-maximizing strategy warranty-incompatible
C-rate	Fast charge/discharge can void or weaken remedies, or accelerate degradation
Data storage	Battery management system data becomes evidence in warranty or safety disputes
Response and repair time	Remedy delays can turn a technical claim into lost revenue or availability

Actual warranty and insurance language is project-specific. The modeling conclusion is general:

```
dispatch consumes contractual and insurability headroom as well as physical
```

DNV also describes more flexible warranty structures where operational metrics feed formulas rather than fixed tables. This matters commercially: the owner may be allowed to cycle harder during a high-value event, but the model must show the degradation, warranty, augmentation, and future-value consequence of doing so.

Illustrative throughput check:

- battery: `30 MW / 60 MWh`
- allowed annual equivalent full cycles in the investment case: `300`
- planned merchant dispatch: `180` cycles/year
- shaped-PPA support: `90` cycles/year
- reserve activation allowance: `50` equivalent cycles/year

```
planned_cycles = 180 + 90 + 50 = 320 cycles/year
```

That exceeds the `300` cycle budget before outages, test cycles, or emergency actions. The model can still choose that operating plan, but it must say what changes: higher degradation cost, reduced merchant activity, a different PPA promise, OEM approval, insurance endorsement, augmentation timing, or lower financeable value.

Project Policies Can Bind The Dispatch Plan

Insurance, lender covenants, OEM warranty, and operating procedures are separate diligence items from the physical degradation model. A project policy may restrict cycling, duty cycle, fire-safety operation, monitoring, approved revenue modes, or claim eligibility. The exact language is negotiated and project-specific, so the course does not use a universal insurance-cycle number.

The modeling rule is:

```
useful_cycles_per_year =
  min(physical_opportunity_cycles,
      warranty_compatible_cycles,
      project_policy_permitted_cycles,
      grid_and_product_eligible_cycles)
```

If the policy allows fewer cycles than the dispatch model needs, the model should not quietly average the difference away. The decision owner has to change one of the inputs: obtain written approval or an endorsement, change coverage, reduce cycling, change the revenue stack, reprice the contract, or lower the asset value.

Degradation Belongs Inside The Optimizer

A common weak model optimizes gross market value first and subtracts degradation later.

That can choose the wrong dispatch.

Suppose the optimizer sees two opportunities:

Opportunity	Gross value	Degradation cost	Net value
Shallow cycle	12 EUR/MWh	4 EUR/MWh	8 EUR/MWh
Deep high-power cycle	25 EUR/MWh	22 EUR/MWh	3 EUR/MWh

If degradation is outside the optimizer, it chooses the deep cycle. If degradation is inside, it may choose the shallow cycle or save the battery for later.

The objective should therefore include:

```
degradation_cost_t = f(throughput_t, depth_t, power_t, temperature_t, soc_t
```

In many commercial models this function is simplified. The simplification is acceptable if it is explicit and stress-tested. The model fails if it quietly assumes cycling is free, or if it optimizes revenue under a dispatch pattern that the warranty, data-retention process, thermal-management system, or repair-time commitment cannot support.

Calendar Aging Changes The “Wait” Decision

If only cycle aging existed, the battery could wait forever for perfect spreads.

But time can also degrade performance. That means there is an option-value balance:

```
cycle now and consume throughput
or wait and consume time
```

This matters for low-spread environments. A battery that never cycles may still lose calendar life while earning too little.

The practical decision is:

```
use the asset when compensated for energy losses, wear, risk, and alternati
```

Augmentation In The Operating Plan

For a project with contracted availability or shaped delivery, degradation can become a future capex event.

If usable capacity falls below what the contract or revenue strategy requires, the owner may need augmentation. That means aggressive early dispatch can create a later cost. A correct model links today’s cycling to tomorrow’s ability to perform.

The simple test is:

```
does today's dispatch bring forward augmentation,
reduce availability,
or weaken contract performance?
```

If yes, degradation becomes a timing and optionality problem as well as a `EUR/MWh` cost.

In Practice

A BESS valuation should show at least five operating-headroom views.

View	Question
Marginal dispatch threshold	What spread is needed before cycling is worthwhile?
Annual throughput budget	How many MWh can be cycled without eroding the plan?
Warranty envelope	Which operations are allowed, excluded, or costly?
Insurance operating covenant	Which operations would threaten coverage, premium, exclusions, or claim eligibility?

View	Question
Augmentation path	When does capacity need to be restored, and at what cost?

For a dispatch desk, the daily rule is:

```
do not bid below degradation-adjusted break-even unless another obligation
```

For an investment case, the capital rule is:

```
do not count revenue that depends on using the battery harder than the warr
insurance policy, availability plan, or augmentation budget can support
```

Recap

Battery dispatch should be priced as the use of a finite asset. The operative metric is degradation-adjusted spread: discharge revenue minus charging energy adjusted for round-trip efficiency, degradation cost, fees, tariffs, opportunity cost, and any warranty, insurance, availability, or augmentation consequences. That turns the opening trade into a different decision. A high gross spread can fail once it consumes too much wear or contractual headroom, while a lower gross spread can be worth taking if it preserves optionality or supports a bankable obligation. Revenue stack, duration, dispatch, degradation, and insurability have to be modeled together.

Chapter 13. Project Valuation Under Uncertainty

References: [11] [24] [15] [39]

An investment committee receives one clean base case: one IRR, one NPV, one debt-service path, one merchant-tail value. The sheet looks settled, but the decision is still exposed. A few MWh may arrive in lower-value hours, a tariff line may bite when coverage is thin, a battery revenue stack may compress, or a lender may ignore upside that equity wants to count. The useful project model makes those weak points visible. It turns physical production, market prices, contracts, grid costs, and financing into a cash-flow argument about what has to be true for the project to work.

Opening Puzzle

An investment pack shows one base case:

- IRR: `8.4%`
- NPV: `+6 million EUR`
- DSCR: above the lender threshold
- merchant tail: positive

The sensitivity tab shows that a `3 EUR/MWh` capture-price miss or a six-month grid delay can erase the equity case.

Which assumption is the investment committee really underwriting?

First-Principles Model

Project valuation is discounted cash flow under uncertainty.

```
NPV = sum_t(FCF_t / (1 + r)^t) - initial_investment
```

Units:

- `FCF_t`: EUR in period `t`
- `r`: discount rate per period
- `NPV`: EUR

Free cash flow is a stack of uncertain drivers:

```
FCF_t =
 energy_revenue_t
 + contracted_revenue_t
 + service_revenue_t
 - grid_and_balancing_cost_t
 - opex_t
 - augmentation_or_repowering_capex_t
 - taxes_and_fees_t
 - debt_service_t
```

The model is a structured argument about physical production, market realization, contract risk, grid economics, operating cost, financing, and residual value.

Definitions

Base case is the central scenario used for decision discussion. It should not be confused with the expected value unless scenario probabilities are explicitly assigned.

Downside case is a plausible adverse scenario that tests whether the project survives bad but relevant conditions.

Sensitivity changes one input or driver at a time to show what the model is most exposed to.

Scenario changes a coherent bundle of inputs together: lower capture price plus higher curtailment, or higher BESS revenue plus faster degradation.

Expected value is the probability-weighted value across scenarios, if probabilities are stated.

Lender view prioritizes contracted cash flow, debt service, downside resilience, security, and covenant headroom.

Equity view prioritizes residual cash flow after debt, upside, optionality, refinancing, repowering, and merchant-tail value.

Lethal assumption is a model input that can change the investment decision if wrong.

Merchant tail is the post-contracted or partially uncontracted period where revenue depends more directly on market prices, capture, curtailment, tariff design, degradation, and residual asset condition.

Numerical Intuition

Use sensitivity as the useful number:

```
annual cash-flow sensitivity = exposed MWh x change in EUR/MWh
```

For project models, pair every simple scale anchor with the assumption it stresses:

- capture price sensitivity stresses market and shape assumptions
- capacity factor sensitivity stresses resource and availability assumptions
- annual tariff sensitivity stresses grid-access assumptions
- degradation sensitivity stresses BESS dispatch and warranty assumptions
- ACER 2025 defines network charges broadly enough to include recurring use-of-network charges and one-off connection charges; actual values are country-, voltage-, user-, tariff-year-, and contract-specific.
- Pexapark 2026 market-outlook figures are dated 2025/January 2026 practitioner market color. Use them as 2025/January 2026 market color rather than valuation constants.

Use this section for sensitivity structure. The actual leverage, DSCR, WACC, capex, opex, IRR target, and tariff assumptions belong in a deal-specific evidence pack.

Worked Estimate

Illustrative solar project:

- Annual generation: `100 GWh/year`
- Base realized price: `50 EUR/MWh`
- Downside realized price: `40 EUR/MWh`
- Ignore inflation, taxes, degradation, curtailment, and discounting for this one sensitivity.

Base annual revenue:

```
100 GWh/year x 1,000 MWh/GWh x 50 EUR/MWh
= EUR 5,000,000/year
```

Downside annual revenue:

```
100,000 MWh/year x 40 EUR/MWh
= EUR 4,000,000/year
```

Annual revenue gap:

```
EUR 1,000,000/year
```

If the same project has a hypothetical annual grid/tariff cost uncertainty of `300,000 EUR/year`, the realized-price downside is larger in this illustrative case. But the tariff line may still be more lethal if debt headroom is thin or if the tariff cost is fixed during low-price years.

In this illustrative case, price moves more euros per year than the tariff uncertainty. In a real lender case, the fixed tariff line may still matter more if it bites exactly when coverage is thin. Rank sensitivities twice: first in euros per year, then in downside coverage impact.

Mechanism

A valuation should move through five layers.

1. Physical Layer

What does the asset physically do?

- solar or wind resource
- capacity factor
- availability
- degradation
- clipping

- curtailment
- BESS state of charge, duration, and efficiency
- grid import/export limits

2. Market Layer

What are those MWh worth?

- price zone
- capture price
- negative price exposure
- intraday and balancing exposure
- ancillary-service or flexibility revenue
- saturation and revenue compression scenarios

3. Contract Layer

Who owns which risk?

- PPA price and tenor
- volume and shape obligation
- floor, cap, indexation, or merchant exposure
- certificate treatment
- curtailment and outage treatment
- dispatch/control rights
- credit and termination mechanics

4. Grid And Cost Layer

What does the location cost?

- connection charge
- use-of-network charges
- capacity or power-based charges
- energy charges
- loss-related charges
- reactive-power charges
- balancing costs

- augmentation or repowering

ACER's 2025 tariff report is useful because it defines network charges as more than a single generic tariff. Valuation should not bury them in one unexplained line.

5. Finance Layer

Who gets paid first, and what happens in downside?

- debt service
- reserve accounts
- distributions
- refinancing assumptions
- tax
- merchant-tail terminal value
- lender downside tests
- equity upside cases

Scenario Design

A weak model has one base case and a few arbitrary plus/minus toggles.

A stronger model has coherent scenarios:

Scenario	Physical case	Market case	Contract/grid case	What it tests
Base	Expected production and availability	Central capture and BESS revenue	Known tariff and PPA terms	Investment committee's reference case
Low-price / low-capture	Same MWh, worse timing value	Lower capture, more negative-price exposure	PPA floor/cap may or may not protect	Solar merchant-tail fragility
Grid-constrained	More curtailment or export limits	Fewer sellable MWh in high-production hours	Connection/tariff constraints bind	Location and connection risk

Scenario	Physical case	Market case	Contract/grid case	What it tests
BESS compression	Battery works physically	Ancillary/arbitrage revenue compresses	Degradation still incurred	Revenue-stack fragility
High-volatility upside	Same asset, more spreads	More arbitrage/flex value	Dispatch rights and warranty limits matter	Optionality value

Scenario weights can be useful, but they are assumptions. A `30%` probability is not evidence unless the method behind it is stated.

The Investment-Case Waterfall

A project model for investment review should report NPV and IRR, and it should show where the cash flow is lost as the project moves from physics to finance.

Illustrative investment case:

Layer	Annual EUR effect	Question it answers
Gross physical generation at reference price	6,000,000	What would the asset earn before shape and constraints?
Capture discount	-900,000	What is the cost of producing in lower-value hours?
Curtailment and availability loss	-350,000	How much physical output cannot be monetized?
PPA shape or firming cost	-600,000	What does the customer product cost to serve?

Layer	Annual EUR effect	Question it answers
Grid, tariff, and balancing cost	-500,000	What does the location and settlement perimeter cost?
BESS or optimizer net contribution	+450,000	Does flexibility add value after efficiency, degradation, and fees?
Opex and asset-management cost	-400,000	What is the recurring operating burden?
Debt service	-2,200,000	What cash leaves before equity receives anything?
Equity cash flow before tax	1,500,000	What remains for the sponsor?

The exact numbers are hypothetical. The structure is the point. Each line has an owner, a model, a source of uncertainty, and a mitigation lever.

Now ask the sharper question:

```
What breaks first in the downside case?
```

Downside	Annual EUR effect	Why it matters
Capture price falls by 4 EUR/MWh on 100 GWh exposed volume	-400,000	Same energy, lower timing value
Curtailment rises by 3% of output at 45 EUR/MWh	-135,000	Grid or system congestion turns MWh into zero revenue

Downside	Annual EUR effect	Why it matters
BESS net revenue falls by 40%	-180,000	Optionality was real, but less valuable than expected
Tariff/cost line increases by 200,000 EUR/year	-200,000	Fixed costs hurt most when revenue is already weak
Debt service unchanged	0	Financing usually remains fixed unless refinancing or covenant terms change
Equity cash-flow reduction	-915,000	Equity takes the first pain after fixed claims

Lender and equity views differ. The lender asks whether contracted and resilient cash flow covers debt under stress. The equity sponsor asks whether the retained upside justifies the residual risk after debt, grid, contract, and merchant-tail exposure.

The Financeability Lens

Debt service coverage ratio, or DSCR, is a simple ratio:

```
DSCR = cash available for debt service / debt service
```

If cash available for debt service is `2.75 million EUR/year` and annual debt service is `2.20 million EUR/year`:

```
DSCR = 2.75 / 2.20 = 1.25x
```

A lender DSCR threshold is a deal input, not a universal constant. The mechanism is enough to change the conversation:

- a price downside may be tolerable for equity while failing debt coverage

- fixed grid charges can hurt coverage more than variable costs
- merchant-tail value may support equity value while receiving little lender credit
- optimizer revenue may be valuable but discounted if it is volatile, short-tenor, or weakly controlled
- a PPA can improve financeability while reducing upside or adding shape obligations

The project is bankable only if the cash flows that lenders trust survive the cases lenders care about.

Debt Sizing Is The Inverse Problem

Equity often asks:

```
What return do we get if this debt package is available?
```

A lender often starts from the other side:

```
How much debt can the project support under the cash-flow case we trust?
```

Debt-service capacity follows:

```
maximum debt service = cash available for debt service / required DSCR
```

Units:

- cash available for debt service: EUR per period
- required DSCR: ratio from lender or scenario
- maximum debt service: EUR per period

The `required DSCR` depends on the term sheet, lender, project, contract, and market.

Illustrative downside case:

- Cash available for debt service in downside year: `2.70 million EUR`
- Required DSCR in the scenario term sheet: `1.25x`

```
maximum debt service = 2.70 million EUR / 1.25
                     = 2.16 million EUR/year
```

Now compare a simple fixed-payment debt package. A standard annuity formula gives the payment per euro borrowed:

```
payment_factor = r x (1 + r)^N / ((1 + r)^N - 1)
```

Where:

- `r`: interest rate per period
- `N`: number of payment periods

For `25 million EUR` of debt, `5%` annual interest, and `15` annual payments, the payment factor is about `9.63%`:

```
annual debt service ~= 25.0 million EUR x 9.63%
                    ~= 2.41 million EUR/year
```

The downside DSCR would be:

```
DSCR = 2.70 / 2.41
     ~= 1.12x
```

In this illustrative case, the project may be attractive to equity and still fail the debt-sizing test because the proposed debt service is too high for the downside cash flow. The solution is not always "reject." It could be less debt, longer tenor, different amortization, stronger contracted revenue, a different PPA structure, lower fixed grid cost, or a clearer split between bankable revenue and merchant upside.

This is debt sculpting in plain language: shape the debt service around the cash flows the lender is willing to count, rather than pretending all modeled upside supports the same amount of debt.

Revenue Quality Ladder

Not every euro in the model has the same financing quality.

Revenue or saving	Typical finance question
Long-term contracted revenue from a strong counterparty	Is it enforceable, settled clearly, and resilient under downside?
Contracted revenue with shape or imbalance exposure	Who owns the mismatch, and how volatile is the residual cost?
Grid/tariff saving from a design choice	Is it locked in by connection terms or exposed to tariff changes?
BESS arbitrage or reserve revenue	Is it contractable, saturating, volatile, or dependent on optimizer performance?
Merchant-tail upside	Is this base-case value, equity upside, or lender-discounted optionality?
Portfolio netting benefit	Is it measurable, controllable, and inside the same economic perimeter?

The financeability question is:

```
Which value can safely support fixed obligations?
Which value belongs to equity upside?
Which value should be haircut or excluded until evidence improves?
```

Assumption-Flip Test

Before approving the project, write the one-line test that would change the recommendation.

```
Recommendation flips from approve to reject if:
  downside DSCR falls below the required lender case,
  or equity IRR depends mainly on unsourced merchant-tail / optimizer upsid
  or grid/tariff exposure cannot be bounded before financial close.
```

For a pure equity decision, the flip may be different:

```
Recommendation flips from reject to approve if:
  the PPA transfers shape risk without destroying upside,
  or BESS dispatch rights convert volatile spreads into controllable value,
  or the project reduces portfolio exposure enough to justify weaker standa
```

The assumption-flip test is often the most honest line in the model. It tells management which piece of evidence is worth buying next.

Example: Project Value Versus Portfolio Value

Illustrative asset A has a standalone base NPV of `EUR 2 million`.

Illustrative portfolio effect:

- It reduces imbalance costs elsewhere by `EUR 200,000/year`.
- It improves hedge fit by `EUR 100,000/year`.
- It adds fixed grid cost of `EUR 150,000/year`.
- Ignore discounting for this illustration.

Net annual portfolio effect:

```
200,000 + 100,000 - 150,000 = EUR 150,000/year
```

The standalone project model and portfolio model answer different questions.

```
project_value = asset cash flows on its own
portfolio_value = asset cash flows + effect on the existing portfolio
customer_value = value of the product to the buyer
```

Do not mix them without saying so.

In Practice

For a solar-plus-BESS project in a zonal market such as the Nordics, valuation should make the following sensitivities first-class:

- price zone and capture-price risk
- curtailment and grid-connection constraints
- network tariff structure and tariff year
- balancing and imbalance exposure
- BESS round-trip efficiency and degradation
- warranty/augmentation assumptions
- contracted phase versus merchant tail
- BRP/BSP or optimizer control rights
- portfolio netting and hedge fit

The practice-relevant deliverable is an investment-committee summary that says:

```
Here is the base case.
Here are the assumptions that matter.
Here is what breaks the project.
Here is what the PPA or BESS changes.
Here is what remains merchant, operational, or partner-execution risk.
```

Misconceptions

“More model detail means more truth.” Detail can hide fragile assumptions.

“The base case is expected value.” Only if scenarios and probabilities are defined.

“Lender value and equity value are the same.” Debt cares about downside resilience; equity also cares about upside and residual optionality.

“A contracted project is de-risked” is incomplete. Contracts can leave volume, shape, balancing, curtailment, credit, tariff, and operational risk exposed.

“Merchant tail is just terminal value” is too narrow. Merchant tail depends on market design, asset condition, repowering, capture erosion, and future contractability.

“Debt is cheaper, so more debt is always better.” Debt can increase equity returns in the good case while making the downside case fail earlier.

Caveats / Current Debates

Pexapark's 2026 outlook is useful market color for the current European repricing debate, but the figures are dated and proprietary/practitioner in nature. Live use should distinguish between source-context market observations and durable mechanisms.

Project-finance inputs travel poorly across deals. DSCR thresholds, leverage, WACC, capex, opex, and target IRR depend on the project, lender, country, contract, tax position, and date.

Tariff assumptions are especially fragile. ACER's definitions help structure the model, but actual tariff values require country, voltage level, network user type, tariff year, and connection terms.

Recap

A project valuation is a risk argument written in cash flows. The base case gives the central scenario, but the decision comes from testing which assumption can change the answer: capture price, curtailment, tariff cost, BESS revenue, debt service, contract protection, or merchant-tail value. Once the model separates lender-quality cash flow from equity upside and shows what breaks first in downside, it becomes useful for judgment rather than false precision.

Chapter 14. Portfolio Modeling And Hedging

References: [7] [20] [27] [6] [39]

The investment memo shows a clean project: attractive standalone economics, a familiar zone, a technology the team knows how to build, and a contract structure that can be explained. The portfolio question is what the company already owns in the same hours, same price area, same grid bottleneck, or same partner perimeter. A new asset changes exposure: it adds dependence on some market conditions and reduces dependence on others. Portfolio modeling measures that marginal change before the project is treated as safer than it is.

Opening Puzzle

A 100 MW solar project has a strong standalone NPV. Should a company already holding 800 MW of same-zone solar be happier to add it than a company with mostly wind, storage, and offtake obligations in other zones?

The project model estimates whether the asset makes money on its own. The portfolio model estimates what exposure the company buys more of. A single project can be attractive and still make the portfolio more fragile. It may increase concentration in sunny-hour prices, same-zone congestion, the same BRP’s forecast errors, the same PPA shape exposure, or the same grid-connection bottleneck. Portfolio modeling makes those duplicate exposures explicit.

First-Principles Model

A portfolio is a bundle of exposures.

For each asset or contract, ask what cash flow moves when a state variable moves:

```
Portfolio cash flow = sum of asset cash flows + contract cash flows + h
```

The relevant state variables are concrete:

- area price in each bidding zone
- solar and wind output shape
- curtailment and grid constraints
- imbalance price and imbalance volume
- battery state of charge and dispatch opportunity
- PPA delivery obligation
- counterparty credit and settlement performance
- BRP/BSP or optimizer behavior

The portfolio question is: when a stress happens, do these terms offset each other or concentrate?

Same-zone solar often concentrates exposure. Solar assets produce in the same hours, face similar capture-price pressure, and may share congestion exposure. Wind plus solar may offset some shape exposure if their production patterns differ. BESS may hedge some price-shape exposure if the owner has dispatch rights, grid rights, and a contract that lets the battery respond. A PPA may hedge price but add shape risk. A hedge may reduce price risk but introduce basis risk if the hedge reference does not match the asset’s realized price.

Definitions

`Portfolio exposure` is sensitivity of portfolio value or cash flow to a market, physical, contractual, credit, or operational driver.

`Hedge` is a position, asset, contract, or operating right that reduces adverse movement in an exposure. Mack's hedging chapter frames hedges as risk-reducing positions and highlights basis risk when the hedging instrument and hedged exposure do not move together.

`Netting` means positive and negative deviations partly cancel inside the same commercial or settlement perimeter. It requires compatible time intervals, settlement rules, data, and responsibility boundaries.

`Diversification` means reducing risk because exposures are imperfectly correlated. Owning more assets is not diversification if they all lose value in the same state of the world.

`Basis risk` is mismatch between the exposure you have and the instrument or reference price you use to hedge it.

`Optionality` is the right, but not the obligation, to change behavior when conditions change. A dispatchable battery has more operational optionality than an as-produced solar plant, but only if the owner or its partner controls dispatch within the relevant market windows.

`Dispatch right` is the contractual and operational authority to decide how an asset is charged, discharged, bid, curtailed, or scheduled.

Numerical Intuition

Use exposure concentration before using market data:

```
concentrated exposure = volume x common risk driver x correlation
```

If a portfolio has 3 TWh/year of highly correlated exposed generation, a recurring 2 EUR/MWh basis or imbalance cost is not a rounding error:

```
2 EUR/MWh x 3,000,000 MWh = EUR 6,000,000/year
```

Use this as a scale check rather than a forecast.

Correlation And Concentration

Suppose an IPP has:

- Asset A: 1 TWh/year of same-zone solar exposure.
- Candidate B: 1 TWh/year of same-zone solar exposure.
- Candidate C: 1 TWh/year of exposure with a different production shape or useful dispatch flexibility.
- Stress case: solar capture value drops by 5 EUR/MWh in the same high-solar hours.

If B shares the same exposure, the stress roughly doubles the affected volume:

```
Incremental stress from B = 5 EUR/MWh x 1,000,000 MWh = EUR 5 million/y
```

If C only has half the same exposure, the same stress affects only about 0.5 TWh:

```
Incremental stress from C = 5 EUR/MWh x 500,000 MWh = EUR 2.5 million/y
```

C may have lower expected return, higher capex, worse grid terms, or weaker contracts. The narrow result is that portfolio value depends on covariance as well as expected project cash flow.

Equations That Earn Their Keep

A minimal exposure model can be written as:

```
Portfolio PnL = Energy PnL + Shape PnL + Hedge PnL + Flex PnL − Imbalance C
```

For a single risk factor `x`, a first-pass sensitivity is:

```
Exposure to x = change in portfolio value / change in x
```

For two exposed cash-flow streams, risk depends on covariance:

```
Portfolio variance = Var(A) + Var(B) + 2 x Cov(A,B)
```

Plain language: if A and B move together in bad states, the last term increases risk. If they move opposite ways, it reduces risk. If they are unrelated, the term is small.

This equation is useful only as a teaching model. Electricity portfolios violate many clean financial assumptions: assets are physical, zones congest, contracts are bespoke, liquidity varies, and dispatch rights matter. Mack's portfolio chapter explicitly warns that electricity portfolio management must handle physical and non-financial constraints alongside financial covariance.

Mechanism: How One Asset Changes The Portfolio

When evaluating a new asset, do not start with a correlation matrix. Start with an exposure cube.

Exposure face	Example question	Why it matters
Geography	Is the asset in a zone where the portfolio is already long?	Same-zone congestion and area-price risk can concentrate.
Technology	Does it produce in the same hours as existing assets?	More MWh can mean more capture discount.
Contract	Is output merchant, pay-as-produced, baseload, shaped, capped, floored, or indexed?	The cash-flow shape may differ from the physical shape.
Hedge	What reference price, tenor, and volume is hedged?	A hedge can reduce price risk while adding basis, maturity, or collateral risk.
Flexibility	Who controls BESS dispatch or curtailment?	Optionality exists only if it can be exercised.
Counterparty	Are buyer, BRP, optimizer, and data provider concentrated?	Operational and credit concentration can hide behind asset diversification.
Liquidity	Could the hedge be adjusted or exited under stress?	A theoretically clean hedge may fail when liquidity is thin.
Data/control	Can performance be measured against a feasible benchmark?	Unobservable decisions cannot be governed.

Only after the cube is visible should you quantify the most important faces.

When evaluating a new asset, run it through seven questions.

1. Price exposure: Does it earn the same reference price as the existing book, or a different area price?
2. Shape exposure: Does it produce or consume in hours where the book is already long or short?

3. Volume exposure: Does forecast error net with or add to current forecast error?
4. Constraint exposure: Does it sit behind the same grid bottleneck, connection rule, or curtailment regime?
5. Contract exposure: Does it create or hedge obligations under PPAs, floors, caps, CfDs, tolls, or merchant tails?
6. Operational exposure: Who has dispatch rights, bidding rights, and imbalance responsibility?
7. Partner exposure: Does it depend on the same optimizer, BRP, BSP, supplier, or data vendor?

A portfolio view is built by stacking these exposures, not by averaging project NPVs.

Marginal Risk Contribution

The marginal-risk question is:

```
What risk does the next asset add at the margin?
```

Illustrative portfolio:

- Existing portfolio: 2 TWh/year, heavily solar-exposed in one bidding zone.
- Candidate solar project: 0.3 TWh/year in the same zone.
- Candidate wind project: 0.3 TWh/year in a different shape and zone.
- Stress: high-renewables hours clear 6 EUR/MWh below the base case.

If the new solar project is fully exposed to the same stress:

```
marginal stress loss = 300,000 MWh x 6 EUR/MWh
                     = EUR 1,800,000/year
```

If the wind project is only 30 percent exposed to that same stress:

```
marginal stress loss = 300,000 MWh x 30% x 6 EUR/MWh
                     = EUR 540,000/year
```

The solar project may still be better. It may have lower capex, a stronger PPA, better grid terms, or higher expected value. But the investment memo must now say:

```
The standalone NPV is one part of the decision.
The marginal portfolio risk is materially different.
```

This connects project development to portfolio strategy.

Hedge-Quality Checklist

A hedge is good only relative to the exposure it is meant to hedge.

Test	Weak answer	Useful answer
Reference price	“We hedged power.”	Which area/system/product price is hedged?
Volume	“We hedged 50 MW.”	Which MWh, in which hours, under which production uncertainty?
Shape	“Annual volume matches.”	Does the hedge match the hourly/monthly shape that drives cash flow?
Tenor	“The project is long term.”	Does the hedge mature before the exposure?
Basis	“Prices are correlated.”	What happens when congestion or weather breaks the relationship?
Liquidity	“We can adjust later.”	Can the hedge be changed when the market is stressed?
Collateral	“Economics are locked.”	What cash must be posted before delivery revenue arrives?

Test	Weak answer	Useful answer
Governance	“The partner manages it.”	Can the owner see the hedge, benchmark it, and approve changes?

The purpose of the checklist is to expose residual costs, basis risk, liquidity pressure, and control limits before the hedge is treated as protection.

Examples

Before examples, put the candidate site inside the existing book.

Portfolio question	Stronger test
Does the site add the same exposure we already have?	Compare incremental MWh by zone, technology shape, settlement perimeter, and stressed hours.
Does the site reduce an existing obligation?	Map the candidate's production or flexibility against current PPA shape, imbalance, and hedge gaps.
Does the site use a scarce internal capability?	Check whether the same BRP, BSP, optimizer, data pipeline, or credit line becomes a bottleneck.
Does the site create an option?	Identify the specific right: export, import, curtailment, storage dispatch, future hybridization, or local flexibility.

A lower standalone margin can be acceptable if the site reduces a portfolio concentration that already costs money in stress. A higher standalone margin can be rejected if it simply adds volume to the same bad state. The investment memo should make that trade visible rather than bury it in a single project IRR.

Same-zone solar plus same-zone solar is simple to build and easy to understand, but it may concentrate capture-price risk. In its 2026 outlook, Pexapark frames declining capture factors and negative-price risk as major issues in renewables-heavy markets.

Solar plus wind may reduce production-shape concentration if wind output is imperfectly correlated with solar output. But it can still share price-zone, grid, balancing, credit, and partner exposure.

Solar plus BESS can hedge some shape and imbalance exposure if the battery can charge or discharge at useful times. BESS works as a hedge only when the commercial rights, grid access, state-of-charge constraints, degradation limits, and market windows allow it to respond.

A financial hedge can reduce price exposure while leaving basis risk. For a Nordic asset, a system-price hedge may not perfectly hedge area-price revenue if the asset is paid an area price that diverges because of congestion. Nord Pool's bidding-area explanation supports the mechanism: bidding areas produce local prices when transmission constraints bind, while the system price is a separate Nordic reference used for many financial contracts.

In Practice

For a solar-plus-BESS IPP with Nordic area-price exposure, the portfolio model should separate:

- same bidding zone versus different bidding zone
- system-price reference versus area-price realized revenue
- as-produced solar volume versus shaped PPA obligation
- battery dispatch right versus third-party optimization right
- balancing responsibility versus balancing-service provision
- grid connection capacity versus commercial offtake volume

The practice-relevant output is a decision memo that says:

```
This asset is attractive on standalone economics, but it increases same-zon
```

Where `X` is not sourced, keep it as scenario volume. Do not invent correlations.

Misconceptions

Diversification means lower co-movement in the states you care about.

A hedge usually trades upside, liquidity, margining, complexity, or basis risk for reduced downside.

Portfolio value is not the sum of project values. Project value ignores how the asset interacts with existing contracts, exposures, and control rights.

BESS becomes a portfolio hedge only when its physical option value survives duration, power, energy, efficiency, degradation, grid rules, market rules, and dispatch-control constraints.

Netting depends on settlement perimeter, timing, measurement, and legal responsibility.

Current Debates And Caveats

Pexapark's 2026 outlook argues that IPPs are moving from passive asset ownership toward active revenue management, portfolio optimization, and flexibility. Read it as 2026 practitioner market color on the direction of travel.

Correlations are unstable. A solar-wind relationship estimated over one year may fail in a drought year, a grid-constrained year, or a policy-shock year.

Hedge liquidity varies by zone, tenor, product, and counterparty. Hedging is used conceptually here; specific instruments need current market validation before use in an investment committee recommendation.

Recap

A portfolio model turns a project decision into an exposure decision. The standalone NPV still matters, but the next asset may concentrate sunny-hour price risk, same-zone congestion, shape obligations, hedge basis, partner dependence, or dispatch constraints. The durable test is marginal risk contribution. Identify which market condition already hurts the book, then check whether the new asset, contract, hedge, or control right offsets that condition or adds more volume to it.

Chapter 15. Trading Partners And Performance Scorecards

References: [3] [4] [12] [20] [43]

A quarterly report shows the trading partner earned less than the benchmark used in the plan. Sales needs an explanation, finance needs a quantified impact, and the asset owner needs to know whether the result came from execution, market conditions, constraints, data quality, or the mandate itself. Realized profit and loss (PnL) cannot answer that alone. The required comparison is a feasible version of the same quarter: schedules, bids, dispatch choices, data, risk limits, contract rights, and market windows that were actually available at the time. Only then can governance separate poor execution from low spreads, bad forecasts, curtailment, missing data, or limits the partner was never allowed to cross.

Opening Puzzle

Your outsourced trading partner earns `2 EUR/MWh` less than the planning benchmark over a quarter.

The gap may reflect poor solar forecasts, low spreads, unexpected curtailment, conservative risk limits, missing meter data, poor market liquidity, contractual rules that prevented useful dispatch, or partner execution. A bad realized result can have causes other than execution.

Governance needs a counterfactual that is feasible under the information and control rights available at the time.

The benchmark must ask: with the information, constraints, forecasts, market access, risk limits, and control rights available at the time, what could a competent actor reasonably have done?

First-Principles Model

Partner oversight is a three-line comparison:

```
Realized result
− Feasible benchmark result
= Execution delta
```

`Feasible` is the controlling word.

A benchmark simulates decisions that were possible before or during delivery, using only data that existed at that moment. It must respect:

- forecasts available then
- market prices and order-book information available then
- asset limits and outages known then
- battery state of charge and degradation limits then
- grid and connection limits then
- PPA obligations and risk limits then
- market gate closures and liquidity then
- contractual control rights then
- data actually visible to the asset owner or partner then

If the partner could not dispatch the battery, the benchmark cannot dispatch it. If the asset owner did not have bid-level data, the scorecard must state that the benchmark is narrower. If a contract forbids grid charging, a benchmark that grid-charges violates the feasible-information test.

Definitions

`Shadow trading` is a counterfactual simulation of feasible trading, dispatch, or bidding decisions used to benchmark actual execution.

`Feasible counterfactual` is the action path that could have been taken with contemporaneous information and real constraints.

`Realized PnL` is the financial result from actual schedules, bids, dispatch, settlement, and costs.

`Benchmark PnL` is the simulated financial result from the feasible benchmark policy.

`Execution delta` is realized PnL minus benchmark PnL, after normalizing for constraints, risk limits, data availability, and control rights.

`BRP` is the party financially responsible for imbalances at relevant points or portfolios. The eSett handbook frames imbalance settlement around BRPs, trades, metered data, and imbalance adjustments.

`BSP` is the party providing balancing services to the TSO where qualified. Svenska kraftnät’s English page describes the Swedish split of actor roles into BSP and BRP from 1 May 2024 and notes that full independent BSP implementation is expected by 2028 at the latest.

`Control right` is the right to decide, instruct, or veto an action: schedule, bid, dispatch, curtail, charge, discharge, accept imbalance exposure, or offer reserves.

`Data right` is the right to receive the data needed to reproduce and audit decisions: forecasts, bids, prices, dispatch instructions, metering, outages, state of charge, settlement line items, and model/risk-limit settings.

Numerical Intuition

The scorecard number is feasible leakage:

```
feasible leakage = feasible benchmark value − actual realized value
```

This turns a governance conversation into a testable counterfactual. If the relevant exposed volume is 250 GWh and the feasible gap is `1 EUR/MWh`, then:

```
1 EUR/MWh x 250,000 MWh = EUR 250,000
```

The calculation is a materiality test, not the scorecard. The follow-up question is whether the `1 EUR/MWh` gap is attributable to execution, data, risk limits, market liquidity, settlement errors, missing control rights, or a benchmark that was never feasible.

When A Small Delta Matters

Suppose a partner manages 500 GWh/year of exposed production and flexibility. A scorecard finds a persistent, feasible execution delta of `1.5 EUR/MWh` after removing unavailable actions and known constraints.

```
500 GWh/year = 500,000 MWh/year
1.5 EUR/MWh x 500,000 MWh = EUR 750,000/year
```

That number might justify better data rights, a revised incentive fee, tighter reporting, or partial internalization of analytics. Building a full trading desk would need a larger and more durable case. The model should inform the operating decision rather than prescribe it.

Mechanism: Building A Shadow-Trading Benchmark

Start with the commercial question:

```
Was the partner's execution worse than a feasible alternative?
```

Then build the benchmark in layers.

1. Reconstruct the asset state.

For solar and wind, reconstruct available generation, actual generation, curtailment, outages, forecasts, metered volumes, and schedules. For BESS, include state of charge, power limit, energy limit, round-trip efficiency assumption, degradation rule, availability, and grid import/export constraints.

2. Reconstruct the obligation.

List PPAs, hedges, reserve commitments, imbalance responsibility, grid constraints, negative-price clauses, floors, caps, and any customer delivery shape.

3. Reconstruct the decision window.

What could be decided day-ahead? What could be adjusted intraday? What was still possible close to delivery? eSett's settlement sources emphasize time-bound data and reporting obligations; market sequence matters because decisions expire.

4. Reconstruct available information.

Use the forecasts, prices, and outage information that existed then. Never use realized prices, realized weather, or final metering as if the trader knew them earlier.

5. Apply risk limits.

If the commercial mandate says "avoid exposure beyond X," the benchmark must obey X. A benchmark that makes money by violating the mandate is not evidence of underperformance.

6. Apply control rights.

If the partner had bidding rights but not physical dispatch rights, or if the asset owner retained curtailment decisions, separate those effects.

7. Calculate benchmark PnL and attribution.

Do not stop at one number. Attribute the delta to forecast error, market movement, liquidity, constraint, data issue, risk-limit effect, contract limitation, or execution choice.

Partner Scorecard

A useful scorecard has three layers: outcome, attribution, and governance.

Outcome metrics:

- realized PnL versus feasible benchmark PnL
- EUR/MWh execution delta over affected volume
- imbalance cost per MWh
- capture price versus relevant benchmark
- reserve revenue versus feasible qualified availability
- missed opportunity cost from unavailable dispatch

Attribution metrics:

- forecast error by horizon and product
- schedule error versus actual generation or load
- imbalance volume and skewness
- unavailable asset hours
- state-of-charge constraint hours
- grid/curtailment constraint hours
- risk-limit binding hours
- market-liquidity limited hours
- data-quality exceptions

Governance metrics:

- data delivery timeliness
- settlement reconciliation completion
- bid and dispatch auditability
- explanation quality for outlier days
- compliance with agreed risk limits
- escalation speed for outages, curtailment, or market incidents
- incentive alignment with asset-owner value

eSett's market-behaviour reporting chapter is a useful official analogy: it monitors BRP performance with KPIs, imbalance metrics, and escalation logic. An asset-owner scorecard should be narrower and contractual, but the principle is similar: repeated imbalance or data problems are governance facts, not anecdotes.

Illustrative escalation rules make the scorecard usable. These are governance thresholds for the example, not standard market rules.

Signal	Watch level	Escalate level	Action
Feasible execution delta	`0.5 EUR/MWh` for one month	`1.0 EUR/MWh` for two months	Request attribution pack and remediation owner.
Settlement unreconciled	`>1%` of monthly MWh after first close	`>3%` or repeated late close	Freeze performance fee calculation until reconciled.
Missing bid or dispatch evidence	Any material event	Repeated events or high-value days	Mark benchmark as unverified and trigger data-right review.
Risk-limit breach	Near breach	Actual breach or late notification	Require incident report and revised mandate.

Data Rights And Control Rights Assumptions

A scorecard is only as strong as its rights.

Minimum data-right assumption for a serious benchmark:

- metered generation/consumption
- final and preliminary settlement data
- schedules/nominations

- bids submitted, accepted, rejected, and activated
- market prices and reference curves used
- forecast versions available at decision time
- curtailment and outage logs
- battery state-of-charge history, if applicable
- risk-limit and mandate settings
- grid/connection limit records
- partner explanations for exceptions

Minimum control-right map:

- who submits day-ahead orders
- who trades intraday
- who controls dispatch
- who can curtail
- who offers reserves
- who changes risk limits
- who carries imbalance cost
- who receives upside from optimization
- who is responsible for data and settlement reconciliation

If these are not in the contract, the benchmark must say so. “We cannot evaluate reserve-bidding quality because we lack bid-level reserve data” is better than pretending.

The settlement and credit evidence chain should be explicit.

Evidence item	Why it matters	Minimum owner question
Metered generation, load, and storage intervals	Determines physical imbalance and delivery.	Which meter version is final, and when does it close?
Schedules, nominations, trades, and activations	Reconstructs what the BRP/BSP actually submitted or delivered.	Can we tie each settlement line to a schedule or activation record?
Bid, acceptance, rejection, and curtailment logs	Separates missed opportunity from unavailable opportunity.	What did the partner know and submit before gate

Evidence item	Why it matters	Minimum owner question
		closure?
Settlement statements and invoices	Converts operational events into cash.	Are imbalance, activation, fees, and corrections reconciled?
Collateral and credit-limit usage	Shows whether good trades were blocked by credit or cash limits.	Did credit terms constrain trading, and was the owner notified?
Exception and incident records	Creates governance memory.	Which exceptions repeated, and what control changed afterward?

If the chain breaks, the conclusion changes from “the partner underperformed” to “we cannot yet distinguish execution, data, settlement, credit, and mandate effects.”

In Practice

In Nordic practice, partner oversight must respect the settlement model. The eSett handbook frames imbalance settlement through trades, metering, reporting, settlement calculations, invoicing, collateral/risk management, and market behaviour monitoring. Svenska kraftnät’s BRP/BSP page adds an important Swedish caveat: BSP and BRP roles are being separated, with transition and IT-system constraints.

For a renewable IPP, the practical work product is a monthly governance pack:

```
1. What happened?
2. What would the feasible benchmark have done?
3. What explains the gap?
4. Which gaps are market conditions?
5. Which gaps are data/control/contract limitations?
6. Which gaps are partner execution?
7. What action follows?
```

The answer may be renegotiate data rights, modify the incentive fee, tighten risk limits, change dispatch permissions, improve forecasts, or accept that the partner performed well in a bad market.

Misconceptions

“The benchmark should use perfect hindsight.” Perfect hindsight measures regret, not execution.

“Low revenue means poor partner performance.” Low revenue can come from market conditions, constraints, risk limits, forecasts, or contract design.

“A single benchmark is enough.” Use benchmark families: conservative, base, and aggressive within the same feasible rights.

“The asset owner can govern without bid-level data.” Sometimes, but only weakly. Without data rights, the scorecard must be modest.

“If the partner is a BRP or BSP, they control everything.” Responsibilities and control rights are contractual and jurisdiction-specific.

Current Debates And Caveats

Advanced stochastic or agent-based models can improve benchmark realism, but they can also become fragile. Shinde’s thesis supports the idea that stochastic models can outperform deterministic approaches in short-term market settings, while also highlighting computational cost and dependence on forecasts.

For governance, transparency matters. A benchmark that can be reproduced from contemporaneous data, constraints, and rights is more useful than an opaque optimizer whose result cannot be attributed after a bad week.

Recap

Performance attribution starts with realized result minus feasible benchmark result. Shadow trading only helps if the benchmark uses the information, constraints, market access, risk limits, and control rights that existed when decisions were made. That turns the opening complaint about underperformance into a governance test: which part of the gap came from market conditions, forecast error, contract design, data gaps, binding limits, or actual execution? A partner scorecard is strongest when it makes that attribution visible enough to guide the next operating decision.

Chapter 16. Analytical Capability Development

References: [3] [20] [27] [49] [43] [6] [39] [50]

A renewable independent power producer (IPP) can spend its first analytics budget many ways: a pricing model, a data platform, an energy trading and risk management (ETRM) system, a forecast vendor, a partner scorecard, a risk pack, a quant hire, or a battery optimizer. Each choice sounds sensible in isolation. The constraint is that year-one budget, headcount, data rights, and management attention are scarce. The roadmap has to choose which decisions receive analytical capability now, which are bought, and which are deliberately left rough until the evidence improves.

Opening Puzzle

Management approves a constrained year-one build:

- budget: `900,000 EUR`
- team: `1 accountable owner`, `1 analyst`, and `0.5 data engineer`
- outsourced balance responsible party (BRP), balancing service provider (BSP), and optimizer execution
- incomplete partner data rights
- three value pools under pressure: shaped power purchase agreements (PPAs), battery energy storage system (BESS) design, partner oversight

The ideal analytical engine can wait. The immediate question is which minimum set of capabilities changes the largest decisions this year without pretending the company has a full trading house.

First-Principles Model

Build the roadmap from decisions:

```
decision -> model -> data rights -> owner -> control -> action -> outco
```

Each candidate capability must pass four gates:

```
roadmap_priority =
  value_pool_size
  x decision_frequency_or_urgency
  x data/control_readiness
  x implementation_feasibility
  / governance_risk
```

The score disciplines the conversation. It prevents the roadmap from starting with the tool that has the best demo.

Budget And Headcount Constraint

Scenario capacity:

Resource	Year-one limit
Cash budget	`900,000 EUR`
Internal team	`1 accountable owner + 1 analyst + 0.5 data engineer`
External execution	BRP/BSP and optimizer partners
Management review cadence	monthly steering group
Delivery tolerance	useful first version in `90 days`; stable operating version in `12 months`

Candidate backlog:

All costs in this backlog are scenario budget assumptions for the exercise, not vendor quotes or market benchmarks. Pexapark and PRMIA support the capability categories and control questions; a live procurement would replace these placeholders with dated vendor proposals, implementation estimates, information-security review, and internal staffing costs.

Capability	Cost	Internal load	Main value pool	Data/control readiness
PPA quote stack and risk map	`160,000 EUR`	medium	shaped PPAs and customer pricing	high
BESS sizing and dispatch model	`140,000 EUR`	medium	project design and investment-committee (IC) support	medium
Partner data-rights remediation	`120,000 EUR`	high	BRP/BSP oversight	low until contracts change
Feasible benchmark scorecard	`180,000 EUR`	high	partner oversight and mandate design	medium after data remediation
Monthly portfolio risk pack	`110,000 EUR`	medium	management exposure decisions	medium
Lightweight data contracts and reconciliation	`130,000 EUR`	high	all value pools	medium

Capability	Cost	Internal load	Main value pool	Data/control readiness
Forecast/vendor evaluation layer	`90,000 EUR`	low-medium	trading, BESS, partner oversight	medium
Full ETRM implementation	`650,000 EUR+`	very high	deal capture, settlement, audit	low for year one
Internal trading desk build	`1.5m EUR+`	very high	execution margin	low without controls

The company has limited year-one capacity. The roadmap should fund the capabilities that change decisions and unlock later capabilities.

Numerical Intuition

Illustrative annual value at stake:

Value pool	Affected volume or decision	Plausible improvement	Gross value
Shaped PPA pricing	`700 GWh/year` offers	`0.80 EUR/MWh` better risk pricing	`560,000 EUR/year`
BESS design decisions	`3` projects	`250,000 EUR/year` avoided design loss each	`750,000 EUR/year`
Partner oversight	`500 GWh/year` outsourced execution	`0.70 EUR/MWh` leakage reduction	`350,000 EUR/year`
Portfolio risk pack	management decisions and hedges	scenario assumption	`250,000 EUR/year`

These are scenario values, not sourced market facts. Their purpose is to compare decision leverage.

The first-year roadmap should favor capabilities that either capture value directly or create the evidence required to capture value later. Partner oversight may have high value, but if bid, forecast, schedule, dispatch, meter, and settlement rights are missing, the first workstream is data rights before a sophisticated benchmark.

Year-One Budget Decision

The roadmap has to fit the `900,000 EUR` budget and the small team. Data-rights work remains in scope; broad scorecard automation waits until the evidence base improves.

Capability	Year-one decision	Budget	Reason
Data contracts and reconciliation	Approve	`130,000 EUR`	Required before quote, risk, settlement, and partner evidence can be trusted.
PPA quote stack and risk map	Approve	`160,000 EUR`	Addresses the `560,000 EUR/year` scenario value in shaped-offer risk pricing.
BESS sizing and dispatch model	Approve	`140,000 EUR`	Addresses the `750,000 EUR/year` scenario value across three project decisions.
Partner data-rights remediation	Approve	`120,000 EUR`	Full scorecards are weak until raw evidence exists.
Monthly portfolio risk pack	Approve	`110,000 EUR`	Gives management one exposure view from the same position data.
Forecast/vendor evaluation layer	Approve	`90,000 EUR`	Supports PPA, BESS, and partner decisions through a time-stamped forecast archive.
Feasible benchmark scorecard	Gate as pilot	`50,000 EUR`	Define method and test one asset/month

Capability	Year-one decision	Budget	Reason
			sample; defer automation.
Contingency and validation	Approve	`100,000 EUR`	Protects delivery and independent review.
Full ETRM implementation	Defer	`650,000 EUR+`	Exceeds year-one budget, team capacity, and control readiness.
Internal trading desk build	Defer	`1.5m EUR+`	Data rights, limits, settlement, and risk controls are prerequisites.

Approved year-one budget:

```
130 + 160 + 140 + 120 + 110 + 90 + 50 + 100 = 900 thousand EUR
```

The roadmap therefore spends `50,000 EUR` on benchmark method and sample testing, not `180,000 EUR` on a platform that cannot yet prove attribution.

Capability Dependencies

The dependencies matter as much as the ranking.

```
data rights
  -> reconciliation
  -> position and settlement truth
  -> PPA quote stack / BESS model / risk pack
  -> feasible benchmark
  -> partner mandate or fee redesign
```

PPA pricing can start before every partner feed is fixed, because term sheets and market curves can be versioned internally. Partner benchmarking cannot. It needs raw evidence from forecast, bid, schedule, dispatch, activation, meter, and settlement files.

BESS design sits between the two. A project model can start with scenario assumptions, but dispatch value should be haircut until optimizer control rights, reserve commitments, grid import/export rights, and degradation policy are explicit.

Minimum Deliverables

Quarter	Deliverable	Acceptance test
Q1	data contract inventory and reconciliation design	every critical feed has owner, timestamp, unit, granularity, validation rule, and gap owner
Q1	PPA quote stack v1	every offer shows volume, shape, basis, certificates, firming, balancing, credit, and residual risk
Q2	BESS sizing and dispatch model v1	IC can compare duration, revenue stack, degradation, grid rights, and control-right assumptions
Q2	monthly portfolio risk pack v1	management sees exposure by zone, technology, contract, counterparty, hedge, merchant tail, and partner
Q3	forecast/vendor evaluation layer	value-weighted forecast error is tracked for day-ahead, intraday, BESS, and partner use
Q4	partner benchmark pilot	one partner or asset class has feasible attribution using

Quarter	Deliverable	Acceptance test
		time-available data

The first versions may be lightweight models or databases. They need owners, versioning, checks, and reconciliation. Unowned production files are a control failure, even when their formulas are clever.

Build, Buy, Outsource, Or Defer

Capability	Year-one choice	Reason
PPA pricing logic	build internally, with external curve inputs	product risk allocation is strategic
Long-term market curves	buy plus challenge	external views are useful; assumptions need internal ownership
BESS optimizer execution	outsource	24/7 execution and market access are operational capabilities
Dispatch benchmark	build methodology internally	partner governance requires independent feasible comparison
Forecast inputs	buy initially	weather modeling is rarely the first proprietary edge
Value-weighted forecast layer	build internally	company-specific contracts and assets determine value
ETRM	defer or buy lightweight deal/position control	full implementation exceeds year-one capacity

Capability	Year-one choice	Reason
Internal trading desk	defer	data rights, limits, settlement, and risk controls are prerequisites

Pexapark's Next-Gen IPP Playbook is useful market commentary on active revenue management, portfolio management, and middle-office control. PRMIA's ETRM material is useful for control logic around deal capture, risk, settlement, invoicing, workflow, and auditability. These sources guide questions; they do not mandate a full trading organization in year one.

Kill Criteria

Every capability gets a kill or redesign rule.

Capability	Continue if...	Kill or redesign if...
PPA quote stack	used in live offers and changes price, structure, or no-bid decisions	commercial team bypasses it or cannot explain residual risk
BESS model	changes IC recommendation or design option	optimizer value cannot be reconciled to constraints and degradation
Data contracts	critical feeds move toward agreed completeness and timeliness	partner contracts block raw evidence after escalation
Risk pack	management uses it to set limits, hedge, escalate, or approve deals	it becomes a passive dashboard with no decisions
Forecast layer	value-weighted error reduction exceeds cost	updates arrive too late or cannot be tied to actions

Capability	Continue if...	Kill or redesign if...
Benchmark pilot	attribution separates data, forecast, constraint, risk-limit, and execution effects	it depends on hindsight or unavailable data

Kill criteria make the roadmap safer. They give management permission to stop funding work that cannot change a decision.

Decision Frame

The constrained recommendation is:

1. Fund common data contracts and reconciliation first.
2. Build the PPA quote stack and BESS model because live commercial and IC decisions need them.
3. Remediate partner data rights before scaling any scorecard.
4. Create a monthly portfolio risk pack from the same position data.
5. Buy forecast inputs, but build the internal value-weighted evaluation layer.
6. Defer full ETRM and internal trading desk until volume, controls, and data quality justify them.

This roadmap spends the full `900,000 EUR` budget and stays inside the year-one team constraint by limiting automation and treating the partner benchmark as a pilot.

Misconceptions

"The roadmap should start with the platform." The platform should follow the decision and data contracts it must support.

"A dashboard is a control." A control has an owner, threshold, evidence trail, escalation path, and consequence.

"Outsourcing execution removes the need for analytics." Outsourcing execution increases the need for mandate design, raw data rights, feasible benchmarks, and settlement checks.

"A quant hire solves the engine." One strong hire without data rights and decision ownership becomes a bottleneck.

“Every value pool needs a model now.” Some value pools should wait until data, controls, and decision frequency justify the fixed cost.

Current Debates And Caveats

Operating-model choices depend on portfolio scale, market access, retained merchant risk, contract complexity, partner quality, data rights, budget, and risk appetite. Practitioner material can identify the direction of travel toward active revenue management, but it should be treated as dated market analysis.

ETRM, optimizer, forecast, and data-platform choices are vendor- and process-specific. A real procurement decision needs current requirements, demos, references, information-security review, and implementation estimates.

Recap

The analytical roadmap is a capital-allocation problem. With `900,000 EUR` and a small team, the company should buy inputs where the market is mature, build the decision layers that express its own contracts and assets, and defer full systems until data rights and controls are ready. The year-one answer is a constrained operating capability: data contracts, PPA pricing, BESS design, portfolio risk, forecast evaluation, and a modest partner benchmark pilot with explicit kill criteria.

Chapter 17. Trading Risk, Credit, Liquidity, And Controls

References: [49] [27]

A portfolio manager opens the morning meeting by saying that the book is hedged. The statement is incomplete until the team knows where the hedge settles, which counterparty owes collateral, whether treasury has cash before revenue arrives, and whether yesterday's curve and meter data are clean enough to trust. The same trade that reduces spot exposure can move risk into collateral, credit, basis, or process failure. For a renewable IPP adding more active energy management, each contract, dispatch choice, and partner handoff creates a position that must be valued, funded, limited, and escalated before stress turns it into a surprise.

Opening Puzzle

A portfolio manager says, "We are hedged."

The follow-up is: hedged against what?

A fixed-price sale can hedge spot-price exposure while creating collateral pressure. A forward purchase can reduce shape risk while increasing counterparty exposure. A battery dispatch strategy can improve expected revenue while making losses worse in rare price patterns. A trading book can look profitable and still run out of cash if margin calls arrive before revenue does.

Trading risk changes form as contracts, hedges, dispatch choices, collateral terms, and settlement processes move exposure between parties and time periods.

First-Principles Model

Trading risk is a stack:

```
trading risk = market risk + credit risk + liquidity risk + operational
```

Market risk is the risk that prices move.

Credit risk is the risk that a counterparty does not perform.

Liquidity risk is the risk that you cannot trade, fund collateral, or exit a position when needed.

Operational risk is the risk that the process, data, system, limit, approval, valuation, or handoff fails.

The control failure is optimizing one layer while ignoring the others.

Definitions

Market risk is exposure to changes in prices, spreads, volatility, basis, or correlations.

Value at Risk, or VaR, is a loss threshold for a chosen probability and horizon. A one-day 95% VaR says, roughly, "in this model, losses exceed this amount in 5% of comparable cases."

Expected shortfall measures the average loss beyond the VaR threshold.

Mark-to-market (MtM) is the current modeled value of a position using current market inputs or curves.

Stress testing asks what happens under a deliberately severe scenario, even if the scenario is rare or outside recent history.

Counterparty credit risk is the risk that the party on the other side of a contract fails to pay or deliver.

Potential future exposure is a forward-looking estimate of how large the exposure to a counterparty might become if prices move.

Collateral is value posted to reduce counterparty exposure.

Margin call is a demand to post additional collateral after prices or exposures change.

Liquidity risk has two useful flavors. Market liquidity is the ability to trade without moving the price too much. Funding liquidity is the ability to find cash when obligations come due.

ETRM means energy trading and risk management system. In practice, it is the controlled record of trades, positions, valuations, limits, settlements, and reporting.

Numerical Intuition

Start with stress PnL on an open position:

```
first-pass stress PnL = open MWh x adverse price move
```

If a position covers 36,000 MWh and the relevant price moves 15 EUR/MWh against it:

```
36,000 MWh x 15 EUR/MWh = 540,000 EUR
```

That number creates a cash and control question. Current mark-to-market and credit exposure then depend on trade price, current curve, netting, thresholds, collateral terms, and termination terms.

VaR and expected shortfall are model outputs, not controls by themselves:

```
95% VaR = loss threshold exceeded in 5% of modeled cases
expected shortfall = average loss when losses exceed that threshold
```

VaR tells you where the tail begins. Expected shortfall tells you something about the tail.

Worked Estimate

Suppose a renewable portfolio has a forward sale covering 50 MW for 30 days.

Approximate the energy:

```
50 MW x 24 h/day x 30 days = 36,000 MWh
```

Now suppose the forward mark moves against the position by `20 EUR/MWh` before delivery:

```
36,000 MWh x 20 EUR/MWh = 720,000 EUR
```

The portfolio may still be economically hedged against future spot prices. But it may need to fund collateral today.

Now ask the risk questions:

- Is the exposure netted against offsetting trades?
- Is collateral due daily, weekly, or only after a threshold?
- Does the organization forecast cash needs under stress?
- Is the price used for valuation independent and reproducible?
- Are limits defined by nominal MW, MWh, VaR, stress loss, liquidity need, or all of them?
- Who can approve more risk when the market is moving?

That separates “we have a hedge” from “we have a controlled portfolio.”

Equations

Market-risk exposure:

```
exposure = volume x price sensitivity
```

For a simple linear position:

```
exposure EUR = MWh x EUR/MWh
```

Liquidity need:

```
liquidity need = collateral due before cash inflows + operating buffer
```

Control value:

```
control value = avoided tail loss + avoided leakage − control cost
```

Expected shortfall:

```
expected shortfall = average loss conditional on loss being beyond VaR
```

The equations are simple. The difficult step is deciding which exposure boundary is real.

Mechanism

Start with a trade.

It creates a position. The position creates market sensitivity. The market sensitivity creates mark-to-market exposure. The exposure creates credit exposure between counterparties. Credit exposure creates collateral or limit usage. Collateral creates liquidity need. Liquidity need creates treasury risk. The trade record creates settlement and operational risk.

Each step can fail differently.

A price move can hurt the economics.

A counterparty default can turn a gain into a loss.

A margin call can force a cash scramble.

A stale curve can make the valuation wrong.

A missing position can make a limit meaningless.

Stale model data may make the risk report wrong.

A trading control system makes risk visible early enough for action.

The Daily Risk Pack

A renewable IPP needs a daily view that connects contracts, assets, markets, partners, and cash, without bank-style complexity on day one.

Section	Minimum question	Example metric
Positions	What are we long or short, by hour, zone, product, and counterparty?	MW/MWh by delivery period

Section	Minimum question	Example metric
Mark-to-market	What changed in value since yesterday?	EUR PnL by curve and position
Market risk	What happens under normal and stressed price moves?	VaR, expected shortfall, stress loss
Credit exposure	Who owes whom if the market moved today?	current exposure and potential future exposure (PFE)
Liquidity	What cash could be needed before revenue arrives?	cash-flow-at-risk (CFaR), liquidity-at-risk (LaR), collateral due
Basis and shape	Where does the hedge fail to match the asset or contract?	unhedged shape MWh, basis EUR/MWh
Imbalance and partner execution	What did forecasts, nominations, dispatch, and settlement imply?	imbalance MWh and EUR, benchmark gap
Limits	Which limits are near breach?	utilization by limit type
Data quality	Can we trust the report?	missing trades, stale curves, unreconciled meters
Actions	Who must decide today?	owner, deadline, decision right

The daily risk pack is a completeness test. If a material position, curve, collateral term, or partner action is missing, the report says so.

The monthly management version is shorter but stricter. It should answer whether the operating model is improving.

Monthly pack section	Management question
Portfolio exposure	Which zone, technology, product, and counterparty exposures grew materially?
Revenue attribution	What came from market conditions, asset availability, optimization, partner execution, and contract design?
Credit and liquidity	Which counterparties or collateral calls used risk appetite, and what cash buffer remains under stress?
Partner governance	Which BRP/BSP/optimizer issues repeated, and which ones changed behavior after escalation?
Control exceptions	Which limits, data feeds, reconciliations, approvals, or incidents failed?
Decisions requested	Which mandate, hedge, contract, data-right, staffing, or system change needs approval?

For a small first-year team, this pack connects analysis to operating control. It lets leadership see whether active revenue management is becoming a governed capability.

Credit And Liquidity Control Loop

Credit exposure and liquidity exposure are cousins, not twins.

Credit asks:

```
If the counterparty fails, what can we lose?
```

Liquidity asks:

```
If the market moves before cash comes in, what must we fund?
```

The loop is:

```
trade or contract
-> mark-to-market
-> current exposure
-> potential future exposure
-> collateral / credit-limit usage
-> cash forecast
-> liquidity buffer or action
```

PRMIA's chapter on credit-risk management for energy and commodity derivatives supports current exposure, potential future exposure, netting, collateral, and settlement information as core credit inputs. PRMIA's chapter on liquidity-risk measurement for energy firms separates funding liquidity from trading liquidity and introduces cash-flow-at-risk and liquidity-at-risk concepts. In plain language: a hedge can be economically useful and still create a cash call at the worst time.

Illustrative stress:

- Forward sale: 36,000 MWh.
- Price moves against mark-to-market by 25 EUR/MWh.
- Collateral rule requires 80 percent of negative mark-to-market (MtM) after a threshold.
- Ignore netting and threshold for the simplified calculation.

```
negative MtM = 36,000 MWh x 25 EUR/MWh
             = 900,000 EUR

collateral need = 900,000 x 80%
                = 720,000 EUR
```

The income statement may not yet show a final loss. Treasury still needs the cash.

Limit And Escalation Ladder

Limits should match the way the book can fail.

Limit	What it catches	Escalation trigger
Nominal MW/MWh	Accidental size creep	Position exceeds approved mandate
Stress loss	Tail exposure not visible in average cases	Stress loss above appetite
VaR / expected shortfall	Modelled distribution risk	Risk metric above limit or model stale
Current exposure	Counterparty exposure today	Credit limit breach
PFE	Future counterparty exposure	Tenor or volatility pushes exposure too high
Collateral / liquidity	Cash pressure before settlement	Forecast liquidity buffer breached
Basis / shape	Hedge mismatch	Unhedged shaped exposure above tolerance
Operational	Process failure	Missing confirmation, stale curve, unreconciled meter
Partner benchmark	Outsourced execution drift	Feasible benchmark gap persists

Escalation is a pre-agreed way to keep decisions from becoming improvised during stress.

Decision Frame: Trading Mandate Readiness

A hedge or dispatch mandate is not ready if the largest modeled collateral call cannot be funded before the related revenue arrives. In the worked stress:

```
36,000 MWh x 25 EUR/MWh x 80% = 720,000 EUR
```

If treasury cannot reserve or source that cash under the collateral timetable, reduce size, change collateral terms, net the exposure, or do not trade.

Gate	Must be visible before approval	If missing
Position	MW/MWh by hour, zone, product, and counterparty	Do not expand mandate.
Valuation	independent curve, mark-to-market, stress PnL	Cap size or require review.
Credit / liquidity	current exposure, PFE, collateral timing, cash buffer	Reduce size or renegotiate terms.
Operations	confirmations, settlement, meter data, stale-curve checks	Treat as control exception.
Partner evidence	forecasts, bids, schedules, dispatch, activation, settlement	Do not attribute underperformance commercially.

This turns risk measurement into an approval test. The mandate is sized by the exposure the organization can value, fund, reconcile, and escalate.

Example

Consider a solar-heavy portfolio with three exposures:

- a fixed-price customer contract
- merchant production outside the contracted shape
- a balancing or imbalance exposure managed by a partner

The owner wants more active management.

Evaluate added trading activity with concrete risk questions:

- What exposure do we already have?
- What part can we actually control?
- Which contracts create collateral?
- Which partner decisions are benchmarkable?
- Which reports reconcile trades, schedules, meter data, and settlement?
- What is the largest plausible cash need before cash comes in?
- Who can change risk limits?

Only after that can “more trading” be evaluated as a controlled activity.

In Practice

For a renewable IPP, the first trading capability is often a disciplined risk view before a live trading seat.

That view should connect:

- physical assets
- contracted shapes
- forecast uncertainty
- market positions
- imbalance exposure
- counterparty exposure
- collateral needs

- partner actions
- settlement data
- decision rights

Without that map, adding trading activity can increase complexity faster than value.

Misconceptions

“Hedged means safe.” A hedge can reduce price exposure while increasing basis, collateral, liquidity, credit, or operational risk.

“VaR measures how much we can lose.” VaR is a threshold under a model. It does not tell you how bad losses are beyond the threshold.

“Stress tests are pessimistic.” Stress tests show whether the organization has enough liquidity, limits, collateral, and governance for a severe case.

“Risk management prevents risk-taking.” Good risk management makes intentional risk-taking possible.

“The ETRM is just software.” A system is only useful if trades, curves, confirmations, limits, valuation, settlement, and approvals are governed.

Current Debates And Caveats

Organizations need risk control.

The debate is proportionality:

- Which controls are needed at a small portfolio scale?
- When does local-file model control become too fragile?
- Which risks should be outsourced, and which risks remain with the asset owner?
- How should tail risk be measured when recent history is not a good guide?
- How much risk infrastructure is needed before internal trading is worth the organizational load?

The answer depends on portfolio size, contract complexity, collateral terms, trading mandate, partner setup, and management appetite for tail events.

Recap

Every hedge or trade starts a chain from position to market sensitivity to mark-to-market exposure, credit exposure, collateral, liquidity need, and operational control. The practical risk question is which exposure has been reduced, which exposure has been created, and who can see the difference in time to act. A useful daily risk pack ties contracts, assets, counterparties, cash, limits, partner execution, and data quality into one decision view, so more active management becomes a governed choice rather than an unobserved cash or control surprise.

Volume III: Market Design, Portfolio Strategy, And Operating Models

Chapter 1. BESS Revenue Saturation

References: [4] [17] [18] [19] [20] [38] [31]

A project model depends on one reserve product for most of the value. The battery can respond fast enough, the product has paid well, and the technical eligibility case is credible. A strategy decision has to go beyond eligibility and last year’s spread: how large is the product, how many similar assets can qualify, what the connection allows the battery to do, and what value remains when the easy scarcity rent is competed away?

The saturation problem is the transition from a scarce premium product to a more competitive revenue stack. A battery may still be valuable after the first premium product compresses, but the investment case has to migrate from a single attractive line item toward a stack that can survive stress: arbitrage, intraday optionality, co-location value, avoided curtailment, grid-tariff value, contracted flexibility, or portfolio balancing. This migration only matters if the site has enough usable import, export, metering, and operating rights to move into the next revenue layer. The capital-allocation test is whether the asset still earns its place when everyone can see the same opportunity.

Opening Puzzle

A battery developer shows you a revenue case where most of the money comes from one reserve product. The product has been lucrative recently. The asset is technically eligible. The optimizer says the battery can do it well.

Ask what happens if everyone builds the same asset to chase the same small product.

First-Principles Model

A reserve product has a finite demand curve. The TSO has a reliability need: keep frequency close to the target, restore balance after deviations, or hold a backup service for larger disturbances. The product may pay for capacity availability, activation energy, or both.

A battery is good at many of these tasks because it can move quickly, reverse direction, and control power precisely. If the TSO needs a finite amount of fast response, additional eligible batteries add supply while demand for that exact response may stay limited.

The simple model is:

```
Scarcity rent = value of the product − cost/opportunity cost of the mar
```

As eligible capacity grows, the marginal provider can become cheaper or more numerous. If procurement volume stays limited, the scarcity rent can fall. Test that mechanism before extrapolating past prices.

Definitions

Ancillary services are services procured to maintain system security and quality. Svenska kraftnät’s ancillary-service material describes frequency containment, frequency restoration, manual restoration, fast frequency response, and disturbance reserves; other Nordic TSOs define their own product details. The exact product rules, response requirements, qualification process, and settlement design are country/product/date-specific.

Reserve capacity means being available to respond. Activation or balancing energy means actually moving power when called. A battery can earn a capacity payment without much activation in some products, but it still must hold state of charge, headroom, and operational availability.

FCR contains frequency deviations. aFRR is automatic frequency restoration. mFRR is manual frequency restoration. The labels are useful only after the reader remembers the physical sequence: contain the deviation, restore the system, replace or relieve earlier actions.

Saturation means the product’s high value is eroded because eligible supply expands relative to the product’s procurement need, activation need, or liquidity. It does not mean “all batteries are unprofitable.”

Numerical Intuition

- `coverage ratio = eligible flexible MW / product procurement MW`. Use this as the first saturation test.
- `deliverable-energy ratio = usable MWh during activation window / required activation MWh`. This prevents MW-only saturation claims.
- `2 h` and `4 h` duration examples are setup, not the finding: they matter only relative to the product's activation duration, state-of-charge rules, and opportunity cost.
- Round-trip efficiency is project-specific. Project work should check the specific technology, warranty, operating mode, and data source before using an efficiency assumption.

Worked Estimate

Suppose a product needs `X MW` of reserved capacity in a bidding area and batteries under development add `Y MW` of eligible capacity. Do not begin with a price forecast. Begin with a ratio:

```
coverage ratio = eligible BESS MW / product procurement MW = Y / X
```

If `Y / X` is small, the product may still be scarcity-priced. If `Y / X` approaches or exceeds 1, the analyst should ask who the marginal provider is, whether bids are mutually exclusive across products, and whether the product has additional binding constraints such as duration, state of charge, prequalification, telemetry, or activation risk.

This is enough to reject a naive case where last year's reserve price is copied forward as if no one else can enter.

Evidence Pack For A Saturation Claim

A saturation claim is credible only when the numerator and denominator are both specified.

Evidence item	Why it matters
product procurement volume by country, bidding area, and period	defines the size of the need
qualified and actively offered flexible MW	separates technical eligibility from actual market supply
endurance and state-of-charge requirements	filters nameplate MW into deliverable MW
mutually exclusive commitments across products	prevents double-counting the same battery headroom
historical capacity prices, activation volumes, and availability penalties	shows whether compression is price, quantity, or performance driven
entry pipeline and connection timing	tests whether supply growth is near enough to affect the investment case
connection capacity, import/export rights, and metering treatment	tests whether the battery can actually shift from the crowded product into arbitrage, shaping, local flexibility, or portfolio balancing

Without this evidence, the test can guide strategy, but it cannot support a current price forecast.

Equations

For a reserve hour:

```
net reserve value = capacity payment + expected activation margin − opportu
```

Capacity payment is usually in a power unit over time, such as EUR/MW/h or EUR/MW/year depending on the market. Activation margin is energy revenue or cost when the asset is actually dispatched. Opportunity cost is what the battery gives up in another market: day-ahead arbitrage, intraday trading, co-located solar shaping, or another reserve product.

The minus terms matter. A reserve award can look profitable before state-of-charge management, degradation, exclusivity, and missed arbitrage are included.

Mechanism

The first BESS projects often find value where the system is short flexibility. That can be an ancillary-service product, a balancing product, intraday volatility, grid congestion, or a co-located PPA shape problem.

But flexibility is mobile in commercial logic even when the asset is fixed in location. Developers, optimizers, and lenders look for the same high-value product. If the product is small, the attractive spread invites entry. Entry reduces scarcity. The revenue stack migrates.

That migration can look like:

1. Early high reserve prices because eligible capacity is scarce.
2. More BESS qualifies for the product.
3. Capacity prices compress or become more volatile.
4. The battery increasingly needs energy arbitrage, intraday optionality, co-location value, avoided curtailment, grid-tariff value, or contracted flexibility.
5. The investment case needs a feasible stack under stress rather than one premium product copied forward.

The site connection decides how far that migration can go. A battery with limited import rights, capped export, restrictive metering, or a queue-delayed connection may lose access to the replacement use case exactly when the first product becomes crowded. Chapter 3, Grid Constraints And Portfolio Strategy, treats that as a portfolio strategy problem; here it is a saturation check: compression in one revenue line is survivable only if another feasible line can absorb the same finite MW and MWh.

The mechanism is durable. The dated debate is which step a given country/product has reached in 2026.

Example

A 50 MW / 100 MWh BESS can either reserve 40 MW for an automatic reserve product or use that headroom for intraday and day-ahead spreads. If reserve capacity prices are high and activation is low, reserving capacity may dominate. If more assets enter and capacity prices fall, the battery must re-optimize.

The asset capability stayed constant; the scarce system need changed.

A simple allocation test makes the saturation point concrete.

Revenue line	Before entry	After entry	Decision implication
Automatic reserve capacity margin	`55 EUR/MW/h`	`28 EUR/MW/h`	The same reserved MW may no longer beat arbitrage and PPA-shaping opportunity cost.
Intraday/arbitrage margin	`180,000 EUR/year`	`220,000 EUR/year`	More volatile residual energy can become the next use case.
Co-located solar shaping value	`120,000 EUR/year`	`120,000 EUR/year`	Contract value may be more stable than the reserve product.
Local grid or tariff value	`0 EUR/year`	`80,000 EUR/year`	Optionality depends on connection rights and local rules.

The numbers are scenario inputs, not market forecasts. The allocation rule is: when a crowded product loses margin, move the same finite MW and MWh toward the next scarce constraint only if the contract, grid rights, and state-of-charge rules allow it.

In Practice

For an IPP moving from solar into solar-plus-BESS, the commercial question is:

- Is the revenue bankable or merchant upside?
- Does the asset owner control dispatch, or does the BRP/BSP/optimizer?
- Which commitments are mutually exclusive?
- Do the connection terms allow the import, export, metering, and operating pattern required by the fallback revenue layer?
- Is the battery preserving state of charge for reserves at the expense of solar-shaping value?
- If reserve revenue falls, what part of the stack remains: arbitrage, avoided curtailment, tariff savings, PPA shape, local flexibility, or portfolio balancing?

In partner oversight, the benchmark should compare actual trading with a feasible counterfactual available at the time. It should not assume perfect hindsight or simultaneous participation in incompatible products.

Misconceptions

“Ancillary services are the battery business model.” They are one possible revenue layer. A product can be technically ideal for batteries and still become commercially crowded.

“Saturation means batteries stop mattering.” Saturation moves the value of flexibility toward the next system need with scarcity.

“A recent high reserve price is bankable.” Only if supported by market size, entry assumptions, product rules, qualification risk, and a financeable view of future spreads.

Current Debates And Caveats

As of 2026-05-16, saturation is treated here product by product and site by site. Country-wide or product-wide claims about Nordic BESS ancillary-service saturation need current market evidence, and project decisions also need current connection evidence.

PICASSO 2025 material can anchor dated discussion of aFRR platform operation, including fast activation-optimization timing. Battery-project revenue forecasting still needs the applicable product rules, response times, settlement design, procurement volumes, eligible capacity, and decision-date market evidence.

For a 2026 analyst, the product-by-product test is reserve-market size, qualification rules, activation patterns, eligible BESS entry, connection capacity, import/export rights, and the opportunity cost of using the same MW and MWh elsewhere.

Recap

A reserve product can make a battery look obvious when flexibility is scarce, but the same visibility that attracts the first projects can erode the premium. The mental model is a finite system need facing a growing pool of eligible capacity, with opportunity cost, degradation, exclusivity, connection rights, and state-of-charge rules deciding what the reserve award is really worth. After saturation, the question returns to the opening model: if the single product stops carrying the case, does the site’s feasible remaining stack still justify the asset?

Chapter 2. Intraday Market Liquidity And Granularity

References: [13] [14] [4] [19] [20] [45]

Shorter market intervals and better intraday access create more chances to adjust a renewable portfolio after the day-ahead auction. For an IPP, the first question is commercial: does finer granularity create enough executable value to pay for better forecasts, data, partner oversight, and possibly 24/7 operations?

A solar forecast moves after day-ahead clearing, a battery has more state of charge than expected, or a partner sees a tradable imbalance in the intraday book. The opportunity matters only if the portfolio can act before gate closure, at enough volume, without consuming a more valuable reserve, PPA, or settlement position. Granularity creates options. Liquidity, control rights, systems, and partner execution decide how many of those options become money.

Opening Puzzle

An optimizer sees a close-to-delivery opportunity: buy in one quarter-hour, discharge in another, and reduce an imbalance that looked unavoidable yesterday.

The model shows a price. The order book shows only a few MW. The gate will close soon. The battery is partly committed to a reserve product. The trading partner’s risk limit may reject the trade.

The decision question is which granular opportunities the portfolio can actually exercise.

The answer determines whether finer market products justify only better oversight, an outsourced 24/7 partner, or an internal trading and forecasting capability.

First-Principles Model

Electricity is balanced continuously, while markets schedule and settle in discrete market time units. A shorter market time unit can expose more shape: forecast errors, ramps, congestion, state-of-charge scarcity, and short price spikes that an hourly block may hide.

That exposed shape has option value. The asset owner has the option to rebalance, charge, discharge, withhold energy for a later obligation, or instruct a partner. The option has value only when it can be exercised.

Use this tradability test:

```
monetizable granularity value
  = gross optionality pool
  x executable liquidity share
  x forecast-and-control confidence
  x execution capture
  - transaction, capability, and risk cost
```

This strategy test separates apparent value from tradable value.

Definitions

Market time unit, or MTU, is the time block used for trading or settlement. Nord Pool’s 15-minute MTU source separates staged Single Intraday Coupling (SIDC) rollout from Single Day-Ahead Coupling (SDAC) day-ahead go-live. The SDAC 15-minute MTU go-live was stated for trading day `2025-09-30`, delivery day `2025-10-01`; 2026 decisions still need current border, product, and liquidity data.

Intraday liquidity is the ability to adjust positions after day-ahead clearing and before delivery at useful volume and price. Practitioner commentary on continuous intraday markets emphasizes access, design, and incentives; use that as debate framing unless supported by official market data.

Order-book depth is the volume available at or near the price where the portfolio wants to trade. A visible price with too little depth may be poor evidence for a dispatch benchmark.

Gate closure is the deadline after which bids or trades can no longer be adjusted for a product. Closer gate closure can make forecast updates more valuable, but only if the participant can act in time.

Execution capture is the share of a feasible opportunity that survives partner process, risk limits, collateral limits, communications, and operational constraints.

24/7 capability means the people, systems, data feeds, limits, handover routines, and partner interfaces needed to forecast, decide, trade, nominate, and monitor outside office hours. It can be internal, outsourced, or hybrid.

Numerical Intuition

Start with value at stake, then apply feasibility haircuts.

Suppose a portfolio has `500 GWh/year` whose realized value is affected by close-to-delivery shape, balancing, and dispatch decisions. An average incremental opportunity of `1.50 EUR/MWh` across that exposed volume would create a gross pool of:

```
500,000 MWh/year x 1.50 EUR/MWh = 750,000 EUR/year
```

Now impose market and operating feasibility. If only `40%` of the pool is executable because of order-book depth, gate closure, state of charge, and reserve/PPA commitments, and the partner captures `70%` of what remains:

```
captured value before capability cost
  = 750,000 x 40% x 70%
  = 210,000 EUR/year
```

If incremental data, benchmarking, partner-control, and transaction costs are `120,000 EUR/year` in this illustrative case:

```
net monetizable value = 210,000 − 120,000 = 90,000 EUR/year
```

That may justify better analytics and partner governance before it justifies a full internal desk. If the same logic applies to a larger exposed volume, deeper books, higher capture, or a product with tighter obligations, the build/buy answer can flip.

The strategic use of the 15-minute fact is that shorter intervals increase the number of possible decisions, while liquidity and execution decide the value of those decisions.

Equations

For a feasible benchmark:

```
granularity uplift
  = revenue under feasible granular decisions
  − revenue under coarser feasible decisions
```

Feasible means respecting power, energy capacity, round-trip efficiency, state of charge, degradation, market access, gate closure, liquidity, reserve commitments, PPA obligations, credit limits, and information available at decision time.

For partner oversight:

```
execution gap
  = feasible benchmark P&L
  − actual partner P&L
```

The benchmark must use tradable volumes and time-available information. Hindsight prices and infinite order-book depth create false accusations.

For capability investment:

```
capability case clears when:
  addressable value at stake x capture improvement
  > incremental cost + operational risk + model risk
```

Mechanism

Renewable portfolios make close-to-delivery decisions more valuable because forecast error shrinks as delivery approaches. A solar forecast made at noon for tomorrow is useful. A forecast made one hour before delivery can be commercially sharper. Intraday markets convert some of that sharper information into trades, nominations, dispatch choices, or imbalance reduction.

Batteries add another layer because their opportunity cost changes quickly. State of charge, degradation, warranty budget, reserve commitments, PPA shape, and the next delivery period all compete for the same MWh. A granular price can reveal a good action in one interval and a poor action in the next.

Market design decides how much of this becomes executable. Shorter MTUs increase the choice set. Deeper order books make the choice set tradable. Later gate closure makes updated forecasts usable. Good partner execution turns feasibility into cash. Poor data rights or slow decision processes can leave the opportunity visible but unexercised.

Decision Frame

An asset owner should choose among four capability levels.

Observe when granular products are live but the value pool is small, liquidity is thin, or the asset has little controllable flexibility. The minimum build is a benchmark that records what could have been done under real constraints.

Partner when 24/7 execution matters but the firm lacks market access, staffing, collateral operations, or operational risk maturity. The contract should require data rights, instruction rights, benchmark cooperation, and clear attribution of missed opportunities.

Build internal analytics when product pricing, BESS valuation, PPA firming, and partner challenge all depend on the same granular view. This is often the first owned capability because it improves decisions without taking full execution risk.

Build or internalize execution only when value at stake, liquidity depth, control rights, risk governance, and operational readiness support the cost. The desk must be able to trade, nominate, monitor limits, handle settlement evidence, and operate through nights, weekends, and abnormal market conditions.

Example

A solar-plus-BESS portfolio clears day-ahead based on yesterday's forecast. By intraday, cloud timing shifts. The portfolio is short in two quarter-hours and long later.

A naive benchmark buys the shortage and sells the surplus at the best observed prices. A useful benchmark asks harder questions:

- Were there enough MW in the book at those prices?
- Would the trade have cleared before gate closure?
- Did the battery have state of charge and export headroom?
- Was the same battery capacity already reserved for another product?
- Did the partner have authority to act?
- Would transaction cost, imbalance treatment, or credit exposure consume the spread?

Only the remaining opportunity belongs in a partner scorecard or capability business case.

In Practice

For outsourced trading, 15-minute granularity changes the questions a scorecard can ask:

- Which opportunities were visible before gate closure?
- Which were executable at observed depth?
- Which were blocked by state of charge, grid export, reserve obligations, PPA commitments, or risk limits?
- Which missed opportunities were partner execution gaps rather than market infeasibility?
- Which repeated misses justify contract changes, data-right changes, or internal analytics?

For PPA design, the same issue appears as residual firming. A shaped product may be sellable only if the seller can trade or dispatch close enough to delivery to manage uncovered intervals. Granular products therefore change both revenue upside and product cost.

Misconceptions

“More granularity always helps.” More granularity creates more decision points. It also creates more data, process, transaction, and imbalance risk.

“A 15-minute product means a battery captures 15-minute value.” The battery captures value only through access, depth, timing, controls, state of charge, and execution.

“Intraday trading is just cleanup after day-ahead.” In renewable-heavy portfolios, close-to-delivery trading can become a core operating layer. The maturity of that layer remains product-, border-, and market-specific.

“Partner execution is a black box.” It stays a black box only if the asset owner accepts reports without time-stamped forecasts, bids, trades, constraints, and settlement evidence.

Current Debates And Caveats

The 2026-05-16 Nord Pool source described the 15-minute MTU transition and included a 30 September 2025 SDAC go-live claim. Implementation status, border-level liquidity, and product depth should be refreshed from official market-operator sources before using them in a live investment or trading decision.

Practitioner analysis on continuous intraday markets is useful for framing the access, design, market-maker, and incentive debate. It should not be used alone to assert current liquidity depth or trading profitability.

By 2026, European and Nordic market-design discussions increasingly treat 15-minute products, close-to-delivery trading, and balancing-platform integration as central to renewable integration. The commercial value remains market-, border-, product-, and capability-specific.

Recap

Granularity is monetizable optionality under constraints. Shorter intervals can reveal the value of better forecasts, faster batteries, and close-to-delivery trading, but the value survives only through liquidity depth, gate closure, control rights, partner execution, and operating capability. The strategic decision is whether to observe, partner, build analytics, or internalize execution based on addressable value at stake rather than on the existence of finer prices alone.

Chapter 3. Grid Constraints And Portfolio Strategy

References: [7] [8] [22] [11] [51]

A site with grid rights is more than a parcel of land and a resource study. It is an option on scarce network capacity: the right to connect by a certain date, export under certain conditions, import for storage or flexible load, and possibly earn local flexibility revenue when the network is constrained. Two projects with the same irradiation and capex can separate completely once connection timing, firmness, tariff treatment, and import rights are priced. The grid is part of the asset because it controls when optionality can be exercised.

Opening Puzzle

Three solar-plus-BESS sites pass the first internal filter.

One can connect quickly, but export is capped. One has stronger export rights, but the queue may delay connection. One has conditional access and a possible local-flexibility route, but the rules are still being clarified.

The decision is which site option the IPP should build, wait on, partner around, or drop.

The generation profile alone cannot answer it. The missing terms sit in the connection agreement, tariff class, import right, curtailment rule, and local flexibility market.

First-Principles Model

A power asset is valuable when it can move useful power through a useful connection at a useful time under a contract the owner can perform.

The grid right is a bundle:

```
grid-right value
  = connection timing value
  + firm export value
  + import-right value
  + flexible-access value
  + local flexibility value
  - grid charges
  - curtailment and constraint losses
  - rule and delay risk
```

The project value becomes:

```
site option value
  = base asset value
  + grid-right value
  + portfolio option value
  - development cost
  - carrying cost of waiting
```

This framing matters because scarce grid rights can create an option even before the plant is built. The firm may choose to secure, stage, partner, or sell a position depending on how the grid right evolves.

Definitions

Connection timing is the expected date when the project can energize and export or import under the agreed conditions. A later date has opportunity cost and may miss contract, subsidy, tariff, or portfolio windows.

Firm export right is the ability to inject up to a defined capacity without routine curtailment beyond the normal market and system rules. A non-firm or conditional export right may still be valuable if the curtailment pattern is predictable and priced.

Import right is the ability of a battery or flexible load to withdraw from the grid. For BESS, import rights can decide whether the battery is only a solar-shaping asset or can also perform arbitrage, reserve recovery, and residual PPA firming from market energy.

Flexible connection agreement means access subject to constraints, such as curtailment or operational limits during network stress. ACER’s 2025 tariff material treats flexible connections as useful for speeding connections and reducing congestion, while warning that they should be governed carefully so they do not substitute indefinitely for needed reinforcement.

Local flexibility revenue is compensation for actions that relieve local network constraints, such as reducing injection, increasing consumption, shifting load, or dispatching storage. It is bankable only when product rules, qualification, metering, dispatch authority, and payment terms are clear.

Numerical Intuition

The first grid numbers are usually timing, curtailed MWh, import-enabled cycles, and qualifying flexible MW.

Use four project tests:

```
delay cost = annual contribution margin x years of delay
curtailment loss = curtailed MWh x realized value of curtailed MWh
import-right value = grid-charged cycles x net spread after losses, tar
local-flexibility value = qualified flexible MW x expected net payment
```

Worked Estimate

Take a 100 MW solar project with an illustrative `1,000` full-load-hour assumption:

```
annual output = 100 MW x 1,000 h = 100,000 MWh/year
```

If the project would earn an average contribution margin of `8 EUR/MWh` after normal operating costs and contracted settlement effects, each operating year is worth:

```
annual contribution margin = 100,000 MWh x 8 EUR/MWh
                           = 800,000 EUR/year
```

A two-year grid delay therefore has a first-order opportunity cost of roughly:

```
delay cost before discounting = 800,000 x 2
                              = 1,600,000 EUR
```

Now add a 50 MW / 100 MWh battery. If import rights allow `200` additional grid-charged cycles per year and the net spread after losses, tariffs, and degradation is `12 EUR/MWh`, the import-right value is:

```
import-right value = 100 MWh/cycle x 200 cycles x 12 EUR/MWh
                   = 240,000 EUR/year
```

These are illustrative decision numbers. They show why a site with slower connection but better import rights, or faster connection with weaker export firmness, may deserve a real option valuation rather than a pass/fail grid note.

Equations

For a constrained generator:

```
curtailment loss = curtailed MWh x realized value of curtailed MWh
```

The realized value may include PPA obligations, certificates, imbalance exposure, avoided negative-price output, or lost co-located battery value.

For co-location:

```
net co-location value
  = avoided curtailment
  + shape and arbitrage value
  + grid and tariff value
  + local-flexibility value
  - capex
  - degradation
  - operational conflicts
```

For a scarce connection option:

```
decision value
  = max(build now, wait, partner, sell/drop)
  - development and carrying costs
```

Exercise now only if the build-now path beats the best feasible alternative after grid-position risk, queue risk, tariff timing, and partnership value are considered.

Mechanism

Bottlenecks turn location and timing into money. A national price forecast can look attractive while a distribution feeder is full at noon, a transmission connection is staged, or a tariff class makes battery imports expensive. A project may be technically possible and commercially weak because its grid right cannot support the product being sold.

Tariff reform is one way regulators expose network costs. ACER’s 2025 network-tariff material describes European debates around power-based charges, time-of-use signals, locational signals, flexible connections, and storage-specific regimes. The same material emphasizes complexity, user predictability, and the risk that short-term flexibility tools delay reinforcement if badly designed.

The site developer should therefore treat the grid right as a living commercial object. Export firmness affects production and PPA delivery. Import rights affect BESS optionality. Flexible access affects curtailment risk and time to revenue. Local flexibility rules affect whether the asset can sell constraint relief rather than merely suffer a constraint.

Decision Frame

Build when the connection date is credible, export and import rights support the intended product, curtailment is bounded or compensated, tariffs are understood, and local flexibility revenue is upside rather than the only path to bankability.

Wait when grid timing, tariff class, curtailment regime, or import rights are too uncertain, but the site has a scarce position that may become more valuable after reinforcement, rule clarification, or a portfolio need.

Partner when the site option depends on a capability the IPP does not own: flexible demand, a local industrial load, a DSO flexibility aggregator, a trading party with qualification access, or a co-developer with queue position and permitting leverage.

Drop when the apparent resource quality depends on rights that are unlikely to arrive, when flexible access creates unpriced curtailment risk, or when the battery cannot import/export in the way the product economics require.

Example

A solar project has 100 MW of generation capacity and 70 MW of firm export. A co-located battery can reduce noon clipping only if it has room to charge, the later export window is available, and the PPA rewards shifted delivery.

Now change one grid-right term: the battery may import from the grid during low-price hours.

The project is no longer only a solar-shaping asset. It may support arbitrage, reserve recovery, imbalance reduction, and shaped PPA delivery when solar is unavailable. The same battery without import rights may still be useful, but its option set is narrower.

The connection answer `yes`, `conditional`, `staged`, or `no import` can change the preferred site even when the solar-yield model is unchanged.

In Practice

For each candidate site, the investment memo should include a grid-right card:

- expected connection date and queue uncertainty
- firm and conditional export MW
- import rights for BESS or flexible load
- curtailment trigger, compensation, and notice process
- network tariff class, tariff year, and loss treatment
- local flexibility products the asset could qualify for
- metering, telemetry, and control requirements
- portfolio value of the site under stressed price, grid, and PPA scenarios

The card should end with a build, wait, partner, or drop recommendation. A site that cannot answer those fields is not ready for a clean valuation.

Misconceptions

"Congestion is always bad." Congestion can destroy value, but it can also create locational spreads, storage value, demand-flexibility value, and strategic entry points.

“The spot price is enough for project value.” Connection rights, curtailment, tariffs, and dispatch constraints can dominate the valuation.

“A battery fixes a constrained site.” A battery helps only when it has the grid rights, state of charge, duration, tariff treatment, and control rights needed for the constraint.

“Flexible access is free acceleration.” Earlier access may be valuable, but the curtailment regime and reinforcement path must be priced.

Current Debates And Caveats

As of the cited 2025 ACER tariff material, European tariff and connection debates include locational signals, time-of-use signals, flexible connections, and storage-specific regimes. ACER does not provide a portable answer for a specific project.

Country, voltage level, network owner, tariff year, metering design, and user class decide the real economics. A Swedish, Finnish, Danish, or Norwegian grid example should be treated as local evidence unless the official rule is checked for the specific site and date.

By 2026, grid bottlenecks and tariff reform are central variables for renewable and BESS strategy. Site value remains connection-, tariff-, and rule-specific.

Recap

Grid scarcity turns a site into an option on connection timing, export firmness, import rights, flexible access, and local constraint relief. The right valuation question asks whether the grid-right bundle lets the IPP build now, wait, partner, or drop the site under realistic curtailment, tariff, and product constraints.

Chapter 4. Zonal And Nodal Market Design

References: [7] [4] [11] [52] [23] [51] [53] [54]

A transmission constraint can appear as an area-price spread, a nodal price difference, redispatch cost, curtailment, congestion rent, network tariff, hedge-basis loss, or a contract clause. For an IPP, zonal and nodal markets decide where locational risk enters the cash-flow model. The practical questions are: where does congestion settle, who carries it, which hedge follows it, and whether the site or portfolio sees the signal early enough to change the decision.

Opening Puzzle

Two solar projects produce the same extra MWh in the same hour.

One is on the surplus side of a constrained interface. The other is near load where expensive local generation is needed.

Physically, those MWh do not have the same marginal value. The market-design question is where that difference becomes visible:

```
energy price?
redispatch cost?
tariff?
curtailment?
congestion rent?
hedge basis?
contract settlement?
```

The decision error is to argue about zonal versus nodal in the abstract before locating the cash-flow exposure.

First-Principles Model

Use a two-node system.

- Node A has cheap generation.
- Node B has load and expensive generation.
- A line connects A to B.
- The line has a capacity limit.

If the line has spare capacity, cheap power from A can serve B. The marginal value of one more MWh is similar across the two nodes.

If the line binds, one more MWh at B cannot be supplied by more cheap imports from A. It must come from local expensive generation, local storage, demand response, or curtailment avoidance. The local marginal value at B rises.

The physical fact is:

```
constraint -> local marginal value difference
```

The market-design choice is:

```
how much of that value difference becomes a visible energy price,
and how much is recovered or managed elsewhere?
```

Definitions

- **Bidding zone:** area where the market generally clears one price for a period, subject to the market design.
- **System price:** Nordic reference price calculated while disregarding available transmission capacity between Nordic bidding areas.
- **Area price:** bidding-area price when transmission constraints affect flows between areas.
- **Zonal pricing:** design where congestion between zones is reflected in prices, while constraints inside a zone are handled by other mechanisms.
- **Nodal pricing:** design where prices are calculated at more granular network nodes, often through locational marginal pricing.
- **Redispatch:** adjustment of physical dispatch after market clearing to respect network constraints.
- **Countertrading:** market action used to relieve congestion by buying and selling in different locations or directions.
- **Congestion rent:** value created by a constrained flow across a price difference.
- **Basis risk:** mismatch between the price used for a hedge or contract and the price or cost that actually drives the asset’s cash flow.

Numerical Intuition

Start with the visible congestion value:

```
congestion value = price spread x constrained flow
```

If a line carries `60 MW` from a `30 EUR/MWh` area to a `100 EUR/MWh` area for one hour:

```
(100 - 30) EUR/MWh x 60 MWh = EUR 4,200/hour
```

In a market-coupled example, spread times constrained flow is visible congestion rent. In other designs, related congestion costs may appear as redispatch, curtailment, tariff recovery, or hedge basis, but those are separate cash-flow channels.

Now put the idea into a portfolio exposure.

A project generates `200 GWh/year`. The project is hedged against a broad area or system price, but its cash flow is exposed to a local congestion effect worth `15 EUR/MWh` in the stressed hours that matter. If `40%` of annual generation is effectively exposed after contract and hedge coverage:

```
200,000 MWh/year x 40% x 15 EUR/MWh
= EUR 1.2 million/year
```

Market design becomes an investment question when the first-order exposure is the part of the spread that the hedge does not follow.

Three Settlement Locations

Take the same physical constraint and place it in three designs.

Design channel	Where the cost/signal appears	Asset-owner question
Zonal spread	Area price differs across bidding-zone boundary.	Does the PPA or hedge settle in the same area as the asset?
Internal redispatch	Broad zone price clears, then system operator fixes internal constraint.	Is the cost socialized, tariffed, curtailed, or allocated to the asset later?
Nodal price	Local price directly reflects node-level congestion and losses.	Can the asset hedge nodal basis, and does the customer accept that basis?

The same wire has created a different commercial problem in each row.

Mechanism

Zonal pricing creates larger price areas. It can support liquidity and simpler financial contracting because more participants share one price. It also averages away constraints inside a zone. If internal constraints are material, the system still needs redispatch, countertrading, curtailment, grid reinforcement, tariff recovery, or local flexibility.

Nodal pricing exposes more granular locational values. It can send sharper siting and dispatch signals, and it aligns energy prices more closely with physical constraints. It also creates more granular basis risk, more complex hedging, more data requirements, and distributional fights over who gains and loses.

Creti and Fontini's transmission-pricing material is useful because it links congestion, nodal prices, transmission rights, and hedging. Brown and Glaum's grid lecture gives the simple congestion-rent anchor: constrained cross-border or inter-area flow creates a price difference times flow. ACER's tariff material adds the European practical point: locational signals can also come through connection charges, use-of-network charges, flexible connection terms, and storage-specific regimes.

Decision rule:

```
find the constraint, then find the settlement channel
```

Example

An IPP considers a 150 MW solar project inside a large bidding zone.

The base case uses the zone price. The grid review says the project sits behind an internal bottleneck during high-solar hours. The market design does not publish a separate nodal energy price for that bottleneck.

The investment memo needs a congestion-exposure check:

- Is output likely to be curtailed in sunny constrained hours?
- Are grid reinforcement or flexible connection terms required?
- Could tariff reform later recover congestion cost from users like this project?
- Does a co-located battery reduce the local constraint or only shift exposure?
- Does the PPA settle on zone price while the asset suffers local curtailment?
- Does the hedge cover the cash-flow driver or only the headline area price?

Now suppose the same project is in a nodal design. The constraint may appear directly in local prices. That improves signal clarity, but the PPA and hedge now need to specify whether settlement follows node, hub, zone, system price, or a basis adjustment.

Decision Frame

For each asset, contract, or portfolio hedge, build a locational-risk test.

Step	Question	If answer is weak
1. Locate the physical constraint	Which line, transformer, zone boundary, or local grid limit can bind?	Revenue forecast is ungrounded until this is known.

Step	Question	If answer is weak
2. Locate the price signal	Area price, node price, redispatch, tariff, curtailment, or contract adjustment?	Add explicit basis or cost-allocation sensitivity.
3. Locate the hedge	Does the hedge settle where the asset is exposed?	Quantify residual basis risk.
4. Locate the control lever	Can BESS, curtailment control, demand response, or connection design reduce exposure?	Do not count flexibility value twice.
5. Locate the investment signal	Would a different site, grid offer, or portfolio mix change the answer?	Escalate before investment committee, not after signing.

For an IPP with Nordic area-price and grid exposure, this test matters for capture price, system-to-area basis, PPA settlement, BESS value near bottlenecks, tariff exposure, and whether a portfolio is diversified across real constraints or only across labels.

Misconceptions

“The market price is natural.” It is a design output built from physical constraints, algorithms, rules, and settlement choices.

“Zonal pricing ignores the grid.” It exposes some constraints at zone boundaries and handles others through redispatch, countertrading, curtailment, tariffs, or planning.

“Nodal pricing removes risk.” It moves more risk into visible locational prices and basis management.

“Congestion outside the energy price is gone.” It reappears as cost, constraint, curtailment, tariff, or hedge basis.

“A portfolio is diversified because it has many zones.” It is diversified only if the zones and assets reduce exposure to the same physical and price drivers.

Current Debates And Caveats

By 2026, European zonal-versus-nodal debate is best treated as an active design question about congestion visibility, liquidity, redispatch cost, hedging, investment signals, and distributional consequences. The mechanism and risk logic here do not recommend a specific Nordic or European market design.

Reform status, redispatch cost allocation, and tariff treatment are date-specific. Pricing or policy claims need the relevant official rule set for the market and date being assessed.

Recap

A transmission constraint creates a local value difference. Zonal pricing, nodal pricing, redispatch, tariffs, curtailment, and hedges decide where that difference enters cash flows. The decision test is to locate the constraint, locate the price or cost channel, locate the hedge, and locate the control lever. If the hedge follows a different price than the asset's physical exposure, market design has become portfolio risk.

Chapter 5. PPA Structures And Market Evolution

References: [6] [32] [33] [34] [35] [36] [37] [25]

A buyer asks for a cleaner and firmer product. That request may contain several priced demands: hedge value, residual firming, a shape closer to load, storage-backed delivery, stronger certificate evidence, lower credit exposure, or control over dispatch and reporting. For a renewable IPP, the product roadmap should begin with priced buyer paths rather than product labels. Each new promise must be matched to a cost stack, evidence stack, control-right stack, and credit stack before it becomes a product.

Opening Puzzle

A renewable IPP can sell a familiar as-produced PPA. It knows the asset, the certificates, the settlement process, and the retained risks.

Now three buyers arrive:

- a price-sensitive buyer wants a simple hedge
- an industrial buyer wants a flatter delivered profile
- a data-center-style buyer wants hourly evidence and cleaner residual supply

The decision is which buyer path can be priced, delivered, evidenced, governed, and financed.

A product name has little value until residual firming, data quality, control rights, credit, and accountability after delivery are priced.

First-Principles Model

A PPA product is a priced promise backed by capabilities.

```
required seller premium
  = residual firming cost
  + balancing and hedge cost
  + evidence and audit cost
  + credit and collateral cost
  + control-right cost
  + risk margin
```

The capability stack is:

```
capability stack
  = assets
  + portfolio
  + storage and flexibility
  + market access
  + certificate and meter data
  + dispatch/control rights
  + partner governance
  + financing and credit support
```

The product clears only if buyer willingness to pay exceeds the seller premium and the independent power producer (IPP) has the rights, evidence, and operating control to perform. If the buyer value is lower than the cost stack, keep the offer simpler. If buyer value is high but the seller lacks evidence or control, partner or defer. If the IPP owns the capability and the buyer pays for it, the next product path may be worth building.

Definitions

Residual firming is the cost of covering the gap between the seller’s renewable output and the buyer’s promised shape after portfolio netting, storage dispatch, hedging, and imbalance settlement.

Evidence quality is the credibility of the data and certificates supporting the buyer’s claim. Hourly or granular evidence is a product-design concept here; compliance treatment depends on the relevant standard, geography, and date.

Control rights are the contractual and operational rights to dispatch storage, nominate energy, trade residual positions, allocate certificates, instruct partners, and receive the data needed to verify performance.

Credit cost is the cost of counterparty risk, collateral, parent guarantees, margining, termination exposure, and lender constraints.

Granular certificate or time-stamped attribute-certificate concepts add temporal, and sometimes locational, detail to the evidence chain. A marketed claim depends on the current standard text, issuer rules, cancellation rules, audit rights, and the buyer’s intended disclosure language.

Numerical Intuition

Use a priced path before any product ladder. Suppose an IPP is considering three offers for the same buyer load:

Offer	Seller promise	Main retained exposure
As-produced PPA	deliver metered renewable output and certificates	buyer keeps shape, volume, and residual market exposure
Shaped PPA	deliver a closer hourly or block profile	seller prices residual firming, imbalance, basis, and dispatch constraints
Evidence-rich product	add stronger time/location evidence and reporting	seller prices certificate data, audit trail, residual supply, credit, and claim control

The shape premium starts with the uncovered hours after portfolio and storage are modeled:

```
residual firming cost
  = uncovered MWh x replacement cost
  + imbalance cost
  + hedge and basis cost
  + battery degradation
  + transaction cost
```

Illustrative path:

```
uncovered volume after portfolio and storage = 25,000 MWh/year
net residual firming cost = 9 EUR/MWh
residual firming cost = 25,000 x 9 = 225,000 EUR/year
```

For a buyer load of `87,600 MWh/year`, add evidence, credit, and control costs:

```
evidence and audit systems = 0.60 EUR/MWh x 87,600 = 52,560 EUR/year
credit and collateral cost = 1.20 EUR/MWh x 87,600 = 105,120 EUR/year
control-right and partner-governance cost = 0.80 EUR/MWh x 87,600 = 70,
```

The incremental cost before risk margin is:

```
225,000 + 52,560 + 105,120 + 70,080
= 452,760 EUR/year
```

Spread across the buyer's load:

```
452,760 / 87,600 MWh = 5.17 EUR/MWh
```

If the buyer's willingness to pay for the improved shape and evidence is below that level, the offer should be narrower. If the buyer pays more and the seller controls delivery, the product path becomes plausible.

Evidence Boundary

EnergyTag's 2026 response is useful debate framing for hybrid PPAs, firming costs, granular evidence, and hourly matching. It reports market observations and cites studies suggesting material value from better matching in selected cases. Those figures frame pricing questions; the underlying study, geography, standard, and period still control any quantitative claim.

Pexapark-style market commentary is useful for the claim that IPPs are being pushed toward more active revenue management, structuring, and portfolio capability. Reported thresholds, market shares, and adoption claims remain dated practitioner color unless independently sourced.

Buyer interest in shaped, hybrid, portfolio-backed, and evidence-aware PPAs appears to be increasing. Product bankability still depends on priced firming, evidence rules, control rights, credit support, and the seller's actual portfolio.

Market-Evolution Evidence

The European and Nordic evidence should be read separately before drawing a product conclusion.

Commission Recommendation (EU) 2026/917 reports that EU corporate PPA contracted electricity grew from `7.4 TWh` in 2020 to `31.4 TWh` in 2024, while the number of contracts rose from `60` to `276`. It also notes that by 2024 solar PV formed the majority of reported corporate PPA contracted electricity, more than `10%` of contracts were hybrid, and information and communications technology (ICT) buyers accounted for more than `40%` of committed electricity. These are dated policy-context figures, useful for sizing the strategic importance of the market.

The same Recommendation cites ACER's 2024 conclusion that existing voluntary PPA templates were sufficient for current market needs. ACER's assessment matters because it pushes the product debate away from "we need another template" and toward barriers such as project development, buyer credit, collateral, accounting, market transparency, and cross-border risk.

The 2025 Nordic Energy Research / AFRY report adds Nordic-specific friction. It reports a `43%` decline in the Nordic PPA market from 2021 to 2024 in AFRY's database, highlights `12` Nordic bidding areas plus Iceland, and treats basis risk, price opacity, contract tenor, credit, accounting treatment, and regulatory change as recurring barriers. A Nordic product roadmap therefore needs a basis-risk view, a certificate view, a credit view, and a structure view for each buyer path.

The product-strategy implication is narrower than "PPAs are rising." Product work has to price risk allocation: shape, basis, credit, collateral, evidence, tenor, control rights, and residual firming.

Product Paths

The roadmap should be built around buyer/product paths.

Buyer path	Product direction	Main priced cost	Evidence and control test
Low-cost hedge	As-produced, fixed/indexed, or	capture discount, imbalance, credit	standard certificate and settlement

Buyer path	Product direction	Main priced cost	Evidence and control test
	simple virtual structure		evidence may be enough
Flat industrial load	Profile, block, or shaped product	residual firming, hedge/basis, imbalance	seller must show how uncovered hours are priced and governed
Storage-aware buyer	Hybrid solar-plus-BESS or portfolio-backed product	degradation, dispatch opportunity cost, grid rights	seller must control or audit dispatch and state-of-charge policy
Hourly/evidence-aware buyer	Time-matched or evidence-rich product	data, certificates, residual clean supply, audit	seller must avoid compliance claims beyond current standards
Flexible buyer	Indexed, collar, load-shifting, or flexibility-linked product	option value, control sharing, performance measurement	buyer flexibility must be measurable, callable, and compensated

The product path should advance only when buyer value exceeds the priced cost and the seller has the rights to perform.

Strategy Decision

Build when the product uses capabilities the IPP needs anyway: portfolio shape pricing, residual firming model, certificate evidence, storage dispatch policy, partner benchmarking, and contract-risk allocation.

Partner when the product needs market access, 24/7 operations, supplier interface, granular certificate infrastructure, or specialized balancing execution that a third party can perform while the IPP keeps pricing and risk ownership.

Defer when buyer interest is real but certificate treatment, data rights, grid access, storage controls, lender treatment, or internal systems are not ready.

Avoid when the product promise depends on dispatch rights the IPP does not hold, certificates it cannot verify, residual firming it cannot price, or credit exposure it cannot finance.

Example

A buyer asks for a "24/7-style" clean product.

Start with the priced path:

- Is the buyer paying for hedge value, cleaner hourly evidence, physical delivery quality, or public-claim credibility?
- Which hours remain uncovered after the IPP's assets, portfolio, storage, and contracts are modeled?
- What is the residual firming cost in those hours under stress prices?
- Are certificates time-stamped and accepted for the buyer's intended claim?
- Who controls battery dispatch, nominations, and residual market purchases?
- What credit support, collateral, and termination exposure does the contract create?
- Can performance be verified after the fact with auditable meter and certificate data?

If those answers clear the cost stack, the product may be worth building. If they do not, use a shaped PPA, an indexed hedge, a portfolio-backed product with disclosed residual exposure, or no bid.

In Practice

The first product-roadmap artifact should be a priced buyer-path sheet. For each candidate offer, it should show:

- target buyer segment and paid value driver
- promised shape, hedge, evidence, and certificate treatment

- residual firming volume and cost under base and stress cases
- balancing, hedge, basis, and imbalance exposure
- storage dispatch rights, degradation budget, and grid rights
- partner role, data rights, and audit rights
- credit exposure, collateral, and lender constraints
- minimum premium or margin required for approval

The roadmap then becomes an investment plan for capabilities rather than a list of branded products.

Misconceptions

“The future PPA is one standard product.” PPA strategy is segmentation by buyer value, risk transfer, evidence need, and control rights.

“Hourly matching automatically means compliance.” Compliance depends on the relevant standard, certificate system, geography, and date.

“Hybrid means premium.” Hybrid means capability and cost. It earns a premium only when the buyer values what it solves.

“Annual green MWh are obsolete.” Annual procurement remains useful for buyers whose main problem is annual volume or price hedging.

“The contract can promise what the portfolio cannot deliver.” The contract can allocate the exposure. The cash flows and evidence trail will show whether it was priced.

Recap

Future PPA strategy is a priced matching problem. A buyer may want cheaper hedging, better shape, hybrid delivery, portfolio support, or stronger evidence, and each request adds residual firming, evidence, control-right, credit, and governance costs. The useful roadmap chooses build, partner, defer, or avoid for each buyer path after the full cost stack and rights stack are visible.

Chapter 6. Storage Duration And Long-Duration Flexibility

References: [38] [5] [46] [41] [47] [11] [15] [55]

A project memo compares two battery sizes, one grid connection, one possible evening customer product, and a trading case for spreads and reserves. Nameplate duration gives the simple comparison: two hours versus four. The investment comparison is capital against scarcity. Extra MWh may underwrite a contract, recover curtailed solar, cover peak net-load hours, or sit idle while better revenue is sacrificed. The decision is whether the next hour of usable energy solves a paying scarcity problem better than a contract, product change, portfolio substitute, or delay.

Opening Puzzle

A project team brings three options for the same site.

- Build a `50 MW / 100 MWh` battery.
- Build a `50 MW / 200 MWh` battery.
- Skip the extra duration; buy flexibility, change the PPA shape, or wait.

The site has good solar, a constrained grid connection, a possible evening product for a customer, and a trader who says the battery can also chase spreads and reserves.

The simple sizing question is:

```
Is 4 hours better than 2 hours?
```

The investment question is:

```
What scarcity problem are we buying insurance against, and is owning the ne
```

First-Principles Model

A storage decision has three curves.

The first is the scarcity-duration distribution:

```
How often does the valuable shortage last 15 minutes, 1 hour, 4 hours,
```

The second is the asset capability curve:

```
What can the storage asset actually do, given MW, MWh, state of charge,
```

The third is the money curve:

```
Who pays for each extra hour of capability, and what revenue or avoided
```

Only after those three curves are visible should the model size the extra energy block:

```
incremental_energy_MWh = incremental_power_MW x scarcity_duration_h
```

That calculation sizes the insurance policy. The value test comes from the scarcity-duration distribution, the revenue or avoided cost, and the alternatives.

Definitions

Short-duration storage: storage mainly used for fast services, intraday shifting, reserves, congestion windows, imbalance control, or daily shape problems. A `2-hour` battery and a `4-hour` battery are common illustrative examples; real categories depend on product and system context.

Long-duration energy storage: storage intended to cover scarcity periods longer than ordinary intraday shifting. The economically relevant threshold is the duration at which the next MWh changes a decision.

Scarcity-duration distribution: a table or curve showing how long the valuable shortages or obligations last.

Value-duration curve: the marginal value of adding each extra hour of usable storage duration.

Marginal duration hurdle: the annualized cost and opportunity-cost threshold that the next MWh of storage energy capacity must clear.

Capacity credit: dependable capacity contribution divided by nameplate capacity. For storage, this depends on event duration, state of charge, availability, and the reliability test.

Firming: using assets, contracts, storage, market purchases, demand flexibility, or portfolio diversity to make uncertain supply meet a target obligation.

Numerical Intuition

Start with the marginal duration hurdle:

```
required margin per MWh
  = annualized incremental duration cost
    / annual incremental discharged MWh from that extra duration
```

If the extra duration costs `1.75 million USD/year` to carry and is used for `100 MWh` in `350` valuable cycles:

```
annual incremental discharged MWh = 100 MWh x 350 = 35,000 MWh/year
required margin = 1,750,000 USD/year / 35,000 MWh/year
                = 50 USD/MWh
```

If the same extra duration is useful only `35` times per year:

```
annual incremental discharged MWh = 100 MWh x 35 = 3,500 MWh/year
required margin = 1,750,000 USD/year / 3,500 MWh/year
                = 500 USD/MWh
```

Rare events require either very high value per MWh, a capacity-style payment, a customer contract, avoided penalties, or a cheaper resource. Otherwise the extra MWh is underused capital.

The IEA battery report gives useful dated scale anchors. In its STEPS scenario, the average duration of the global battery storage fleet rises from around `2 hours` in its 2023 baseline to `2.7 hours` in `2030` and `3.4 hours` by `2050`. For four-hour utility-scale battery storage, it projects average total upfront costs falling from `290 USD/kWh` in `2022` to `175 USD/kWh` in `2030`. Use these as global scenario anchors; a site quote still needs project-specific engineering, procurement, and construction (EPC), grid, warranty, insurance, and contracting terms.

The Extra Two Hours

Take the second and third hours of a battery seriously as a capital decision.

Suppose the first option is:

```
50 MW / 100 MWh = 2 hours
```

The larger option is:

```
50 MW / 200 MWh = 4 hours
```

The extra duration is:

```
100 MWh
```

Use the IEA `2030` four-hour cost anchor of `175 USD/kWh` only as a scale check, not a project quote or marginal-duration cost. The marginal cost of extending this project needs vendor/EPC quotes separating cells, containers, power-conversion-system sharing, balance of plant, augmentation, warranty, grid, tax, and financing effects. As an order-of-magnitude test, the additional 100 MWh of duration represents roughly:

```
100 MWh x 1,000 kWh/MWh x 175 USD/kWh = 17,500,000 USD
```

If the annual carrying charge is simplified to `10%`:

```
annual hurdle = 17,500,000 USD x 10% = 1,750,000 USD/year
```

Use this as a sanity check. An EPC estimate or marginal energy-block quote needs project design, supplier terms, site conditions, taxes, and financing.

Now the decision becomes practical:

- If the extra two hours can earn `50 USD/MWh` net after losses, degradation, tariffs, and opportunity cost, it needs about `350` fully useful cycles per year.
- If it can earn `100 USD/MWh`, it needs about `175` fully useful cycles per year.

- If it is used only during `20` rare stress events, it needs value closer to `875 USD/MWh`, or it needs to be paid as dependable capacity, contracted firming, avoided curtailment, avoided grid reinforcement, or insurance against a penalty.

The decision question is:

```
Where are the paying events on the duration curve?
```

Equations

The prerequisite identity:

```
duration_h = usable_energy_MWh / power_MW
```

The decision equation:

```
marginal duration value
  = added revenue
  + avoided cost
  + added contract value
  + added capacity/adequacy value
  + option value
  - incremental capex and opex
  - degradation and augmentation cost
  - grid and tariff cost
  - opportunity cost
  - added operational risk
```

The extra duration is worth owning only if this is positive under credible scenarios.

For adequacy:

```
dependable MW during event
  = min(power rating, usable MWh available for event / event duration)
    x availability/confidence factor
```

If a `100 MWh` battery faces a `4-hour` stress event, the energy-limited average is:

```
100 MWh / 4 h = 25 MW
```

If the stress event lasts `20 hours`, the same stored energy averages:

```
100 MWh / 20 h = 5 MW
```

Storage adequacy therefore needs an energy-duration check alongside nameplate MW.

Mechanism

Adding duration creates four possible kinds of value.

First, it can capture longer price spreads. If the high-price window lasts four hours instead of two, extra MWh may earn real arbitrage value.

Second, it can make a contract more credible. A shaped PPA, evening product, or customer-load product may require more deliverable energy than a short battery can provide.

Third, it can increase dependable capacity during peak net-load hours. The IEA battery report describes batteries becoming important providers of secure capacity in high-demand, low-renewable hours, especially as more projects move toward four-to-eight-hour duration.

Fourth, it can create strategic option value. Extra duration may let the IPP switch between PPA firming, intraday optimization, reserves, imbalance management, curtailment recovery, and local congestion services.

Each value has an opportunity cost.

The battery can spend the same MWh only once. If state of charge is reserved for an evening PPA, it may not be available for a reserve product. If it is held for a rare stress event, it may skip daily spreads. If it imports from the grid, tariffs and connection rights may change the answer. If it cycles deeply, degradation and warranty constraints matter. If the connection is export-limited, stored energy may not be deliverable in the valuable hour.

Longer duration is most attractive when the valuable events frequently last longer than the current battery and when the market or contract pays for the extra capability. It is least attractive when the long events are rare, poorly paid, impossible to deliver through the grid connection, or better handled by a portfolio substitute.

Example: Site X

Consider a solar site with:

- `100 MW` solar export potential at noon
- `80 MW` firm export capacity
- a proposed customer product of `40 MW` from 18:00 to 22:00
- optional grid charging, but with uncertain tariffs and import capacity
- a trading partner who can optimize the battery, subject to contract rules

The evening product needs:

```
40 MW x 4 h = 160 MWh
```

A `40 MW / 80 MWh` battery can cover `40 MW` for two hours, or `20 MW` for four hours. It can support the product but cannot fully underwrite it.

A `40 MW / 160 MWh` battery can cover the four-hour shape in a clean simplified model. But it has now been partially converted from a merchant flexibility asset into a contract-performance asset. The owner must ask what revenue stack has been sacrificed.

There are at least five possible answers:

Option	What it says	Failure mode
Build 2h	Most value is in short spreads, reserves, imbalance control, or local congestion.	Customer product overpromises firming.
Build 4h	The extra two hours are paid by contract value, recurring spreads, avoided curtailment, or dependable capacity.	Extra MWh sit idle or block better revenue.

Option	What it says	Failure mode
Contract flexibility	Rare long events are better bought from the market, hydro, demand response, or a portfolio counterparty.	Counterparty or liquidity fails in stress.
Change the PPA	Sell a shape the asset can actually support.	Sales loses a product the customer values.
Wait or stage	Preserve option value until tariffs, connection rights, or market rules are clearer.	Competitors secure the site, customer, or grid capacity first.

The best answer depends less on the label “2h” or “4h” than on which row survives the data.

In Practice

A useful investment memo should make the duration question more precise:

1. What is the scarcity-duration distribution for this site, zone, customer shape, and portfolio?
2. What does the value-duration curve look like after losses, degradation, tariffs, and opportunity cost?
3. Which revenue streams are mutually exclusive?
4. Which duration can be contracted, financed, or underwritten?
5. What do the grid connection, import rights, export limit, flexible connection terms, and network tariffs allow?
6. What portfolio substitutes exist: wind, hydro exposure, market purchases, demand flexibility, interconnectors, tolling, optimizer contracts, or hedges?
7. What evidence would make us change from build to buy, from 2h to 4h, or from ownership to contract?

ACER’s 2025 tariff report is a useful warning here. Locational signals, flexible connection agreements, and storage-specific tariff treatment can change the economics of the same physical battery. A storage project is also a grid-rights instrument.

The storage substitute can be something other than a battery. A DNV/Nyrstar Netherlands case study describes an industrial "virtual battery" at a zinc electrolysis site: roughly `70 MW` of adjustable load over `3–4 days`, created by using process buffers rather than electrochemical storage. The same paper estimates direct economic benefit at `190,000 EUR/MW/year` in its Netherlands case context. Use those numbers only inside that case context; the portable mechanism is the decision logic. For multi-day scarcity, compare longer battery duration with demand flexibility, process buffers, grid-service contracts, and connection agreements.

For multi-day or seasonal scarcity, the hurdle changes again. The IEA report says batteries are poorly suited to balancing over weeks or seasons. Jenny Chase makes the same economic point in plain language: a battery that cycles once a year has to recover its capital from very few uses. Seasonal problems may point toward reservoir hydro, interconnection, demand flexibility, thermal backup, hydrogen or derivatives, contracts, or accepting some curtailment and residual risk.

Misconceptions

"Four hours is twice as good as two." Only if the extra two hours solve paid events often enough.

"Long-duration energy storage (LDES) is just a bigger battery." Longer scarcity durations change technology, utilization, contract, finance, and siting logic.

"A battery solves firm clean supply." A battery helps inside its MW, MWh, state-of-charge, efficiency, degradation, grid, and contract limits.

"Rare events are too rare to matter." Rare events can matter enormously, but they may need a different payment mechanism than daily arbitrage.

"If a model has a high merchant case, the duration decision is settled." Merchant upside, contract firming, capacity value, partner incentives, and grid rights can point in different directions.

Current Debates And Caveats

Battery duration is moving. The IEA scenario material points toward a larger share of four-to-eight-hour battery systems as solar shares grow, while weekly or seasonal balancing usually points beyond batteries alone.

Technologies and substitutes for longer duration may include pumped hydro, compressed air, flow batteries, iron-air batteries, thermal storage, hydrogen, flexible industrial demand, process buffers, interconnection, and firm low-emission generation. No one of these is automatically the answer. Each has a different maturity, siting problem, efficiency, capex structure, permitting risk, response time, control-right problem, and bankability profile.

The practical debate for an IPP is:

```
Which scarcity durations are we paid to cover, and is ownership the best wa
```

Recap

Storage duration should be judged at the margin. The same extra MWh can be contract insurance, a way to reach longer price spreads, dependable capacity in a bounded stress event, or idle capital that blocks better uses of state of charge and grid rights. The opening choice between two and four hours now becomes a test: map the scarcity-duration distribution, price the next MWh against its marginal duration hurdle, and compare ownership with buying flexibility, changing the PPA shape, using portfolio substitutes, or waiting for better evidence.

Chapter 7. Data Centres, 24/7 Procurement, And Flexibility

References: [6] [56] [57] [58] [37] [59]

A data-center buyer can bring a large load, long horizon, strong credit, and a visible climate claim. The risk starts when the sales phrase covers more obligations than the product defines. Annual renewable energy, hourly clean matching, physical supply quality, price hedging, buyer flexibility, and public evidence are different promises. If they are bundled under broad 24/7 language, the IPP may inherit the expensive hours, the grid constraint, the reporting burden, and the missing control right all at once. The strategic work is to identify the stress-hour obligation before the term sheet assigns it.

Opening Puzzle

A hyperscaler asks for clean power for a large data center.

Annual renewable energy, hourly clean matching, physical delivery quality, price hedging, flexibility, and evidence are different products. The product definition should answer:

```
Which promise are we actually willing to underwrite:
annual certificates, hourly clean matching, physical supply quality,
price hedging, flexibility, or some bounded combination of these?
```

For a renewable IPP, a data center is a large load, a grid-connection event, a reliability-sensitive customer, a possible flexibility resource, and a public evidence claim. The strategic decision is whether to offer a product, partner for it, narrow it, or refuse it before the seller inherits an uncapped firming problem.

First-Principles Model

Separate three quantities that are often blurred together.

The first quantity is the load:

```
load_h = MWh consumed in hour h at the buyer site
```

The second quantity is the clean portfolio:

```
deliverable_clean_h =
  generation_h
  + contracted_supply_h
  + discharged_storage_h
  + market_or_portfolio_firming_h
```

The third quantity is the claim:

```
claim_h = what the buyer is allowed to say about hour h
```

A priced product keeps load, supply, and claim aligned. A loose product sells one component while leaving the others undefined.

The product-risk question is:

```
What is the residual obligation in the hard hours,
who controls the actions that reduce it,
and who pays when the evidence or delivery does not match the promise?
```

Definitions

Annual matching: comparing clean-energy procurement and consumption over a year.

Hourly matching: comparing clean supply and consumption in each hour.

Locational matching: comparing supply and consumption inside a defined grid, bidding zone, node, or certificate boundary. The boundary is rule-specific.

24/7 clean energy: a procurement ambition to match consumption with clean energy across all hours. In this chapter it is treated as a product-design concept.

Firming: using a portfolio, storage, flexible load, contracts, or market purchases to reduce mismatch between variable clean supply and the load obligation.

Residual obligation: the hourly MWh and EUR exposure left after the clean portfolio, storage, hedges, and certificates have done their work.

Granular certificate: a certificate concept tied to more detailed time and sometimes location attributes. Compliance treatment is not settled here.

Evidence standard, for product design here, means the commercial evidence a product must specify before anyone prices or markets the claim. Legal or compliance treatment still comes from the relevant standard.

Evidence element	Product question
Buyer load data	What interval, meter, site boundary, and data owner define consumption?
Supply data	Which generator, storage, contract, or firming source is eligible in each interval?
Certificate attributes	Which vintage, location, technology, cancellation, and claim rules apply?
Matching interval	Annual, monthly, hourly, or more granular?

Evidence element	Product question
Locational boundary	Same country, bidding zone, grid region, certificate region, or contract-defined boundary?
Residual firming record	Which hours were not matched, who bought firming, and at what rule or price?
Audit rights	Who can inspect source data, corrections, and certificate cancellation?
Claim language	What can the buyer say publicly, and what language is explicitly excluded?

If these fields are not priced and controlled, the seller has not defined the product well enough to price or market it.

Certificate And Claim Mechanics

The GHG Protocol Scope 2 Guidance and certificate standards make one point operationally important for a 24/7-style product: physical energy, contractual attributes, and public claims have to be tracked separately.

In the Scope 2 Guidance, companies in markets with contractual instruments report both location-based and market-based Scope 2 totals. The location-based method uses average grid emissions factors. The market-based method uses contractual information such as energy attribute certificates, direct contracts including PPAs where applicable, supplier-specific emission rates, and residual mix. If a contractual instrument does not meet the Scope 2 Quality Criteria, the company uses fallback data in the market-based hierarchy.

Energy attribute certificates represent attributes of generated electricity. They can be bundled with a PPA or sold separately. Once attributes are conveyed away from the underlying electricity, the remaining electricity becomes “null power” for market-based claims and should be treated with residual-mix logic. This matters for a seller because a PPA can transfer energy economics while leaving the evidence claim incomplete, or transfer certificates while leaving hourly physical supply quality unresolved.

EnergyTag's Granular Certificate Scheme Standard v2 adds the temporal layer. A granular certificate records a production or storage-discharge interval with timestamps in UTC, with the standard using a maximum production/storage discharge interval of one hour. Its storage treatment also matters for hybrid products: storage charge records, storage discharge records, storage discharge granular certificates, and loss allocation must be defined before stored energy can support a time-matched evidence claim.

For a term sheet, the evidence schedule should therefore specify:

- which certificate or attribute instrument is transferred
- whether the certificate is bundled with energy or sold separately
- production interval, cancellation, beneficiary, and registry evidence
- residual-mix treatment for uncovered or attribute-stripped energy
- storage charge/discharge evidence if BESS is used in the claim
- statement language the buyer may use, and language reserved for legal/accounting review

Numerical Intuition

The decision number is residual firming-at-risk:

```
firming_at_risk =
  sum over stress hours(
    residual_MWh_h x stress_firming_premium_EUR_per_MWh
  )
```

This turns the 24/7 claim into a product-risk number. A product with a small annual mismatch can still be expensive if the mismatch appears in the same cold, dark, high-price, congested hours.

ENTSO-E's 2026 data-centre report gives a second set of dated system-scale anchors. It maps data-centre flexibility resources to products: uninterruptible power supply (UPS) batteries are most suitable for very fast response and primary reserve, while cooling flexibility, thermal storage, on-site generation, and workload shifting fit different durations and control constraints. The same report cites a 2030 European short-duration/ramping flexibility need of roughly `15–30 GW`. It also cites a BloombergNEF/Eaton 2021 five-market estimate for Germany, Ireland, the Netherlands, Norway, and the United Kingdom: `16.9 GW` of technical data-centre flexibility potential, with up to `3.8 GW` realistically available after participation willingness and operational constraints.

Those numbers do not price a PPA. They tell the strategist that data centers may become controllable grid participants after reliability, tenant consent, operating policy, market access, and control architecture are solved.

Stress-Hour Residual Cost

Suppose a buyer wants a clean product for a flat `100 MW` load. Pair annual energy with the stress-hour residual.

Illustrative assumptions:

- stress hours in the year: `200 h`

- buyer load in each stress hour: `100 MWh`
- deliverable clean portfolio in those hours: `35 MWh/h`
- residual obligation: `65 MWh/h`
- stress firming premium above the base product price: `150 EUR/MWh`

The stress-hour residual is:

```
residual_MWh = 65 MWh/h x 200 h = 13,000 MWh
firming_at_risk = 13,000 MWh x 150 EUR/MWh
                = 1,950,000 EUR/year
```

If the buyer pays a `5 EUR/MWh` premium on the whole product volume, the premium may look large in annual terms. But if only `2 EUR/MWh` of that premium is available to cover stress-hour risk on a roughly `876,000 MWh/year` load, the annual risk budget is:

```
risk_budget = 876,000 MWh/year x 2 EUR/MWh
            = 1,752,000 EUR/year
```

In this illustrative case, the risk budget is smaller than the stress-hour residual cost before collateral, evidence systems, imbalance, credit, and partner fees. The offer has to change:

- raise the premium
- lower the hourly coverage promise
- exclude or index stress-hour firming
- add portfolio or hydro/wind diversification
- require buyer flexibility
- partner with a supplier or trader
- sell annual or shaped clean energy instead of a 24/7-style promise

The product threshold is now visible: the offer needs a higher premium, a narrower promise, a partner, or a residual-risk transfer.

Equations

Hourly residual:

```
residual_MWh_h =
  max(0, load_MWh_h - deliverable_clean_MWh_h)
```

Plain meaning: the amount the product still needs in hour `h` after clean supply, storage, portfolio netting, and eligible firming.

Product margin:

```
product_margin =
  customer_premium
  - expected_firming_cost
  - evidence_and_reporting_cost
  - imbalance_and_basis_cost
  - credit_and_collateral_cost
  - risk_capital_charge
```

The product is ready only when the remaining margin is positive under the stress cases the firm is willing to own.

Hourly matching rate:

```
hourly_match_rate =
  sum_h min(load_MWh_h, eligible_clean_MWh_h)
  / sum_h load_MWh_h
```

Illustrative hourly-matching scenario: if a 40 MW flat load consumes `350,400 MWh/year`, moving from `58%` hourly matching to an `85%` target requires:

```
additional_matched_MWh = (85% - 58%) x 350,400
                       = 94,608 MWh/year
```

That extra matched volume is concentrated in the hours where the existing portfolio does not match the load. Those are the hours that drive storage, portfolio, supplier, flexibility, or no-bid decisions.

Data-centre flexibility available to the grid:

```
available_flex_MW =
  technical_flex_MW
  x participation_willingness
  x operational_availability
  x control_right_factor
```

The ENTSO-E difference between technical potential and realistically available potential matters because a UPS battery, cooling system, or workload scheduler becomes a grid product only when someone can command, measure, qualify, and compensate it without violating the data center's primary mission.

Mechanism

A data-center product has two sides.

On the buyer side, the load is unusually valuable because it is large, creditworthy, continuous, and often strategically committed to clean power. That can justify long-tenor products, portfolio PPAs, hybrid structures, and evidence-aware offers.

On the risk side, the same load is operationally constrained. A data center cannot be treated like a casual flexible demand block unless the operator has agreed to that behavior. Its UPS is built first for uptime. Cooling flexibility is bounded by thermal comfort and equipment limits. IT workload shifting depends on workload type, tenant agreements, latency, redundancy, and internal policies. On-site generation may help resilience or connection management, but it can complicate emissions claims.

So there are different commercial products under the phrase "data-center clean energy":

Business	What is sold	Main risk
Annual renewable procurement	Annual MWh and certificates	Buyer thinks annual evidence means hourly supply quality.
Shaped or portfolio PPA	Better match to load shape	Seller underprices stress-hour residuals.

Business	What is sold	Main risk
24/7-style clean product	Hourly/evidence-aware clean supply target	Compliance, certificate, firming, and grid promises exceed capability.
Data-center flexibility	UPS, cooling, workload, or generation response	Uptime mission, tenant consent, and qualification block delivery.
Grid-connection partnership	Flexible connection, staged ramp, congestion management	Commercial value depends on local grid rules and operator acceptance.

The strategy is to choose what the product actually promises before a term sheet combines the obligations implicitly.

Example: Product Decision Test

A data-center buyer asks for “24/7 clean power” in a constrained zone.

The product team should write one page before pricing:

Question	If yes	If no
Is the buyer asking for annual reporting, hourly evidence, physical delivery quality, or price hedge?	Price the right product.	Do not quote; clarify the decision the product must support.
Is residual hourly exposure measured under stress weather and price cases?	Add firming-at-risk to quote stack.	The product is not ready.
Can the IPP control storage, portfolio supply, nominations, and partner actions?	Sell a stronger shape.	Sell a bounded product or partner.
Are certificates and evidence accepted for the buyer’s intended claim?	Include reporting cost and audit trail.	Avoid compliance language.
Does the buyer provide flexibility or accept residual exposure?	Share residual exposure.	Price the residual or narrow the promise.
Does the local grid support the load and product structure?	Proceed to site economics.	Model grid as a binding constraint in the product.

The test ends in one of four decisions:

- offer a simple annual PPA
- offer a shaped or portfolio-backed product with disclosed residuals

- partner for supplier, certificate, or trading execution
- no-bid the 24/7 claim until evidence, control, or pricing improves

In Practice

For a renewable IPP, the first-year data-center playbook should build a repeatable product test:

- load shape and siting: what does the customer actually consume, where, and under what connection limits?
- product promise: annual, shaped, hourly, local, certificate-backed, or price hedge?
- residual map: which hours remain uncovered, and how expensive are they in stress?
- control rights: who dispatches BESS, nominates positions, buys firming, and owns imbalance?
- evidence chain: which data, certificates, meter intervals, and audit trail support the claim?
- buyer flexibility: can UPS, cooling, workload shifting, or on-site generation reduce grid/product risk without threatening uptime?
- decision: build, partner, defer, narrow, or avoid

Practitioner commentary is useful for identifying 2026 debates around hyperscaler load, artificial-intelligence-driven flexibility, grid constraints, and 24/7 claims. Product claims still need sourced physics, current official rules, standards, and contract evidence.

Misconceptions

“A data center is just a flat load.” It is a large connection event with reliability architecture, workload composition, cooling systems, UPS behavior, and public procurement claims.

“Annual clean energy is harmless if the buyer understands it.” If the sales language implies hourly supply quality, the seller has created a product-risk gap.

“UPS batteries are automatically grid batteries.” They are uptime assets first. Grid use depends on duration, control, qualification, operator consent, tenant obligations, and the ability to preserve protection.

“A 24/7-style product is a certificate problem.” It also involves stress-hour firming, grid, control-right, credit, and data exposure.

“Data-center flexibility reduces PPA risk for free.” Flexibility has an owner. If the owner will not let the IPP call it during the valuable hours, it is not part of the product.

Current Debates

Dated 2026 sources support treating data centers as several power-system questions:

- passive load that stresses connection queues and local grids
- premium clean-power buyer that pays for shaped and evidence-aware products
- flexible resource through UPS, cooling, workload shifting, or on-site generation
- partner in staged or flexible grid-connection arrangements
- source of new public scrutiny around clean-energy claims

Detailed Scope 2, RE100, Association of Issuing Bodies (AIB), European Energy Certificate System (EECS), guarantee of origin (GO), and granular-certificate compliance treatment remains outside this product-design test. For dated product work, use the applicable primary standards and certificate rules. The commercial rule is narrower: do not sell a cleaner claim than the portfolio, evidence, and control rights can support.

Recap

A data-center clean-power product depends on what it can measure, control, and price in the hard hours. Annual MWh may satisfy one buyer need; hourly matching, local evidence, physical reliability, and flexibility create different residual obligations. Keep load, clean portfolio, and claim separate until the term sheet says exactly who controls firming actions and who pays when delivery or evidence falls short. That separation turns a broad 24/7 request into a decision: annual PPA, shaped or portfolio-backed offer, partnership, narrower promise, deferral, or no-bid.

Chapter 8. Grid-Forming Inverters And System Services

References: [22] [23] [18] [11] [60] [61]

A project team can price the battery, the transformer, the grid connection, and the expected energy-market stack. Then an inverter option appears in the design. It promises stronger behavior in a weak grid, voltage support, faster response, or readiness for future grid-forming requirements. The investment question is whether this package removes a binding constraint, satisfies a requirement, creates a paid service, preserves a valuable option, or simply adds cost. Grid-forming capability becomes commercial only after the path through specification, qualification, control, measurement, settlement, and enforcement is visible.

Opening Puzzle

A battery vendor offers a more advanced inverter package. The engineer says it may support weak-grid operation, voltage control, fast response, and future grid-forming requirements. The commercial team asks whether the project should pay for it. The investment question is:

```
Is system-service capability a compliance cost, a connection advantage,
a revenue option, or a distraction for this site and portfolio?
```

Grid-forming capability can buy or preserve optionality in a grid whose missing services may no longer arrive automatically from large synchronous machines.

First-Principles Model

The electricity system needs more than energy.

Energy balance asks:

```
Are MWh supplied over time?
```

System security asks:

```
Can voltage, frequency, protection, and restoration survive disturbance
```

Historically, synchronous machines bundled several useful behaviors with energy production: rotating inertia, fault-current contribution, voltage/reactive-power support, and familiar protection behavior. Inverter-based resources reproduce those behaviors only when hardware, controls, settings, grid strength, testing, and connection rules support them.

The strategic model is:

```
system-service option value =
  connection value
  + compliance value
  + service revenue
  + avoided curtailment or constraint cost
  + strategic option value
  - incremental capex and opex
  - qualification/testing cost
  - dispatch and state-of-charge opportunity cost
  - performance and penalty risk
```

If the left side is positive under credible scenarios, capability is worth buying. If not, it may be a research option or a procurement requirement to watch.

Definitions

Inverter-based resource: a resource connected through power electronics, such as many solar, wind, and battery projects.

Grid-following inverter: an inverter that normally synchronizes to an existing grid voltage and frequency reference.

Grid-forming inverter: an inverter controlled to help establish or support voltage and frequency reference behavior. Exact technical criteria depend on standards, equipment settings, tests, and grid context.

System service: a service needed for secure operation, such as frequency response, voltage support, reactive reserve, short-circuit power, inertia-like behavior, black-start or restoration support, fault ride-through, or weak-grid operation.

Qualification chain: the path from technical capability to commercial value: specify, test, qualify, control, measure, settle, and enforce.

Reactive power: AC power component associated with voltage support, commonly measured in var or MVAr.

Short-circuit power or fault level: measures related to system strength and fault behavior. They matter for protection and stability, but exact values are system-specific.

Numerical Intuition

Start with the paid-or-required capability gap:

```
capability_gap =
  required_or_valuable_service_capability
  − qualified_available_capability
```

Then ask what the gap costs:

```
gap_value =
  avoided_connection_delay
  + avoided_curtailment
  + service payment
  + avoided_remedial cost
  − capability cost
  − opportunity cost
```

Two sourced anchors show why this belongs in a strategy chapter.

First, Kirschen and Strbac's power-system operation chapter uses a two-bus voltage example where remote voltage control can keep voltage inside limits only over a bounded transfer range. In the example, transfers below `49 MW` or above `172.5 MW` violate a voltage limit, and local reactive support becomes more effective than remote control, especially after contingencies. The transferable claim is that voltage support is locational. A service can have high value at one node and limited value elsewhere.

Second, Energinet's 2025 Danish ancillary-service tender conditions for DK1 and DK2 list stability properties such as short-circuit power, inertia, reactive reserves, and voltage control. In that material, Energinet describes checking load flow, short-circuit power, N-1 situations, and reactive reserves after operational schedules and again when operating conditions change. Procurement timing is service-specific: the tender conditions describe monthly, weekly, day-ahead, post-spot, and intraday routes where required. Read this as dated evidence that stability properties can be operational requirements or procurement items; it does not establish that a generic grid-forming inverter package is qualified, payable, or needed at a specific site.

Third, Fingrid's 2026-2035 grid development plan treats inverter-dominated areas as a stability problem requiring technical solutions such as grid-forming storage, synchronous compensators, static synchronous compensators (STATCOMs), and updated grid-code work. That is a planning signal, not a generic revenue forecast. For an IPP it means the commercial file should track whether stability capability affects connection timing, connection terms, curtailment exposure, or a separately procured service.

The Option Premium

Suppose a `100 MW` BESS project can buy an enhanced inverter/control package.

Illustrative assumptions:

- incremental capability cost: `4 EUR/kW`
- project power rating: `100,000 kW`
- upfront option cost: `400,000 EUR`
- simple annual carrying charge: `10%`
- annual cost hurdle: `40,000 EUR/year`

Now test three scenarios.

Scenario	Value logic	Decision signal
No requirement, no payment, no connection benefit	`0 EUR/year` benefit versus `40,000 EUR/year` hurdle	Do not buy unless the option protects a larger strategic uncertainty.
Capability helps avoid `1 GWh/year` curtailment or constraint loss worth `50 EUR/MWh`	`50,000 EUR/year` benefit	Barely clears before testing, warranty, and opportunity cost.
Capability unlocks a tender, connection approval, or	`250,000 EUR/year` benefit	Strong candidate if technical delivery and qualification are credible.

Scenario	Value logic	Decision signal
avoided delay worth `250,000 EUR/year`		

The `4 EUR/kW` is a scenario assumption. The decision shape matters more. System-service capability is valuable only when it changes a binding constraint, a required compliance path, a paid service, or a strategic option.

Equations

Qualification-adjusted service value:

```
service_value =
  technical_capability
  x qualification_probability
  x availability
  x controllability
  x payment_or_avoided_cost
```

If any multiplier is close to zero, do not underwrite the revenue line.

Dispatch opportunity cost:

```
opportunity_cost =
  best_feasible_energy_or_reserve_value_lost
  + degradation_or_warranty_cost
  + state_of_charge_constraint_cost
```

For a battery, a system-service commitment may require headroom, state of charge, inverter capacity, telemetry, or operating mode. It can compete with arbitrage, FCR, aFRR, mFRR, PPA firming, and curtailment recovery.

Connection-option test:

```
buy_capability if:
  expected avoided delay or avoided constraint cost
  + expected service value
  >
  annualized capability cost
  + qualification and performance risk
```

The commercial translation from engineering to strategy is this qualification-adjusted value test.

Mechanism

System services become strategic when the grid changes faster than the market product catalog.

Energy markets are good at pricing MWh. They are less clean at pricing local voltage support, short-circuit power, black-start readiness, or weak-grid behavior. Some of these services may be:

- mandatory connection requirements
- procured through tenders or bilateral instructions
- embedded in grid codes
- compensated through cost-plus or regulated arrangements
- valuable indirectly because they reduce curtailment, connection delay, or reinforcement need
- excluded from base-case revenue while still relevant as synchronous machines retire

For an IPP, the mistake is to treat all of these as either “free future upside” or “irrelevant engineering.” They are options with different exercise conditions.

Service question	Commercial interpretation
Required for connection?	Compliance cost or development enabler.
Procured by TSO/grid owner?	Potential revenue, subject to qualification and availability.

Service question	Commercial interpretation
Reduces curtailment or connection constraint?	Project-value uplift, even without a market product.
Requires SoC, headroom, or operating restrictions?	Opportunity cost against existing stack.
Lacks testing or measurement?	Keep it outside bankable revenue.
Depends on future grid-code change?	Strategic option, not base-case revenue.

Example: Commercial Proposal Test

The project team proposes grid-forming-ready inverters for a solar-plus-BESS site near a weak part of the network.

The proposal is investment-ready only after seven questions are answered:

1. What local system problem are we solving: voltage, fault level, frequency response, weak-grid stability, restoration, or connection capacity?
2. Is the capability required, paid, or merely hoped for?
3. Which document defines performance: grid code, tender, connection agreement, original equipment manufacturer (OEM) specification, or pilot protocol?
4. What is the incremental cost and what revenue or avoided cost clears it?
5. What dispatch freedom do we lose?
6. Who carries non-performance risk if the service fails?
7. Does this capability make the site more valuable as a portfolio platform?

The answer may be:

- buy now because it unlocks connection or avoids a known constraint
- buy the hardware-ready option but defer certification spend
- require the EPC/OEM to preserve upgrade paths
- partner with a specialist aggregator or system-service provider
- decline because the service is not defined, paid, or binding at this site

In Practice

For a renewable IPP, the practical workflow is a four-bucket classification.

Bucket	What to do
Mandatory	Treat as connection/compliance cost; include in capex and schedule.
Monetized	Model as revenue only after qualification, control, availability, and settlement rules are

Bucket	What to do
	clear.
Indirect value	Model through avoided curtailment, avoided delay, better connection terms, or better site ranking.
Future option	Preserve upgrade rights, data, and OEM capability, but do not let it finance the project.

This is especially relevant for batteries. A technically fast BESS still needs service-specific qualification. Voltage support, short-circuit behavior, grid-forming controls, FCR, aFRR, mFRR, black-start, and adequacy each have different qualification and opportunity-cost logic.

Misconceptions

“Grid-forming means extra revenue.” Only if a service is defined, qualified, controllable, measured, and paid or otherwise valuable.

“No market product means no value.” The value may sit outside a standard revenue line: connection approval, avoided curtailment, avoided remedial action, or future optionality. Keep it out of base-case revenue until the path is documented.

“Fast battery equals system-service battery.” Fast response is one property. The service may need reactive capability, fault behavior, state of charge, controls, telemetry, or restoration procedures.

“Reactive power can be ignored in the commercial model.” ACER’s tariff review shows reactive-energy charges and requirements appear widely across Europe, and Kirschen and Strbac show why local voltage support can bind transfer capability.

“Grid-forming is the same as synthetic inertia.” Keep the words separate. Different services need different controls, tests, and commercial treatment.

Current Debates

As of 2026, the active question is how inverter-heavy systems define and pay for the services that synchronous machines used to bundle with energy production.

Textbook and official material establish the durable mechanism: voltage, frequency, reactive support, fault behavior, and restoration are real system needs. Energinet’s 2025 Danish tender material is a dated example of short-circuit power, inertia, reactive reserves, and voltage control appearing as operationally checked, procured stability properties. Practitioner analysis is useful for the frontier debate around grid-forming and grid-following controls, but named-market claims require current TSO, grid-code, equipment, and tender documents.

Recap

Grid-forming and system-service capability should be valued like an option whose exercise conditions matter. The inverter package is worth paying for when it changes a real connection path, compliance obligation, curtailment constraint, paid service, or portfolio option. It is weaker when it is only an attractive technical label. The decision should move through the qualification chain: what service is specified, who tests it, how it is controlled and measured, what payment or avoided cost it creates, and what dispatch freedom or performance risk it consumes.

Chapter 9. Data And Machine Learning In Energy Management

References: [43] [46] [62]

A forecast arrives before the day-ahead gate closes, or after the trader has already acted. A dispatch recommendation names a battery action, but the contract may give that control right to someone else. A partner report shows a gap, but the benchmark may be using information no one had at the time. For a renewable IPP, artificial intelligence and machine learning (AI/ML) belong inside a decision system before they are a technology category. Their value depends on which decision changes, who owns that decision, how much value survives timing and constraints, and how much loss a wrong recommendation can create before anyone catches it.

Opening Puzzle

A vendor shows a strong-looking result: the solar forecast is 12 percent better than the incumbent model.

The day-ahead trader receives the forecast after the gate has closed, the battery is already committed to reserve, and the partner contract does not give the IPP dispatch rights.

How much of the model improvement can become portfolio value?

First-Principles Model

An energy-management model is a sequence:

```
data -> forecast -> decision -> market or asset action -> feedback -> r
```

Break one link and the economics shrink. Bad metering contaminates the model. A good forecast arriving late cannot affect a closed market. A dispatch recommendation without control rights is a suggestion. A strategy without a time-consistent backtest remains unsupported. An automated action without limits can turn model error into portfolio loss.

The word “AI” is used for several different tasks:

- a forecaster that maps weather and plant state into generation, load, price, imbalance, or congestion scenarios
- an optimizer that chooses nominations, dispatch, reserve participation, or hedges under constraints
- a monitor that detects anomalous metering, availability, partner actions, or settlement results
- a scenario generator that makes stress cases for PPA, BESS, and portfolio decisions
- a governance system that tracks drift, overrides, model ownership, and model retirement

These are not interchangeable. A forecast model answers “what might happen?” An optimizer answers “what should we do, given what we know?” A control system asks “are we allowed to do it, and how badly can it go wrong?”

Wind-forecasting material in *Renewable Electricity and the Grid* makes the same point from the physical side: the useful forecast horizon depends on the decision. Very short horizons support dispatch and balancing; day-ahead horizons support scheduling and market positions; longer horizons support planning and valuation. Accuracy tends to decay with horizon, and different methods need different inputs. Physical models need site and asset detail; statistical and learning models need long, clean histories. The right model is matched to the decision horizon.

Definitions

Actionability-adjusted uplift is the raw model improvement after reducing for timing, tradable volume, liquidity, constraints, control rights, and adoption.

Model-risk charge is a governance allowance for losses or control burden introduced by relying on a model. It includes the risk the model adds to the decision.

Model value at risk is the value exposed to a wrong model-driven action: affected MW, duration, adverse price error, and frequency of bad events.

Model inventory is the register of models, owners, uses, inputs, limits, validation status, and retirement rules.

Decision owner is the person or function accountable for accepting, rejecting, or overriding a recommendation.

Backtest means a historical replay using only information available before each decision time. A perfect-foresight simulation is useful as an upper bound; an operational backtest must use the information that was available before the decision.

Value-weighted error is forecast error weighted by the economic consequence of being wrong in that interval.

Numerical Intuition

Start with value-weighted error.

```
20 MWh error x 5 EUR/MWh consequence  =   100 EUR
20 MWh error x 80 EUR/MWh consequence = 1,600 EUR
```

The physical error is the same. The commercial error is 16 times larger in the second interval. A model can improve average accuracy and still miss the hours that matter.

Then calculate captured value:

```
captured value
  = raw uplift x actionability factor - model cost - model-risk charge
```

Use the actionability factor as a disciplined way to avoid overestimating model value:

```
actionability factor
  ~= timing share
   x tradable-volume share
   x control-right share
   x liquidity-and-constraint share
   x adoption share
```

Now suppose a model finds 5 MWh of better action in each of 400 high-value intervals per year, and the relevant spread is 70 EUR/MWh.

```
raw uplift = 5 x 400 x 70 = 140,000 EUR/year
```

If only 35 percent survives timing, liquidity, control rights, constraints, and adoption:

```
captured value before cost and risk = 140,000 x 0.35
                                    = 49,000 EUR/year
```

This tells management where the value is being lost. The higher-value project may be earlier data delivery, intraday access, a cleaner partner interface, or a contract amendment that gives the asset owner the right to act.

Evidence Pack For Funding A Model

A model investment case needs evidence that the recommendation can reach an action.

Evidence item	Metric	Decision use
Timing coverage	recommendations received before gate closure / total recommendations	Buy or build only if the model can still change the action.
Tradable exposure	MWh with market access, liquidity, and control rights	Size the value pool before funding model complexity.
Benchmark validity	replay uses only information known before each decision	Reject hindsight backtests for partner governance.
Model downside	MW affected x hours x adverse EUR/MWh x detection count	Decide automation limit, human approval, and drift monitoring.

A 12 percent forecast improvement is only the first test. The investment case should state how many MWh are still tradable when the forecast arrives, who has the right to act, and what loss limit applies if the recommendation is wrong.

Model Risk

Model value at risk asks how badly an automated action can fail.

Suppose a model can affect 40 MW for two hours. In a bad regime it can be wrong by 100 EUR/MWh. If that happens 15 times before the error is caught:

```
model value at risk = 40 MW x 2 h x 100 EUR/MWh x 15
```

```
                    = 120,000 EUR
```

This operating-control number tells you whether the model can be left to act alone, whether it needs a limit, whether it needs human approval, and how fast monitoring must catch drift.

The governance question arrives early because a model that cannot explain the binding constraint, log overrides, or show which data were known at decision time may be unfit for commercial control even if the test score looks good.

Equations

A decision-value equation:

```
model value
  = value(decision with model)
  - value(decision without model)
  - model cost
  - model-risk charge
```

A fair partner-monitoring equation:

```
execution leakage
  = feasible benchmark value - actual realized value
```

The word “feasible” carries the weight. The benchmark must use the same information timing, asset state, market access, transaction costs, risk limits, and contractual control rights available at the time. Otherwise the benchmark is hindsight-biased.

A build/buy/outsource test:

```
own internally when:
  decision value is high
  x strategic control is high
  x observability is high
  x vendor lock-in risk is high
  > cost and governance burden
```

Use this as a decision rule rather than a procurement rule. Some models belong inside the firm. Some belong with a specialist. Some should be bought but independently benchmarked. Some should stay out of scope.

Mechanism

The capability ladder usually climbs like this.

First, build the data layer. Asset availability, meter data, forecasts, market prices, nominations, dispatch instructions, imbalance outcomes, reserve commitments, contract obligations, and partner actions must be time-aligned. Without this, the company cannot tell whether a model failed, a partner failed, the asset failed, or the market simply moved.

Second, benchmark the decisions already being made. For each asset or portfolio, ask what a feasible forecast-based action would have done. Do not jump to fully automated trading. Start by reconstructing the decision surface.

Third, improve the highest-value forecast horizon. If the value is in day-ahead nomination, improve day-ahead forecast and scenario quality. If the value is in intraday repair, improve short-horizon updates and liquidity timing. If the value is partner oversight, improve attribution and counterfactuals.

Fourth, introduce optimization support. The optimizer must know state of charge, grid limits, degradation, PPA obligations, bid rules, minimum sizes, transaction costs, and reserve commitments. Optimization that ignores the asset is an incomplete calculation.

Priyanka Shinde’s work on efficient trading in short-term electricity markets for renewable integration supports this discipline: stochastic intraday and virtual-power-plant models can improve decisions, but only when market inputs, transaction costs, computation, risk preferences, and gate timing are represented. For an IPP, the practical implication is narrower: fund models where the operating decision is observable and executable.

Fifth, automate only where the loss from delay is larger than the model-risk charge. Sometimes the right design is a decision-support model with a human owner and a narrow mandate.

Example

Consider a solar-plus-BESS portfolio with outsourced market execution.

Three AI projects are proposed:

Proposal	Evidence required before funding	Default first-year choice
Price model for merchant arbitrage	tradable MWh, liquidity, transaction cost, risk limits, and time-consistent backtest	buy or observe until owned actionability is proven
Partner-monitoring model	forecast, bid, schedule, dispatch, meter, activation, and settlement files with timestamps	build benchmark method internally
Short-horizon solar forecast	delivery before intraday action window, control rights, and value-weighted error tracking	buy forecast input; build value layer internally
Battery optimizer	SoC, grid limits, reserve commitments, degradation cost, warranty constraints, PPA obligations	outsource execution if needed; own objective and audit logic

The decision question is:

```
Which model changes a decision we actually own or can govern?
```

If the answer is “none,” the right action is to change the operating model before buying more model complexity.

In Practice

A renewable IPP should ask four questions before funding an AI project:

- What decision does this model change?
- What is the actionability-adjusted value, not the raw performance uplift?
- What loss can the model create before detection?
- Who owns the decision, the override, the benchmark, and the retirement rule?

The first high-value AI system may be a clean data model, forecast diagnostics by horizon, imbalance attribution, feasible dispatch benchmarks, partner scorecards, model drift alerts, and a small inventory of governed models.

At that point, advanced analytics becomes part of the commercial operating system.

Misconceptions

"AI trading edge is the goal." For an asset owner, the first high-value uses may be pricing, dispatch support, risk control, and partner governance.

"A better forecast is automatically valuable." A forecast creates value only when it changes a feasible action.

"A profitable backtest proves the model." A profitable backtest may contain future information, omit costs, ignore liquidity, overfit a regime, or assume rights the firm did not have.

"Explainability is only compliance." Explainability is how the organization learns whether a loss came from weather error, price error, constraint handling, partner action, or model design.

"Automation is the final form." The target is reliable decision ownership under uncertainty. Implementation can be automated or deliberately manual.

Recap

AI/ML earns a place in energy management only when the chain closes: data becomes a forecast, the forecast changes a feasible decision, the decision reaches a market or asset action, and the result feeds back into risk control. The same uplift can be valuable, idle, or loss-making depending on timing, tradable volume, control rights, liquidity, adoption, and model risk. The build/buy/outsource decision should be made after those constraints are priced into the capability.

Chapter 10. Trading Capability Design

References: [3] [12] [43]

The contract is signed, the partner has the market systems, and nominations, balancing, and dispatch can now move through someone else’s desk. The remaining question is what the IPP can still see, decide, and challenge when portfolio value, PPA margin, imbalance cost, degradation, and risk appetite are affected by those actions. Market access, execution, analytics, decision rights, risk ownership, and governance are different capabilities. Treating them as one outsourced package can make the operation lighter while making the economics harder to control.

Opening Puzzle

An IPP signs with a trading partner. The partner handles market access, nominations, balancing, and perhaps battery optimization.

The IPP has outsourced execution tasks. It still owns the economics of bad forecasts, bad dispatch, wrong risk appetite, weak contract design, or a partner incentive misaligned with the asset owner. Those effects still land somewhere: project value, PPA margin, imbalance cost, degradation, or portfolio risk.

The decision is which capabilities must stay inside the company so outsourced execution remains observable, benchmarkable, and aligned with the owner’s risk appetite.

First-Principles Model

Break the trading layer into separate capabilities:

- market access: the legal, technical, and operational ability to participate in markets
- execution: submitting bids, nominations, trades, and dispatch instructions
- analytics: forecasting, optimization, risk measurement, and benchmarking
- decision rights: who is allowed to choose dispatch, bids, risk limits, or hedges
- risk ownership: who bears imbalance, price, shape, degradation, penalty, and opportunity-cost exposure
- governance: how actual behavior is measured against a feasible standard

Build, buy, or outsource is a bundle of decisions across these capabilities.

Definitions

BRP: a balance responsible party, defined here at a general level as the actor responsible for balancing or imbalance responsibility for a delivery point or portfolio. Nordic implementation details are market- and date-specific.

BSP: a balancing service provider, defined here at a general level as an actor providing balancing services to the TSO. Product rules and qualification are country/product-specific.

Market access: the ability to participate in a market through the required contracts, systems, qualifications, collateral, and operational processes.

Execution: the act of placing bids, nominations, trades, dispatch instructions, or activation responses.

Decision rights: contractual or organizational authority to choose an action.

Risk appetite: the limits defining acceptable exposure, downside risk, liquidity risk, imbalance risk, degradation, and operational risk.

Shadow trading: a feasible counterfactual simulation used to benchmark actual decisions. It must avoid perfect hindsight.

Benchmark: a comparison standard. A useful benchmark is feasible, information-timed, contract-aware, and cost-aware.

Numerical Intuition

Start with the observability gap.

Using a 15-minute operating interval, one asset produces `96` delivery intervals per day. A 25-asset portfolio creates:

```
25 assets x 96 intervals/day = 2,400 asset-intervals/day
```

If the owner receives only monthly revenue totals, most operational decisions are invisible. If it receives interval-level forecasts, bids, dispatch instructions, SoC, prices, and settlement line items, the same partner relationship becomes governable.

So the first build/buy/outsource metric is:

```
observable decision share
  = decisions with enough data to benchmark / total economically releva
```

Outsourcing execution can be sensible. Outsourcing observability leaves the asset owner unable to benchmark retained risk.

Partner fees, headcount thresholds, market share, and optimizer-volume thresholds are date- and firm-specific. They should be treated as deal inputs, not general rules.

Worked Estimate

Internalization value can be sketched as:

```
internalization value
 = avoided leakage
 + new optionality
 - people cost
 - systems cost
 - operational and compliance risk cost
```

Suppose a portfolio has only `40%` of economically relevant intervals observable enough to benchmark. A better data-rights and analytics setup raises that to `80%`.

```
benchmarkable exposure gain = 80% - 40% = 40 percentage points
```

The 15-minute interval is illustrative; the live interval depends on the relevant market, product, and implementation date. The gain in observability creates the ability to separate market conditions, forecast error, asset constraints, risk-limit choices, partner execution, and missing data.

The structure matters more than the exact percentage: observability, controllability, and risk maturity determine whether to build, buy, or outsource.

Equations

A simple capability ROI expression:

```
capability ROI
  = value pool addressed
  x observable decision share
  x control share
  - people/systems/risk cost
```

The numerator is the portion of market revenue the capability can actually improve.

For partner oversight:

```
partner performance gap = feasible benchmark - actual outcome
```

But the benchmark must include:

- information available at the time
- market access and gate closure
- transaction costs
- asset constraints

- contract rights
- risk limits
- degradation and availability

Without those, the benchmark cannot govern the relationship because it relies on actions that were not actually available.

Mechanism

Outsourcing works best when the outsourced partner has scale, market infrastructure, operational coverage, and product expertise that the asset owner does not yet need to replicate.

Outsourcing fails when the asset owner cannot answer basic questions:

- What risks did we retain?
- What did the partner optimize for?
- Did the partner have the right data?
- Were actions constrained by contract, market access, asset limits, or risk limits?
- How would a feasible benchmark have behaved?
- Is underperformance persistent, explainable, and material?

Building in-house can make sense when the value at stake, decision speed, portfolio complexity, and internal maturity justify the cost and operational risk. Buying tools can make sense when internal judgment exists but needs better data, forecasts, optimization, or reporting. Outsourcing can make sense when execution complexity is high and the company can still govern the partner intelligently.

The sequence usually matters: own the question before owning the execution.

Capability-Contract Matrix

The practical tool is a matrix. It separates capability from contract.

Capability	Own internally	Buy as tool	Outsource as service	Non-negotiable control right
Market access	Sometimes later	Rarely enough alone	Often early	Visibility into positions, fees, collateral, nominations, and settlement
Forecasting	Own methodology and error tracking	Buy weather/data/models	Use partner forecast as input	Forecast archive and performance diagnostics
Dispatch optimization	Own objective function and constraints	Buy optimizer/model stack	Outsource execution/24-7 coverage	Ability to audit actions against feasible benchmark
BESS degradation policy	Own risk appetite and warranty economics	Buy degradation model	Outsource monitoring support	Dispatch must respect agreed degradation cost and warranty constraints
PPA shape pricing	Own retained-risk model	Buy data and curves	Get quotes/market color	Final pricing assumptions and risk margin remain internal
Hedging	Own risk appetite and hedge objective	Buy analytics/ETRM	Outsource execution	Approve hedge mandate, tenor, basis, collateral terms, and limit usage

Capability	Own internally	Buy as tool	Outsource as service	Non-negotiable control right
Credit and liquidity	Own limits and cash-stress view	Buy systems	Outsource admin, not accountability	Current exposure, potential future exposure (PFE), collateral, and liquidity reports
Partner performance	Own benchmark design	Buy data pipeline	Partner supplies execution data	Time-stamped actions, constraints, prices, forecasts, and settlement line items
ETRM / system of record	Own data model and governance	Buy platform	Managed service possible	Audit trail, workflow, access controls, and exportable data

The matrix prevents a common mistake: outsourcing the action while accidentally outsourcing the evidence needed to judge the action.

Data Rights And Service Levels

The partner agreement should make the evidence trail explicit.

Area	Required evidence from partner	Cadence	Internal right retained
Nominations and balancing	forecasts, nominations, accepted bids, activations, prices, volumes, imbalance settlement lines	interval-level plus settlement close	benchmark and dispute material deviations
BESS dispatch	state of charge, availability, reserve commitments, dispatch instructions, degradation cost, warranty constraint	near-real-time or daily	approve objective function and risk limits
Credit and liquidity	collateral, current exposure, fees, limit usage, settlement cash timing	daily or monthly by exposure type	set limits and escalation rules
Data quality	missing files, stale curves, unreconciled meters, changed assumptions	exception-based plus monthly review	require remediation before attribution claims

A small leakage number can justify a serious control layer only after the organization can observe and influence it. Suppose an outsourced optimizer governs `500 GWh/year`. A feasible benchmark finds a recurring `0.60 EUR/MWh` avoidable gap, but only `40%` is observable and controllable after data-right, mandate, and constraint filters:

```
addressable value = 500,000 MWh/year x 0.60 EUR/MWh x 40%
                  = 120,000 EUR/year
```

If the data-rights and benchmark layer costs `80,000 EUR/year`, it can clear the investment test if the gap is recurrent and the evidence improves. That supports data rights and benchmark governance before it supports a full internal trading desk.

Build-Buy-Outsource Decision Tree

For each capability, ask four questions in order.

```
1. Is the decision strategically material?
2. Does the asset owner retain the downside?
3. Can the decision be observed and benchmarked?
4. Is internal execution worth the cost, risk, and operational burden?
```

The answer can be:

- own the judgment and outsource execution
- buy tooling and keep decision rights internal
- outsource both execution and first-line analytics, but require audit data
- internalize execution only after value, scale, data, people, and controls are ready

The minimum internal control layer is smaller than a trading desk:

- position and exposure view
- PPA/product-pricing model
- BESS dispatch objective and degradation economics
- partner benchmark
- credit and liquidity view
- risk appetite and escalation path
- data-rights contract language

Once that exists, the firm can decide what to internalize from evidence rather than organizational fashion.

Example

A solar-plus-BESS IPP has outsourced BRP/BSP and optimization services. A month later, the battery shows lower revenue than expected.

Incomplete diagnosis: “The optimizer failed.”

Audit questions:

- Were market prices different from the scenario?
- Was the battery unavailable or constrained?
- Did the PPA require preserving energy for a shaped delivery obligation?
- Was state of charge reserved for a reserve product or imbalance risk?
- Did the partner have the forecast and control rights needed?
- Was the benchmark feasible at the time, or did it use future prices?
- Did risk limits intentionally avoid a high-upside action?

Only after that can the IPP separate market conditions from partner underperformance.

In Practice

In Nordic practice, eSett’s Nordic Imbalance Settlement Handbook v5.3 is useful for settlement mechanics: agreements, imbalance settlement, balancing-service settlement, invoicing, collateral, and risk management. Sweden provides a dated implementation example for the BSP/BRP actor-role split: the contractual split took effect on 2024-05-01, and the cited Svenska kraftnät material described full independent BSP implementation as staged toward 2028. Product-specific participation still depends on the relevant country, product, rulebook, and date.

The practice-relevant framework is portable:

- outsource execution where external scale and 24/7 operations matter
- retain internal data ownership and economic understanding
- build shadow-trading benchmarks before making accusations
- define risk appetite before judging partner conservatism

- connect PPA product design to dispatch rights and imbalance exposure
- let portfolio scale decide when internal capability pays for itself

For a renewable IPP moving into solar-plus-storage and structured PPAs, maturity starts with knowing which risks remain on the owner's balance sheet.

Misconceptions

"A BRP/BSP contract removes the asset owner's risk." It may transfer specific duties or services, but commercial exposure depends on the contract and market design.

"Building a trading desk proves maturity." Not necessarily. It can create cost, compliance burden, operational risk, and distraction before the value pool is large enough.

"Buying an optimizer solves dispatch." It helps only if objectives, constraints, data, rights, and incentives are correct.

"A benchmark should show the best possible historical trade." A useful benchmark shows a feasible action using the information available at that time.

"Partner governance is mistrust." Good governance can make the relationship clearer, fairer, and less emotional.

Current Debates And Caveats

Current optimizer models, fixed-price or floor-style structures, trading-desk economics, and partner market shares are date-specific. Use current evidence before turning any of them into a 2026 market-practice claim.

The durable claim is narrower: market access, execution, analytics, risk ownership, decision rights, and governance are separate. Outsourcing one leaves the others to be governed explicitly.

Recap

Build, buy, or outsource is a control-design question. Separate the trading layer into market access, execution, analytics, decision rights, risk ownership, and governance, then ask which pieces must remain observable, benchmarkable, and owned internally. A partner can press the buttons, a tool can improve forecasts or optimization, and a contract can allocate duties, but the asset owner still needs to know which risks sit on its balance sheet and whether actual actions matched a feasible standard.

Chapter 11. IPP Operating Models For Active Portfolio Management

References: [50] [27] [62]

An active renewable IPP has more recurring choices than a simple asset owner: quote a shaped PPA, size a battery, decide who controls dispatch, challenge a trading partner, hedge a merchant tail, explain a settlement variance, and update a project model after the market has moved. The operating-model decision is a constrained capital-allocation problem. The firm must decide whether the next scarce euro and management hour should buy internal analytics, stronger partner governance, market systems, or an execution desk.

Opening Puzzle

A renewable IPP owns solar assets, has several batteries in development, sells PPAs, and outsources balancing or trading execution. Customer requests are becoming more shaped. Merchant exposure is growing. The trading partner sends reports, but the IPP cannot always tell whether a result came from market conditions, asset constraints, or execution.

Management asks:

```
What should we build internally this year?
```

The answer should rank capabilities by value at stake, controllability, observability, cost, risk, and sequence. A trading desk may eventually be right. So may a small analytics layer plus much tougher partner governance. The decision depends on which constraints bind first.

First-Principles Model

An IPP converts scarce resources into risk-adjusted cash flow:

```
sites + grid rights + capital + assets + contracts + data + market acce
  -> decisions under constraints
  -> cash flow, risk, and option value
```

When revenue depends mostly on long-term contracted production, the organization can run as a handoff chain:

```
develop -> finance -> build -> operate -> invoice
```

When revenue depends on shape, flexibility, imbalance, tariffs, merchant tails, partner execution, and portfolio interaction, the organization needs a feedback loop:

```
market evidence
-> product and project decisions
-> contract and dispatch rights
-> partner or internal execution
-> settlement and attribution
-> updated pricing, valuation, and limits
```

The next operating-model investment should strengthen the weakest high-value part of that loop.

Definitions

Internal analytics means owned models, data, assumptions, benchmarks, and decision support for PPA pricing, BESS valuation, portfolio exposure, grid constraints, and partner challenge.

Partner governance means the data rights, benchmark methods, meeting cadence, service levels, audit rights, escalation rules, and commercial incentives used to manage BRPs, BSPs, optimizers, traders, suppliers, and software vendors.

Execution desk means in-house authority and capability to trade, nominate, dispatch, manage collateral, monitor positions, run limits, and handle operational incidents across relevant hours.

ETRM means energy trading and risk management infrastructure: deal capture, position records, risk reporting, workflow, access controls, settlement, and audit trail.

Capability sequence means the order in which the firm builds, buys, partners, or postpones capabilities so that later complexity rests on working data, controls, and decision rights.

Evidence Boundary

A 2025 Pexapark practitioner playbook describes IPPs moving toward active revenue management, with capability needs in structuring, pricing, short-term bidding, dispatch, portfolio management, risk, analytics, contract management, settlement, and data. It also reports practitioner scale color around in-house short-term or physical-dispatch capability at roughly `0.5–1 GW` for some IPPs and `1–2 GW` as a critical operating scale.

Use those numbers as market-color framing, not thresholds. A smaller portfolio with complex BESS, structured PPAs, and retained imbalance exposure may need serious analytics early. A larger portfolio with simple wrapped contracts may rationally outsource more.

Energy trading risk-management sources add a durable warning: active commercial surfaces need reliable records of deals, positions, limits, settlement, workflow, audit trail, and physical constraints. The exact software stack changes over time, but unauditable handoffs create governance risk.

Numerical Intuition

Rank capabilities by addressable value rather than by perceived sophistication.

```
capability priority
  = value at stake x controllability x observability x urgency
    / cost-and-risk
```

Value at stake is the plausible annual EUR affected by better decisions.

Controllability asks whether the IPP can actually change the action.

Observability asks whether results can be measured and attributed.

Urgency asks whether waiting destroys option value, creates unmanaged risk, or locks in a poor contract.

Cost-and-risk includes people, systems, vendor lock-in, operational incidents, model error, compliance burden, collateral, and distraction.

Worked Estimate

Suppose `500 GWh/year` is exposed to shaped-PPA pricing and residual firming decisions. A recurring `2 EUR/MWh` error in pricing, hedge cost, or risk allocation creates:

```
gross value at stake = 500,000 MWh/year x 2 EUR/MWh
                     = 1,000,000 EUR/year
```

If the firm can influence `60%` of that exposure and observe enough to improve half of the decision quality:

```
addressable value = 1,000,000 x 60% x 50%
                  = 300,000 EUR/year
```

Now compare a partner-governance capability. Suppose `300 GWh/year` is nominated, dispatched, or optimized by a partner. A realistic avoidable leakage of `0.50 EUR/MWh`, after fair constraints, creates:

```
gross leakage pool = 300,000 MWh/year x 0.50 EUR/MWh
                   = 150,000 EUR/year
```

That pool is smaller than the PPA-pricing example, but it may be easier to observe and may improve contract renegotiation, fee design, and future internalization decisions.

Choose the capability that clears its own cost and risk first while preparing the next one.

Equations

Operating-model quality:

```
decision quality
  = model quality
  x data quality
  x decision-right clarity
  x feedback quality
  x risk-control quality
```

Use this as a failure detector. If any factor is close to zero, the operating loop weakens.

Internalization test:

```
internalize when:
  retained value at stake
  x need for strategic control
  x ability to govern execution
  >
  people + systems + liquidity + collateral + operational + model-risk cost
```

This test often supports internal analytics before internal execution, benchmark ownership before market access, and risk ownership before trading autonomy.

The Three Main Choices

Internal analytics is the first candidate when the firm cannot price shaped products, value BESS optionality, explain portfolio exposure, or challenge partner performance from its own facts. It gives the IPP a common model language across development, commercial, finance, asset management, and risk.

Partner governance is the first candidate when the firm already relies on external execution but lacks data rights, feasible benchmarks, attribution, escalation triggers, or contract incentives. It reduces blind dependence while preserving the benefits of outsourced market access.

An execution desk is the first candidate only when the retained value pool is large, liquidity is deep enough, controls are mature, systems can support positions and settlement, and management is willing to own operational risk. A desk without reliable positions, limits, collateral process, and settlement evidence can create more risk than value.

The practical sequence may be:

Binding constraint	First build	Next option
Product mispricing	PPA shape and residual-firming analytics	portfolio hedge and risk review
Unclear partner performance	feasible benchmark and data rights	incentive redesign or partial internalization
Settlement and position confusion	deal/position records and controls	ETRM or workflow upgrade
High-value intraday opportunity	24/7 partner contract with benchmark	internal desk after controls mature
Complex BESS portfolio	dispatch, degradation, and grid-right model	direct optimizer governance or execution

Mechanism

The operating model must connect four decisions that otherwise drift apart.

Product decisions translate buyer needs into shape, hedge, evidence, credit, and residual-risk promises.

Asset decisions translate sites, grid constraints, tariffs, BESS duration, degradation, and dispatch rights into feasible physical options.

Portfolio decisions translate a single project into zone exposure, shape exposure, hedge position, merchant tail, liquidity need, and optionality.

Governance decisions translate forecasts, bids, schedules, state of charge, imbalances, settlements, and partner actions into attribution and learning.

The energy-management function exists to keep those decisions using the same facts.

Year-One Sequence

First, write the decision inventory. It should name the recurring choices that move value: shaped-PPA quotes, BESS sizing, grid-right assumptions, dispatch-right allocation, partner instructions, imbalance exposure, hedges, settlement checks, and model assumption changes. For each decision, name the owner, data source, model, approval limit, partner dependency, and feedback timing.

Second, build the analytical layer. Start with an asset-and-contract table, production and price data, curve assumptions, residual-firming model, BESS dispatch constraints, partner data feed, and model inventory. The operating gain is that PPA, BESS, partner, and hedge decisions use the same version of reality.

Third, harden partner governance. Require time-stamped forecasts, nominations, trades, bids, state-of-charge traces, constraint flags, settlement data, and explanations for material gaps versus a feasible benchmark. Add escalation triggers and rights to review the model or process when gaps repeat.

Fourth, decide whether execution should remain outsourced, become hybrid, or move partly inside. This decision should be made only after the firm can reconstruct positions, run limits, explain settlement, and measure partner performance.

Example

A buyer wants a shaped renewable product: flatter than solar, cleaner than generic market power, and priced for several years.

Sales asks whether the product can be sold.

Development asks whether the project can support it.

Trading or the partner asks who manages residual shape and imbalance.

Finance asks whether credit, collateral, and merchant-tail exposure are acceptable.

Risk asks which stress case breaks the margin.

The operating model turns those questions into one decision:

```
What product can the portfolio sell,
at what price,
with which residual exposures,
under whose control,
with what partner evidence,
and with what feedback after delivery?
```

If the IPP cannot answer that question, it should build the answer before expanding the product promise.

In Practice

The first control pack can stay small:

- position and exposure register
- PPA quote log with residual-firming and credit assumptions
- BESS dispatch, state-of-charge, degradation, and warranty view
- partner performance attribution against feasible benchmarks
- liquidity, collateral, and settlement exceptions view

- model and assumption change log

Each item should have an owner, cadence, data source, and escalation trigger. The purpose is commercial control rather than paperwork.

Misconceptions

“Operating model means org chart.” The org chart is a consequence. The operating model is the decision system underneath it.

“Build a trading desk to become sophisticated.” Sophistication starts with knowing retained risk, feasible benchmarks, data rights, limits, and settlement evidence.

“A battery makes the company active.” A battery creates choices. The company becomes active when it can price, govern, dispatch, and learn from those choices.

“Data infrastructure is back-office plumbing.” In active energy management, data quality is commercial control.

“Scale decides everything.” Scale matters, while retained risk, complexity, control rights, and observability can move earlier than headline MW.

Recap

The next-generation IPP operating model is a constrained sequence of capability investments. Internal analytics, partner governance, market systems, and an execution desk should compete for capital against addressable value at stake, controllability, observability, cost, operational risk, and timing. The strongest first build is usually the one that improves many recurring decisions and creates evidence for the next build/buy/partner choice.

Chapter 12. Decision Memos For Power-Market Strategy

References: [13] [3] [4] [6] [11] [38] [50] [43] [62]

A customer asks for clean power in a shape the solar site does not naturally produce. The battery concept can cover part of the shape, the trading partner can execute part of the market action, finance needs bankable cash flow, and strategy needs to know whether the offer can become a repeatable portfolio product. Each team may be correct within its own remit while the combined recommendation fails: the MWh may not cover the promise, the contract may leave residual exposure with the seller, the partner may lack auditable control, or the model may miss the first binding constraint. The memo has to combine physics, market action, contract risk, portfolio effect, and governance in one decision.

Opening Puzzle

An investment committee receives a one-page recommendation: approve a shaped clean-power offer for a large customer. The packet contains a solar site, a 50 MW / 100 MWh battery concept, an optimizer, and a target price. Each ingredient is plausible on its own.

The committee still lacks the decision. Which constraint binds first: physical coverage, stress-hour firming, grid access, dispatch control, credit support, or partner evidence? Which assumption would change the recommendation?

First-Principles Model

Every power-market decision in this course has seven layers:

```
physics
-> grid and connection
-> institutions and obligations
-> market sequence
-> contract transfer
-> asset and portfolio optionality
-> organization and governance
```

The layers are practical failure modes.

Physics fails when MW and MWh are confused, or when an energy-limited asset is treated as firm capacity.

Grid fails when injection, withdrawal, curtailment, connection rights, or tariffs bind.

Institutions fail when no one knows who is BRP, BSP, optimizer, market participant, contract owner, or decision owner.

Markets fail when the plan assumes liquidity, products, gate timing, or settlement rules that do not exist.

Contracts fail when an apparently simple price leaves shape, volume, balancing, credit, or merchant-tail exposure unassigned.

Assets and portfolios fail when a project model ignores correlation, cannibalization, optionality, degradation, or substitute flexibility.

Organizations fail when the model says “act” but the company has no data, right, limit, partner evidence, or owner.

Definitions

A binding constraint is the first limit that prevents the preferred action from being feasible or economic.

Residual exposure is the part of an obligation not covered by physical output, storage, hedge, contract transfer, or portfolio offset.

Coverage ratio is feasible deliverable energy divided by promised shape energy for the relevant time block.

Assumption-flip test asks: what is the smallest plausible assumption change that reverses the recommendation?

Decision memo is a short recommendation that states the decision, evidence, retained risk, owner, next measurement, and assumption-flip test.

Numerical Intuition

The Capstone Case

Use a deliberately simple shaped-PPA case.

The customer asks for 30 MW from 18:00 to 22:00 every day. The hourly obligation is:

```
promised shape = 30 MW x 4 h = 120 MWh/day
```

The project team proposes a battery with 100 MWh nameplate energy. Suppose the operating plan leaves `80 MWh` of deliverable evening energy after state-of-charge policy, round-trip and auxiliary losses, degradation margin, availability, reserve commitments, and grid export limits:

```
deliverable battery energy = 80 MWh/day
```

Before market purchases, portfolio offsets, or contract flexibility, the daily residual exposure is:

```
residual exposure = 120 MWh − 80 MWh = 40 MWh/day
```

The first conclusion may be “the battery is too small.” The decision conclusion is more specific:

```
The proposed asset does not physically cover the promised shape by itse
The product can still be viable only if residual exposure is priced, he
contractually transferred, or covered by portfolio/market actions under
```

The memo then prices or assigns the residual exposure.

The Diagnostic Ratios

Diagnostic ratios:

```
coverage ratio
  = feasible deliverable MWh / promised shape MWh

residual exposure
  = promised shape − physical hedge − contractual hedge − portfolio hedge

control ratio
  = exposure under reliable decision control / total retained exposure

observability ratio
  = benchmarkable decisions / economically material decisions

timing ratio
  = decision latency / action window
```

If the coverage ratio is low, the product is not physically hedged. If the control ratio is low, the firm may have sold a product it cannot govern. If observability is low, partner performance will be argued rather than measured. If timing is poor, a forecast may arrive after the useful decision has passed.

The ratios point to the first place the recommendation can break.

The Decision Memo

Here is the memo structure.

Decision: Should the firm offer the 30 MW evening shaped product from the solar-plus-BESS site under the proposed structure?

Recommendation: Do not sell it as a physically covered product from this site alone. Offer it only as a shaped portfolio product with explicit residual-exposure pricing, defined market-purchase rights, partner data/control requirements, and a downside case that survives the residual 40 MWh/day exposure.

Reason: The proposed battery concept covers only part of the evening obligation after reserving energy for uncertainty and operational risk. The remaining exposure must be handled by market purchases, portfolio offset, contract flexibility, larger storage, or reduced product firmness. Treating the product as “solar plus battery firming” would hide the retained risk.

Required conditions:

- The PPA must state who bears shape, imbalance, and residual market-purchase risk.
- The optimizer or partner must provide time-stamped forecasts, schedules, dispatch actions, constraints, and settlement data.
- The project model must price the residual exposure under stress alongside average spreads.
- The battery dispatch policy must state when it serves the customer shape, arbitrage, reserves, or imbalance control.
- The portfolio model must show whether other assets genuinely offset the exposure or merely share the same weather and price regime.
- The risk review must define the limit that stops the product from being repeated if realized leakage exceeds the expected margin.

Next measurement: After the first operating period, compare actual actions with a feasible forecast-based benchmark. Attribute deviations to weather, price, liquidity, grid constraints, SoC limits, partner choices, contract design, or model error.

Binding-Constraint Table

The memo should include a binding-constraint table before the recommendation is debated.

Layer	Candidate binding constraint	Evidence needed	Decision consequence
Physics	battery usable MWh below promised shape	tested SoC policy, degradation margin, availability	reduce firmness, increase storage, or price residual exposure
Grid	import/export or curtailment limit	connection agreement, tariff,	change site design or reject site-specific promise

Layer	Candidate binding constraint	Evidence needed	Decision consequence
		congestion/curtailment history	
Market	residual evening purchases too expensive or illiquid	forward/intraday liquidity, stress prices, hedge availability	change price, hedge, tenor, or product shape
Contract	PPA language leaves shape/imbalance risk with seller	term sheet risk allocation	reprice, amend terms, or walk away
Asset	BESS needed for competing revenue stack	dispatch opportunity-cost model	reserve capacity, split use, or change product
Portfolio	residual exposure is correlated with existing book	exposure cube and stress test	require portfolio hedge or concentration limit
Partner	optimizer lacks data/control/audit rights	service-level agreement (SLA), data fields, benchmark protocol	add data rights or do not rely on outsourced execution
Organization	no owner for risk limits and evidence	decision inventory and governance calendar	delay product until owner/control exists

A missing right to act can be the first binding constraint, even when it is not the largest visible EUR exposure.

Evidence Request List

The decision meeting should produce a short evidence list:

- physical coverage: hourly generation, SoC, availability, degradation, and curtailment assumptions

- grid economics: connection rights, import/export limits, tariff structure, and loss treatment
- market evidence: hedge availability, residual purchase liquidity, stress prices, balancing exposure
- contract evidence: who owns shape, volume, imbalance, curtailment, credit, and termination risk
- portfolio evidence: marginal exposure, concentration, hedge fit, and substitute flexibility
- partner evidence: forecasts, bids, schedules, dispatch actions, constraints, settlement line items, benchmark rights
- finance evidence: lender treatment of contracted revenue, merchant tail, optimizer value, and downside cash flow
- governance evidence: owner, limit, escalation, model version, and post-delivery attribution

This list is concrete because an evidence request has to name the missing data, rule, right, or assumption.

Retained-Risk Table

Even an approved memo must say what risk remains.

Retained risk	Who owns it	First warning signal	Mitigation
residual shape cost	seller or portfolio desk	evening market purchases exceed risk margin	reprice, hedge, soften shape, add flexibility
battery opportunity cost	asset owner	reserve/arbitrage value exceeds PPA-use value	dispatch policy and product repricing
imbalance and forecast error	depends on BRP/PPA terms	systematic deviations from benchmark	better forecast, data rights, risk-sharing terms
basis risk	seller/hedging function	area/system spread widens under stress	basis hedge, zone limit, risk margin

Retained risk	Who owns it	First warning signal	Mitigation
liquidity/collateral	treasury/risk	collateral forecast breaches buffer	liquidity line, smaller hedge, different contract
partner execution	commercial/risk owner	unexplained benchmark gap persists	SLA escalation, fee change, partner change
model error	model owner	realized drivers outside model range	model review and assumption update

The table makes the recommendation easier to test.

Assumption-Flip Test

The recommendation changes if one of these assumptions changes enough:

```
Assumption                         If it changes...
---------------------------------  ------------------------------------
Usable battery energy              More usable MWh can physically cover mor
Evening market-purchase cost        Lower or hedgeable cost may make residu
Customer flexibility                A softer shape may reduce the firm obli
Portfolio offset                    Diversified assets may cover the residu
Control rights                      Better rights may make intraday or BESS
Partner data and auditability       Better observability may make outsource
Tariff or grid constraints          A binding import/export rule may destro
Financing treatment                 Lenders may discount merchant or optimi
```

The assumption-flip test tells management what to investigate next.

Equations

The basic equations are simple; the boundary conditions carry the decision.

```
promised shape MWh = contracted MW x delivery hours
```

```
coverage ratio = feasible deliverable MWh / promised shape MWh
```

```
residual exposure
  = promised shape
  − physical asset hedge
  − contracted hedge
  − portfolio hedge
```

```
expected product margin
  = customer price
  − raw generation value
  − shaping cost
  − balancing and imbalance cost
  − degradation cost
  − grid and tariff cost
  − partner and transaction cost
  − risk margin
```

```
decision value
  = value(recommendation)
  − value(next best feasible alternative)
  − implementation cost
  − added risk
```

The calculation depends on specifying every term.

Mechanism

The course began with power and energy because dimensional mistakes create wrong commercial promises. It then moved to markets because energy is traded through time: day-ahead planning, intraday repair, balancing correction, and imbalance settlement. It moved to actors because obligations need owners. It moved to contracts because risk is transferred through words as well as wires. It moved to storage because flexibility has MW, MWh, efficiency, degradation, and opportunity cost. It moved to portfolios because one asset changes the value of another. It moved to organization because models need owners, limits, data, and decision rights.

The capstone decision combines those layers in one recommendation.

A shaped product promises delivery in time. A battery is an option with competing uses. A trading partner creates decision-right and evidence arrangements. A portfolio has a correlation structure. A model becomes useful when it becomes a governed action pathway.

The result is a compact decision method.

In Practice

When facing a new investment or product decision, write the first page before opening the model.

```
1. What decision is being made?
2. What physical constraint binds first?
3. Who owns each obligation?
4. What market action is feasible before the deadline?
5. What does the contract transfer, and what remains?
6. What optionality are we spending or preserving?
7. What is the portfolio effect?
8. What data and control rights prove performance afterward?
9. What assumption would flip the recommendation?
```

Then build the model to answer those questions. Do not let the model decide which question was important.

Misconceptions

“Expertise is knowing the answer.” In practice, expertise often means knowing which assumption must be checked before the answer is reliable.

“The battery solves shape.” It may help within its power, energy, efficiency, degradation, grid, market-timing, and contract limits.

“The partner handles optimization.” The partner may execute. The asset owner still needs to understand retained exposure, benchmarks, data rights, and risk appetite.

“The contract fixes the economics.” The contract allocates risk. Some risks may simply move to a place where they are harder to see.

“The model is the strategy.” The model is a reasoning instrument. Strategy is the decision about what to build, sell, hedge, outsource, govern, and measure.

Recap

Power-market judgment is the ability to translate a commercial promise into the constraints that can make or break it. In the shaped-product case, the memo has to show whether the promised shape is physically covered, where the residual exposure sits, who can act before the market window closes, and what evidence will prove performance afterward. The memo names the decision, the binding constraint, the retained risk, the owner, the next measurement, and the assumption that would reverse the recommendation.

Chapter 13. Resource Adequacy, Missing Money, And Capacity Markets

References: [23] [52] [46] [38]

A renewable IPP is offered three ways to respond to scarcity: build a four-hour battery, invest in longer-duration storage, or buy a firming contract that covers the hard hours. The annual energy forecast is insufficient on its own. The decision depends on the scarcity-duration distribution, the derating curve, bankable capacity revenue, performance penalties, and the price of contracting around the exposure. Adequacy becomes a capital-allocation problem when the firm must choose which scarce reliability attribute it wants on its balance sheet.

Opening Puzzle

A portfolio can produce plenty of clean MWh over the year and still fail a customer or capacity obligation during a cold, dark, low-wind week.

The decision question is:

```
Should the IPP build BESS, build longer-duration storage, buy a contract, o
```

The answer turns on the length and frequency of scarcity events, how the market derates each resource, and whether revenue for dependable capacity is bankable enough to finance the asset.

First-Principles Model

Adequacy is about dependable MW in stressed periods. For an IPP, the strategic object is the scarcity exposure:

```
scarcity exposure
  = promised MW during stress
  - dependable owned MW during stress
  - contracted firming MW
```

The capital-allocation test is:

```
risk-adjusted adequacy value
  = energy and flexibility value
  + bankable capacity revenue
  + contract firming value
  - capex and fixed opex
  - degradation and performance cost
  - penalty, rule, and liquidity risk
```

The scarce attribute may be one hour of fast capacity, four hours of evening coverage, overnight duration, multi-day resilience, fuel-backed flexibility, interconnector access, or demand response. The resource should be matched to the scarcity-duration distribution rather than to a generic capacity label.

Definitions

Scarcity-duration distribution shows how often shortage or obligation events last for different lengths, such as 1 hour, 4 hours, overnight, multi-day, or seasonal.

Derating curve shows how many dependable MW a resource receives as event duration increases. A storage asset’s dependable MW falls when the event lasts longer than its usable MWh can support.

Capacity credit is dependable capacity divided by nameplate capacity under a defined reliability method.

Bankable capacity revenue is revenue that lenders and investment committees can reasonably underwrite because qualification rules, payment terms, penalties, counterparty, and duration are clear enough.

Missing money is the concern that energy and ancillary-service revenues may fail to finance needed capacity because scarcity prices are capped, politically constrained, rare, risky, or incomplete.

Capacity remuneration mechanism is a market design that pays for availability, qualified capacity, or reliability contribution in addition to energy.

Numerical Intuition

Use a derating curve before comparing technologies.

Take a 100 MW / 200 MWh battery and assume all 200 MWh are usable for this simple test:

```
dependable MW over event = min(100 MW, 200 MWh / event duration)
```

Stress duration	Dependable MW	Derated share of 100 MW
1 hour	100 MW	100%
2 hours	100 MW	100%
4 hours	50 MW	50%
8 hours	25 MW	25%
24 hours	8.3 MW	8.3%

Now compare an 8-hour storage design, 100 MW / 800 MWh:

Stress duration	Dependable MW	Derated share of 100 MW
1 hour	100 MW	100%
4 hours	100 MW	100%
8 hours	100 MW	100%
24 hours	33.3 MW	33.3%

The second design is more useful for longer events, but it also costs more and may cycle less often. The question is whether the market or contract pays for that extra duration.

Worked Estimate

Suppose a planning study or contract stress test says scarcity events are distributed as:

```
60% of stressed hours in events up to 2 hours
30% in events around 6 hours
10% in events around 24 hours
```

Using the simplified derating logic:

```
2-hour battery weighted dependable MW
  = 100 x 60%
  + (200 MWh / 6 h) x 30%
  + (200 MWh / 24 h) x 10%
  = 60 + 10 + 0.8
  = 70.8 MW
```

This toy stress-coverage score is only a teaching test. A financeable adequacy value needs chronological stress events, state of charge at event start, recharge assumptions, availability, correlated weather and outages, deliverability, the market’s actual derating rule, and formal metrics such as effective load-carrying capability (ELCC) or loss-of-load expectation (LOLE) where relevant.

For the 8-hour design:

```
8-hour storage weighted dependable MW
  = 100 x 60%
  + 100 x 30%
  + (800 MWh / 24 h) x 10%
  = 60 + 30 + 3.3
  = 93.3 MW
```

This calculation defines the next question before the investment choice is made:

```
extra bankable revenue from longer duration
  >= extra capex + fixed opex + degradation + technology + performance
```

If capacity revenue pays only nameplate MW or short-duration performance, the longer asset may not recover its cost. If the product derates short duration sharply during multi-hour stress and pays a bankable premium for longer coverage, the decision can flip.

Equations

Capacity revenue:

```
capacity revenue
  = qualified dependable MW
  x capacity price
  x availability factor
  − expected penalties
```

Bankable capacity revenue:

```
bankable revenue
  = capacity revenue
  x rule confidence
  x counterparty confidence
  x tenor confidence
```

Build versus buy:

```
build if:
  owned−resource value
  − owned−resource cost and risk
  >
  firming contract cost
  − avoided risk
```

Avoid:

```
avoid if:
  premium for selling the obligation
```

```
<
expected firming cost + penalty risk + management cost
```

Mechanism

In an energy-only design, scarcity prices are supposed to rise during tight conditions and pay resources that are available when they are needed. Kirschen and Strbac explain the investment logic: rare price spikes can provide essential revenue for capacity, while also creating volatility and consumer-risk concerns.

Real markets add frictions. Demand may be price-insensitive. Price caps may limit scarcity rents. Market-power mitigation may suppress bids. Investors may distrust rare revenue events. Policy can change over an asset life. Creti and Fontini frame the missing-money problem around these market failures and show why capacity remuneration can theoretically replace some missing scarcity revenue.

For an IPP, the mechanism becomes practical: identify the scarcity duration, estimate the derating curve, check the revenue source, then decide whether the firm should own the asset, buy the obligation from someone else, or avoid selling the obligation.

Decision Frame

Build short-duration BESS when scarcity events are short enough, the battery also earns energy/reserve/imbalance value, grid rights allow the dispatch, and capacity accreditation gives meaningful qualified MW.

Build longer-duration storage when the scarcity-duration distribution has enough long events, capacity or contract rules reward duration, technology and performance risk are manageable, and the asset adds portfolio value beyond the capacity payment.

Buy a firming or capacity contract when the obligation is valuable to sell but owned resources would be poorly matched, under-accredited, too expensive, or too risky. The counterparty, tenor, collateral, and performance terms then become part of the product cost.

Avoid the obligation when scarcity duration is uncertain, derating rules are unfavorable, capacity revenue is unbankable, penalties are asymmetric, or the buyer premium does not cover firming cost.

Example

A data-center buyer wants a clean product that is firm during winter evening stress. The IPP owns solar, a four-hour BESS, and some market hedges.

The first model should begin with stress coverage rather than annual clean MWh:

- how many stress events last 1-4 hours, 4-12 hours, and multi-day
- how much dependable MW the BESS receives under the relevant derating method
- whether the battery will be full when scarcity begins
- whether grid export rights bind during the event
- whether capacity revenue or buyer premium is bankable
- what a third-party firming contract would cost
- what penalty applies if the product fails during scarcity

The answer may support a four-hour battery, longer-duration storage, a contract with a firm supplier, a narrower PPA shape, or no bid.

Evidence That Changes The Decision

The build/buy/avoid answer should change when the evidence changes.

Build BESS becomes stronger if stress events are short, capacity accreditation rewards short-duration storage, intraday/reserve revenue remains material, grid rights are firm, and degradation cost is manageable.

Build longer-duration storage becomes stronger if planning studies show frequent overnight or multi-day scarcity, derating rules pay for duration, capacity or customer contracts provide revenue floors, and technology cost/risk improves.

Buy contracts becomes stronger if scarce hours are rare but severe, counterparties can supply firming more cheaply, collateral is manageable, and the IPP can preserve upside without owning an ill-matched asset.

Avoid becomes stronger if revenue depends on rare uncapped price spikes, capacity rules are unstable, performance penalties are large, or the buyer will not pay the firming premium.

In Practice

An adequacy-aware investment memo should include:

- scarcity-duration distribution by market, zone, or customer obligation
- derating curve for each candidate resource
- bankable revenue stack: energy, reserves, capacity, contract premium, and merchant tail
- penalty and performance regime
- grid rights and fuel or state-of-charge assumptions during stress
- cost of third-party firming
- sensitivity that flips BESS, longer-duration storage, contract, and avoid decisions

That memo is more useful than a generic adequacy primer because it turns reliability language into capital allocation.

Misconceptions

"Enough annual MWh means enough reliability." Reliability can fail in specific hours even when annual energy is abundant.

"Nameplate MW equals dependable MW." Dependable MW depends on availability, weather correlation, fuel, state of charge, grid access, and duration.

"Capacity revenue is automatically financeable." Financeability depends on qualification rules, tenor, counterparty, penalties, and rule-change risk.

"Long duration always wins adequacy." Long duration wins only when the scarcity-duration distribution and payment rules reward the extra MWh.

"Batteries solve adequacy." Batteries solve the adequacy events they are sized, charged, qualified, and paid to cover.

Current Debates And Caveats

Capacity-market and missing-money debates remain design-specific. Kirschen and Strbac support the general logic that scarcity prices can fund capacity but create volatile revenues and consumer concerns. Creti and Fontini support the logic that price caps and other market failures can create missing money and motivate explicit capacity remuneration. Neither source provides a current country-specific rule for a specific storage investment.

Capacity accreditation, derating, penalty, and capacity-price rules are market-specific inputs. For storage, the bankable number depends on rule date, duration, state-of-charge assumptions, performance testing, and the penalty design attached to non-delivery.

Recap

Adequacy becomes an IPP strategy question when scarcity obligations, capacity revenue, and customer firming promises enter the investment case. The core artifact is a scarcity-duration distribution paired with a derating curve and bankable revenue test. That evidence decides whether the firm should build short-duration BESS, build longer-duration storage, buy firming, narrow the product, or avoid the exposure.

Chapter 14. Market Power, Scarcity Pricing, And Monitoring

References: [52] [23] [53]

A high-price hour needs an evidence pack. The system may have been genuinely short of energy, capacity, ramping, reserves, or network access. It may also have become locally dependent on one owner because a bottleneck removed alternatives. The evidence pack should show what was scarce, which suppliers were feasible, who was pivotal, whether the bid or outage had an economic explanation, and how the result changed the participant's wider portfolio.

Opening Puzzle

A constrained area has `1,200 MW` of load in a winter evening hour.

Feasible imports are `500 MW`. Other local suppliers can provide `450 MW`. One supplier controls `300 MW` of remaining local capacity.

The price clears very high.

First test whether the area could meet load without that supplier:

```
residual demand facing supplier = 1,200 - 500 - 450 = 250 MW
```

At least `250 MW` of the supplier's `300 MW` is needed. In that hour and area, the supplier is pivotal. Pivotality by itself is evidence to investigate, not evidence of abuse.

First-Principles Model

Scarcity pricing and market power sit next to each other.

Scarcity pricing means a real constraint has made one more MWh, MW, ramp, reserve, or network path valuable.

Market power means a participant can profitably move price away from the competitive level by changing its own behavior.

Electricity makes the distinction hard because:

- demand may not respond inside the delivery interval
- storage is limited and state-dependent
- supply has ramp, availability, fuel, and start constraints
- transmission constraints can shrink the relevant market
- portfolio owners can gain on inframarginal output when price rises

The disciplined question is:

```
Was the high price explained by real scarcity, by strategic behavior, o
```

Definitions

- **Market power:** ability to profitably influence price away from the competitive level.
- **Structural market power:** market position created by ownership, concentration, location, scarcity, or network constraints.
- **Behavioral market power:** actions such as physical withholding, economic withholding, scheduling choices, outage timing, or dispatch choices.
- **Physical withholding:** making capacity unavailable when it could physically produce.
- **Economic withholding:** offering available capacity at a price high enough to avoid clearing unless scarcity, opportunity cost, risk, or rules justify it.
- **Residual demand:** demand left for one supplier after other feasible supply and imports are removed.
- **Pivotal supplier:** supplier whose capacity is needed to meet demand under the relevant constraints.
- **Residual Supply Index:** a test comparing supply available without the supplier to demand. Values below 1 indicate the supplier is pivotal in the simplified formulation used here.
- **Scarcity pricing:** high price caused by genuine scarcity of energy, capacity, flexibility, reserves, or network capability.

Numerical Intuition

Use three tests. None proves misconduct by itself; together they organize the investigation.

1. Residual Demand

```
residual demand = load − feasible imports − other available supply
```

For the opening hour:

```
1,200 MW − 500 MW − 450 MW = 250 MW
```

The supplier is needed for at least `250 MW`.

2. Residual Supply Index (RSI)

```
RSI = (total available supply − supplier available supply) / load
```

Suppose total available supply is:

```
500 MW imports + 450 MW other local + 300 MW supplier = 1,250 MW
```

Then:

```
RSI = (1,250 − 300) / 1,200 = 0.79
```

In this illustrative hour, supply without the supplier covers only `79%` of load. The result is a pivotal-supplier warning.

3. Portfolio Uplift

If withholding `20 MWh` raises the price by `35 EUR/MWh` on `180 MWh` of the owner's remaining generation, the gross portfolio uplift is:

```
35 EUR/MWh x 180 MWh = EUR 6,300
```

If the withheld unit would have earned `90 EUR/MWh` of margin on the `20 MWh`, the lost margin is:

```
90 EUR/MWh x 20 MWh = EUR 1,800
```

Illustrative incentive before rules, penalties, reputation, uncertainty, and opportunity cost:

```
6,300 - 1,800 = EUR 4,500
```

This portfolio uplift is why market monitors care about portfolio ownership as well as the marginal unit.

Evidence Pack For A High-Price Hour

The investigation should move in this order.

Evidence layer	Question	Typical evidence
Physical scarcity	What constraint bound?	Load, generation, reserves, interconnector capacity, outage and derating data, weather, fuel or water constraints.
Market structure	Who had feasible alternatives?	Available capacity by owner, import limits, storage SoC, demand response, reserve commitments.
Pivotality	Could demand be met without one owner?	Residual demand, RSI, local constraint model, sensitivity

Evidence layer	Question	Typical evidence
		to one asset or interconnector.
Bid and outage rationale	Was behavior explainable?	Bid logs, cost and opportunity-cost model, start constraints, fuel limits, outage records, dispatch instructions.
Portfolio economics	Who gained and where?	Remaining generation, hedge position, balancing exposure, congestion exposure, customer contracts.
Counterfactual	Would price remain high under competitive behavior?	Re-run with cost-based offers, feasible imports, storage and demand response, and actual constraints.
Rule boundary	Did the behavior violate rules or obligations?	Applicable market rules, Regulation on Wholesale Energy Market Integrity and Transparency (REMIT)-style reporting and monitoring evidence, TSO instructions, internal approvals.

Creti and Fontini distinguish structural market power from behavioral market power and discuss physical withholding, economic withholding, the Herfindahl-Hirschman Index (HHI), Lerner-style margins, Pivotal Supplier Index, and Residual Supply Index. Kirschen and Strbac emphasize why high demand and inelastic short-run conditions make electricity vulnerable to market power and why ex ante mitigation and ex post monitoring both exist. Those sources support the evidence structure for analyzing a market participant; a named-market claim still needs the participant, period, data, rule boundary, and conduct evidence.

Mechanism

Start with a normal hour. Many suppliers can meet demand, imports are available, and storage or demand response can substitute at the margin. No single participant controls the last feasible MWh.

Now add a constraint. Load rises, a line binds, storage is empty, an interconnector is derated, a generator is unavailable, or reserves are already committed. The relevant market becomes smaller. Residual demand facing the remaining supplier rises.

If the supplier is pivotal, high prices may still be legitimate. Scarcity can be real. The marginal unit may have high opportunity cost, start cost, fuel risk, reserve obligation, or reliability risk. The same facts can also create an incentive to withhold. The evidence pack separates those explanations instead of letting price alone decide.

Broad concentration metrics are weak on their own. HHI can describe ownership concentration, but electricity market power is often local and interval-specific. Pivotality and residual supply tests are closer to the operational question: under the actual constraint, who was needed?

Example

An owner has two units behind a bottleneck.

- Unit A: already running, low marginal cost, `180 MWh` in the hour.
- Unit B: expensive peaker, `20 MWh` available.
- Constraint: imports are maxed out.
- Price effect if Unit B does not clear: `+35 EUR/MWh`.

If Unit B is offered high and does not clear, Unit A may earn more on all `180 MWh`. A monitor then asks:

- Was Unit B physically available?
- Was the high offer tied to real cost, risk, or opportunity value?
- Did the owner have reserve obligations or fuel limits?
- Did the owner benefit on Unit A, hedges, or other portfolio positions?
- Would a cost-based counterfactual still produce scarcity pricing?
- Were market rules, disclosure obligations, and internal approvals followed?

Now change one fact. Suppose demand response or batteries can add `300 MW` in the same hour. The supplier is no longer pivotal. The same high bid has a different probability of moving price. Flexibility can reduce local market-power exposure by creating alternatives at the margin.

Decision Frame

For a renewable IPP or trading oversight function, the practical task is to govern one's own exposure and conduct.

Decision	What to do
Asset siting	Check whether the site sits behind constraints where scarcity and market-power concerns will recur.

Decision	What to do
Contracting	Avoid writing PPA or hedge terms that ignore local scarcity, basis, or curtailment exposure.
BESS strategy	Value flexibility partly by the alternatives it creates in tight constrained hours.
Trading governance	Keep bid rationale, outage records, dispatch logs, risk approvals, and opportunity-cost logic readable after the fact.
Partner oversight	Require BRP/BSP evidence that distinguishes market conditions from execution choices.
Compliance culture	Require an evidence trail for high-price hours.

The practical test is simple: could the organization explain the hour later using data available at the time?

Misconceptions

“High prices prove market power.” High prices can be the correct scarcity signal.

“Market power requires a huge company.” A small participant can be pivotal behind a local bottleneck.

“Economic withholding means every high bid is bad.” A high bid can reflect real opportunity cost, fuel, start, reliability, reserve, or risk constraints.

“HHI settles the question.” Concentration measures help, but local constraints and time-specific pivotality often matter more.

“Price caps solve the problem.” Caps can protect customers in some conditions, but they can also weaken scarcity and investment signals if used without a replacement mechanism.

Current Debates And Caveats

The design problem is the boundary between scarcity pricing, investment adequacy, and market-power mitigation. Too little scarcity pricing can underpay flexibility and capacity. Too little monitoring can let weak competition become expensive. Different markets place the burden on energy prices, balancing markets, capacity mechanisms, redispatch, transmission expansion, local flexibility, demand response, or bid mitigation.

Claims about current abuse, REMIT status, mitigation action, or named-market conduct need official current evidence.

Recap

The residual-demand test is the anchor. A high price becomes interpretable only after the analyst knows what was scarce, which alternatives were physically and contractually available, who was pivotal, and how the participant’s wider portfolio was affected. Scarcity pricing can be healthy; strategic withholding can be harmful; the same hour can contain traces of both. The discipline is to build the evidence pack before reaching for a label.

Casebook: Applied Decision Memos And Models

Case 1. Shaped Solar-Storage Power Purchase Agreement Quote

References: [6] [32] [33] [37] [39] [50]

A buyer asking for “solar plus storage” is usually asking for several things at once: price certainty, evidence of clean supply, a volume shape, and someone to carry the mismatch between production and consumption. The seller can accept that mismatch, sell a simpler product, or price the shape explicitly. The work product is a quote stack and a decision memo that make the residual obligation visible before the term sheet turns it into risk.

Decision Question

Should a renewable independent power producer (IPP) offer a shaped solar-plus-battery energy storage system (BESS) power purchase agreement (PPA), a simpler as-produced PPA, or no-bid?

Situation

A corporate buyer wants a 10-year clean-power product for a stable industrial load. The buyer prefers a flat hourly shape because its procurement team wants budget certainty and less exposure to evening prices.

The seller owns a solar project and can add a co-located battery. The commercial team asks for a quote that is attractive to the buyer, acceptable to investment committee, and explicit about who owns residual risk.

Scenario Data

These are scenario numbers for the case, not market quotes.

Item	Value
Buyer annual demand	100 GWh/year
Product requested	100 gigawatt-hours (GWh)/year flat profile, about 11.4 megawatts (MW) average
Solar project output	90 GWh/year

Item	Value
Co-located BESS	25 MW / 50 megawatt-hours (MWh)
Battery round-trip efficiency	88%
As-produced solar offer floor	45 EUR/MWh
Buyer maximum acceptable shaped price	68 EUR/MWh
Stress firming premium in hard hours	120 EUR/MWh
Residual hard-hour volume after solar+BESS	20 MWh/h
Number of hard hours in stress case	250 h/year

For the base quote, use these illustrative quote-stack inputs. They are scenario assumptions for the exercise, not market benchmarks.

Quote component	EUR/MWh
Raw as-produced clean energy	45.0
Shape and firming cost	9.0
Expected imbalance and forecast error	2.0
Grid, tariff, and settlement administration	1.5
Certificates and evidence handling	1.0
Credit and tenor allowance	1.5
Battery opportunity cost	3.0
Risk margin	2.0

Commercial Workbench Companion

Use the Commercial Workbench alongside this case to explore the same logic interactively: how much delivery is covered, where gaps remain, what indicative premium is being tested, when the battery may be worth more elsewhere, and what evidence you would want before taking the idea into a real project.

Terms To Pin Down Before Pricing

Before naming a number, the seller needs to pin down what the offer actually promises. A shaped PPA bundles clean MWh with settlement, evidence, and fallback obligations.

Term	Case Boundary	Why It Matters
Quantity	about 11.4 MW flat hourly profile, 100 GWh/year commercial target	fixes the shape being sold
Physical source	seller solar output plus co-located BESS where available	separates asset-backed delivery from market firming
Settlement	fixed shaped price against the agreed hourly profile	determines who pays when production and load differ
Residual firming	seller covers expected residuals; hard-hour tail must be capped, indexed, or passed through	prevents an uncapped short-volatility position
Imbalance	balance responsible party (BRP)-managed, with agreed data and performance reporting	connects contract economics to operational evidence
Certificates and matching evidence	defined certificate vintage, geography, delivery period, and data package	avoids selling a clean-energy claim without proof rules

Term	Case Boundary	Why It Matters
Dispatch rights	seller/optimizer retains rights needed to deliver the shape and manage the BESS	stops the same flexibility being promised elsewhere
Data and audit rights	hourly meter data, certificate evidence, residual procurement records, dispatch logs, and settlement files are deliverable	makes the product testable after delivery
Credit and collateral	buyer credit support and seller exposure limit stated in the term sheet	prevents tenor and collateral from being buried in the energy price

Tasks

1. Build a quote stack for the shaped product.
2. Identify which risks remain with the seller.
3. Decide whether to offer the shaped product, offer only as-produced energy, or no-bid.
4. State the first assumption that would change the decision.

Worked Solution

The quote stack must start with the energy the asset can naturally provide, then add the cost of reshaping that energy into the buyer's requested profile. In this scenario, the buyer asks for `100 GWh/year` flat while the solar project produces `90 GWh/year`; the annual clean-energy shortfall and any certificate-backed replacement supply have to be priced separately from hard-hour firming. The quote is usable only if the term sheet assigns the residual mismatch to a party, a price, a cap, or a reopener.

Component	EUR/MWh	Risk Owner If Included In Fixed Price
Raw as-produced clean energy	45.0	Seller
Shape and firming cost	9.0	Seller

Component	EUR/MWh	Risk Owner If Included In Fixed Price
Expected imbalance and forecast error	2.0	Seller / BRP contract
Grid, tariff, and settlement administration	1.5	Seller
Certificates and evidence handling	1.0	Seller
Credit and tenor allowance	1.5	Seller
Battery opportunity cost	3.0	Seller
Risk margin	2.0	Seller
Indicative shaped quote	**65.0**	

At 65 EUR/MWh, the product sits below the buyer’s stated 68 EUR/MWh ceiling. The tail risk still needs an explicit boundary.

Numerical Intuition

The stress firming cost is:

```
20 MWh/h x 250 h/year x 120 EUR/MWh = 600,000 EUR/year
```

The fixed risk margin in the quote is:

```
100,000 MWh/year x 2 EUR/MWh = 200,000 EUR/year
```

The quoted margin falls short of the illustrated stress case by `400,000 EUR/year`.

```
Uncovered stress cost = 600,000 - 200,000 = 400,000 EUR/year
Premium needed to absorb the full stress = 400,000 / 100,000 MWh = 4 EU
Full stress-inclusive fixed quote = 65 + 4 = 69 EUR/MWh
```

That price exceeds the buyer's `68 EUR/MWh` ceiling. The shaped product is commercial at `65 EUR/MWh` only if the term sheet caps, indexes, passes through, or reopens hard-hour firming beyond the quoted risk margin.

Decision Memo

Recommendation: offer the shaped product at 65 EUR/MWh only with a residual-firming collar.

The collar should be explicit. In this scenario the fixed price includes `200,000 EUR/year` of risk margin. The seller can cover hard-hour firming inside that budget; volumes or costs beyond it need a pass-through, index, hard cap, or reopener. If the buyer wants the seller to carry the full illustrated stress case inside the fixed price, the quote rises to about `69 EUR/MWh`, above the stated buyer ceiling.

The investment-committee test is whether the price, cap, credit terms, evidence package, data rights, and dispatch rights describe the same product. If the buyer wants the seller to carry unlimited residual firming, the quote must move up or the product must narrow.

The as-produced fallback should remain on the table at 45 EUR/MWh. It is less valuable to the buyer, but it avoids making a fixed-price promise that the asset cannot physically support under stress.

Assumption Flip

The recommendation changes if the residual hard-hour volume rises from 20 MWh/h to 35 MWh/h.

```
35 MWh/h x 250 h/year x 120 EUR/MWh = 1,050,000 EUR/year
```

At that point the seller either needs a much higher fixed price, a portfolio hedge, a third-party firming partner, a softer shape, or a no-bid decision.

Failure Mode To Avoid

Approve the fixed shaped quote because the expected quote stack is below the buyer’s ceiling, then leave hard-hour residuals to be absorbed by the portfolio without a cap, index, or reopener.

That decision leaves the portfolio carrying an unbounded residual exposure. The product promises shape, timing, settlement, and fallback procurement alongside annual clean energy.

Links Back To The Lectures

- Load Is Not Energy
- Solar Production
- PPA Risk Allocation
- PPA Structure Catalog
- PPA Pricing
- BESS Revenue Stacks
- Dispatch Optimization
- Project Valuation
- Future Of PPAs

Case 2. The 1h, 2h, Or 4h Battery Decision

References: [41] [47] [5] [38] [11] [42] [21] [39]

Battery duration is an investment claim: the next MWh of energy capacity must earn more than it costs after efficiency, degradation, tariffs, grid rights, and contract obligations. This case asks the duration question as an investment test rather than as a technology preference.

Decision Question

For a constrained solar site, should the project build a 1-hour, 2-hour, or 4-hour battery?

Situation

A solar project has secured land and a grid position. The grid connection can export the solar plant, but adding a battery energy storage system (BESS) requires choosing a battery duration before the power purchase agreement (PPA) and optimizer contracts are finalized.

The commercial team wants a simple answer. The answer depends on how scarcity, grid access, contract value, warranty limits, and insurance limits change from the first hour to later hours.

Scenario Data

These are scenario assumptions for the case.

Item	Value
Solar project size	80 megawatts peak (MWp)
Grid export limit	60 MW
Battery power rating	30 MW
Candidate durations	1h, 2h, 4h
Carrying charge used for the test	10% of incremental capex/year
Round-trip efficiency	88%

Item	Value
Degradation allowance	included in net margin
Insurance policy	scenario assumes the modeled operating plan is insurable; if a policy caps cycles or duty cycle below the modeled plan, rerun the test under that cap

Incremental Block	Incremental Energy	Incremental Capex	Annual Hurdle
First hour	30 MWh	5.0 million EUR	0.50 million EUR/year
Second hour	30 MWh	3.5 million EUR	0.35 million EUR/year
Third and fourth hours	60 MWh	7.0 million EUR	0.70 million EUR/year

Incremental Block	Usable Cycles/Year	Net Value Per MWh Moved	Extra Curtailment/Contract Value
First hour	310	55 EUR/MWh	0.05 million EUR/year
Second hour	240	45 EUR/MWh	0.15 million EUR/year
Third and fourth hours	120	25 EUR/MWh	0.07 million EUR/year

Duration Gates

The duration decision needs more than an arbitrage calculation. Each incremental hour must pass the commercial, technical, and contractual gates that make the model value collectible.

Gate	What To Check	Failure Signal
Marginal value	the next MWh of energy capacity clears its own hurdle	average spread from the first hours is applied to all hours
Cycle, warranty, and insurance budget	modeled throughput, average state of charge, C-rate or charge/discharge rate, temperature envelope, data-retention obligation, degradation allowance, warranty terms, and insurance policy are mutually compatible	optimizer earns value by consuming life, evidence, or coverage headroom not priced in the case
Grid import and export rights	connection agreement permits the charging and discharging pattern	battery cannot access the hours that justify the duration
Product eligibility	extra duration qualifies for the intended reserve, firming, or PPA product	extra MWh exists physically but earns no contracted premium
Control rights	dispatch rights remain with the party whose economics justify the investment	PPA, optimizer, and reserve obligations all assume priority
Augmentation option	site layout, transformers, inverters, and permits leave room for later expansion	management must choose all duration now under weak evidence

DNV's storage-warranty review makes the second gate concrete. A capacity warranty may care about usage/cycling degradation, calendar degradation, temperature, average state of charge, C-rate, data storage, and repair response. The case therefore treats "cycles/year" as a commercial shorthand. Before live use, replace it with the warranty-compatible operating envelope and the actual insurance-policy limits.

The product-eligibility gate should use named product requirements. In the Nordic transmission system operator (TSO) battery paper, frequency containment reserve for disturbances (FCR-D) needs at least `20 min` endurance, frequency containment reserve for normal operation (FCR-N) needs `1 h` in each direction, automatic frequency restoration reserve (aFRR) can require `1–4 h`, and manual frequency restoration reserve (mFRR) uses a different activation and endurance logic. Those examples are product checks for this case: 1h, 2h, and 4h should be matched to a specific revenue product before they are compared as asset labels.

Tasks

1. Calculate the annual value of each duration block.
2. Compare each block with its annual hurdle.
3. Recommend 1h, 2h, or 4h.
4. State what evidence would change the decision.

Worked Solution

The first hour value:

```
30 MWh x 310 cycles/year x 55 EUR/MWh + 50,000 EUR/year
= 561,500 EUR/year
```

The first hour clears its 500,000 EUR/year hurdle by about 61,500 EUR/year.

The second hour value:

```
30 MWh x 240 cycles/year x 45 EUR/MWh + 150,000 EUR/year
= 474,000 EUR/year
```

The second hour clears its 350,000 EUR/year hurdle by about 124,000 EUR/year.

The third and fourth hour value:

```
60 MWh x 120 cycles/year x 25 EUR/MWh + 70,000 EUR/year
= 250,000 EUR/year
```

The third and fourth hours miss their 700,000 EUR/year hurdle by about 450,000 EUR/year.

Numerical Intuition

The best duration is the last block whose marginal value exceeds its marginal cost.

Option	Annual Value	Annual Hurdle	Margin To Hurdle
1h	0.56 million EUR/year	0.50 million EUR/year	+0.06 million EUR/year
2h	1.04 million EUR/year	0.85 million EUR/year	+0.19 million EUR/year
4h	1.29 million EUR/year	1.55 million EUR/year	-0.26 million EUR/year

The 4-hour option creates more gross value than the 2-hour option, but it destroys value at the margin in this scenario.

Decision Memo

Recommendation: build the 2-hour battery, preserve design optionality for later augmentation, and do not commit to 4 hours until the site has stronger evidence of longer-duration scarcity or contracted value.

The binding constraint is the weak marginal value of hours three and four. The 2-hour asset captures the strongest short-duration price and curtailment value while avoiding a large capital block with too few high-margin cycles.

The investment decision should approve the 2-hour build and require the design team to preserve practical augmentation options where the cost is modest: space, cable routes, transformer headroom, inverter/control compatibility, and permitting language. Preserving those options keeps the project from paying today for duration that the evidence does not yet support.

The commercial team should ask for six pieces of evidence before revisiting 4 hours:

- a contract that pays for longer sustained delivery
- tariff or grid rules that reward longer import/export management
- historical or forecast price data showing enough multi-hour scarcity events after degradation and fees
- warranty, insurance, and augmentation terms showing that the value can be captured without consuming battery life or coverage headroom faster than priced
- battery management system data-access and storage arrangements strong enough to support warranty, availability, and safety disputes
- dispatch-control rights showing how PPA, reserve, merchant, and emergency uses are prioritized against the same state of charge

Assumption Flip

The 4-hour decision changes if the third and fourth hours can earn 70 EUR/MWh net value for 160 cycles/year.

```
60 MWh x 160 cycles/year x 70 EUR/MWh + 70,000 EUR/year
= 742,000 EUR/year
```

It can also change in the other direction if an insurance policy caps eligible operation below the modeled cycle need. In that case, replace the modeled useful cycles with the policy-compatible cycle count before comparing duration blocks. The project may still proceed, but only after the owner obtains different coverage, changes the operating plan, or prices the lost optionality.

That clears the 700,000 EUR/year hurdle. The case for 4 hours depends on the frequency and value of longer scarcity windows, and on whether the operating plan remains compatible with warranty, insurance, tariff, and control-right constraints.

Failure Mode To Avoid

Choose four hours because the incremental capex per MWh is lower than the first two hours, then apply the average short-duration spread to all four hours.

That decision uses the economics of the valuable first hours to justify weaker later hours. The third and fourth hours need their own scarcity evidence, contract value, tariff benefit, or portfolio hedge value.

Links Back To The Lectures

- Frequency And Reserves
- Battery Storage And Time Shifting
- BESS Revenue Stacks
- Battery Duration Selection
- Battery Degradation, Warranties, And Operating Constraints
- Storage Duration And Long-Duration Flexibility

Case 3. Grid-Constrained Site Assessment

References: [11] [9] [10] [28] [29] [22] [51] [30]

Solar development often begins with land and irradiation. Project value often ends with grid access, tariffs, congestion, curtailment, and what the connection agreement actually permits. The work product is a grid-aware investment assessment for three attractive sites.

Decision Question

Which site should the independent power producer (IPP) prioritize when irradiation, grid rights, tariffs, and portfolio fit point in different directions?

Situation

The development team has three candidate sites for a new solar-plus-storage project. All three can be developed, but only one can receive near-term investment resources.

The highest-yield site must still be tested against grid economics and commercial optionality.

Scenario Data

These are scenario assumptions. They are not tariff quotes or connection terms. The named grid and tariff sources support the categories to diligence: firm timing, reserved capacity, import/export rights, limited-access terms, curtailment treatment, capacity-tariff exposure, and producer-payment geography.

Metric	Site A	Site B	Site C
Solar size	100 megawatts peak (MWp)	90 MWp	110 MWp
Expected PV yield	1,060 kWh/kWp/year	1,000 kWh/kWp/year	1,030 kWh/kWp/year
Earliest firm connection	2030	2028	2028
Export limit	65 MW	80 MW	55 MW

Metric	Site A	Site B	Site C
Expected curtailment	5%	1%	8%
Grid/tariff/loss cost	8 EUR/MWh	4 EUR/MWh	11 EUR/MWh
Battery energy storage system (BESS) import allowed	conditional	yes	no
Portfolio zone fit	weak	strong	medium
Permitting complexity	medium	medium	low

Assume the merchant value before grid costs is 48 EUR/MWh for all sites. Ignore capex differences for this first-pass comparison.

Tasks

1. Calculate net annual MWh after curtailment.
2. Calculate net value per MWh after grid/tariff/loss cost.
3. Estimate annual gross margin before capex and opex.
4. Add a grid-rights and timing assessment.
5. Recommend the priority site.

Worked Solution

Annual MWh before curtailment:

Site	Calculation	Gross Output
A	100,000 kWp x 1,060 kWh/kWp	106,000 MWh/year
B	90,000 kWp x 1,000 kWh/kWp	90,000 MWh/year

Site	Calculation	Gross Output
C	110,000 kWp x 1,030 kWh/kWp	113,300 MWh/year

After curtailment:

Site	Net MWh/year
A	100,700
B	89,100
C	104,236

Net value per MWh:

Site	Value Before Grid Cost	Grid/Tariff/Loss Cost	Net Value
A	48 EUR/MWh	8 EUR/MWh	40 EUR/MWh
B	48 EUR/MWh	4 EUR/MWh	44 EUR/MWh
C	48 EUR/MWh	11 EUR/MWh	37 EUR/MWh

Annual gross margin before capex and opex:

Site	Net MWh/year	Net Value	Annual Gross Margin
A	100,700	40 EUR/MWh	4.03 million EUR/year
B	89,100	44 EUR/MWh	3.92 million EUR/year
C	104,236	37 EUR/MWh	3.86 million EUR/year

Grid Rights Assessment

Site	Invest / Hold / Drop	Binding Reason	First Evidence That Would Change The Treatment
A	Hold	2030 firm connection and conditional BESS import rights make the higher annual margin hard to use	firm 2028 connection, confirmed import rights, and a priced grid-cost/security package
B	Advance	2028 connection, low curtailment, full BESS import, and strong portfolio fit make the cash flow executable	unexpected capacity-tariff exposure, queue slippage, or weaker metering/import rights
C	Drop from near-term priority	no BESS import, high curtailment, and high grid/tariff/loss cost weaken shaped-product optionality	faster-than-assumed permitting path plus changed import/export rights

The tariff and connection question includes EUR/MWh, reserved capacity, geography, limited access, and producer payments. Nordic tariff work increasingly separates these items. A site assessment should therefore ask what capacity is being reserved, when it is firm, whether limited or interruptible access is acceptable, and whether import rights let the battery charge when the commercial case requires it.

Numerical Intuition

The top-line spread between sites is small after grid effects, while the timing penalty is large.

Timing Test	Calculation	Result	Decision Meaning
Site A annual uplift over Site B	4.03 - 3.92 million EUR/year	0.11 million EUR/year	small recurring advantage
Opportunity cost of waiting for Site A instead of advancing Site B	2 years x 3.92 million EUR/year	7.84 million EUR	near-term margin deferred
Simple catch-up period before discounting	7.84 / 0.11 years	about 71 years	Site A does not beat Site B unless the connection and import-rights assumptions change

Use this as an undiscounted comparison rather than a project net present value (NPV). It keeps a small annual yield advantage from hiding a large timing and optionality penalty.

Decision Memo

Recommendation: prioritize Site B.

Site A has the highest simple annual gross margin, but Site B has the best decision quality. It connects earlier, has lower curtailment, lower grid cost, stronger portfolio fit, and full BESS import rights. Those features make it more useful for shaped PPAs, reserve participation, and portfolio balancing.

Site A remains a development hold, not a near-term investment priority, unless firm connection timing and battery import rights change.

The first diligence request should be the grid-rights evidence pack: firm connection date, export capacity, import capacity, connection contribution, reinforcement cost, securities or deposits, queue milestones, curtailment rules, capacity-tariff exposure, producer-payment geography, limited-access terms, metering boundary, and whether BESS operation is allowed under the same connection agreement or needs a separate approval.

The second diligence request should be the development evidence pack: site access, land title or lease route, geotechnical conditions, environmental restrictions, grid interconnection feasibility study, permitting path, construction logistics, engineering, procurement, and construction (EPC) allocation, operations and maintenance (O&M) assumptions, and resource-assessment basis. The IFC solar guide treats these as core bankability inputs rather than late-stage documentation.

Site A should remain in development if the connection timing improves or if a buyer values the location enough to pay for the grid uncertainty. Site C is not a priority unless its low permitting complexity creates a materially faster path than assumed.

Assumption Flip

The recommendation changes if Site A obtains firm 2028 connection and full BESS import rights.

Then the main penalty is grid cost rather than timing and optionality. Site A's higher output may justify prioritization, especially if the portfolio already has enough exposure in Site B's zone.

Failure Mode To Avoid

Rank the sites by base-case P50 generation value, then treat connection timing, import rights, curtailment, metering boundary, and tariff exposure as diligence items.

That decision puts the grid outside the economics even though the grid determines when the MWh is deliverable, what the battery can do, and which commercial products remain feasible.

Links Back To The Lectures

Case 4. Battery Revenue Stacks Under Saturation

References: [4] [17] [18] [19] [41] [5] [42] [44] [21] [39] [48]

A battery revenue stack is a set of coupled obligations on the same MW, MWh, state of charge, grid rights, cycle budget, insurance policy, and control system. When a service saturates, test which obligations remain valuable after opportunity cost and downside are counted.

Decision Question

Should the asset owner contract part of the battery's flexibility, keep it merchant, or reserve optionality while the revenue stack evolves?

Situation

A 50 MW / 100 MWh battery can participate in energy arbitrage and reserve products. A utility offers a five-year floor contract for part of the asset. An optimizer offers full merchant dispatch with no guaranteed payment. A customer team asks whether part of the battery should be reserved for shaped PPA delivery.

The project cannot sell the same flexibility three times.

An optimizer can schedule within the conflict; it cannot remove the conflict. DNV's hybrid-storage optimizer material is useful here because it frames dispatch as a constrained problem over market products, loss trees, interconnection limits, cycling and throughput requirements, warranty strategies, and capacity maintenance. A revenue stack is credible only if the same constraint set governs every product in the stack.

Scenario Data

These are scenario assumptions.

Item	Value
Battery	50 MW / 100 MWh
Round-trip efficiency	88%

Item	Value
Expected merchant stack value	1.70 million EUR/year
Downside merchant stack value	0.70 million EUR/year
Floor contract fixed payment	0.90 million EUR/year
Expected upside retained under floor	0.30 million EUR/year
Shaped PPA flexibility premium	0.75 million EUR/year
Merchant value left after reserving PPA capacity	0.80 million EUR/year
Lender-counted share of pure merchant value	20% scenario assumption
Lender-counted share of fixed floor	100% scenario assumption
Insurance cycle limit	scenario assumes no binding limit beyond the modeled cycle budget; verify policy terms before using the stack

Commercial Workbench Companion

Use the Commercial Workbench after the worked solution in Case 1. Its BESS value view mirrors this case's main tension: whether the battery should support a shaped customer product, back a floor-like contract, or keep more merchant optionality.

Tasks

1. Compare expected value and financeable value.
2. Identify which option has the strongest downside protection.
3. State the opportunity cost of reserving capacity for a shaped PPA.
4. Recommend an allocation.

Contracting Routes

Recent BESS offtake discussions use several structures rather than one generic “battery revenue” line. The case reduces them to four routes the investment committee can compare.

Route	What Is Sold	Control-Rights Consequence	When It Fits
Full merchant optimization	optimizer pursues energy, reserve, and imbalance opportunities	asset owner keeps upside and downside; lender credit is limited	owner accepts volatility and has strong controls
Floor or revenue-share contract	floor, toll, revenue swap, or minimum guaranteed payment	some upside and dispatch discretion are transferred; availability, reporting, termination, and exclusivity terms matter	project needs financeability or downside protection
Partial hybrid PPA	defined battery flexibility supports a customer profile, residual remains merchant or contracted separately	part of the battery is reserved for the PPA product	customer premium exceeds opportunity cost and risk boundary is clear
Combined hybrid PPA	renewable energy and BESS are packaged into one customer product	larger share of flexibility is locked to the customer	buyer values firmness enough to pay for it and evidence/control rights are explicit

For Nordic reserve routes, add a product-specific appendix before approving the stack. The Nordic transmission system operators′ (TSOs′) 2025 battery paper separates fast frequency reserve (FFR), frequency containment reserve for disturbances (FCR-D), frequency containment reserve for normal operation (FCR-N), automatic frequency restoration reserve (aFRR), and manual frequency restoration reserve (mFRR) by response time, endurance, bid size, and prequalification route. It also warns that the same capacity cannot be bid in multiple markets for the same delivery period. A revenue-stack proposal should therefore show the exact delivery intervals where reserve capacity, activation energy, PPA delivery, arbitrage, and state-of-charge buffers overlap.

Worked Solution

Route	Expected Value	Downside Value	Financeable Value In This Scenario	Main Constraint
Full merchant	1.70 million EUR/year	0.70 million EUR/year	0.34 million EUR/year	Volatile and weakly financeable
Floor contract	1.20 million EUR/year	0.90 million EUR/year	0.90 million EUR/year	Gives up some upside
Shaped PPA plus residual merchant	1.55 million EUR/year	depends on residual firming	0.75 million EUR/year plus residual credit	Ties up control rights

The full merchant route has the highest expected value, but the lowest financeable value. The floor contract has lower expected value but stronger downside and lender readability. The shaped PPA route is attractive only if the PPA premium is high enough to compensate for lost merchant optionality.

Numerical Intuition

The opportunity cost of using battery capacity for the shaped PPA is:

```
full merchant expected value − residual merchant value
= 1.70 million − 0.80 million
= 0.90 million EUR/year
```

The shaped PPA pays 0.75 million EUR/year for that flexibility. On expected value alone, it is short by 0.15 million EUR/year before considering customer value, contract tenor, and financeability.

Decision Memo

Recommendation: contract a defined tranche of the asset under a floor-like structure and keep the residual slice merchant. A first term-sheet version could reserve `25 MW / 50 MWh` for the floor-backed obligation and leave the other `25 MW / 50 MWh`, plus any unused state-of-charge headroom under the priority rules, available for merchant optimization. The exact split can move, but the tranche must state MW, MWh, state-of-charge band, cycle budget, priority rights, and reporting obligations. Allocate battery capacity to the shaped PPA only if the PPA premium rises or the PPA materially improves project financeability.

The binding constraint is control-right allocation. Every MW reserved for a customer profile is a MW unavailable for reserve capacity, arbitrage, or response to a new market product. The term sheet should therefore state the priority order for reserve obligations, customer shape delivery, merchant arbitrage, degradation limits, project-policy limits, minimum state of charge, availability reporting, termination triggers, and emergency override.

It should also state the evidence layer: battery management system data access, data retention, performance-test method, availability calculation, response/repair clock, and who may override dispatch during safety, grid, or warranty events. Without that evidence layer, the floor provider, optimizer, customer, insurer, and owner can each believe they own the same operating headroom.

The commercial team should price shaped PPA flexibility at its opportunity cost. If the customer cannot pay the opportunity cost, use a narrower shaped PPA plus a separate floor or optimizer agreement for the battery.

The decision pack should show two values side by side: expected value and financeable downside value. A route that maximizes expected merchant upside may still be the wrong first financing structure if it leaves the project with weak debt sizing, high collateral exposure, or no lender-readable revenue floor.

Assumption Flip

The recommendation changes if merchant revenues saturate and the expected merchant stack falls by 40%.

```
1.70 million EUR/year x 60% = 1.02 million EUR/year
```

If the residual merchant value after reserving PPA capacity falls by the same 40%, it becomes:

```
0.80 million EUR/year x 60% = 0.48 million EUR/year
```

The repriced opportunity cost is then:

```
1.02 - 0.48 = 0.54 million EUR/year
```

Against a `0.75 million EUR/year` shaped-PPA premium, the PPA tranche now clears expected opportunity cost before financeability and risk adjustments. The recommendation changes only if the control-rights, cycle, state-of-charge, and insurance constraints can be written into the term sheet.

Failure Mode To Avoid

Use last year's reserve revenue, an arbitrage forecast, and a shaped-PPA premium as three additive upside lines, then haircut the total for prudence.

The same failure appears when the stack ignores project-specific policy constraints. If the actual insurance, warranty, lender, or operating documents allow fewer cycles, different duty cycles, or fewer operating modes than the optimizer assumes, part of the stack is unavailable unless the owner obtains approval, changes coverage, or changes operation.

Both versions double-count flexibility. The same MW, MWh, state of charge, control rights, cycle budget, warranty envelope, data evidence, and grid connection must serve every product. Revenue stacking is constrained co-optimization; simple revenue addition double-counts the asset.

Links Back To The Lectures

- Frequency And Reserves
- Battery Storage And Time Shifting
- BESS Revenue Stacks
- Dispatch Optimization
- Valuation Under Uncertainty
- BESS Revenue Saturation

Case 5. Balance-Responsibility And Balancing-Service Performance Audit

References: [3] [4] [12] [20] [43] [49]

A trading or optimization partner can execute the market interface, but the asset owner still needs evidence before it changes fees, mandates, incentives, or partners. The main work is attribution. A monthly revenue gap may come from market conditions, missing data, asset constraints, forecast error, risk limits, credit limits, settlement corrections, or dispatch choices. Partner action is justified only after the evidence separates those causes.

Decision Question

Should the asset owner change partner terms now, require evidence remediation first, or treat the gap as market and constraint noise?

Situation

A renewable independent power producer (IPP) outsources balance responsible party (BRP) and balancing service provider (BSP) execution work. The partner sends monthly revenue reports. The asset owner wants to know whether the partner created value, leaked value, or operated within realistic constraints before opening a commercial renegotiation.

The partner contract allows performance review, but the current data annex is weak. Forecast files, bid files, schedule revisions, optimizer recommendations, manual overrides, meter data, activation records, and settlement corrections are delivered inconsistently.

Scenario Data

These are scenario numbers for one month. The revenue row is already net of imbalance, curtailment, and reserve effects. The lower rows are raw evidence inputs for attribution, not extra lines to add or subtract from revenue.

Metric	Actual Partner Result	Feasible Benchmark
Net energy and reserve revenue	5.90m EUR	6.40m EUR
Imbalance cost	−0.42m EUR	−0.31m EUR

Metric	Actual Partner Result	Feasible Benchmark
Curtailed or unavailable energy value	`−0.18m EUR`	`−0.17m EUR`
Data completeness	`92%`	`99% required`
Forecast files delivered on time	`88%`	`98% required`
Bids matching agreed risk limits	`100%`	`100% required`

The feasible benchmark uses only information available before gate closure, actual grid and asset limits, contracted control rights, agreed risk limits, and market products the partner was allowed to access. The percentage thresholds in this case are contract-scenario thresholds, not eSett rules or generic market standards.

Raw attribution pack:

Evidence Item	Observation	Initial EUR Impact
Missing or late data	forecast and bid files missing for several high-spread intervals	`0.12m`
Grid and asset constraints	export limit and outage periods reduced feasible dispatch	`0.08m`
Forecast error inside agreed range	weather-driven production miss within tolerance band	`0.09m`
Risk-limit conservatism	partner stayed inside owner-approved risk limits during volatile hours	`0.10m`
Residual bidding/dispatch gap	unexplained difference after known constraints	`0.11m`

Evidence quality flags:

Required Evidence	Status	Consequence
Forecast versions and timestamps	incomplete	cannot prove what was knowable
Bid submissions and revisions	partial	cannot fully test partner action
Optimizer recommendations	missing for 18% of intervals	cannot separate tool advice from manual override
Asset availability and outages	complete	physical constraints can be attributed
Meter and settlement data	provisional, corrections pending	revenue result may move
Collateral and credit-limit evidence	partial	cannot tell whether otherwise feasible trades were blocked
Contract rights and risk limits	complete	feasible benchmark boundary is known

Tasks

1. Calculate the headline gap.
2. Attribute the gap using the raw evidence.
3. Decide which amount can be used for partner action now.
4. Draft a scorecard and governance recommendation.

Worked Solution

Headline gap:

```
benchmark_revenue - actual_revenue
= 6.40m EUR - 5.90m EUR
= 0.50m EUR
```

Attribution:

Driver	EUR Impact	Governability	Can Support Partner Action Now?
Missing or late data	`0.12m`	shared governance / contract data rights	yes, for data remediation
Grid and asset constraints	`0.08m`	mostly non-controllable	no
Forecast error inside agreed range	`0.09m`	model improvement	no, unless tolerance is renegotiated
Risk-limit conservatism	`0.10m`	owner policy choice	no, unless mandate changes
Residual bidding/dispatch gap	`0.11m`	partner execution candidate	only after evidence file is complete
Total	**`0.50m`**		

The only clean commercial action this month is data remediation. The `0.11m EUR` residual may be partner execution leakage, but the missing forecast, bid, and optimizer files make a fee claim premature.

Numerical Intuition

The headline gap overstates the partner-action amount.

```
residual_bidding_gap_share = 0.11 / 0.50 = 22%
data_and_observability_share = 0.12 / 0.50 = 24%
```

Nearly a quarter of the gap is an evidence problem, and only about a fifth is a candidate execution problem. A strong scorecard prevents the owner from negotiating a `0.50m EUR` accusation when the supported execution claim is closer to `0.11m EUR` and still evidentially weak.

Evidence Required Before Partner Action

The owner should require raw files before using the scorecard for fee clawback, termination, incentive reset, or mandate reduction.

Evidence Layer	Required Files Or Data	What It Separates
Forecast and schedules	forecast versions, schedules, revisions, timestamps, gate-closure state	model error versus late information
Bids and dispatch	submitted bids, accepted bids, activations, optimizer recommendations, manual overrides	partner choice versus market acceptance
Asset and grid constraints	availability, outages, import/export limits, reserve commitments	controllable execution versus physical constraint
Meter and settlement	metered volumes, imbalance settlement, reserve settlement, corrections	economic result versus provisional estimate

Evidence Layer	Required Files Or Data	What It Separates
Contract rights	data rights, control rights, risk limits, permitted products	what the partner was allowed to do
Credit and collateral	collateral calls, credit limits, settlement-bank or counterparty notices	revenue result versus liquidity constraint
Service-level agreement (SLA) and escalation trail	dated exceptions, response times, remediation owners, and repeated defects	one-off error versus governable pattern

This mirrors the logic in Nordic settlement governance: performance and imbalance outcomes depend on reported data, settlement quantities, participant obligations, and corrections. A private partner scorecard should be at least as disciplined about raw evidence.

Commercial Workbench Companion

Use the Commercial Workbench as a compact version of the evidence discipline in this case. The partner-evidence view is useful when the question is: what evidence would make a commercial action fair and defensible?

Scorecard

Use a scorecard that separates result, evidence, and attribution.

Scorecard Area	Month Result	Status	Action
Revenue versus feasible benchmark	`−0.50m EUR`	watch	track with attribution; no fee action on headline gap
Data completeness	`92%` versus `99%` requirement	fail	remediation plan with dated file-delivery obligations

Scorecard Area	Month Result	Status	Action
Forecast timeliness	`88%` versus `98%` requirement	fail	daily timestamp feed and exception report
Risk-limit compliance	`100%`	pass	keep current limits until management reviews risk appetite
Constraint attribution	`0.08m EUR`	accepted	exclude from partner execution claim
Residual execution candidate	`0.11m EUR`	unresolved	retest after complete raw files and settlement corrections

The scorecard thresholds are scenario contract requirements for this exercise. They should be replaced by the actual service-level agreement before a live partner review. The scorecard should be repeated for at least three comparable months before commercial penalties are applied, unless the contract already defines automatic data-delivery breaches.

Decision Memo

Recommendation: do not renegotiate fees or accuse the partner based on the `0.50m EUR` headline gap. Issue a formal data-rights and reporting remediation notice, agree the feasible-benchmark methodology, and put the partner on a three-month evidence watch.

Immediate actions:

- amend or enforce the data annex for forecast, schedule, bid, dispatch, optimizer, meter, activation, and settlement files
- require timestamped delivery and exception reporting
- freeze the feasible-benchmark method before seeing next month’s results

- review whether owner-approved risk limits or credit limits are too conservative for volatile periods
- retest residual bidding/dispatch gaps only where evidence is complete

Partner commercial action becomes supportable if, for three comparable months:

- data completeness meets the `99%` requirement
- forecast timeliness meets the `98%` requirement
- settlement corrections are reconciled
- the residual bidding/dispatch gap remains material against an agreed EUR threshold or percentage of monthly benchmark revenue
- the gap occurs in intervals where action was feasible and permitted

Assumption Flip

The recommendation changes if complete raw files show that the partner ignored feasible profitable bids, violated the agreed strategy, or failed to act inside clear market windows while limits and asset constraints allowed action.

In that case, the `0.11m EUR` residual becomes an execution claim rather than an unresolved attribution bucket, and the owner can move to fee adjustment, incentive redesign, tighter mandate, or replacement process.

Failure Mode To Avoid

Use a perfect-hindsight dispatch and attribute the full `0.50m EUR` gap to the partner.

Perfect hindsight can estimate an upper bound, but it cannot govern a BRP/BSP relationship. The governance benchmark must use forecasts, market prices, liquidity, grid limits, asset status, control rights, and risk limits available at the time.

Links Back To The Lectures

- Market Participants And Responsibilities
- Market Clock
- Forecast Error And Imbalance
- Frequency And Reserves
- Forecasting, Error, And Commercial Value

- Trading Partners And Performance Scorecards
- Analytical Capability Development

Case 6. Data-Centre 24/7 Product Test

References: [6] [56] [57] [58] [37] [59] [39] [32]

Large electricity buyers increasingly care about both when clean energy is produced and how many clean MWh are bought over a year. The seller must separate physical supply, financial settlement, certificate evidence, residual firming, credit, and control rights. The decision is whether a renewable independent power producer (IPP) should offer a 24/7-style product, narrow the promise, partner, or walk away.

Decision Question

Should the seller offer hourly matched clean power, a narrower shaped product, a partnered offer, or no-bid?

Situation

A data-centre buyer requests a clean-power product for a flat 40 MW load. The buyer asks for high hourly matching, budget stability, and clear evidence treatment.

The seller has solar assets, access to a battery project, and a trading partner. It does not own enough diverse clean generation to self-supply every hour.

Scenario Data

These are scenario assumptions.

Item	Value
Buyer load	40 MW flat
Annual demand	350.4 GWh/year
Seller-controlled clean supply	260 GWh/year
Annual clean-energy coverage	74%
Hourly matching before extra firming	58%
Target hourly matching for premium product	85%

Item	Value
Residual stress-hour shortfall	22 MWh/h
Stress hours	300 h/year
Firming premium in stress hours	150 EUR/MWh
Buyer premium for stronger hourly matching	4 EUR/MWh

Tasks

1. Calculate the annual buyer demand.
2. Calculate stress-hour firming-at-risk.
3. Compare the buyer premium with the stress exposure.
4. Define what the offer promises and how the evidence will be checked.
5. Recommend the offer shape.

Worked Solution

Annual demand:

```
40 MW x 8,760 h/year = 350,400 MWh/year
```

Stress-hour firming-at-risk:

```
22 MWh/h x 300 h/year x 150 EUR/MWh
= 990,000 EUR/year
```

Buyer premium:

```
350,400 MWh/year x 4 EUR/MWh
= 1,401,600 EUR/year
```

The premium covers only the illustrated stress-hour procurement cost before collateral, credit, evidence handling, operational costs, margin, and any additional clean supply needed to move from the initial `58%` hourly matching level toward the `85%` threshold. The case still needs careful structuring before rejection or approval.

Hourly matching gap:

Matching Gap	Calculation	Result	Decision Meaning
Clean MWh currently matched hourly	350.4 GWh/year x 58%	203.2 GWh/year	starting point for the premium claim
Clean MWh needed for 85% hourly matching	350.4 GWh/year x 85%	297.8 GWh/year	target product obligation
Incremental time-matched clean gap	297.8 - 203.2 GWh/year	94.6 GWh/year	must come from partner supply, storage, buyer flexibility, or narrowed promise
Minimum annual clean-supply gap versus target	297.8 - 260.0 GWh/year	37.8 GWh/year	even annual volume is short before timing is solved

Premium headroom:

Premium Headroom	Calculation	Result
Buyer premium	350,400 MWh/year x 4 EUR/MWh	1.40 million EUR/year
Stress-hour firming cost	22 MWh/h x 300 h/year x 150 EUR/MWh	0.99 million EUR/year

Premium Headroom	Calculation	Result
Remaining headroom before credit, evidence, operations, margin, and non-stress matching	1.40 - 0.99 million EUR/year	0.41 million EUR/year
Remaining headroom spread across load	0.41 million EUR / 350,400 MWh	about 1.17 EUR/MWh

What The Offer Must Promise

The product should be split into claims that can be evidenced and controlled.

Claim Or Service	Required Evidence Or Control	Seller Exposure If Vague
Annual clean-energy coverage	generation volumes, certificates, delivery period, geography, allocation rules	buyer receives a green volume claim but not an hourly product
Hourly matching threshold	hourly load, hourly clean supply, storage dispatch, residual procurement records	seller may be judged against a standard it did not price
Residual firming	cap, index, pass-through, partner supply, or reopener for hard hours	uncapped procurement exposure in scarce hours
Budget stability	settlement reference, volume tolerance, collateral and credit terms	clean-energy product becomes an unpriced financial hedge
Certificate treatment	certificate vintage, cancellation, location, matching period, audit trail	environmental claim can fail after the commercial deal is signed
Buyer flexibility	whether uninterruptible power supply (UPS), cooling,	load flexibility is assumed in the model but unavailable in

Claim Or Service	Required Evidence Or Control	Seller Exposure If Vague
	workload shifting, or on-site generation can be used and who controls it	operation
Reliability boundary	priority between buyer uptime, backup policy, grid services, and seller firming obligations	a flexibility resource is counted in the product when the buyer will not release it

For a data-centre buyer, flexibility resources can be valuable, but reliability constraints come first. Do not count buyer-side flexibility as seller supply unless the buyer grants defined availability, control, telemetry, settlement rights, and an uptime-compatible operating policy.

Add the accounting and certificate evidence pack before the offer is approved.

Evidence item	Source-backed reason	Contract implication
Dual Scope 2 boundary	GHG Protocol separates location-based and market-based Scope 2 reporting where contractual instruments exist	avoid language that treats annual certificates as physical hourly delivery
Energy attribute certificate status	certificates transfer attributes; attributes can be bundled or unbundled from the electricity	state whether guarantees of origin (GOs) or other instruments are included, transferred, canceled, or retained
Residual mix and null power	attribute-stripped energy cannot support the same renewable claim	define fallback treatment for uncovered or certificate-separated MWh
Granular time stamp	EnergyTag v2 uses UTC timestamps and production or	align load, clean supply, and storage evidence at the same

Evidence item	Source-backed reason	Contract implication
	storage-discharge intervals up to one hour	interval
Storage evidence	storage charge and discharge records are needed to preserve attribute accounting through storage	require battery energy storage system (BESS) charge source, discharge record, losses, and cancellation trail

Numerical Intuition

Annual coverage can look acceptable while hourly matching is weak.

Metric	Result	Why It Matters
Annual clean-energy coverage	74%	Useful annual signal; timing exposure remains
Hourly matching before firming	58%	Shows residual hourly procurement need
Stress-hour firming-at-risk	0.99 million EUR/year	Main seller exposure in hard hours
Buyer premium	1.40 million EUR/year	Commercial headroom before other costs

Decision Memo

Recommendation: do not offer a broad self-supplied 24/7 clean-energy claim. Offer the partnered hourly-matching product only if the partner contract covers the 94.6 GWh/year time-matched clean gap, caps or passes through stress-hour procurement, and provides the evidence trail required by the selected certificate/accounting framework. Otherwise offer the narrower shaped product.

Any broad 24/7 clean-energy claim needs the applicable accounting and certificate framework in the contract. The commercial product can still be valuable if it says exactly what is being delivered: hourly matched supply to a defined threshold, financial settlement against a defined reference, and residual procurement above a defined boundary.

The recommended offer is therefore a product ladder:

Approval Gate	Minimum Condition	If Missing
Hourly matching	load, generation, storage, and residual procurement measured on the same interval	sell shaped product, not hourly claim
Residual firming	cap, pass-through, index, or named partner supply	no-bid on broad claim
Evidence framework	certificate status, cancellation rule, storage treatment, and audit trail defined before signing	remove 24/7-style language
Buyer flexibility	telemetry, control rights, availability, and uptime-compatible policy	exclude buyer flexibility from seller supply

Offer	Seller Commitment	Buyer Benefit	When To Use
As-produced plus certificates	clean energy volume and evidence	lowest complexity	buyer accepts timing exposure
Shaped clean product	defined hourly shape with residual boundary	budget and profile improvement	buyer values shape but not full hourly claim
Partnered hourly-matching product	threshold-based hourly matching with named firming	stronger 24/7-style claim	premium covers firming, evidence,

Offer	Seller Commitment	Buyer Benefit	When To Use
	partner and evidence pack		credit, and operational cost
No-bid on broad claim	seller rejects uncapped 24/7 language	avoids mispriced obligation	offer cannot be made auditable

Assumption Flip

The recommendation changes if stress-hour shortfall rises to 40 MWh/h.

```
40 MWh/h x 300 h/year x 150 EUR/MWh
= 1,800,000 EUR/year
```

That exceeds the buyer premium before other costs. The seller should then require a higher premium, a partner portfolio, a narrower matching threshold, or a no-bid decision.

Failure Mode To Avoid

Offer the product as a premium clean-power contract because annual clean-energy coverage is high, while leaving hourly matching rules, certificate evidence, and residual firming treatment to later documentation.

That decision sells the label before defining the product. Annual matching can hide the hours that drive firming cost, evidence disputes, and residual procurement exposure. A flat load needs a time-indexed delivery promise alongside the annual energy total.

Links Back To The Lectures

- Load Is Not Energy
- Market Clock
- Solar Production
- PPA Structure Catalog
- PPA Pricing
- Forecasting

- Portfolio Modeling And Hedging
- Trading Risk, Credit, Liquidity, And Controls
- Future Of PPAs
- Data Centres, 24/7 Procurement, And Flexibility

Case 7. Year-One Energy-Management Operating Model

References: [50] [49] [27] [43] [6] [39]

An independent power producer (IPP) becomes an energy manager when it decides which revenue and risk decisions it must own, which execution activities it can outsource, what raw data it has rights to inspect, and which controls stop a model from becoming an unmanaged production process. This case forces the year-one resource allocation.

Decision Question

Given limited budget, headcount, partner data gaps, and several competing value pools, what should the company build, buy, outsource, audit, or postpone in year one?

Situation

The company has operating solar assets, a growing development pipeline, several structured power purchase agreement (PPA) discussions, and first battery projects entering commercial design. Trading execution and balancing are mostly outsourced. Partner reporting is useful but incomplete. Management wants a credible year-one operating model without approving a large desk or full energy trading and risk management (ETRM) implementation.

The operating model must support four decisions:

- shaped PPA pricing
- battery energy storage system (BESS) sizing and dispatch value
- balance responsible party (BRP), balancing service provider (BSP), and optimizer oversight
- portfolio exposure and risk escalation

Scenario Data

These are scenario assumptions.

Item	Value
Operating renewables	`600 MW`

Item	Value
Development pipeline under commercial review	`1.8 gigawatts (GW)`
First BESS projects	`3 projects`
Annual PPA offers under active discussion	`700 GWh/year`
Outsourced execution volume	`500 GWh/year`
Current internal analytics team	`1 accountable owner + 1 analyst`
Additional hire allowed	`0.5 data engineer` contractor
Year-one budget	`900,000 EUR`
External execution model	BRP/BSP and optimizer partners
Management constraint	no internal trading desk in year one

Partner data-right gaps:

Data Right	Current Status	Operating Consequence
Forecast versions and timestamps	partial	hard to judge what was knowable
Bid and schedule revisions	partial	partner actions cannot be fully reconstructed
Optimizer recommendations	limited	manual override versus model decision is unclear
Meter and settlement files	available with delay	monthly attribution lags
Reserve activation and imbalance detail	inconsistent	revenue stack cannot be reconciled cleanly
Control-right map	contract language exists, operational map missing	dispatch priority conflicts are hard to prove

Data Right	Current Status	Operating Consequence
Credit, collateral, and liquidity evidence	fragmented	cannot tell whether feasible actions were blocked by limits or cash timing

Value pools:

Value Pool	Scenario Value At Stake	Evidence Quality
Shaped PPA risk pricing	`560,000 EUR/year`	medium-high
BESS design and revenue stack	`750,000 EUR/year`	medium
Partner leakage reduction	`350,000 EUR/year`	low until data rights improve
Portfolio risk and hedge decisions	`250,000 EUR/year`	medium

Control failures observed:

- commercial offers use inconsistent shape, basis, certificate, and residual-firming assumptions
- BESS cases double-count merchant and PPA use of the same MWh in some sensitivities
- partner reports show results without complete raw files
- management sees portfolio exposure after settlement rather than before decisions
- model versions and assumption owners are unclear

Tasks

1. Allocate the `900,000 EUR` budget across year-one capabilities.
2. Assign headcount ownership and name what remains outsourced.
3. Prioritize a 90-day plan and a 12-month roadmap.
4. State which partner data-right gaps block partner action.
5. Define kill criteria for the roadmap.

Worked Solution

Year-one resource allocation:

Capability	Build, Buy, Outsource, Audit, Or Postpone	Budget	Owner	Reason
Data contracts and reconciliation	build/audit	`130,000 EUR`	accountable owner + data engineer	unlocks all other controls
PPA quote stack and risk map	build	`160,000 EUR`	accountable owner + commercial lead	shaped offers need consistent risk pricing
BESS sizing and dispatch model	build with optimizer input	`140,000 EUR`	analyst + investment owner	investment committee (IC) decisions are live
Partner data-right remediation	audit/renegotiate	`120,000 EUR`	accountable owner + legal/commercial	raw evidence is missing
Portfolio risk pack	build	`110,000 EUR`	analyst + risk owner	management needs one exposure view
Forecast/vendor evaluation layer	buy inputs, build evaluation	`90,000 EUR`	analyst	value-weighted errors support multiple decisions
Feasible benchmark pilot	build small pilot	`50,000 EUR`	accountable owner	method first, automation

Capability	Build, Buy, Outsource, Audit, Or Postpone	Budget	Owner	Reason
				later
Contingency and independent review	buy	`100,000 EUR`	steering group	avoids brittle models and weak validation
Total		**`900,000 EUR`**		

Postpone:

- internal trading desk
- full ETRM implementation
- automated benchmark platform across all assets
- proprietary weather model

Outsource:

- BRP/BSP market access
- 24/7 dispatch execution
- base weather/price forecast feeds
- some legal/accounting/certificate advice

Own internally:

- PPA risk allocation and quote logic
- BESS investment assumptions
- data-right requirements
- feasible benchmark methodology
- monthly portfolio risk view
- model register and assumption ownership

Forced Tradeoff

Building a polished partner scorecard immediately would spend money before the evidence layer is ready. Partner data-right gaps make a full scorecard expensive and weak. The stronger year-one allocation funds the data annex and only a narrow benchmark pilot.

Route	Spend First	What It Buys	Why It Wins Or Loses
Full scorecard now	automation and dashboards	broad reporting surface	weak attribution if raw forecasts, bids, dispatch logs, and settlement files remain incomplete
Data rights plus pilot	data annex, reconciliation, narrow benchmark	fewer assets covered at first	stronger governance because the scorecard rests on evidence the owner can inspect

The PPA quote stack and BESS model receive more year-one funding because live commercial and IC decisions can use them with available internal data. Partner commercial action waits for raw evidence.

First 90 Days

Month	Deliverable	Decision It Enables
1	data-rights inventory and gap notice to partners	shows what can be measured, audited, and renegotiated
1	credit, collateral, and liquidity evidence map	shows whether trading constraints are economic, operational, or treasury-driven
1	model register and assumption owner map	prevents uncontrolled local models

Month	Deliverable	Decision It Enables
2	shaped-PPA quote stack v1	quote, redesign, no-bid, or pass-through decision
2	BESS duration/revenue-stack model v1	1h/2h/4h, grid-right, degradation, and control-right decisions
3	first monthly portfolio risk pack	management view of price, shape, basis, imbalance, credit, liquidity, and partner exposure
3	partner benchmark methodology note	agrees feasible benchmark rules before automation

The first 90 days should produce usable decision artifacts, not a finished system.

Twelve-Month Roadmap

Quarter	Milestone	Acceptance Test
Q1	common data contract inventory	every critical feed has source, owner, timestamp, unit, granularity, validation rule, and gap owner
Q2	PPA and BESS models used in live decisions	at least one quote or IC recommendation changes because of the model
Q2	risk pack live	management uses it for limits, hedge questions, partner escalation, or deal approval
Q3	forecast evaluation layer	day-ahead, intraday, BESS, and partner forecast errors

Quarter	Milestone	Acceptance Test
		are value-weighted
Q3	partner data annex amended or escalation logged	missing raw evidence has a commercial owner
Q4	benchmark pilot completed	one asset class or partner has feasible attribution using time-available data

Risk Pack

The recurring risk pack should be short and decision-linked.

Section	Minimum Content	Escalation Trigger
Executive exposure	price, shape, basis, merchant tail, and liquidity summary	exposure outside approved range
PPA pipeline	quote stack, residual firming, certificates, credit, basis	unpriced risk in term sheet
BESS and dispatch	revenue stack, state-of-charge constraints, degradation, reserve/PPA conflicts	double-counted MW/MWh or cycle budget
Partner scorecard	data completeness, forecast timeliness, feasible benchmark gap	raw evidence below requirement
Settlement and reconciliation	meter, imbalance, reserve, invoice exceptions	unreconciled material variance
Credit and collateral	counterparty exposure, collateral calls, guarantees	liquidity stress or credit breach

Section	Minimum Content	Escalation Trigger
Model changes	version changes, assumption moves, validation exceptions	unapproved model used in decision
Partner governance	missing files, late explanations, service-level agreement (SLA) breaches, unresolved incident owners	repeated defect without remediation owner

The pack is a control only if management uses it to approve, stop, redesign, hedge, escalate, or request evidence.

Data Rights Before Partner Action

Partner action is blocked where raw evidence is missing.

Action	Evidence Required First
fee reduction or clawback	complete bids, schedules, dispatch recommendations, activations, and settlement data
mandate change	proof that current limits caused avoidable loss or excess risk
optimizer replacement	evidence that feasible recommendations were materially worse than benchmark
BRP/BSP replacement	repeated residual execution gap after data, constraints, forecast, and risk-limit attribution
incentive redesign	clear split between market conditions, constraints, and controllable partner decisions

Until those files exist, the company can demand data remediation and better reporting. It should be cautious about claiming execution leakage.

Kill Criteria

Workstream	Continue If...	Kill Or Redesign If...
PPA quote stack	used in live offers and changes price or structure	commercial team bypasses it or outputs cannot explain residual risk
BESS model	changes IC decisions or preserves valuable design option	dispatch value cannot be tied to grid rights, degradation, and control rights
Data remediation	partner completeness and timeliness improve against dated milestones	contract rights remain blocked after escalation
Risk pack	management takes decisions from it	it becomes a passive report
Forecast layer	value-weighted error improvement exceeds cost	updates arrive too late for action
Benchmark pilot	attribution uses time-available data and settlement reconciliation	it relies on hindsight or missing files

Decision Memo

Recommendation: build a small control-oriented energy-management function in year one while keeping market execution outsourced.

Approve the full `900,000 EUR` allocation shown above. The first year should produce a PPA quote stack, BESS design model, data-right remediation, monthly portfolio risk pack, forecast evaluation layer, and a narrow feasible-benchmark pilot. Do not approve an internal trading desk or full ETRM implementation until data rights, independent settlement checks, risk limits, and evidence-based partner attribution are working.

Add a monthly energy-risk forum with named authority for limits, credit and collateral exceptions, model approvals, partner escalations, and stop/redesign decisions. A small team can run the operating model only if decision rights are explicit.

The operating model is deliberately constrained:

- one accountable owner coordinates decisions, partner governance, and steering
- one analyst owns models, risk pack production, and value-weighted forecast analysis
- the half-time data engineer builds data contracts, reconciliation checks, and repeatable feeds
- partners continue to execute, but their raw evidence obligations become explicit

The first-year system should be deliberately small but auditable. Every model that affects a quote, dispatch view, partner scorecard, or investment recommendation needs an owner, version, source-data list, change log, and decision record. That minimum control layer should be in place before a larger trading platform or internal desk is considered.

Assumption Flip

The recommendation changes if partners refuse or cannot provide forecast, bid, schedule, dispatch, meter, activation, and settlement files on a reliable timeline.

In that case, shift more budget from model expansion to contract remediation, data interfaces, and replacement-option analysis. Building more models before the company can observe partner decisions would create false control.

Failure Mode To Avoid

Internalize market execution before the company has data rights, independent settlement checks, feasible benchmarks, risk limits, and escalation rules.

That sequence creates operational and control risk. Execution capability only helps when the organization can see the decisions being made, attribute outcomes, enforce limits, and settle disputes with evidence.

Links Back To The Lectures

- Forecasting, Error, And Commercial Value

- Trading Partners And Performance Scorecards
- Analytical Capability Development
- Data And Machine Learning In Energy Management
- Trading Capability Design
- IPP Operating Models For Active Portfolio Management

Glossary

An alphabetical reference for the main terms used in the course.

Actionability factor

Fraction of a model improvement that can become value after information timing, market access, liquidity, control rights, and constraints.

Actionability-adjusted uplift

Raw model or capability uplift after reducing for timing, tradable volume, liquidity, asset constraints, control rights, and organizational adoption.

Activation

Automatic or instructed conversion of reserve capability into physical response.

aFRR

Automatic Frequency Restoration Reserve: reserve automatically activated to restore frequency/control area balance after disturbance.

Aggregator

Entity that combines distributed assets or loads so they can participate commercially or technically as a portfolio.

Allocated capacity

Grid capacity assigned or reserved for a connection or network user under a specific process or agreement.

Analytical engine

Connected set of models, data flows, controls, and decision rights that turns market and asset information into repeatable commercial decisions.

Annual matching

Comparing clean-energy procurement and consumption over a year.

Apparent power

Combined active and reactive loading of AC equipment, commonly represented as S where $S^2 = P^2 + Q^2$.

As-produced PPA

PPA where buyer takes generation as produced, inheriting shape/profile exposure.

Assumption-flip test

Search for the smallest plausible change in an assumption that would reverse a recommendation.

Augmentation

Addition or replacement of battery equipment to restore or maintain usable capacity.

Back office

Function responsible for confirmations, settlements, invoicing, contract administration, reporting, and operational control records.

Backtest

Historical simulation of model or policy performance using only information available before each decision time.

Balance obligation

Obligation to plan for balance and financially settle imbalances under market rules.

Balancing capacity

MW held available for balancing service under TSO product rules, whether or not activation occurs.

Balancing energy

Energy delivered or absorbed when balancing bids or reserves are activated.

Balancing market

Real-time or near-real-time mechanism used by the TSO to maintain system balance.

Bankable revenue

Revenue treated as dependable enough for financing, depending on contract, counterparty, tenor, and downside protection.

Baseload PPA

PPA with a flat delivery profile over a defined period.

Basis risk

Mismatch between the exposure being hedged and the instrument/reference used to hedge it.

Battery duration

Battery energy capacity divided by power rating, usually expressed in hours.

Behavioral market power

Actual conduct that uses market-power potential through bidding, withholding, scheduling, or dispatch choices.

BESS

Battery Energy Storage System.

BRP

Balance Responsible Party: entity financially responsible for imbalances at a delivery point or portfolio.

BSP

Balance Service Provider: entity providing balancing services to the TSO.

C-rate

Charge or discharge rate relative to battery energy capacity; 1C roughly means full usable charge or discharge in one hour.

Calculation period

Time period over which a reference price or settlement amount is calculated.

Calendar aging

Battery degradation associated with time, temperature, and state-of-charge conditions, even without heavy cycling.

Cannibalization

Value erosion when similar generators produce at the same time and depress prices during their own production hours.

Cap

Contractual maximum price or value above which upside is limited or shared, depending on structure.

Capability priority score

Heuristic ranking of a capability by value at stake, controllability, observability, urgency, cost, and risk.

Capability-contract matrix

Build/buy/outsource table that separates internal judgment, tools, outsourced services, and non-negotiable control/data rights.

Capacity credit

Dependable capacity contribution of a resource divided by its nameplate capacity.

Capacity factor

Actual energy produced over a period divided by energy that would be produced at full nameplate output over that period.

Capacity remuneration mechanism

Market or regulatory mechanism that pays for capacity, availability, or reliability contribution in addition to energy.

Capacity warranty

Warranty or performance guarantee specifying retained usable battery capacity over time, usually conditional on operating conditions and measurement method.

Capture price

Generation-weighted average market price received by an asset or technology.

Capture rate	
Capture price divided by average spot price for the same market/period.	
Cash available for debt service	
Cash flow available to pay scheduled debt service after operating revenue, operating costs, and allowed deductions/reserves according to the model or financing terms.	
Cash Flow at Risk	
Liquidity metric focused on potential cash demands versus available sources over a horizon.	
Certificate contract price	
Price or embedded value assigned to certificates such as GOs or EECS certificates in a PPA.	
CfD	
Contract for Difference: financial contract settling the difference between a strike/reference price and a market/reference price.	
Closed-loop latency ratio	
Feedback latency divided by decision cadence; high values mean the organization learns slower than it acts.	
Collateral	
Value posted to reduce credit exposure between counterparties or clearing participants.	
Congestion rent	

Value associated with a price difference across a constrained transmission interface multiplied by feasible flow, in simplified market-design examples.

Connection agreement

Agreement governing how an asset connects to the grid, including technical, capacity, cost, and operational conditions.

Connection charge

Charge, often one-off, covering part or all of connecting a new user or upgrading a connection.

Constraint

Physical, market, contractual, or operational limit a model or decision must obey.

Construction contribution

One-off payment or cost contribution for grid works required to connect or serve a network user.

Contract quantity

Electricity quantity specified by a PPA or related contract, such as all metered output, part of metered output, or a fixed delivery schedule.

Control right

Authority to decide, instruct, or veto scheduling, bidding, dispatch, curtailment, hedging, or reserve actions.

Cost of capital

Required return for the capital supplied to a project, including debt and equity costs.

Counterparty credit risk

Risk that the counterparty to a contract does not pay, deliver, or otherwise perform as required.

Credit support

Collateral, guarantee, parent support, letter of credit, or similar instrument used to reduce counterparty credit risk.

Cross-border PPA

PPA where buyer and generation asset are in different national markets or bidding zones, often creating basis and transmission-right questions.

Curtailment

Reduction of output below available generation, for economic, grid, or system-flexibility reasons.

Customer value

Buyer's benefit from a product, such as hedge value, budget certainty, decarbonization claim, hourly alignment, or procurement simplicity; distinct from seller cost.

Customer willingness-to-pay

Maximum price or premium a buyer may rationally accept based on hedge value, budget certainty, decarbonization value, evidence needs, and procurement alternatives.

Cycle aging

Battery degradation associated with charge-discharge use, including throughput, depth of discharge, rate, and operating pattern.

Daily risk pack	Recurring report connecting positions, MtM, market risk, credit exposure, liquidity/collateral, basis/shape, partner execution, limits, data quality, and actions.
Data contract	Definition of a data feed's source, owner, timestamp, unit, granularity, validation rule, and decision use.
Data right	Right to receive data needed to reproduce, audit, and govern decisions.
Day-ahead market	Auction before delivery day where hourly or sub-hourly electricity volumes are cleared for the next day.
Debt	Money lent to a project or company, usually paid before equity and therefore usually cheaper but less tolerant of downside cash-flow weakness.
Debt sculpting	Shaping debt service over time to fit the cash-flow profile a lender is willing to count.
Debt sizing	Process of estimating how much debt a project can support from lender-counted cash flow, coverage requirements, tenor, interest rate, reserves, and amortization structure.
Decision latency	

Time between the arrival of useful information and the latest moment a feasible action can still be taken.

Decision memo

Short recommendation document stating the decision, core evidence, retained risk, binding assumptions, owner, next measurement, and assumption-flip test.

Decision owner

Person or function accountable for accepting, rejecting, or overriding a model recommendation or operational action.

Decision variable

Quantity the optimization model is allowed to choose.

Degradation

Loss of battery capacity or performance over time due to calendar aging and cycling.

Degradation-adjusted spread

Discharge price minus charging cost adjusted for efficiency, degradation, fees, tariffs, and opportunity cost.

Delivery schedule quantity

Fixed quantity or quantities to be delivered or settled during defined periods under a contract schedule.

Depth of discharge

Fraction of stored energy practically used in a discharge.

Dispatch

Operational choice of when/how much an asset charges, discharges, generates, or bids.

Dispatchable

Controllable by an operator within technical and fuel/resource constraints.

Distribution

Lower-voltage network delivery between the transmission system and end users / distributed resources.

Diversification

Risk reduction from imperfectly correlated exposures rather than asset count alone.

DSCR

Debt service coverage ratio: cash available for debt service divided by required debt service over a period.

DSO

Distribution System Operator: entity responsible for operating a distribution network.

Duration hedge

Extra battery energy capacity bought partly to handle uncertainty in spreads, delivery obligations, reserve calls, curtailment windows, or grid constraints.

Economic withholding

Offering capacity at a price high enough that it is unlikely to clear, even though it is technically available.

Energy

Quantity accumulated or consumed over time, usually Wh, kWh, MWh, TWh.

Energy attribute certificate

Instrument that represents attributes of generated electricity, such as renewable attributes, and can be transferred or canceled separately from physical electricity depending on the scheme.

Energy management

Capability connecting forecasts, market positions, dispatch, contract obligations, imbalance exposure, partner execution, and portfolio risk.

Energy-limited

Able to provide power only until a stored-energy, fuel, water, or demand-flexibility constraint binds.

Energy-management operating system

Set of recurring decisions, data flows, models, limits, partner controls, feedback loops, and review rhythms that turns assets and contracts into governed commercial action.

Energy-only market

Market design where investment recovery relies mainly on energy and ancillary-service revenues, including scarcity prices.

Equity

Ownership capital that receives residual cash flows after operating costs, debt service, reserves, and other senior claims.

Equivalent full cycle

One full charge-discharge equivalent measured by throughput relative to usable energy capacity; partial cycles can add up to equivalent full cycles depending on the convention used.
ETRM
Energy trading and risk management system used to record, value, monitor, and control trades, positions, limits, settlement, and reporting.
Expected shortfall
Average loss conditional on losses exceeding a selected tail threshold such as VaR.
Expected unserved energy
Expected MWh of demand not served because supply is insufficient.
Exposure cube
Portfolio map that classifies risk by geography, technology, contract, hedge, flexibility, counterparty, liquidity, and data/control rights.
Fault level
Measure related to system strength and available fault current at a network point.
FCR
Frequency Containment Reserve: reserve intended to contain frequency deviations quickly after an imbalance.
Feasible counterfactual
Simulated action path that could have been taken with contemporaneous information, constraints, market access, risk limits, and control rights.

Feedback latency

Time between a decision and the reliable measurement or settlement data needed to evaluate it.

Final position

Market position carried into delivery after accepted trades, nominations, or adjustments.

Financeability lens

View of project cash flows focused on resilient cash for debt and covenant/downside requirements alongside equity upside.

Firming cost

Cost of transforming variable renewable output into a firmer or more useful delivery shape through storage, portfolio netting, market purchases, hedging, or flexibility.

Firming-at-risk

Stress measure for the cost of residual firming exposure: residual MWh in hard hours multiplied by the relevant firming premium or stress price.

Flexibility potential

Technical or practical ability of a load, generator, storage asset, or portfolio to change injection/withdrawal in response to grid or market needs.

Flexible connection agreement

Connection or network-use arrangement that allows limited, conditional, or interruptible injection or withdrawal, often to accelerate connection or reduce reinforcement needs.

Floor

Contractual minimum price or value below which settlement is protected, depending on structure.

Forecast-based dispatch

Dispatch optimized using information available at decision time.

FPA

Flexibility Purchase Agreement or flexibility-style arrangement; exact meaning must be defined by source/context before use.

Front office

Commercial function that originates, structures, prices, trades, dispatches, or manages market-facing portfolio decisions.

Generation adequacy

Generation-resource component of system adequacy.

Governance calendar

Recurring cadence for position review, partner performance, PPA pricing, portfolio exposure, model assumptions, capability strategy, and investment gates.

Granular certificate

Certificate or attribute-tracking concept that adds temporal and sometimes locational granularity to energy attribute tracking.

Grid tariff

Charge paid for use of the grid, potentially including fixed, capacity, energy, loss, and reactive-power components.

Grid-following inverter

Inverter that synchronizes to an existing grid voltage/frequency reference.

Grid-forming inverter

Inverter controlled to establish or support voltage/frequency reference behavior more actively.

Guarantee of Origin

European energy attribute certificate used to document electricity attributes within the relevant issuing and cancellation rules.

Hedge-quality checklist

Test of whether a hedge matches the underlying exposure by reference price, volume, shape, tenor, basis, liquidity, collateral, and governance.

Hourly matching

Comparing clean supply and consumption within each hour.

Imbalance

Difference between scheduled/nominated and actual generation or consumption.

Incremental duration

Additional MWh added to a battery design while keeping MW approximately fixed.

Incremental duration value

Additional value from adding MWh to a battery after additional revenue, capex, degradation, tariffs, and opportunity cost.

Independent price validation

Process of checking marks, curves, or valuations independently from the team that originated or owns the position.

Inertia

Resistance of rotating mass to sudden frequency change.

Insurance operating covenant

Condition in an insurance policy or endorsement that limits how an asset may be operated while keeping coverage, pricing, or claim eligibility intact. For BESS this may include cycle, throughput, duty-cycle, safety, monitoring, or approved-use restrictions.

Intraday market

Market after day-ahead clearing where participants adjust positions closer to delivery.

Investment-case waterfall

Path from gross physical value to equity cash flow that shows capture, curtailment, contract, grid, BESS, opex, debt, and residual effects layer by layer.

IPP

Independent Power Producer: company that owns or operates generation assets and sells electricity or related products without being the incumbent utility.

IRR

Internal rate of return: discount rate that makes project NPV equal zero.

Limited grid access

Connection or access arrangement where injection or withdrawal is constrained under specified conditions.

Liquidity at Risk

Risk measure for liquidity pressure under market, credit, collateral, or trading-liquidity stress.

Liquidity risk

Risk that a firm cannot trade, exit, or fund positions when needed without unacceptable cost or disruption.

Load

Electrical demand at a point, portfolio, zone, or system over time.

Load shape

Pattern of demand across time.

Location-based Scope 2 method

Scope 2 accounting method based on average emissions factors for the grid or location where electricity consumption occurs.

Long-duration energy storage

Storage intended to address longer scarcity periods than short intra-day shifting.

Loss charge

Charge or adjustment intended to recover electrical losses associated with network use.

Loss of load expectation

Expected duration of insufficient supply over a planning horizon.

Loss of load probability

Probability that available resources cannot meet load in a specified period.

Loss tree

Engineering model that tracks energy losses through a renewable, storage, or hybrid system so dispatch and valuation use deliverable energy rather than ideal nameplate energy.

Margin call

Demand to post additional collateral after exposure or collateral requirements change.

Marginal duration hurdle

Annualized cost and opportunity-cost threshold that the next MWh of storage energy capacity must clear before extra duration is worth buying or building.

Marginal risk contribution

Incremental stress, variance, liquidity, or concentration added by the next asset, hedge, contract, or partner dependency.

Market access

Ability to participate in a market through required contracts, systems, qualifications, collateral, and operations.

Market monitoring

Monitoring of bids, outages, constraints, behavior, and outcomes to distinguish normal scarcity, opportunity cost, and problematic conduct.

Market power

Ability to profitably influence market price away from the competitive level for a meaningful period.

Market risk

Exposure to losses from movements in prices, spreads, volatility, basis, or correlations.

Market time unit

Time granularity of traded or settled energy, such as hourly or 15-minute periods depending on market and date.

Market-based Scope 2 method

Scope 2 accounting method based on contractual instruments such as energy attribute certificates, qualifying PPAs, supplier-specific emission rates, and residual mix where available.

Merchant tail

Post-contracted or partially uncontracted period where project revenue depends more directly on market prices and exposed risks.

Merchant upside

Uncontracted or weakly contracted expected value from market optimization.

mFRR

Manual Frequency Restoration Reserve: reserve manually or centrally activated to restore balance and manage system needs.

Middle office

Independent or semi-independent function that measures risk, validates prices, monitors limits, benchmarks performance, and challenges front-office decisions.

Missing money

Concern that market revenues may be insufficient to finance needed capacity because scarcity or investment signals are muted, risky, capped, or incomplete.

Mixed-integer linear programming

Optimization method where some variables are continuous and others are integer or binary, useful when dispatch choices include on/off, mutually exclusive, or commitment-style decisions.

Model inventory

Register of models, owners, purposes, inputs, outputs, limits, validation status, and retirement criteria.

Model register

Inventory of models, owners, inputs, outputs, validation status, and decisions supported.

Model value at risk

Scenario estimate of value exposed to a wrong model-driven action across affected MW, duration, adverse price error, and bad-event count.

Model-risk charge

Scenario or governance allowance for losses, overrides, monitoring, and control burden introduced by relying on a model.

Multi-buyer PPA

PPA where several buyers aggregate demand to contract with one generator or supplier structure.

Nameplate capacity

Rated maximum power under defined conditions.

NEMO

Nominated Electricity Market Operator: exchange/operator designated for day-ahead and intraday market coupling functions.

Netting

Partial cancellation of positive and negative deviations inside compatible commercial or settlement boundaries.

NPV

Net present value: discounted value of future cash flows minus investment cost.

Null power

Electricity from which energy attributes have been separated, so it cannot support the same market-based renewable claim without a valid contractual instrument.

Numerical weather prediction

Physics-based weather model used as an input to power, load, price, and system forecasts.

Objective function

Mathematical expression a model seeks to maximize or minimize.

Offtaker

Buyer of electricity or environmental attributes under a commercial contract such as a PPA.

Opportunity cost

Value of the best feasible alternative action forgone when an asset is committed to a different action.

Paid-or-required capability gap

Difference between system-service capability that is required or valuable at a site and capability that is already qualified and available.

Partner benchmark

Feasible counterfactual used to compare actual BRP/BSP/optimizer performance with actions available under real constraints and rights.

Pay-as-produced PPA

PPA where the buyer purchases electricity as generated, so the buyer generally takes more generation-profile exposure.

Peak load

Maximum load over a defined period and geography.

Perfect foresight dispatch

Dispatch optimized with actual future information; useful as an upper bound for operational benchmarks built from tradable information available at decision time.

Physical withholding

Making capacity unavailable even though it could physically produce.

Pivotal supplier	Supplier whose capacity is needed to meet demand under the relevant constraints.
Portfolio exposure	Sensitivity of portfolio value or cash flow to a market, physical, contractual, credit, or operational driver.
Portfolio PPA	PPA backed by a portfolio of assets rather than a single identified plant, potentially spreading resource and operational risks.
Potential future exposure	Forward-looking estimate of how large counterparty exposure may become under future market movements.
Power	Rate of energy flow, usually measured in W, kW, MW, or GW.
PPA	Power Purchase Agreement: contract for purchase/sale of electricity, often long-term.
Private-wire PPA	PPA where generation and consumption are connected through a direct/private wire rather than only through ordinary grid supply arrangements.
Product-readiness score	Heuristic assessment of whether a firm has the assets, portfolio, data, control rights, risk limits, and evidence needed to offer a more advanced PPA product.

Project policy constraint	Project-specific insurance, lender, warranty, or operating-policy limit on cycling, throughput, duty cycle, approved revenue modes, data retention, or claim eligibility.
Qualification chain	Sequence that turns technical capability into commercial value: specify, test, qualify, control, measure, settle, and enforce.
Quote stack	Pricing structure that decomposes an offered PPA price into raw energy, shape/firming, imbalance, grid/tariff, certificates, credit, tenor, optionality, and risk-margin components.
Reactive power	Component of AC power associated with voltage support and electric/magnetic fields, commonly measured in var/MVAr.
Reference price	Market price, index, or formula used to settle a financial PPA, CfD, floor/cap, or indexed price component.
Regional upscaling	Converting asset-level forecasts or samples into a portfolio, bidding-zone, or control-area forecast.
Reserve capacity	MW held available for possible TSO activation under product rules.
Reserve headroom	

MW or MWh capability held back so a battery can deliver a reserve obligation if called.

Residual demand

Demand left for a supplier after subtracting other available supply and feasible imports.

Residual load

Load remaining after subtracting variable renewable generation or another specified supply category.

Residual mix

Emissions or attribute mix left for electricity consumption after contractual instruments such as certificates have been removed or claimed.

Residual obligation

Hourly MWh, EUR, evidence, or operational exposure left after contracted clean supply, storage, hedges, certificates, and control actions have been applied.

Residual Supply Index

Market-power diagnostic comparing supply available without a supplier to demand; values below one indicate pivotality in the simple form.

Retained-risk table

Decision-memo table naming risks that remain after the recommendation, their owners, warning signals, and mitigations.

Revenue quality ladder

Course tool ranking modeled value by how confidently it can support fixed obligations versus equity upside or unproven optionality.

Revenue stack

Set of revenue sources a flexible asset may pursue, subject to mutual exclusivity, opportunity cost, and operational constraints.

Risk limit

Quantitative or qualitative boundary on positions, losses, exposures, stress losses, collateral needs, or permitted strategies.

Round-trip efficiency

Energy discharged divided by energy charged over a cycle.

Scarcity duration

Time length of the shortage or obligation a resource must cover.

Scarcity pricing

High pricing that reflects real scarcity of energy, capacity, flexibility, or network capability.

Scarcity-duration distribution

Distribution showing how often scarcity or obligation events of different lengths occur, for example 1-hour, 4-hour, overnight, multi-day, or seasonal events.

Security of supply

Broad ability of the power system to serve demand reliably across operating and planning horizons.

Shadow price

Marginal value of relaxing a binding constraint by one unit in an optimization problem.

Shadow trading	Counterfactual simulation of feasible trading/dispatch decisions used to benchmark actual execution or partner performance.
Shape cost	Cost of converting asset output into a promised contract profile, including market purchases, storage losses, degradation, hedging, balancing, grid/tariff costs, and risk margin.
Shape risk	Risk that production or consumption timing differs from the contracted or hedged delivery shape.
Short-circuit ratio	Grid-strength indicator comparing available short-circuit power with connected converter capacity; detailed definition depends on method.
State of charge	Battery energy currently stored, usually expressed as MWh or percent of usable capacity.
Storage discharge granular certificate	Granular certificate associated with storage discharge, requiring matching storage charge/discharge records and loss treatment under the relevant scheme rules.
Stress test	Scenario analysis of severe but plausible or intentionally challenging market, credit, liquidity, or operational conditions.

Structural market power	Market-power potential arising from market structure, location, ownership, concentration, constraints, or pivotality.
Subscribed capacity	Capacity level contracted for grid use or tariff purposes.
System adequacy	Longer-term ability of the system to have enough resources and network capability to meet demand under credible conditions.
System margin	Difference between available dependable resources and demand or required reserve/adequacy target, depending on definition.
System security	Short-term ability of the power system to remain balanced and withstand disturbances.
System-service option	Capability that may create value through connection approval, compliance, avoided curtailment, paid services, or future grid requirements, but is not yet base-case revenue.
Terminal value	Value assigned to the final state of a storage model, often to avoid artificial emptying at the end of the optimization horizon.
Termination amount	

Amount due on early contract termination, often linked to mark-to-market or alternative valuation methods.

Throughput budget

Allowed or planned cumulative charged/discharged energy over a period or asset life.

Transmission

High-voltage bulk power transport over the main grid.

TSO

Transmission System Operator: entity responsible for secure operation of the transmission system and system balance within its mandate.

Use-of-network charge

Recurring charge for grid use, potentially covering infrastructure, losses, system services, metering, or reactive charges depending on tariff design.

Value at Risk

Loss threshold for a specified probability and horizon; it does not describe the severity of losses beyond the threshold.

Value of lost load

Estimate of consumer willingness to pay to avoid involuntary interruption of electricity supply.

Value pool

Source of possible economic improvement, such as better pricing, reduced imbalance, improved dispatch, or reduced leakage.

Value-duration curve

Curve or table showing the marginal value of adding each extra hour of usable storage duration after efficiency, degradation, grid, tariff, contract, and opportunity-cost constraints.

Value-weighted error

Forecast or model error weighted by the economic consequence of being wrong in that interval or scenario.

Virtual battery

Controllable demand or process flexibility that can absorb, defer, or release electricity demand over time in a way that mimics some storage services without storing electricity electrochemically.

Voltage

Electrical potential difference; grid voltage must remain within operational limits.

WACC

Weighted average cost of capital: combined cost of debt and equity after weighting by capital structure and tax treatment where applicable.

Warranty envelope

Operating conditions under which a battery's warranty, performance guarantee, availability commitment, or remedy structure remains valid.

Weather-dependent

Output materially depends on weather conditions outside operator control.

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